

Photonic Cooling of Microelectronics

Thermodynamics, Electro-Optics, and Material Platforms
of a Thin-Film Anti-Stokes Fluorescence Cold Plate

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compiled from internal design notes and published literature

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Part I

Thermodynamic Foundations

Chapter 1

Thermodynamics of a Self-Recycling Photonic Cooler

1.1 The problem: heat out of a hot spot, and where the entropy goes

A modern high-performance CPU / GPU / ASIC dissipates in excess of a few hundred watts over a die area of $\sim 1 \text{ cm}^2$, with localized hot spots reaching power densities in the range $10\text{--}10^3 \text{ W/mm}^2$ [1]. Conventional convective cooling — air, liquid, and two-phase — is limited by the conduction bottleneck through the lid, interface, and cold plate; control is through only two handles (flow rate, coolant temperature), and the net electrical overhead grows superlinearly with die power density. This motivates a solid-state, spatially addressable refrigeration mechanism.

Anti-Stokes fluorescence (ASF) cooling provides one. A narrow-line pump at frequency ω_p is absorbed by an emitter whose mean fluorescence frequency ω_f is *higher* than ω_p because the upper-manifold population is thermalized by phonons before radiating (Figure 1.3). Each emitted photon carries away on average an energy $\hbar(\omega_f - \omega_p) = k_B T_h \cdot \bar{n}_{ph}$ from the lattice phonon bath. The pump power does work on the electromagnetic field; the fluorescence carries the net heat outward, and everything but the pump enters the optical channel as a radiation flux. The device in this book adds two refinements to this core idea:

1. A thin-film *photonic network* pair — one channel delivering the pump, a second collecting fluorescence — that replaces free-space optics with on-chip waveguiding and resonant enhancement.
2. A *laser power converter* (LPC) on the fluorescence channel that recycles the radiated optical power back to electricity, so the system's input is electrical, not optical, and the loop can in principle generate net power while cooling a sufficiently hot chip.

This chapter establishes the bookkeeping that governs that loop: the first law (energy), the second law (entropy), and the compact efficiency / coefficient-of-performance (COP) expressions to which every later chapter reduces. The treatment follows directly from a sequence of Q&A-style arguments preserved throughout the chapter; readers familiar with the material may skip to the boxed feasibility condition (1.17) and the design map of Figure 1.4.

What an extractor could deliver in principle

Before any of the bookkeeping, it is worth fixing the scale of what the three extractor families could deliver if every photonic lever the later chapters identify were pulled. The volumetric cooling power

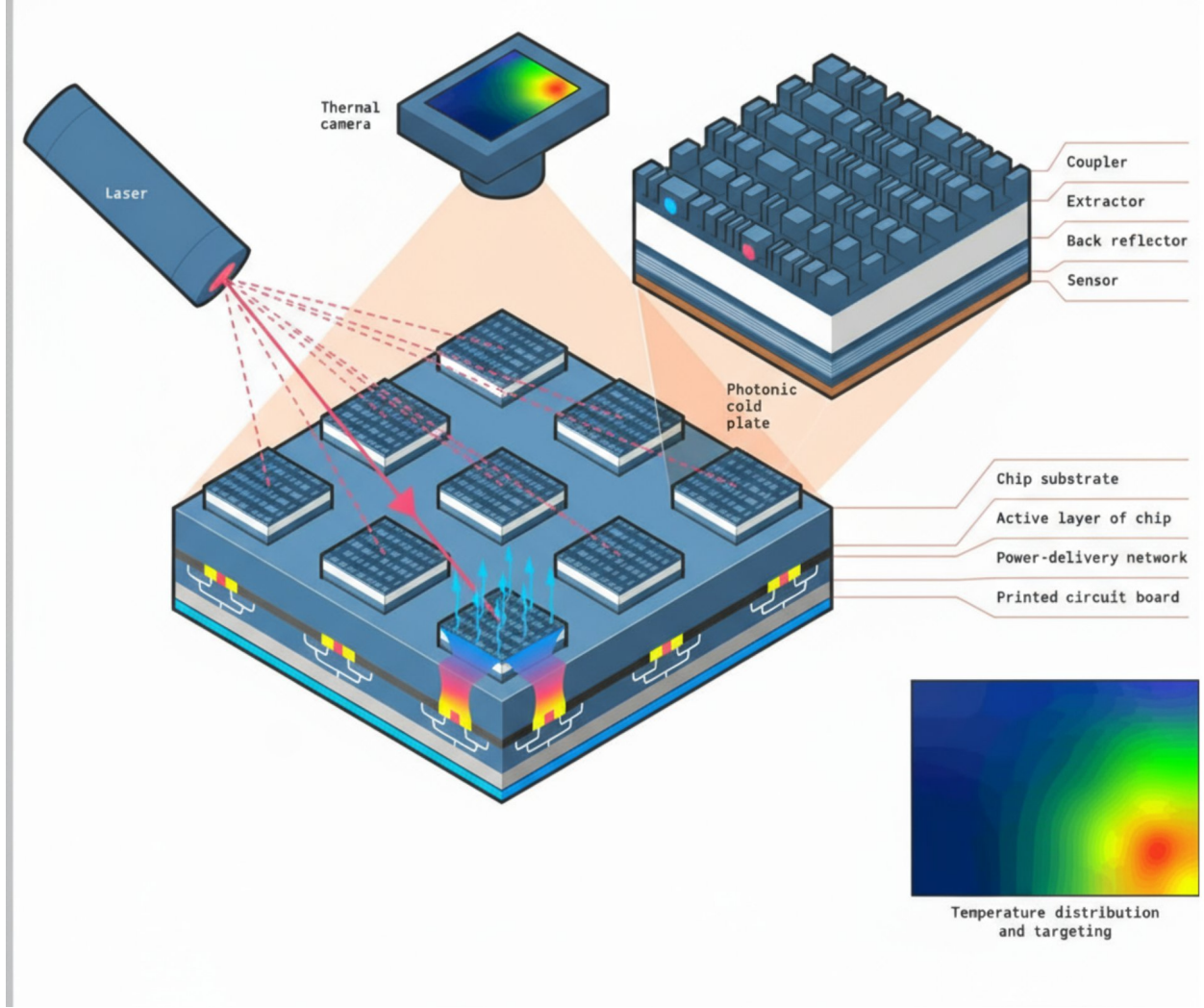
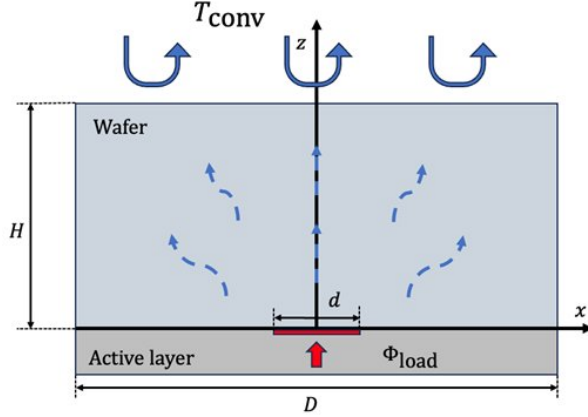


Figure 1.1: Photonic cold plate in operation: a tiled array of micro-scale coolers on a chip, each tile exposing the layered coupler / extractor / back-reflector / sensor stack to a scanning pump laser. A thermal camera closes a spatial control loop onto the hottest tiles, matching cooling power to instantaneous chip power density.

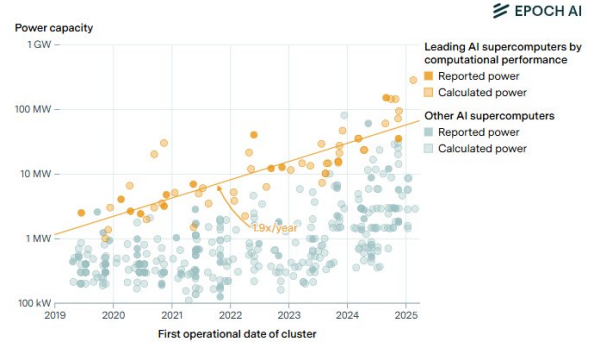
of any anti-Stokes extractor has the form derived in Section 8.3.3,

$$p_{\text{cool}}^{\text{max}} = n_t x_{\text{max}} \Gamma_{\text{tot}} \hbar \omega_p \eta_q, \quad \frac{P_{\text{cool}}}{A} = p_{\text{cool}}^{\text{max}} d, \quad (1.1)$$

where n_t is the emitter density, x_{max} the largest excited fraction the pump can sustain against its own stimulated emission (set by how many $k_B T$ below the emitter's zero line the pump sits), $\Gamma_{\text{tot}} = F_p / \tau$ the Purcell-enhanced cycling rate, and $\eta_q = \bar{\omega}_f / \omega_p - 1$ the quantum defect that bounds the single-stage efficiency. Table 1.1 evaluates it for each family with parameters at the edge of what photonic engineering could plausibly provide; the numbers are ceilings, not predictions, and the chapters that follow are about how far each can be approached. Three features of the table motivate everything downstream: the doped crystals are efficient but slow (millisecond lifetimes make them low-density even at Purcell factors of 10^2); the semiconductors are the densest by two orders of magnitude but at a quantum defect of a few percent, so they circulate 30–100 $\times$ their



(a) Hot-spot model: a power flux Φ_{load} of width d is dissipated at the active layer and must leave the wafer of thickness H against convection at T_{conv} .



(b) Cluster-scale power capacity of leading AI supercomputers, growing at $\sim 1.9\times/\text{year}$ and now approaching the 10^2 MW range. Source: Epoch AI.

Figure 1.2: Why hot-spot density, not total site power, is the binding constraint on next-generation compute. The heat-flux problem (a) is set at the micron scale within a chip; total draw (b) sets the energy-recovery target.

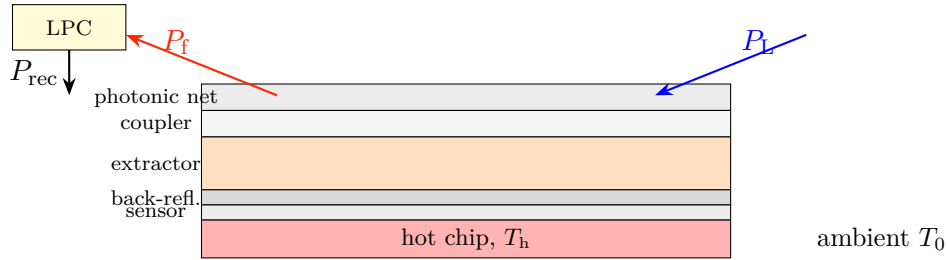


Figure 1.3: Functional schematic of the photonic cold plate: pump laser (P_L) enters through the photonic network, drives ASF in the extractor at hot-spot temperature T_h ; the blue-shifted fluorescence (P_f) is collected by the network and converted to electrical power by the LPC. The back reflector forces a second pump pass and redirects escaping fluorescence; the sensor layer closes a local thermal control loop.

cooling power as pump; and the molecular dispersions sit between the two in density but are the only single-stage family whose efficiency reaches 10%, and the natural family for the cascaded stages that reach 30–60%. Two organic emitters added in Sections 8.3.1 and 8.3.6 sharpen the picture. A near-infrared tricarbo-cyanine pumped at 980 nm carries an anti-Stokes shift of 225 meV — $\eta_q = 18\%$, three to four times the rhodamine figure — so that under the same photonic assumptions it matches the R640 row in density while circulating only $7\times$ its cooling power as pump, at a pump wavelength where diode-laser efficiency is highest and the fluorescence lands on a GaAs LPC; its handicap, a bare IQE of ~ 0.2 , is exactly what a Purcell factor on the radiative rate repairs ($0.2 \rightarrow 0.96$ at $F_P = 100$), and two or three such stages reach the 30–60% tiers where the rhodamine cascade needs six to ten. A superradiant J-aggregate is the opposite case: its 234 ps lifetime and 10^{20} cm^{-3} density give a volumetric ceiling $10\times$ the semiconductor's, but its 6 nm line puts the pump within a few $k_B T$ of the zero line, so $\eta_q \simeq 3\%$ and it circulates $33\times$ — a dense, low-defect emitter like GaAs, useful as the bluest stage of a cascade rather than on its own. Neither changes the tiers; both change which family is asked to reach them.

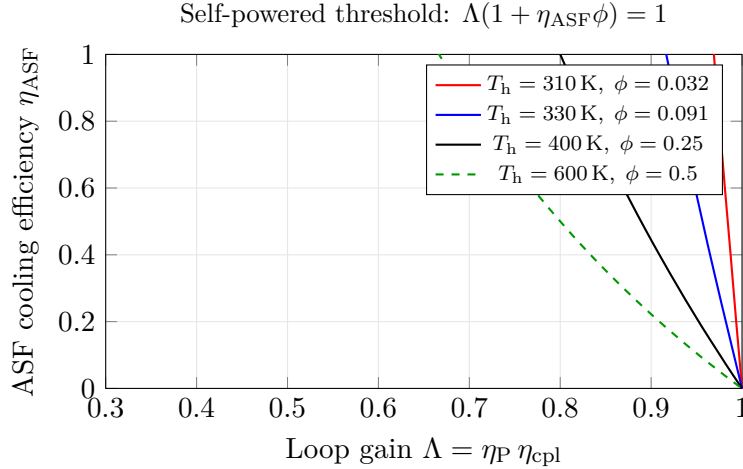


Figure 1.4: Self-powered threshold curves in the loop-gain Λ vs. ASF-efficiency η_{ASF} plane, for four chip temperatures. Below each curve the loop requires external electrical power; above it the loop is self-powered. The Carnot factor $\phi = 1 - T_0/T_h$ sets the steepness: even state-of-the-art rare-earth ASF ($\eta_{\text{ASF}} \sim 0.03$) can cross the threshold only on the $T_h = 600\text{ K}$ curve and only with $\Lambda \gtrsim 0.97$, which is why the device targets isolated chip hot spots rather than die averages.

1.2 The thermodynamic question

The framing question that drives every equation that follows:

Is it possible by the second law of thermodynamics to run a refrigeration cycle in which work is done to cool a part of a system that is generating heat (a chip at a high temperature) while simultaneously using that heat to convert to electrical power, so as to have not only refrigeration but net power generation — where the energy coming out of the chip compensates for the work needed to run the refrigeration cycle and also generates net electrical output?

The system that prompts the question is exactly the one of Figure 1.1: a laser cools the chip via ASF, the emitted fluorescence (which contains both the input laser power that drove the ASF and the up-converted heat from the chip) is sent through an LPC, and the LPC’s electrical output is fed back to the laser to drive many cycles.

The short answer is: *yes, in principle, but only under a specific thermodynamic condition* — the chip must be acting as a genuine hot reservoir relative to some colder entropy sink, so that part of its heat can be converted to work while the rest is removed. In that case, what the device is is not a forbidden “refrigerator that powers itself from a single thermal bath,” but a heat-engine + cooling transducer hybrid. What is *not* allowed is net work extraction while cooling a system if there is no temperature gradient, no external free-energy source, and no entropy sink.

Building toward that statement requires careful bookkeeping of energy (the first law) and of entropy (the second law) around every block in the loop. Sections 1.5–1.6 develop the first-law accounting, sections 1.8–1.10 pin down where the second law actually enters, and sections 1.11–1.15 combine them into the sharp feasibility test.

Platform	In-principle parameters	$x_{\max}$	η_q (single stage)	$p_{\text{cool}}^{\max}$ (W/mm ³)	P_{cool}/A at d (W/mm ²)	Pump per W lifted	η_{ASF} route to $\geq 30\%$
Doped crystal, Yb:YLF	10% Yb (10^{21} cm^{-3}), $\tau = 1.8 \text{ ms}$, $F_P = 100$, pump 1020 nm	0.08	2%	~ 20	0.2 (10 μm); (1 mm)	20 50 $\times$	none (zero Stokes shift already exploited)
Doped crystal, Cr:LiSAF	10^{21} cm^{-3} , $\tau = 67 \mu\text{s}$, $F_P = 30$ –100, pump 900 nm	4×10^{-3}	6%	20–80	0.2–0.8 (10 μm)	17 $\times$	cascade ($\eta_{\text{EQE}} > 0.97$ per stage) (needs graded-gap cascade)
Semiconductor, GaAs DH	$\eta_e = 0.9$, $F_P(B) = 10$, $N \simeq 2 \times 10^{19} \text{ cm}^{-3}$, 300 K	(Auger-limited)	3.5%	10^5 – 10^6	10^3 – 10^4 (10 μm)	30 $\times$	
Molecular dispersion, R640-SMILES	10^{-1} M , 400 K, blue-edge emission, $F_P = 100$, 50 μm	3×10^{-3}	10%	10^5	6×10^3 (50 μm)	10 $\times$	N -stage dye cascade: 47% ($10 \times 5\%$, $\eta_{\text{EQE}} = 0.99$), 67% ($6 \times 10\%$)
Molecular, NIR tricarbo-cyanine	10^{-1} M isolated, 400 K, pump 980 nm 200 meV below zero line, $\text{IQE}_0 = 0.2$, $\tau = 0.6 \text{ ns}$, $F_P = 100$ ($\text{IQE} \rightarrow 0.96$), 50 μm	3×10^{-3}	18% ($\eta_{\text{cool}} = 13\%$)	1.7×10^5	8×10^3 (50 μm)	7 $\times$	2–3 stages: 36% ($2 \times 18\%$), 59% ($3 \times 18\%$, $\eta_{\text{EQE}} = 0.99$)
Organic solid, J-aggregate (silica-TDBC)	10^{20} cm^{-3} , $\tau = 234 \text{ ps}$ (super-radiant), $F_P = 10$, 400 K, pump 60 meV into hot band, 1 μm	0.15; ~ 0.02	3%	10^6 – 6×10^6	10^3 –6 ($1 \mu\text{m}$)	10^3 33 $\times$	terminal (bluest) stage of a cascade; not a single-stage route

Table 1.1: In-principle cooling-power ceilings of the three extractor families from (1.1), evaluated at $\eta_{\text{EQE}} = 1$ with photonic parameters at the edge of plausibility (derivations and the bulk values they start from: Tables 8.2 and 8.4; Cr:LiSAF tail from Section 8.1.2; GaAs from Section 9.3.3). “Pump per W lifted” is $1/\eta_q$, the optical power the network must circulate and the LPC recover for every watt of heat removed; it is the quantity that makes power recovery (Chapter 7) inseparable from cooling. The book’s design point — $\gtrsim 10^3 \text{ W/mm}^2$ at $\eta_{\text{ASF}} \geq 10\%$, rising to 30–60% by cascading — lives in the molecular rows. The last two rows are the organic emitters of Sections 8.3.1 and 8.3.6: for a low-IQE emitter the Purcell factor acts on the radiative rate only, so $\text{IQE}(F_P) = F_P \text{IQE}_0 / [F_P \text{IQE}_0 + 1 - \text{IQE}_0]$ and the same photonics that raises the rate rescues the yield; the J-aggregate row is capped by exciton–exciton annihilation rather than by the pump. Rows evaluated with `calc/ceiling_rows.py`.

1.3 The two relevant temperatures

Two temperatures appear in every expression to follow, and only two qualify as *thermodynamic reservoirs*:

- T_h — temperature of the *hot reservoir*: the chip / cooled body / die region in contact with the extractor layer.
- T_0 — temperature of the *cold reservoir*: the ultimate ambient sink that absorbs all rejected heat and the entropy carried by emitted radiation.

Ordinary ASF refrigeration operates with $T_h \leq T_0$ (you want to drop the sample below ambient). The device under study typically operates in the *opposite* regime: $T_h > T_0$, because the hot spot is already well above the ambient air / water temperature. That sign flip is what allows simultaneous cooling and net power extraction, as we show in Section 1.13. Why no other temperature qualifies (e.g. the laser’s diode-junction temperature, the LPC’s photovoltaic-cell temperature) is the subject of Section 1.8.

1.4 Notation: loop efficiencies at a glance

Before the first-law accounting below, we fix the symbols the rest of the book uses for the loop’s efficiencies and coefficients of performance. Two conventions are in circulation for these quantities (one from the classical ASF literature, one from the Maxwell component-level bookkeeping); the notation adopted here reconciles them and is used consistently from this chapter onward.

Symbol	Name	Definition
<i>Loop-level efficiencies (each dimensionless, in $[0, 1]$ unless noted):</i>		
η_P	pump laser wall-plug efficiency	optical pump power out of the laser / wall electrical power in
η_{cpl}	photonic-network coupling efficiency (aggregate)	fraction of laser power delivered to the extractor plus fluorescence power delivered to the LPC and/or environment; equal to $\eta_{\text{cpl},\text{in}} \cdot \eta_{\text{cpl},\text{out}}$
$\eta_{\text{cpl},\text{in}}$	pump-side coupling	pump-channel routing efficiency: laser $\rightarrow$ network $\rightarrow$ extractor
$\eta_{\text{cpl},\text{out}}$	fluorescence-side coupling	fluorescence-channel routing efficiency: extractor $\rightarrow$ network $\rightarrow$ LPC or sky-facing radiator
η_{ASF}	anti-Stokes fluorescence cooling efficiency	chip heat extracted per unit absorbed pump power, Q_c/P_{abs} ; equals $\eta_{\text{abs}} \eta_{\text{EQE}} (\bar{\omega}_f/\omega_p) - 1$ at the extractor level (the “ -1 ” sits <i>outside</i> the product: a pump photon not re-emitted through the useful channel is pure heat), so $\eta_{\text{ASF}} \leq \bar{\omega}_f/\omega_p - 1 \simeq 0.03\text{--}0.15$ for any practical anti-Stokes shift
η_{LPC}	laser power converter opt $\rightarrow$ elec efficiency	recovered electrical power out of the LPC / fluorescence optical power in
$\eta_{\text{LPC},\text{max}}$	LPC exergy ceiling	second-law upper bound on η_{LPC} from (1.15)
η_c	overall wall-plug cooling efficiency (gross)	Q_c/P_{in} , before LPC recycling; composes as $\eta_c = \eta_P \cdot \eta_{\text{cpl}} \cdot \eta_{\text{ASF}}$
COP_{net}	net coefficient of performance	$Q_c/(P_{\text{in}} - P_{\text{rec}})$, accounting for LPC-recycled electrical return; can exceed unity or go negative in the self-powered / net-generation regime
<i>Extractor sub-efficiencies (compose into η_{ASF}):</i>		
η_{abs}	pump absorption fraction	$\alpha_{\text{dye}}/(\alpha_{\text{dye}} + \alpha_{\text{bg}})$: fraction of absorbed pump power actually taken up by the active emitter rather than parasitic background absorption
η_e	escape efficiency	fraction of radiatively emitted photons that leave the extractor before being reabsorbed
η_{IQE}	internal quantum efficiency (IQE)	radiative branching at the emitter, $w_r/(w_r + w_{\text{nr}})$
η_{EQE}	external quantum efficiency (EQE)	net emit-and-escape probability, $\eta_e \cdot w_r/(\eta_e \cdot w_r + w_{\text{nr}})$; equivalently $\eta_e \cdot \eta_{\text{IQE}}/[1 - (1 - \eta_e)\eta_{\text{IQE}}]$

Table 1.2: Notation for the photonic cooling loop’s efficiencies and coefficient of performance. The loop-level entries define the four independent knobs the design must control; the extractor sub-entries define what a materials or photonic inverse-design run actually manipulates to move η_{ASF} . η_c (gross) is always ≤ 1 ; COP_{net} (with recycling) can exceed unity in the self-powered regime of Section 1.13.

Composition. With this notation the first-law loop balance below decomposes cleanly:

$$P_{\text{abs}} = \eta_{\text{cpl},\text{in}} \eta_P P_{\text{in}}, \quad Q_c = \eta_{\text{ASF}} P_{\text{abs}}, \quad P_{\text{rec}} = \eta_{\text{LPC}} \eta_{\text{cpl},\text{out}} P_f, \quad (1.2)$$

so the gross wall-plug cooling efficiency of Table 1.2 is $\eta_c = Q_c/P_{\text{in}} = \eta_P \eta_{\text{cpl},\text{in}} \eta_{\text{ASF}}$, and the self-powered condition of Section 1.13 is the sharpening of this composition once the recycled term

P_{rec} closes the loop.

Relation to the classical ASF literature. Older papers often use η_c for the per-absorbed-photon anti-Stokes cooling efficiency (what we call η_{ASF} here), η_{ext} for what we call η_{EQE} , and η_p for what we call η_{abs} ; a few use η_L for the pump wall-plug efficiency in place of our η_P . The mapping is straightforward but the symbol clash on η_c is the reason we adopt the convention of Table 1.2 for the entire book. Where citations to older work introduce the alternate symbols, the surrounding text translates them.

1.5 First law: energy accounting around the loop

Define the per-stage power flows:

- P_{in} — external electrical power supplied to the laser.
- η_P — laser wall-plug efficiency.
- $P_L = \eta_P P_{\text{in}}$ — laser optical output power.
- η_{cpl} — optical coupling/absorption efficiency: the fraction of P_L that is actually absorbed by the cooling medium. $P_{\text{abs}} = \eta_{\text{cpl}} P_L$.
- $\eta_{\text{ASF}} \equiv Q_c/P_{\text{abs}}$ — anti-Stokes cooling efficiency, defined as the ratio of extracted chip heat to absorbed pump power. So $Q_c = \eta_{\text{ASF}} P_{\text{abs}} = \eta_{\text{ASF}} \eta_{\text{cpl}} P_L$.
- P_f — emitted anti-Stokes fluorescence power.
- η_{LPC} — LPC optical-to-electrical efficiency.

The earlier literature in this book also uses $\eta_P \equiv \eta_P$, $\eta_{\text{LPC}} \equiv \eta_{\text{LPC}}$, and $\eta_{\text{ASF}} \equiv \eta_{\text{ASF}}$; both notation sets refer to the same quantities and are used interchangeably.

Why Q_c is not a free parameter

A subtle point that deserves explicit airing: if η_{ASF} is treated as a multiplier on Q_c , then both Q_c and η_{ASF} appear as independent dials, which leads to confusion. The cleanest framing is that, once the anti-Stokes physics is described by a *cooling efficiency relative to absorbed pump power*, the heat extracted is fixed by η_{ASF} and P_{abs} :

$$Q_c = \eta_{\text{ASF}} P_{\text{abs}} = \eta_{\text{ASF}} \eta_{\text{cpl}} P_L. \quad (1.3)$$

With $\eta_{\text{cpl}} = 1$ (no optical loss), $P_{\text{abs}} = P_L$ but Q_c does *not* equal P_L — it equals $\eta_{\text{ASF}} P_L$. Zero optical loss alone gives $P_{\text{abs}} = P_L$, not $Q_c = P_L$. Achieving $Q_c = P_L$ requires $\eta_{\text{ASF}} = 1$, which is a much stronger statement about the anti-Stokes physics itself.

Energy balance of the anti-Stokes stage

Energy conservation around the extractor in steady state gives

$$P_f = P_{\text{abs}} + Q_c = \eta_{\text{cpl}} P_L (1 + \eta_{\text{ASF}}), \quad (1.4)$$

because the emitted fluorescence carries both the absorbed pump photons *and* the upconverted thermal energy extracted from the chip. The factor $(1 + \eta_{\text{ASF}})$ is the key scaling: every joule of pump that is absorbed becomes $(1 + \eta_{\text{ASF}})$ joules of fluorescence, with the extra η_{ASF} coming directly from the chip.

For comparison with the conventional ASF literature, the same balance written for parasitic non-radiative heating P_{nr} (impurity absorption, fluorescence escaping outside the engineered mode, etc.) is

$$P_{\text{abs}} + P_{\text{cool}} = P_{\text{f}} + P_{\text{nr}}, \quad (1.5)$$

or equivalently

$$P_{\text{cool}} = P_{\text{f}} - P_{\text{abs}} - P_{\text{nr}}, \quad (1.6)$$

where $P_{\text{cool}} \equiv Q_{\text{c}}$ is the *useful* cooling power. In the idealized single-mode, unity-escape, unity-radiative-efficiency limit ($P_{\text{nr}} \rightarrow 0$) this reduces to the familiar anti-Stokes balance

$$P_{\text{cool}} \simeq P_{\text{abs}} \left(\frac{\omega_{\text{f}} - \omega_{\text{p}}}{\omega_{\text{p}}} \right), \quad (1.7)$$

which identifies η_{ASF} microscopically as $(\omega_{\text{f}} - \omega_{\text{p}})/\omega_{\text{p}}$ in the ideal limit.

Recovered electrical power and net input

The LPC converts the captured fluorescence to electricity at efficiency η_{LPC} :

$$P_{\text{rec}} = \eta_{\text{LPC}} P_{\text{f}} = \eta_{\text{LPC}} \eta_{\text{cpl}} P_{\text{L}} (1 + \eta_{\text{ASF}}). \quad (1.8)$$

The laser draws $P_{\text{in}} = P_{\text{L}}/\eta_{\text{P}}$ from the wall, so the *net external electrical power* required to run the loop is

$$P_{\text{net}} = P_{\text{in}} - P_{\text{rec}} = P_{\text{L}} \left[\frac{1}{\eta_{\text{P}}} - \eta_{\text{LPC}} \eta_{\text{cpl}} (1 + \eta_{\text{ASF}}) \right]. \quad (1.9)$$

Equation (1.9) is the central first-law bookkeeping equation of the chapter. The entire question of whether the device is a refrigerator, a self-powered cooler, or a net generator boils down to the sign of the bracketed quantity.

1.6 Net COP and the self-powered threshold

By definition the net coefficient of performance is the heat extracted divided by the net electrical input:

$$\text{COP}_{\text{net}} \equiv \frac{Q_{\text{c}}}{P_{\text{net}}}. \quad (1.10)$$

Substituting (1.3) and (1.9),

$$\text{COP}_{\text{net}} = \frac{\eta_{\text{ASF}} \eta_{\text{cpl}}}{\frac{1}{\eta_{\text{P}}} - \eta_{\text{LPC}} \eta_{\text{cpl}} (1 + \eta_{\text{ASF}})} = \frac{\eta_{\text{P}} \eta_{\text{ASF}} \eta_{\text{cpl}}}{1 - \eta_{\text{P}} \eta_{\text{LPC}} \eta_{\text{cpl}} (1 + \eta_{\text{ASF}})}. \quad (1.11)$$

Both forms are useful: the second exposes the loop-gain product $\eta_{\text{P}} \eta_{\text{LPC}} \eta_{\text{cpl}} (1 + \eta_{\text{ASF}})$ directly. The denominator is what controls the regime of operation:

Three regimes.

- $\eta_P \eta_{LPC} \eta_{cpl} (1 + \eta_{ASF}) < 1$: $P_{\text{net}} > 0$, and the loop is a refrigerator that consumes external electrical power. This is the regime of all current room-temperature ASF demonstrations.
- $\eta_P \eta_{LPC} \eta_{cpl} (1 + \eta_{ASF}) = 1$: $P_{\text{net}} = 0$, $\text{COP}_{\text{net}} \rightarrow \infty$. The loop is at the *self-powered threshold*: it consumes its own recovered fluorescence to drive the laser and needs no external electrical input.
- $\eta_P \eta_{LPC} \eta_{cpl} (1 + \eta_{ASF}) > 1$: $P_{\text{net}} < 0$, and the loop *generates* net electrical power, exporting $P_{\text{gen}} = -P_{\text{net}}$ on top of cooling the chip.

The threshold condition is therefore

$$\boxed{\eta_P \eta_{LPC} \eta_{cpl} (1 + \eta_{ASF}) = 1} \quad (\text{self-powered threshold; first-law}). \quad (1.12)$$

Why diverging COP is not a paradox. A diverging COP_{net} does *not* mean “infinite cooling from nowhere.” It means $P_{\text{net}} \rightarrow 0^+$ while Q_c stays finite: nearly all the laser drive is being recycled, and the chip’s own dissipated heat is making up the rest. At the threshold the loop is just self-powered; above it the loop behaves as a waste-heat recovery engine, with the chip playing the role of the hot reservoir. No second-law problem appears *provided* (i) the chip is a genuine higher-temperature source than the environment, (ii) the total entropy production of the loop + environment remains nonnegative, and (iii) the net electrical output does not exceed the exergy of the removed heat. Pinning down condition (iii) quantitatively is the work of Section 1.11.

1.7 Why the “low efficiency” of ASF does not kill the loop

A common skeptical objection: if $\eta_{ASF} \sim 2\text{--}5\%$ for Yb:YLF at room temperature, how can the loop possibly approach break-even? The answer is that η_{ASF} is not the end-to-end efficiency of the cooling cycle; it is a *branching ratio* (cooling power per unit absorbed pump) that, by virtue of the anti-Stokes shift, cannot be larger than $(\omega_f - \omega_p)/\omega_p$ when the shift is small relative to the pump photon energy. The loop closes for a different reason: the LPC returns the recycled pump *plus* η_{ASF} times the chip-derived heat, so each percentage point that η_{ASF} gains is compounded by the full $\eta_{LPC} \eta_{cpl}$ when the recovered electricity displaces direct wall draw. In typical numbers ($\eta_P \sim 0.7$, $\eta_{LPC} \sim 0.85$, $\eta_{cpl} \sim 1$), the loop crosses (1.12) once $0.7 \cdot 0.85 \cdot (1 + \eta_{ASF}) \geq 1$, i.e. $\eta_{ASF} \gtrsim 0.68$ — which no *single-stage* anti-Stokes extractor can meet: η_{ASF} is bounded by the quantum defect $\bar{\omega}_f/\omega_p - 1$, i.e. $\sim 0.02\text{--}0.03$ for Yb:YLF and $\lesssim 0.1$ for a red-tail-pumped dye in bulk (Section 8.3.5). Purcell enhancement and photon recycling raise η_{EQE} toward unity and bring η_{ASF} *up to* that ceiling; lifting the ceiling itself requires emission-side photonic selection of the blue edge and, for $\eta_{ASF} \geq 0.3$, cascaded stages (Section 8.3.4). The reading of (1.12) is therefore tiered: at $\eta_{ASF} \simeq 0.1$ the loop is an efficient *recycler* of pump energy that needs $\eta_P \eta_{LPC} \eta_{cpl} \gtrsim 0.9$ to break even; at $\eta_{ASF} \simeq 0.3$ it breaks even at 0.77; at $\eta_{ASF} \simeq 0.6$ — the book’s Tier-III design point — it becomes a net generator at 0.63, within reach of present diode lasers and multi-junction converters. The second-law refinement of Section 1.11 sharpens this estimate by adding the Carnot factor on the heat-derived contribution.

1.8 Which temperatures set the Carnot limit?

A common mistake when reading any Carnot-style bound is to substitute device temperatures — the physical temperature of the laser diode or of the LPC semiconductor — for T_0 or T_h . The bound

depends on *reservoir* temperatures, not on any particular subsystem that happens to sit at a given temperature.

What is, and is not, a thermodynamic reservoir. A thermodynamic reservoir is a system large enough that it absorbs energy and entropy without changing temperature. In the loop only two objects qualify: the chip hot spot (at T_h) and the external environment (at T_0). Neither the laser nor the LPC qualifies:

Object	Role in the loop	Sets Carnot limit?
Chip	hot reservoir	✓
Ambient environment	cold (entropy) reservoir	✓
Laser	work source / coherent field	✗
LPC	energy converter	✗

A diode laser at 300 K with a junction at 320 K and an LPC at 310 K all have physical temperatures, but those values appear in the Carnot expression only if the corresponding component *exchanges thermal radiation with the loop* as a true heat bath — which by construction these components do not. They are *energy converters* driven by work (electricity, coherent photons), not heat baths absorbing and emitting blackbody radiation.

The laser is not a heat source — even if its hardware is warm. Even if the laser hardware sits at 300 K, the optical field it produces is *not* a 300-K blackbody field; it is a low-entropy, high-exergy resource. That is precisely why a laser can drive cooling *below* ambient (no heat bath would let you do that): laser light behaves as a work input, not a heat input.

When device temperatures *would* matter. There are cases where component temperatures matter — a blackbody radiator exchanging incoherent radiation, a thermoelectric junction between two temperatures, or a photovoltaic illuminated by thermal radiation. In that special case the radiation source temperature would replace T_h in the bound. But the loop under study uses a coherent laser input, so the only thermodynamic temperatures are the chip’s and the environment’s.

The correct Carnot statement. Combining the exergy of heat $\dot{Q}$ drawn from a body at T_h into a sink at T_0 with the Kelvin-Planck bound,

$$P_{\text{net}}^{\text{gen}} \leq Q_c \left(1 - \frac{T_0}{T_h} \right). \quad (1.13)$$

Everything else (laser, LPC, optics, photonic network) determines *how close* you can get to this bound, not the bound itself. Equation (1.13) is what Section 1.13 and the refined exergy analysis below Section 1.11 must comply with.

1.9 Where is the cold reservoir?

A natural follow-up question: in a fully packaged device, where *physically* is the cold reservoir? Is it just the “inefficiency heat” — the leftover heat dumped by the laser, the LPC, and any non-radiative channels in the extractor? The answer needs a bit of care because this is where second-law accounting is easiest to mis-place.

The reservoir is the ultimate entropy sink. The cold reservoir is the external environment that absorbs (i) all unrecovered energy, and (ii) all entropy exported via radiation and heat. Physically that means the surrounding air, substrate, building, and ultimately Earth, all approximately at T_0 . Even if energy leaves the device as photons, once those photons hit walls, get absorbed, and thermalize, they end up as heat at T_0 . The optical field is just a transport mechanism; the absorber that thermalizes the radiation is the actual cold reservoir.

Inefficiency heat feeds the reservoir, but does not define it. Laser, LPC, and non-radiative-decay losses all dump additional heat into the same environment, increasing entropy production. But even in an *ideal* loop with no inefficiency, you still need a cold reservoir, because extracting work from heat *requires* exporting entropy. Entropy has to go somewhere even in the reversible limit; that “somewhere” is the cold reservoir, by definition. So the cold reservoir is not defined by inefficiency — it is defined by where entropy ends up, even when nothing is “lost.”

Entropy flow in the loop. Tracking the entropy explicitly: the laser provides low-entropy energy (work-like input). The chip heats the extractor at T_h . The anti-Stokes process converts absorbed laser photons + thermal phonons into higher-energy fluorescence photons — but the emitted photons have higher entropy *per unit energy* than the laser photons (broader spectrum, larger angular spread, spontaneous emission). This is where the entropy from the chip is carried away. The fluorescence then propagates into free space (or into the LPC); whatever the LPC does not convert to work, plus the entropy of imperfect conversion in laser and LPC, is dumped as heat into the environment at T_0 .

One-line takeaway. The cold reservoir is the ultimate entropy sink, which in practice is the environment at T_0 — *not* merely the inefficiency sinks, although in a packaged device those sinks all couple to the same environment and the distinction does not affect the bound (1.13).

Aside: T_0 is set by where entropy actually thermalizes. The preceding paragraphs collapse the entire environment onto a single T_0 , which is a fine working assumption in any deployment where everything that leaves the device gets absorbed by the local chassis, air, or coolant loop at room temperature. Worth noting in passing, though, that the ultimate sink temperature for any optical energy that exits the photonic network upward is set by whatever finally absorbs that radiation — and the clear sky at the pump / fluorescence wavelengths is, in principle, a much colder reservoir than the chassis: a clear-night sky in the 8–13 μm atmospheric window has a brightness temperature near 200 K, and in transparent visible / near-IR bands the effective sink at night is essentially the cosmic microwave background [2, 3]. This is a thermodynamic observation, not a near-term design recommendation: the back-reflector spec, the photonic-network routing matrix, the LPC’s role as the system’s primary work recovery path, and the realities of view factor, solar loading, and packaging all argue for $T_0 \sim T_0^{\text{conv}}$ in the loop bookkeeping for any realistic indoor deployment. The point survives as an interesting limit case — in a space asset or in an outdoor installation with an explicit sky-facing output, the effective sink seen by the small fraction of unconverted radiation that the network chooses to route outward could in principle be *much* colder than ambient. We do not exploit this anywhere in the design analysis that follows.

1.10 Closing the optical loop: where does entropy still have to go?

A sharper version of the previous question: if $\eta_{\text{cpl}} = 1$ and 100% of the emitted fluorescence ends up in the LPC and is recycled into the laser drive, no radiation escapes the device — so what plays the

role of the cold reservoir? Are the only thermal losses those inside the laser and LPC?

The answer is yes, but the second law still bites: even if you capture *all the fluorescence energy*, you cannot capture *all the entropy* in a way that lets you fully recycle it as work. Internal entropy generation in the converters is what couples the closed optical loop to the cold reservoir.

Why perfect optical recycling is not enough. Compare the nature of the radiation streams:

- **Laser light:** coherent, narrow spectrum, low entropy per unit energy.
- **Anti-Stokes fluorescence:** spontaneous emission, broader spectrum, many modes (angles, frequencies, polarizations); higher entropy per unit energy than the pump.

The anti-Stokes process is precisely (low-entropy photons)+(thermal energy) $\rightarrow$ (higher-entropy photons), which is how it removes heat from the chip in the first place. The LPC must then attempt the inverse: (higher-entropy radiation) $\rightarrow$ (low-entropy electrical work) — fundamentally a heat-engine problem. The second law requires that any such conversion reject some heat to dispose of entropy, regardless of how perfect the optical capture is.

Where the cold reservoir physically lives in the closed loop. Inside the laser and LPC, but thermally connected to the environment. The relevant entropy-producing processes are:

- carrier thermalization in the LPC (each absorbed photon’s excess energy above the LPC bandgap thermalizes to the lattice as heat);
- radiative recombination at finite voltage in the LPC and laser (each emitted/absorbed photon carries entropy that does not balance the work flow);
- resistive losses in the laser drive electronics;
- non-radiative recombination anywhere in the extractor that is *not* captured by the LPC.

All of that heat is conducted out to the environment at T_0 ; the total entropy export is what makes the second law balance.

What the closed-loop picture replaces. In the open-loop picture, “escaped fluorescence” was a candidate entropy carrier. In the closed-loop picture, the same role is played by the *internal* irreversibilities of the LPC and laser — chiefly the mismatch between the photon distribution coming in and the electrical degrees of freedom going out. Even an ideal photovoltaic cannot convert all photon energy into work (a generalized Shockley-Queisser-style argument); it must reject heat, which is the cold-reservoir coupling.

One-line takeaway. Even with perfect optical recycling, the loop cannot be thermodynamically closed: converting high-entropy fluorescence back into low-entropy electrical work necessarily generates heat that must be rejected to an external cold reservoir. This is what Section 1.11 now quantifies.

1.11 Refined exergy bound: pump-photons vs heat-derived photons

The compact COP expression (1.11) is exact first-law bookkeeping but treats the LPC efficiency η_{LPC} as a single number. The second law puts an additional ceiling on η_{LPC} that distinguishes *which* parts of the fluorescence are fundamentally recoverable as work and which are not. This is

the move from a necessary energy-balance condition to a necessary *and sufficient* thermodynamic feasibility condition.

The two parts of the fluorescence

Energy conservation around the extractor gives $P_f = P_{\text{abs}} + Q_c$, and each part has a different exergy character:

- The pump-derived part P_{abs} is work-like (the pump beam was coherent), so its exergy equals its energy: $B_{\text{in}}^{\text{pump}} = P_{\text{abs}}$.
- The heat-derived part Q_c is Carnot-limited because it was lifted from the chip at T_h , so $B_{\text{in}}^{\text{heat}} \leq Q_c (1 - T_0/T_h)$.

Therefore the maximum exergy carried by the fluorescence is

$$B_f \leq P_{\text{abs}} + Q_c \left(1 - \frac{T_0}{T_h}\right) = \eta_{\text{cpl}} P_L \left[1 + \eta_{\text{ASF}} \left(1 - \frac{T_0}{T_h}\right)\right]. \quad (1.14)$$

The full fluorescence energy itself is $P_f = \eta_{\text{cpl}} P_L (1 + \eta_{\text{ASF}})$, so the exergy fraction $B_f/P_f < 1$ unless $T_h \rightarrow \infty$ or $\eta_{\text{ASF}} \rightarrow 0$.

Second-law bound on the LPC

Since electricity is pure work, the recovered electrical power cannot exceed the fluorescence exergy: $P_{\text{rec}} \leq B_f$, i.e. $\eta_{\text{LPC}} P_f \leq B_f$. Substituting both expressions and cancelling $\eta_{\text{cpl}} P_L$,

$$\eta_{\text{LPC}} \leq \eta_{\text{LPC,max}} = \frac{1 + \eta_{\text{ASF}} \left(1 - \frac{T_0}{T_h}\right)}{1 + \eta_{\text{ASF}}} = 1 - \frac{\eta_{\text{ASF}}}{1 + \eta_{\text{ASF}}} \frac{T_0}{T_h}. \quad (1.15)$$

This is the cleanest general thermodynamic upper bound on the LPC in this notation. It says the LPC cannot convert all fluorescence energy to electricity, because part of the fluorescence energy came from the chip's heat and only the Carnot fraction of that part is convertible to work. The pump-derived part is, in principle, fully recoverable.

Carnot-like factorization of the LPC

It is sometimes useful to separate the thermodynamic ceiling from the internal irreversibility of the LPC:

$$\eta_{\text{LPC}} = \eta_{\text{P,int}} \eta_{\text{LPC,max}}, \quad 0 \leq \eta_{\text{P,int}} \leq 1, \quad (1.16)$$

where $\eta_{\text{LPC,max}}$ is the exergy ceiling (1.15) and $\eta_{\text{P,int}}$ is the practical irreversibility factor. The laser does *not* have an analogous Carnot factor: it is a work-to-radiation converter driven by electricity, so its reversible upper limit is just unity. Real-device $\eta_{\text{P}} < 1$ reflects practical losses (electrical-to-optical conversion inefficiency, heat dissipation in the diode), not a Carnot ceiling.

Sharp feasibility condition

Self-powering requires $P_{\text{rec}} \geq P_{\text{in}}$, i.e. $\eta_{\text{LPC}} \eta_{\text{cpl}} (1 + \eta_{\text{ASF}}) \geq 1/\eta_{\text{P}}$. Imposing (1.15) eliminates the LPC degree of freedom in favour of the chip-temperature-dependent ceiling: substitute $\eta_{\text{LPC}} = \eta_{\text{LPC,max}}$ into the loop condition. The factor $(1 + \eta_{\text{ASF}})$ cancels and one obtains

$$\eta_{\text{P}} \eta_{\text{cpl}} \left[1 + \eta_{\text{ASF}} \left(1 - \frac{T_0}{T_h}\right)\right] \geq 1. \quad (1.17)$$

This is the most useful feasibility test of the chapter. Rearranging to expose the required cooling efficiency,

$$\eta_{\text{ASF}} \geq \frac{1/(\eta_{\text{P}} \eta_{\text{cpl}}) - 1}{1 - T_0/T_{\text{h}}}. \quad (1.18)$$

Two qualitative reads of (1.18) carry the device-economics story:

- **Small $T_{\text{h}} - T_0$ is hostile.** When the chip is only slightly above ambient, $(1 - T_0/T_{\text{h}})$ is small, so even modest gaps in $1/(\eta_{\text{P}} \eta_{\text{cpl}})$ demand unrealistically high η_{ASF} . Self-powering of nearly-ambient chips is thermodynamically very hard.
- **Large $T_{\text{h}} - T_0$ is generous.** When the chip runs hot, the Carnot factor opens up and the required η_{ASF} drops by orders of magnitude. This is why the device's natural target is isolated chip hot spots, not die averages.

Maximum net-generated power

The corresponding upper bound on net electrical output is

$$P_{\text{gen}}^{\text{max}} = P_{\text{L}} \left(\eta_{\text{cpl}} \left[1 + \eta_{\text{ASF}} \left(1 - \frac{T_0}{T_{\text{h}}} \right) \right] - \frac{1}{\eta_{\text{P}}} \right). \quad (1.19)$$

When the right-hand side is negative, no LPC — not even an ideal one — can close the loop into net generation.

1.12 Numerical examples

Take $T_0 = 300 \text{ K}$ throughout.

Example 1 — modest chip overheating, $T_{\text{h}} = 330 \text{ K}$. $1 - T_0/T_{\text{h}} \approx 0.0909$, $\eta_{\text{P}} = 0.7$, $\eta_{\text{cpl}} = 1$. The self-powering condition (1.17) demands $0.7[1 + 0.0909 \eta_{\text{ASF}}] \geq 1$, giving $\eta_{\text{ASF}} \geq 4.71$ — impossible ($\eta_{\text{ASF}} \leq \bar{\omega}_{\text{f}}/\omega_{\text{p}} - 1 \lesssim 0.15$ for any practical anti-Stokes shift; the value $\eta_{\text{ASF}} = 1$ used below is a deliberately generous bound that makes the argument *a fortiori*). The LPC exergy ceiling (1.15) for $\eta_{\text{ASF}} = 1$ is $\eta_{\text{LPC,max}} \approx 0.545$; even an ideal LPC could recover at most $\sim 54.5\%$ of the fluorescence energy. At a 30-K chip-over-ambient difference, the thermal exergy is too small for the loop to be self-powered.

Example 2 — hotter chip, $T_{\text{h}} = 600 \text{ K}$. $1 - T_0/T_{\text{h}} = 0.5$, $\eta_{\text{P}} = 0.7$, $\eta_{\text{cpl}} = 1$. Self-powering requires $0.7[1 + 0.5 \eta_{\text{ASF}}] \geq 1$, i.e. $\eta_{\text{ASF}} \geq 0.857$ — now plausible. The LPC ceiling at $\eta_{\text{ASF}} = 1$ is $\eta_{\text{LPC,max}} = (1 + 0.5)/2 = 0.75$; an ideal LPC could in principle convert up to 75% of the fluorescence energy to electricity. Once the chip is much hotter than ambient, the loop has enough thermal exergy to approach self-powered operation.

Example 3 — realistic coupling losses. Same as Example 2 but with $\eta_{\text{cpl}} = 0.8$ (20% optical loss in the photonic network or the back-reflector). Now self-powering requires $0.7 \cdot 0.8 \cdot [1 + 0.5 \eta_{\text{ASF}}] \geq 1$, i.e. $\eta_{\text{ASF}} \geq 1.57$ — infeasible. The loop is very sensitive to η_{cpl} : a single 20% front-end loss roughly doubles the cooling-efficiency target.

Example 4 — reversible best case. $\eta_P = 0.8$, $\eta_{\text{cpl}} = 1$, $\eta_{\text{ASF}} = 1$, $T_h = 600\text{ K}$, $T_0 = 300\text{ K}$. Then $P_{\text{gen}}^{\text{max}}/P_L = [1 + 1 \cdot 0.5] - 1/0.8 = 1.5 - 1.25 = 0.25$: the loop could in principle export up to 25% of the pump optical power as net electricity. Real-world irreversibilities in laser, LPC, and the anti-Stokes stage all lower this.

Headline. At low chip overtemperature the Carnot factor $(1 - T_0/T_h)$ is small, so the heat extracted from the chip carries very little exergy, and the loop mostly just recycles pump energy — unable to become self-powered unless η_{ASF} is unrealistically high. At high chip temperature the chip heat carries enough exergy that the loop can in principle support self-powering and net generation. In equation form: higher η_P and η_{cpl} help linearly, higher η_{ASF} helps too but is weighted by the Carnot factor $(1 - T_0/T_h)$ in (1.17). The thermal contribution is discounted by $(1 - T_0/T_h)$, and *that is the single binding constraint of the entire device.*

1.13 Carnot-like bound and the power-generation regime

The divergence of COP_{net} raises an immediate second-law flag. The resolution is that power generation is only permitted when the chip acts as a true hot reservoir: $T_h > T_0$. Combining the exergy of heat (1.20) with the Kelvin–Planck bound, the thermodynamic limit on the net electrical output is just (1.13), which is the original form of the constraint that (1.17)–(1.19) sharpened by including the LPC bookkeeping. The ASF loop is simultaneously a refrigerator (pulling Q_c out of the chip) and a heat engine (between the chip and the ambient sink) — not a perpetual-motion machine extracting work from a single bath. The bound (1.13), together with (1.11) and (1.17), defines the design envelope of every later chapter.

Two cases. **Case A:** chip hotter than ambient, $T_h > T_0$. Heat already “wants” to flow out, and part of it can be converted to work while cooling the chip. This is the regime that (1.17) describes; it is allowed and is what makes the device interesting. **Case B:** chip cooled below ambient, or no temperature gradient. Then the device is doing pure refrigeration against the natural heat flow, and net power generation is impossible without additional free-energy input (external work, chemical free energy, another hotter reservoir, or a low-entropy radiation source not already accounted for). Otherwise the cycle would amount to extracting work while refrigerating from a single bath, which is forbidden.

Exergy of the extracted heat

For completeness, the exergy of heat $\dot{Q}$ drawn from a body at T_h into a reference environment at T_0 is

$$\mathcal{E}_{\dot{Q}} = \dot{Q} \left(1 - \frac{T_0}{T_h} \right), \quad (1.20)$$

which is positive when $T_h > T_0$ (Case A) and negative when $T_h < T_0$ (Case B, where work has to be *added* to extract heat). The sign is precisely what separates the two operating regimes and is what Equation (1.17) encodes.

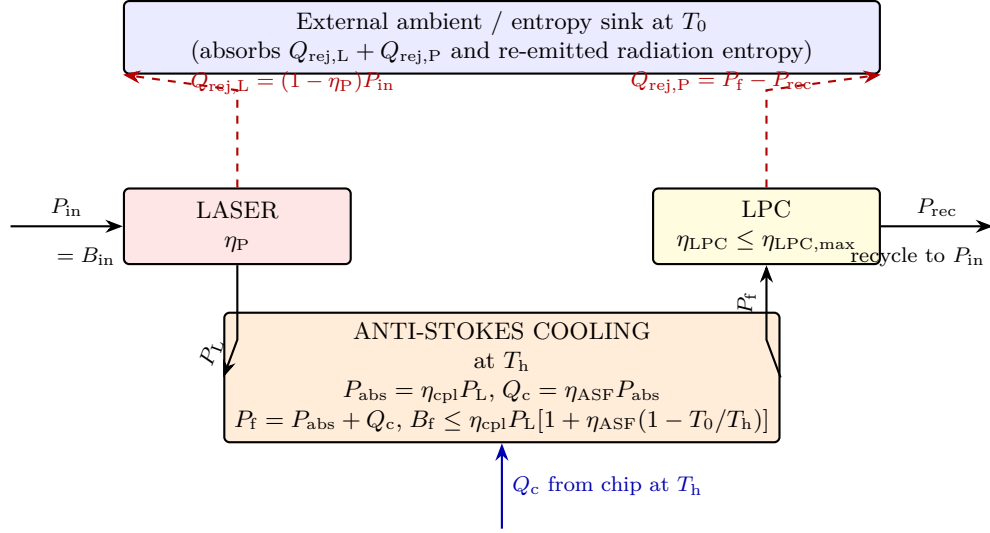


Figure 1.5: Energy and exergy flow of the self-powered thermophotonic cycle. Solid arrows: useful (work-like or recoverable-by-LPC) flows; dashed red arrows: irreversible heat rejection to ambient. The LPC’s recoverable electricity is bounded above by the fluorescence exergy ceiling (1.14), which separates the pump-derived (fully recoverable) part from the heat-derived (Carnot-bounded) part. Self-powering corresponds to $P_{\text{rec}} \geq P_{\text{in}}$, or equivalently to (1.17).

Exergy of fluorescence in the many-mode limit

Spontaneous, incoherent, isotropic fluorescence at lattice temperature T_h carries entropy tied to that temperature. In the naive many-mode limit its exergy satisfies the bound

$$\mathcal{E}_f \lesssim P_f \left(1 - \frac{T_0}{T_h} \right), \quad (1.21)$$

i.e. a Carnot-like fluorescence bound. But *bandwidth alone is not the control parameter*: the discriminator is entropy per unit energy, equivalently phase-space density / étendue / brightness. When the emission is engineered into a small number of low-entropy modes (narrow angular cone, narrow spectrum, coherent), $\mathcal{E}_f$ approaches P_f and the Carnot factor in (1.21) relaxes. The quantitative statement is derived in Chapter 6; the refined bound (1.15) of this chapter is what survives at the loop level once that single-mode engineering is done.

1.14 Exergy flow of the full loop

Figure 1.5 draws the energy and exergy budget of the loop in one place. Electrical input P_{in} (all exergy) enters the laser, which delivers $P_L = \eta_P P_{\text{in}}$ of optical power and rejects $Q_{\text{rej,L}} = (1 - \eta_P) P_{\text{in}}$ to ambient. The pump crosses the photonic network with coupling/absorption efficiency η_{cpl} , reaches the extractor, picks up the chip heat Q_c at T_h , and exits as fluorescence at total energy $P_f = \eta_{\text{cpl}} P_L (1 + \eta_{\text{ASF}})$ with exergy bound $B_f \leq \eta_{\text{cpl}} P_L [1 + \eta_{\text{ASF}} (1 - T_0/T_h)]$. The LPC converts the fluorescence to recovered electricity $P_{\text{rec}} = \eta_{\text{LPC}} P_f \leq B_f$; the difference $Q_{\text{rej,P}} = P_f - P_{\text{rec}}$ is dumped to ambient. The net electrical input to the wall is $P_{\text{net}} = P_{\text{in}} - P_{\text{rec}}$, and the cycle is self-powered when $P_{\text{net}} \leq 0$.

1.15 Design map: when is the loop feasible?

Collapse the feasibility condition (1.17) into two parameters: a *loop-gain* factor $\Lambda \equiv \eta_P \eta_{\text{cpl}}$ (everything that does not depend on chip temperature) and a Carnot factor $\phi \equiv 1 - T_0/T_h$. The threshold is

$$\Lambda(1 + \eta_{\text{ASF}} \phi) = 1 \quad \Leftrightarrow \quad \eta_{\text{ASF}} = \frac{1/\Lambda - 1}{\phi} \equiv \eta_{\text{ASF}}^{\min}. \quad (1.22)$$

Figure 1.4 plots the threshold curves for several chip temperatures. Reading the map:

- **Below the curve:** loop requires net external electrical input. This is the regime of all room-temperature ASF cooling demonstrations to date.
- **On the curve:** self-powered threshold, $P_{\text{net}} = 0$.
- **Above the curve:** self-powered operation with margin; in the reversible limit the surplus is exportable as net electrical generation.

The three curves correspond to T_h ranging from 310 K (just-above-ambient, $\phi = 0.032$) through 400 K (data-centre hot-spot regime, $\phi = 0.25$) to 600 K ($\phi = 0.5$). The horizontal compression of the threshold under increasing T_h is the entire design margin available to the device: hotter chips make a self-powered loop progressively easier.

A few quick reads from the plot:

- If $\phi = 0.1$ (chip ~ 30 K above ambient), even $\Lambda = 0.8$ needs $\eta_{\text{ASF}} \approx (1/0.8 - 1)/0.1 = 2.5$ — impossible.
- If $\phi = 0.5$ (chip at ~ 600 K), the same $\Lambda = 0.8$ only needs $\eta_{\text{ASF}} \approx 0.5$ — a target Purcell-enhanced ASF can plausibly meet.

T_h is a design knob, not a fixed chip property. A reading of Figure 1.4 that is easy to miss on a first pass, but which the rest of the book relies on: the four T_h curves are not four different chip families with four different intrinsic operating temperatures. They are four points along a *single* design axis that the architect can traverse deliberately. Today’s chips inherit $T_h \sim 330$ K not from device physics but from the reliability envelope of Cu / low- κ interconnects operated under convective cooling; a convection-limited package cannot deliver spatially selective heat removal, so the whole die must be held at whatever the worst-case reliability spec allows, and that spec is set by metallization and dielectric aging near room temperature. Photonic microrefrigeration breaks this coupling. Because it extracts heat locally at each hot-spot tile (Section 2.7), it does not require the surrounding regions to be at the same temperature; a compute region can be run at $T_h = 500$ – 600 K — inside the operating window of SiC / GaN devices and refractory-metal interconnects but well outside that of standard Cu / silicon — while the SRAM, control, and I/O regions sit at 200–300 K where subthreshold leakage collapses (Section 10.2). Reading Figure 1.4 through that lens: the $T_h = 600$ K curve is not describing an exotic chip; it is describing an architectural choice to *localize* the compute region into a hot tile that the photonic cold plate can service, in return for a Carnot factor ϕ that is ~ 3 – $4\times$ wider than the one an all-silicon convection-cooled chip can access. This is what the loop-gain condition rewards, and it is one of the primary architectural consequences of the whole photonic-cooling stack; the concrete design template is developed in Section 10.9.

1.16 What to call this cycle

The device does not fit the textbook taxonomy. A standard refrigerator runs on work and moves heat ($W \rightarrow \dot{Q}$); a standard heat engine takes heat and delivers work ($\dot{Q} \rightarrow W$); this device does

$\dot{Q} \rightarrow (W + \text{more effective heat removal})$ — both at once. The closest relatives are absorption refrigeration (heat-driven rather than work-driven) and thermophotovoltaics (photon-mediated heat-to-work), but neither captures the dual role. Defensible names range from conservative (*hybrid heat-engine-refrigeration cycle*, *self-powered optical refrigerator*, *refrigeration cycle with internal work recovery*) to optics-oriented (*thermophotonic cooling and power-generation system*, *anti-Stokes fluorescence heat engine*, *optically mediated heat-recovery refrigeration cycle*). The most compact publishable phrase — and the one we adopt throughout the rest of the book — is

a self-powered thermophotonic refrigeration cycle with internal heat-to-work conversion,

or telegraphically, a *thermophotonic heat-engine-refrigerator hybrid*. What the device is *not* is equally worth spelling out: it is not a perpetual-motion machine (the chip is a real high-temperature reservoir and the ambient is a real cold one), not a single-reservoir heat engine (those are forbidden by Kelvin–Planck), and not “free energy” — it harvests waste heat, it does not create it.

1.17 Caveats and dominant failure modes

Three caveats bound the validity of (1.17) and (1.11) and deserve explicit mention before we proceed into device-level chapters.

You cannot arbitrarily “purify” thermal fluorescence into work. Sections 1.9–1.10 and Chapter 6 show that photonic engineering can lower the entropy of the extracted fluorescence stream and therefore raise its exergy fraction — but the reduction is bounded. A passive structure that merely filters light reduces total extracted power without changing per-photon exergy; a structure that recycles and re-emits pays phonon entropy into the bath at T_h (the bath we are cooling) each cycle. The second law is saved at the full-system level: any entropy reduction of the extracted optical stream is compensated elsewhere in the energy balance.

$\eta_P, \eta_{\text{cpl}}, \eta_{\text{ASF}}, \eta_{\text{LPC}}$ **are not all independent *thermodynamic* bounds.** The first-law master equation (1.11) is exact bookkeeping, but the values that enter it are not all freely chooseable: η_{LPC} is constrained by (1.15), and η_{ASF} is constrained both microscopically (the spectral / mode entropy of the emission) and macroscopically (by exergy of the chip heat). The chapter-level derivations in Chapters 4 to 6 produce these engineering numbers; the thermodynamic master equation tells you what to do with them once they are in hand.

Heat-exchange at the boundary is not free. The Q_c in (1.10) presumes that heat can be extracted from the hot spot at T_h without a conduction bottleneck. In an actual chip the thermal boundary between the extractor and silicon (through the sensor layer, adhesion layers, and micro-bumps) drops the effective extractor temperature below the hot-spot temperature, so the effective Carnot factor available to the cooling cycle is smaller than the naive $(1 - T_0/T_h)$. Mitigation is a thin-film packaging problem (Section 1.20 and Section 5.15) and is what makes the extractor’s physical thickness $\sim 10\ \mu\text{m}$ — thick enough to absorb the pump, thin enough that the thermal-resistance penalty is bounded by a few kelvin.

1.18 Thermally-limited architecture design points

The feasibility test (1.17) bounds whether the photonic loop can self-power; it does not yet say what *architectural* consequence that has for the chip the loop is attached to. Before the HPC

application of Section 1.19, this section makes the bridge: the chip’s electrical power equation, its frequency-scaling regime, and the thermal wall together define a small set of architectural design points that classical cooling can reach and that photonic cooling extends.

The fundamental chip power equation

The total electrical power dissipated by a CMOS chip decomposes into a dynamic (switching) part, a short-circuit part, and a static (leakage / standby) part:

$$\boxed{P_{\text{chip}} = P_{\text{dynamic}} + P_{\text{short}} + P_{\text{static}}.} \quad (1.23)$$

The standard expansions are

$$P_{\text{dynamic}} = \alpha N_{\text{tr}} C_{\text{eff}} V_{\text{dd}}^2 f, \quad (1.24)$$

$$P_{\text{short}} = \alpha N_{\text{tr}} V_{\text{dd}} I_{\text{sc}} t_{\text{sc}} f, \quad (1.25)$$

$$P_{\text{static}} = N_{\text{tr}} V_{\text{dd}} I_{\text{leak}}(V_{\text{dd}}, V_{\text{th}}, T_{\text{j}}), \quad (1.26)$$

where α is the activity factor (fraction of transistors switching per cycle), N_{tr} the transistor count, $C_{\text{eff}} = C_{\text{gate}} + C_{\text{junction}} + C_{\text{wire}}$ the effective per-transistor switching capacitance (gate + drain-junction + interconnect), V_{dd} the supply voltage, f the clock frequency, I_{sc} the short-circuit current that briefly flows during a CMOS rise / fall transition (duration t_{sc}), V_{th} the threshold voltage, and I_{leak} the per-transistor leakage current (subthreshold + gate-oxide + reverse-biased junction). In modern processes P_{short} is typically small ($\lesssim 5\%$) and is folded into P_{dynamic} hereafter except where the distinction matters.

Drive current and the alpha-power MOSFET model

The clock frequency f is set by gate-delay considerations. In the alpha-power MOSFET model

$$I_{\text{drive}} \propto \frac{W}{L} (V_{\text{dd}} - V_{\text{th}})^{\alpha_{\text{MOS}}}, \quad 1 \leq \alpha_{\text{MOS}} \leq 2, \quad (1.27)$$

where $\alpha_{\text{MOS}} \rightarrow 2$ recovers Shockley’s square-law (long-channel limit) and $\alpha_{\text{MOS}} \rightarrow 1$ the velocity-saturation regime of short-channel devices. Modern ≤ 7 nm FinFETs sit at $\alpha_{\text{MOS}} \simeq 1.1$ –1.3. Gate delay scales as

$$t_{\text{d}} \propto \frac{C_{\text{eff}} V_{\text{dd}}}{I_{\text{drive}}} \propto \frac{C_{\text{eff}} V_{\text{dd}}}{(V_{\text{dd}} - V_{\text{th}})^{\alpha_{\text{MOS}}}}, \quad f \sim \frac{1}{t_{\text{d}}}. \quad (1.28)$$

Two limits matter. Far above threshold ($V_{\text{dd}} \gg V_{\text{th}}$), $t_{\text{d}} \propto V_{\text{dd}}^{1-\alpha_{\text{MOS}}}$; for $\alpha_{\text{MOS}} = 2$ this gives $f \propto V_{\text{dd}}$ and, combined with (1.24), the canonical $P_{\text{dynamic}} \propto V_{\text{dd}}^2 V_{\text{dd}} \propto f^3$ of cubic frequency scaling (Section 1.18). For $\alpha_{\text{MOS}} \sim 1$ (short-channel), the frequency scales sub-linearly with V_{dd} , and the P_{dynamic} vs f curve has a knee determined by V_{th} : above the knee, $P \propto f^3$ (cubic regime); below the knee, $P \propto f$ (linear regime). The knee itself sits at the V_{dd} where the device transitions from velocity-saturated drive to threshold-limited drive — exactly where $V_{\text{dd}} - V_{\text{th}}$ becomes comparable to the saturation voltage.

As $V_{\text{dd}} \rightarrow V_{\text{th}}$, the gate delay diverges (the $(V_{\text{dd}} - V_{\text{th}})$ factor in (1.28) vanishes), so attempting to push frequency further at fixed V_{th} requires either raising V_{dd} (back into the cubic regime) or lowering V_{th} (which exponentially raises leakage; see below). This $V_{\text{th}}/V_{\text{dd}}$ trade defines the bottom of the voltage-scaling envelope.

The leakage / threshold-voltage trap

Subthreshold leakage in the alpha-power model is

$$I_{\text{leak,sub}} \propto \frac{W}{L} (\nu_T)^2 \exp\left(-\frac{V_{\text{th}}}{n \nu_T}\right), \quad \nu_T \equiv k_B T_j / q, \quad (1.29)$$

where $n \sim 1.0\text{--}1.5$ is the subthreshold ideality factor and $\nu_T = k_B T_j / q$ is the thermal voltage (~ 27 mV at $T_j = 350$ K). Equation (1.29) is exponentially sensitive to *both* V_{th} and T_j . At a 60 mV/decade subthreshold slope (typical at room temperature), a 60 mV reduction in V_{th} raises $I_{\text{leak,sub}}$ by an order of magnitude; a 10 K rise in T_j roughly doubles $I_{\text{leak,sub}}$. Two consequences for chip design:

- **Lowering V_{th} is dangerous.** It raises f per (1.28) but raises I_{leak} per (1.29); the leakage / performance trade is what sets the operating V_{th} for each process node. Multi- V_{th} designs (different threshold voltages on different gates) are the architect's response.
- **Raising T_j is also dangerous.** Even at fixed V_{th} , I_{leak} doubles every 10 K. For a chip already at $P_{\text{static}} \sim 30\%$ of total power, a 20 K thermal excursion brings the static fraction above 50% — a thermal-runaway hazard if not actively managed.

Pushing T_j downward (the photonic-cooling lever) does the inverse on both counts: it suppresses P_{static} exponentially *and* relaxes the V_{th} lower bound, freeing the architect to use lower- V_{th} cells where they pay off. Conventional cooling cannot fully exploit this positive feedback because pushing T_j lower demands disproportionately more pump / fan power (Section 2.4); microrefrigeration breaks that constraint, and the architectural consequence is the focus of Chapter 10.

Hot spots emerge from the local power-equation imbalance

In a real chip, every term in (1.23)–(1.26) is spatially non-uniform: α is high in active functional units, low in clock-gated cores; N_{tr}/A is high in dense logic, low in cache and pad rings; I_{leak} tracks $T_j(\mathbf{x})$ which is itself the steady-state solution of the local thermal balance. The hot spot is exactly where the local $P_{\text{chip}}(\mathbf{x})/A$ integrated over the active layer exceeds the local thermal removal capacity — where (1.32) is tightest. The 2D model of Section 2.6 treats the resulting peak-to-bulk ratio $\alpha_{\text{geo}} \equiv \max_i q_i / \bar{q}$ as the geometric crisis parameter (we use α_{geo} in this paragraph to distinguish it from the activity factor α in (1.24)); in the language of this section, α_{geo} is set by *which functional units are active* on a given cycle, and is therefore workload-dependent. The photonic-network temporal targeting of Section 2.7 is the response: route cooling to the locally-elevated α_{geo} regions in real time, rather than sizing the bulk cooler for the worst-case sum.

Cubic vs. linear frequency scaling

The P_{dynamic} form (1.24) is linear in f at fixed V_{dd} ; but increasing f past the device's supported point requires raising V_{dd} to keep gate transitions fast. Empirically, at the device-physics limit, $V_{\text{dd}} \propto f$ (modulo threshold-voltage offsets), so the “DVFS-limited” frequency-scaling law becomes

$$P_{\text{dynamic}} \propto f V_{\text{dd}}^2 \propto f^3 \quad (\text{cubic regime, } V_{\text{dd}} \propto f). \quad (1.30)$$

Two qualitatively different operating regimes therefore exist:

- **Linear regime** (V_{dd} fixed, f varied within the device tolerance): $P_{\text{dynamic}} \propto f$; doubling clock frequency doubles dynamic power. This is the regime within a single DVFS step.

- **Cubic regime** (V_{dd} tracking f): $P_{\text{dynamic}} \propto f^3$; doubling frequency *octuples* dynamic power. This is the regime once the device-physics floor on V_{dd} is exhausted and additional headroom is bought by raising voltage.

Modern AI / HPC chips operate near the boundary: the highest boost clocks sit deep in the cubic regime, while sustained workloads sit closer to the linear regime. The thermal envelope decides which.

The thermal wall as architectural constraint

Combining (1.23)–(1.30) with the conduction wall (2.2), $q_{\text{wall}} = \Delta T_{\text{lim}}/R_{\text{die}}$, the maximum chip operating point under classical convection is

$$P_{\text{chip}}^{\text{max,conv}} = q_{\text{wall}} A_{\text{die}} = \frac{\Delta T_{\text{lim}} A_{\text{die}}}{R_{\text{die}}}. \quad (1.31)$$

This is the wall in chip-electrical units: the chip can dissipate no more than $P_{\text{chip}}^{\text{max,conv}}$ without exceeding T_{lim} , regardless of how it is partitioned between P_{dynamic} and P_{static} . Equivalently, in terms of (1.24),

$$\alpha N_{\text{tr}} C_{\text{eff}} V_{\text{dd}}^2 f + P_{\text{static}}(V_{\text{dd}}, T_j) \leq \frac{\Delta T_{\text{lim}} A_{\text{die}}}{R_{\text{die}}}. \quad (1.32)$$

Equation (1.32) is the *architectural budget inequality*: every architectural choice (how many transistors, how much wire density, how aggressive a clock target) must trade against the thermal cap on the right. A chip that wants to scale on any of $(N_{\text{tr}}, f, V_{\text{dd}}, \alpha)$ must either reduce C_{eff} (process node), reduce P_{static} (lower T_j via better cooling), or move the right-hand side (improve cooling, i.e. either reduce R_{die} or raise ΔT_{lim}).

What microrefrigeration does to (1.32). Per Section 2.5, photonic microrefrigeration at fraction s of the chip flux moves the wall to $q_{\text{wall}}^{(\text{hyb})} = \Delta T_{\text{lim}}/[(1-s)R_{\text{die}}]$, which translates to an architectural budget

$$P_{\text{chip}}^{\text{max,hyb}}(s) = \frac{\Delta T_{\text{lim}} A_{\text{die}}}{(1-s)R_{\text{die}}}. \quad (1.33)$$

At $s = 0.5$, the architectural cap doubles; at $s = 0.9$, it grows ten-fold. In the cubic frequency-scaling regime, doubling the power cap permits a $2^{1/3} \sim 1.26\times$ increase in f holding N_{tr} fixed; tenfold growth permits $\sim 2.15\times$. Architecturally, this is the difference between a chip whose top boost clock is throttled by the package and a chip whose top boost clock is set by transistor physics. The practical implications — transistor-density and wire-density budgets, voltage-scaling targets, latency-vs-throughput allocation — are taken up in Chapter 10.

Hot spots, dark silicon, and architectural decisions

Equation (1.32) bounds the *average* chip flux. The 2D / multi-node analysis of Section 2.6 sharpens this: the worst-case node forces the convective system into a high-flow, low-COP regime even when the average is well within budget. This gives rise to two well-known architectural artefacts:

- **Thermal throttling.** When $q_{\text{peak}} >$ the sustainable flux at the operating point, the chip drops f (and often V_{dd}) until the hot spot cools below T_{lim} . This is the “boost-clock cliff” visible in modern CPUs and GPUs.

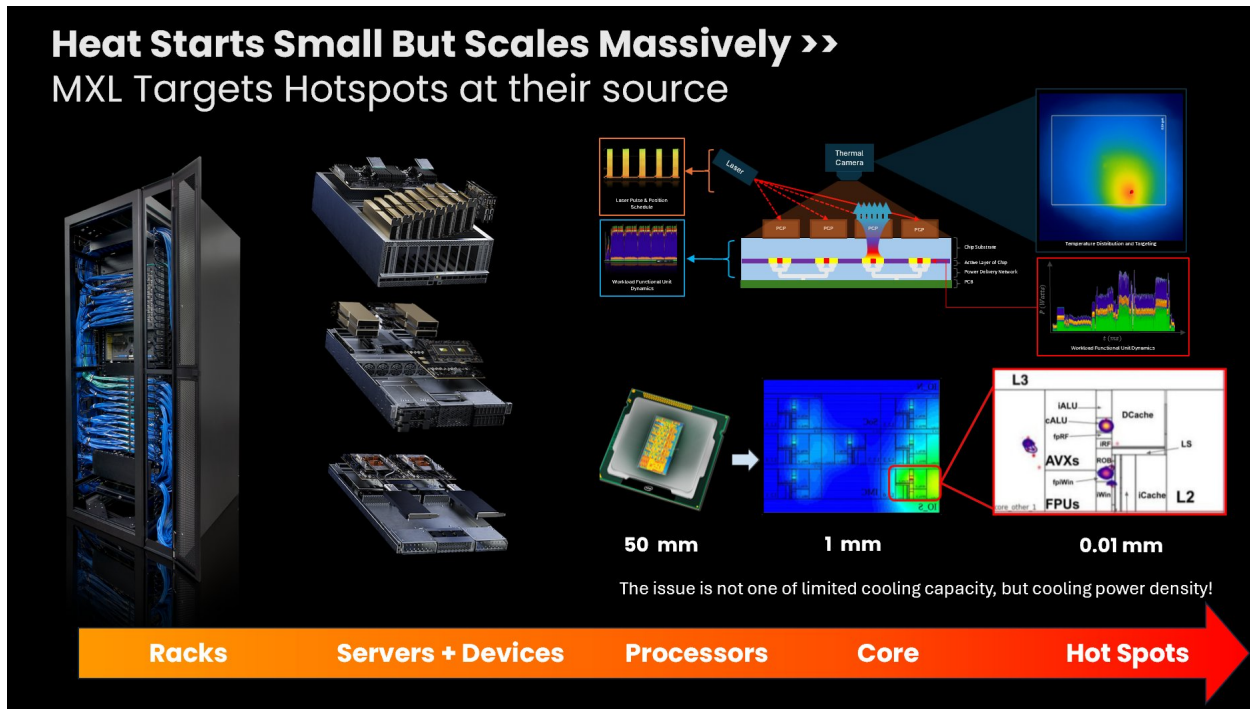


Figure 1.6: Thermal hierarchy in modern data centres. Rack-level total power is within reach of conventional coolants, but the binding constraint is at the $\sim 10\text{ }\mu\text{m}$ -scale hot spot (rightmost panel), where conduction bottlenecks and power densities reach $10^3\text{--}10^4\text{ W/cm}^2$ ($10\text{--}10^2\text{ W/mm}^2$). The issue is not cooling capacity but cooling power *density*.

- **Dark silicon.** When the per-transistor power density exceeds what the package can remove on average, the architect must leave a fraction of the die unpowered (“dark”) at any given time — typically by gating idle functional units — so that $\langle q \rangle_{\text{die}} < q_{\text{passive}}$. Dark silicon is an architectural *response* to (1.32); it is exactly what microrefrigeration’s spatial / temporal targeting (Sections 2.6 and 2.7) makes unnecessary.

The architectural reorganisation that becomes possible once the thermal wall is moved by photonic cooling is the subject of Chapter 10. At the level of this chapter, the take-away is: (1.32) is the boundary condition; ASF microrefrigeration moves the boundary.

1.19 Application: hot-spot cooling in HPC

1.20 System integration

The photonic cold plate does not operate as an isolated device: it is part of a closed optical loop whose end-to-end energy balance (1.4) defines the economics of the entire installation. Figure 1.10 shows that loop at its simplest: a single pump laser, a wavelength-division splitter separating the pump channel from the returned fluorescence, the photonic cold plate (“PCP”) sitting on the chip, and an LPC closing the electrical loop. Packaged, the same topology scales from a single die (Figure 1.11) to a whole rack (Figure 1.12); the exploded view of Figure 1.13 identifies the mechanical subassemblies (heat-sink fin stack, microfluidic manifold, PCP tile carrier, chip socket) that must be co-designed with the optical stack.

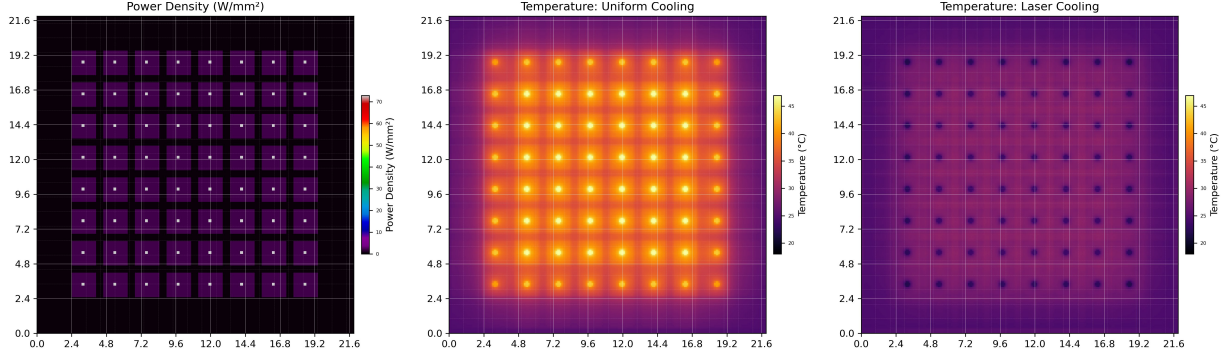


Figure 1.7: Finite-element thermal simulation of a die with a patterned power map (left). Under uniform cold-plate cooling (centre) the hot spots rise $\sim 25^\circ\text{C}$ above the coolant. Under laser-targeted photonic cooling (right) each hot spot is driven locally, clamping the whole die close to the coolant set point. The gap between the two panels is exactly the Carnot margin that (1.13) opens for net power generation when $T_h > T_0$.

1.21 Chapter summary

The compact set of equations carried by the rest of the book:

- Master fluorescence balance: $P_f = \eta_{\text{cpl}} P_L (1 + \eta_{\text{ASF}})$ (1.4), with $Q_c = \eta_{\text{ASF}} \eta_{\text{cpl}} P_L$ (1.3).
- Sharp feasibility (self-powered loop): $\eta_P \eta_{\text{cpl}} [1 + \eta_{\text{ASF}}(1 - T_0/T_h)] \geq 1$ (1.17).
- Refined LPC exergy bound: $\eta_{\text{LPC}} \leq [1 + \eta_{\text{ASF}}(1 - T_0/T_h)] / (1 + \eta_{\text{ASF}})$ (1.15).
- Compact COP: $\text{COP}_{\text{net}} = Q_c / (P_{\text{in}} - P_{\text{rec}})$ (1.11), with the Carnot-bounded net-generation criterion of Section 1.13.

The four loop efficiencies $\eta_P, \eta_{\text{cpl}}, \eta_{\text{ASF}}, \eta_{\text{LPC}}$ are the only dials in this bookkeeping; the rest of the book derives each from underlying physics:

- η_P (laser wall-plug) and η_{cpl} (delivery / coupling) are unpacked in Chapter 3.
- η_{ASF} (cooling efficiency relative to absorbed pump) is the subject of the electro-optic chapters Chapters 4 and 5 (microscopic origin) and the limits chapter Chapter 6 (upper bounds).
- η_{LPC} (LPC) is unpacked in Chapter 7, with material choices for the extractor laid out in Chapter 8.

The architectural reach of these results — what changes for chip design under microrefrigeration, what does not — is taken up in Chapter 10, with the integration / inverse-design loop in Chapter 9 closing the device-level discussion.

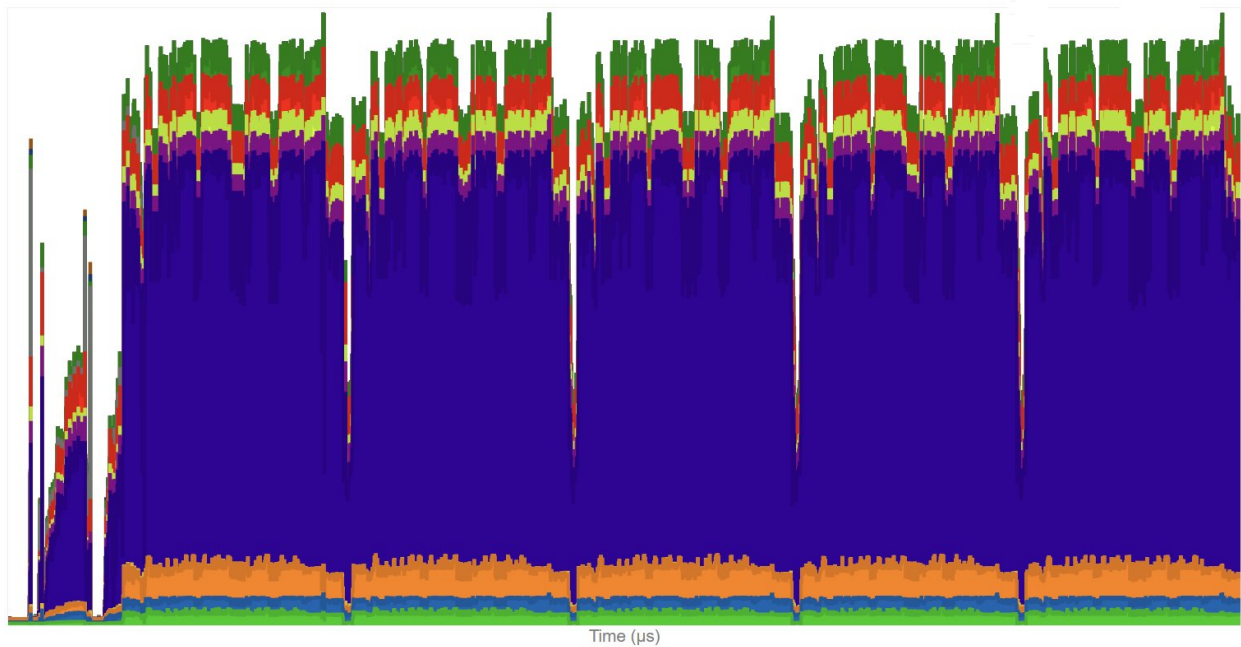


Figure 1.8: Operating cycle of the cooling loop. A Q-switched laser pre-cools the active region between $t_{q,start}$ and $t_{hs,start}$; thereafter a CW pump sustains saturated cooling power $P_{cool,sat}$ against the heat load P_{heat} until the hotspot clears at $t_{hs,end}$. The dashed temperature reference at 225 K is the Yb:YLF cryogenic set point of Chapter 8.

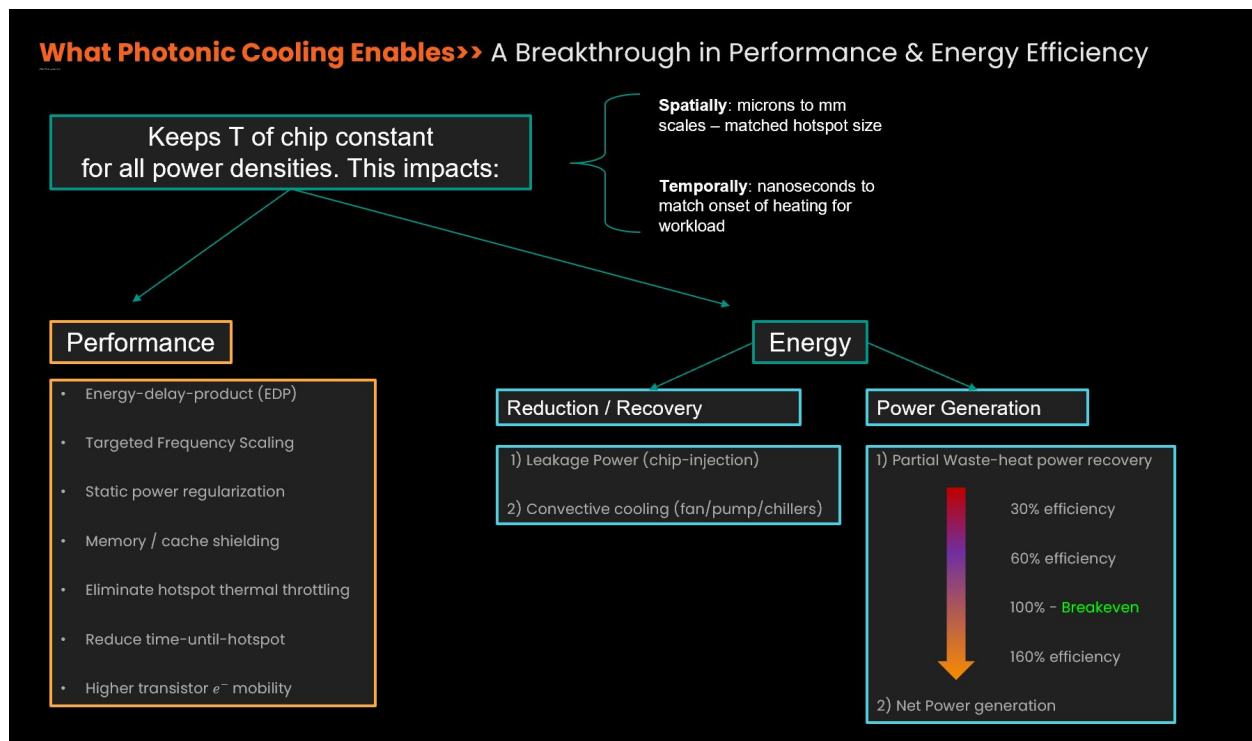


Figure 1.9: Performance and energy consequences of spatially matched hot-spot cooling: removal of thermal throttling, recovery of leakage power, and — when the loop crosses break-even, (1.17) — net power generation from the chip's own dissipated heat.

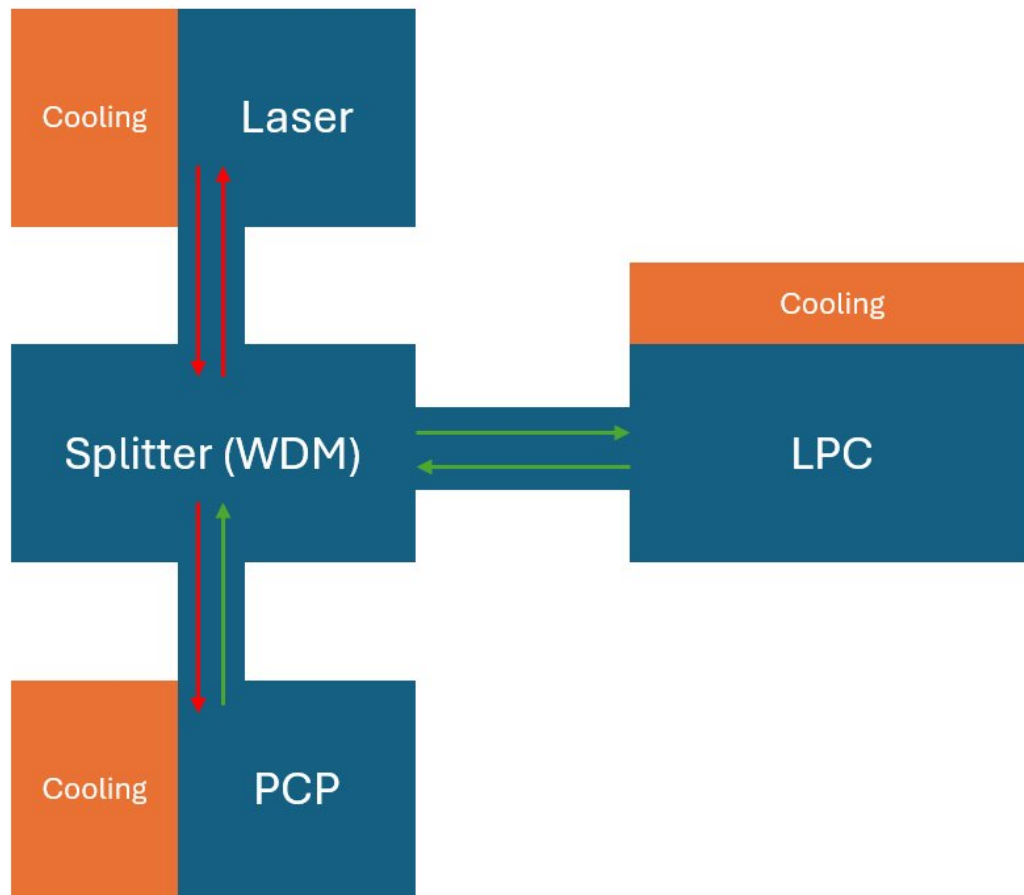


Figure 1.10: Closed-loop block diagram of the photonic cooling system: pump laser $\rightarrow$ wavelength splitter $\rightarrow$ photonic cold plate (PCP) $\rightarrow$ recovered fluorescence returned through the same splitter to an LPC that feeds the pump-drive bus. Each active block carries its own thermal management (shaded orange). Red arrows indicate pump frequency, green arrows fluorescence.

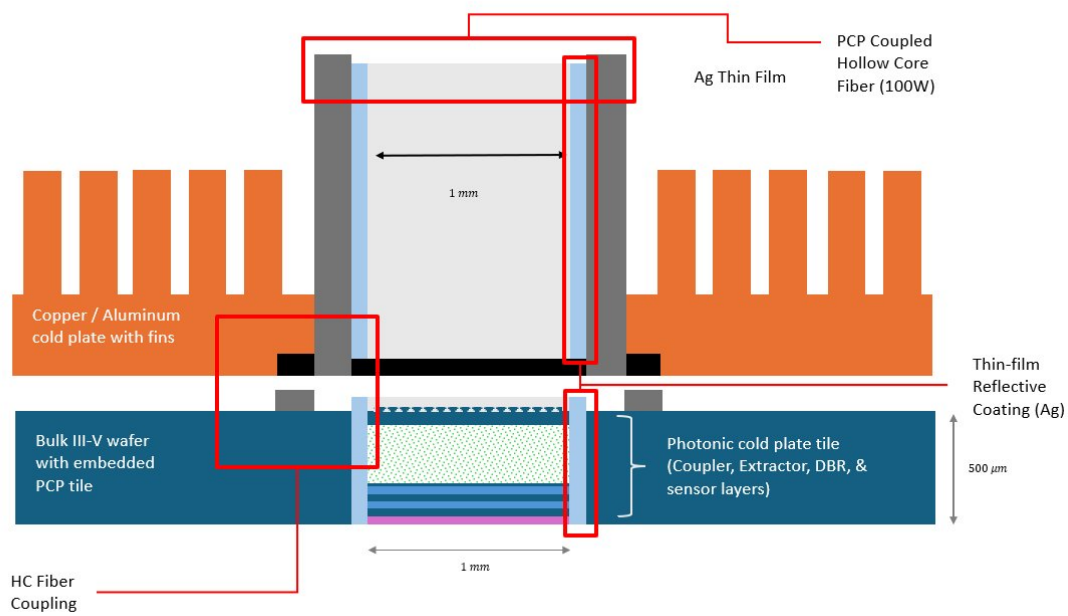


Figure 1.11: Mechanical integration of a single PCP tile with the chip cold plate. The pump is delivered through a 100 W-rated Ag thin-film hollow-core fiber; the 1 mm-pitch PCP tile (coupler / extractor / DBR / sensor) sits embedded in the bulk III-V wafer under a copper-fin cold plate.

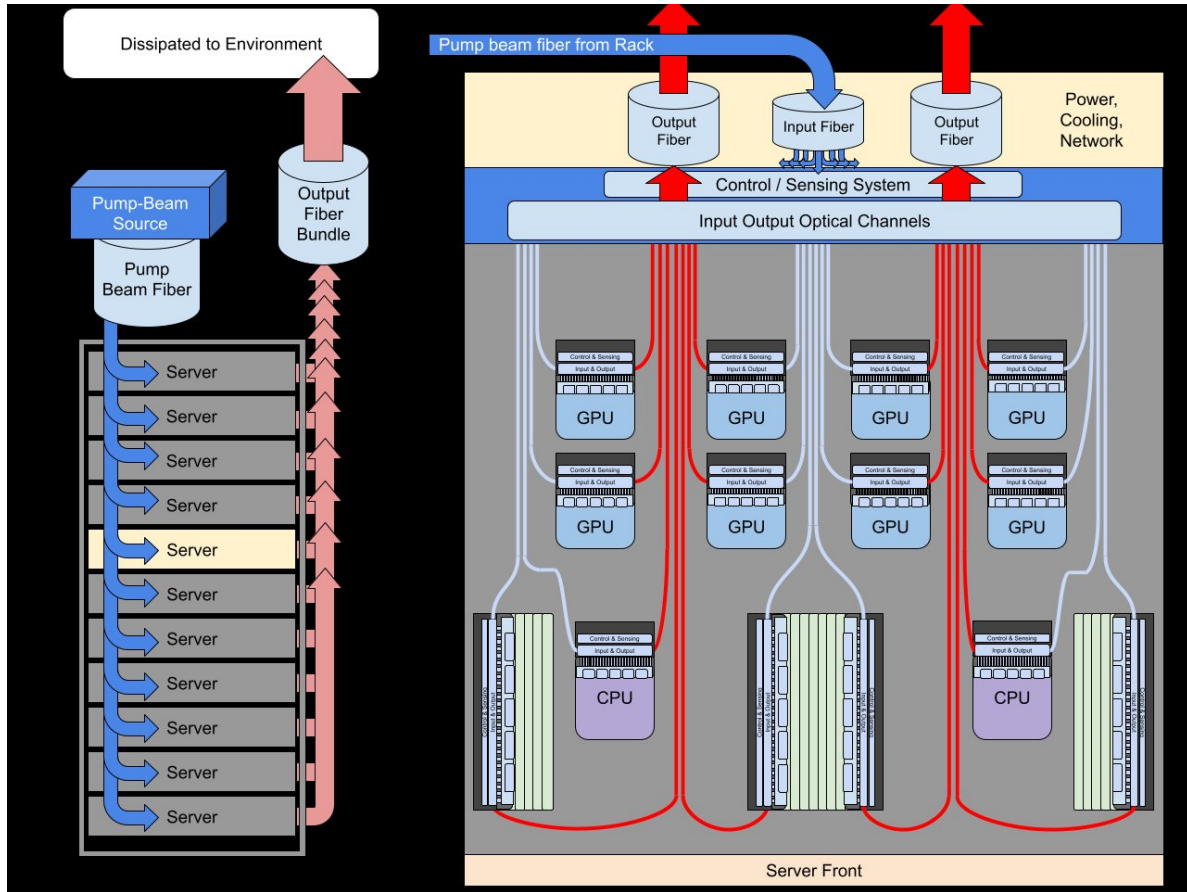


Figure 1.12: Rack-scale deployment. A single high-power pump laser feeds a fiber bundle that distributes pump light to every server in the rack; each server returns its fluorescence on a parallel output fiber. At the top-of-rack, fluorescence is converted back to electrical power and fed into the pump-drive bus. The loop's wall-plug inefficiency ($1 - \eta_{LPC}\eta_{ASF}\eta_{cpl}\eta_P$) is dissipated as low-grade heat handled by conventional air cooling.

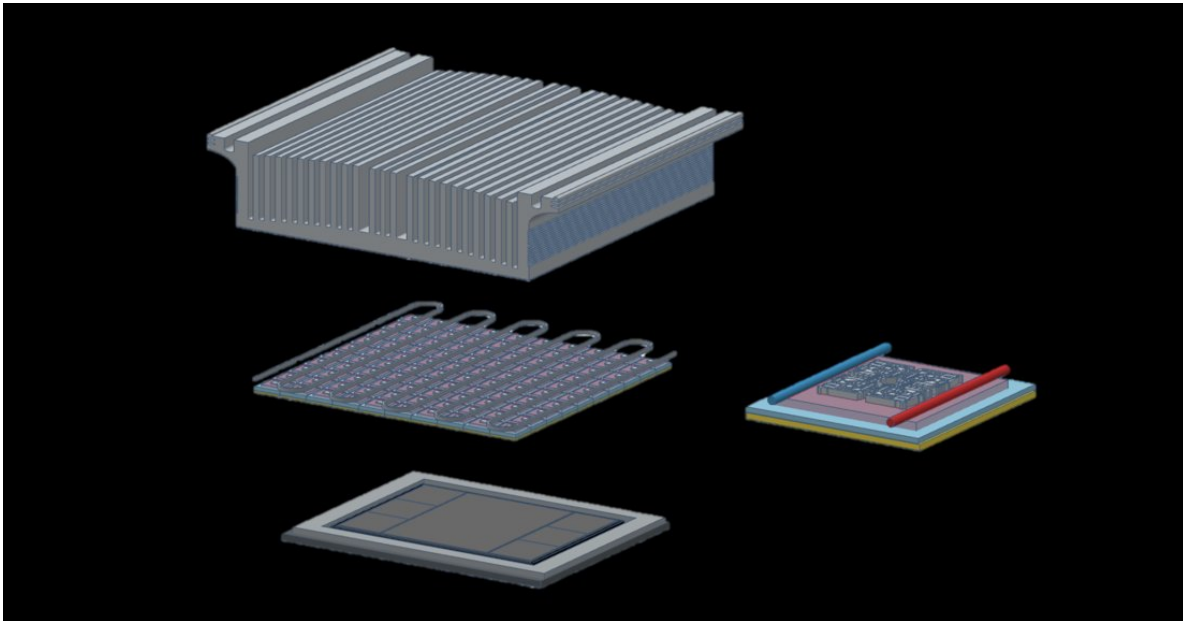


Figure 1.13: Exploded view of the cooling module: copper-fin heat sink, microfluidic manifold, tiled PCP array, and chip carrier. All four layers are independent but share the same mechanical foot-print; the optical elements add $< 500\ \mu\text{m}$ to total stack height.

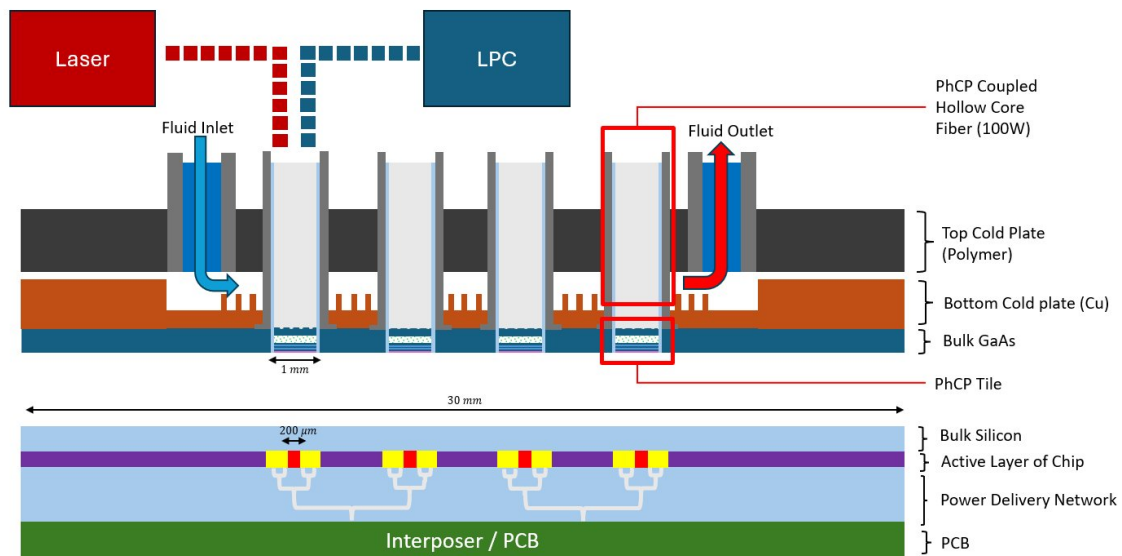


Figure 1.14: Module-scale integration: a 30 mm-span carrier with multiple PCP tiles, each driven by an independent hollow-core fiber feed and returning fluorescence to a shared LPC. The stack below the tiles (bulk silicon / active layer / power-delivery network / interposer / PCB) is the unchanged processor substrate; the photonic cold plate is an add-on layer.

Chapter 2

Microrefrigeration and the Conduction Bottleneck

2.1 Why a separate chapter

Chapter 1 treated the photonic cold plate as a thermophotonic heat-engine–refrigerator hybrid and derived the loop-level COP master equation (1.11) together with the sharp feasibility condition (1.17). That treatment is correct, but it leaves out a question that decides whether the device is interesting at all: *what is photonic cooling competing against, and where does classical convective cooling actually fail?* This chapter fills that gap. We develop two companion thermal models — a 1D zero-resistance picture in which the chip is a single thermal node, and a 2D / multi-node picture that exposes the geometric concentration of heat at hot spots — and we plot COP vs. chip power flux q for both classical convective cooling and laser microrefrigeration on the same axes. The chapter closes by showing how the temporal targeting capability of laser cooling — matching laser pulse rate to the workload’s temporal dynamics — changes the comparison once again. The content is taken from the design slides `slides/LaserCoolingMicrorefrigeration(1).pptx` and is consistent with the loop-level bookkeeping of Chapter 1.

2.2 The conduction bottleneck and the power wall

The history of chip cooling is a sequence of escalating heat-flux ceilings:

Era	Technology	Heat flux (W/mm ²)	Method
1950s–60s	Discrete transistors	< 0.01	Natural air convection
1980s	Early CMOS / VLSI	0.01–0.1	Passive Al heat sinks
2000s	Pentium 4 / multi-core	0.1–0.5	Heat pipes, active air
2020s	Modern AI chips (H100/H200)	0.5–2.0	Vapor chambers, direct liquid
2025+	High-density SiC / GaN	5–10+	Microchannels, jet impingement

Standard air cooling fails above ~ 1 W/mm² because the thermal resistance of the air interface becomes too high; modern manifold microchannel systems can handle ~ 10 W/mm² — roughly 15% of the heat flux at the surface of the Sun (~ 63 W/mm²). The macro-scale and micro-scale problems have evolved with different priorities: macro systems (buildings / industry) optimise the global coefficient of performance COP; micro systems (electronics) optimise heat-flux removal and

minimise thermal resistance, even at the cost of local COP. ASF microrefrigeration sits inside the micro-scale problem but uses thermodynamic levers from the macro-scale toolbox.

2.3 The Zero-Resistance Die

The defining conceptual claim of laser microrefrigeration:

ASF laser cooling creates a “thermal shunt.” By extracting phonon energy at the hot spot and converting it to photons that exit the package at the speed of light, we are not cooling the chip — we are optically screening (even cancelling) the thermal resistance of the die. This allows operation at a “null resistance” point where the hot spot is thermally transparent, preventing the bulk system from ever entering its inefficient, high-flow regime.

To make this quantitative we need two thermal resistances and one temperature limit.

2.4 1D thermal model: classical convection only

Treat the chip as a single thermal node. Let

- q — chip-surface heat-flux density (W/mm^2) — the device dissipation we want to extract,
- R_{die} — die thermal resistance from junction to the back-side / cold-plate interface ($\text{K mm}^2/\text{W}$),
- $R_{\text{ext}}(\dot{V})$ — external (downstream) thermal resistance from the cold-plate interface to the coolant inlet, controlled by the coolant volumetric flow rate $\dot{V}$,
- T_{c} — coolant inlet temperature,
- T_{lim} — maximum allowed junction temperature, $\Delta T_{\text{lim}} = T_{\text{lim}} - T_{\text{c}}$.

At steady state, junction temperature $T_{\text{j}} = T_{\text{c}} + (R_{\text{die}} + R_{\text{ext}}(\dot{V}))q$. The *passive thermal limit* at fixed $\dot{V}$ is

$$q_{\text{passive}}(\dot{V}) = \frac{\Delta T_{\text{lim}}}{R_{\text{die}} + R_{\text{ext}}(\dot{V})}, \quad (2.1)$$

i.e. the largest heat flux the convective system can sustain at a given fan / pump operating point. Increasing $\dot{V}$ reduces R_{ext} and pushes q_{passive} upward, but never past the *ultimate thermal wall*

$$q_{\text{wall}} = \lim_{R_{\text{ext}} \rightarrow 0} q_{\text{passive}}(\dot{V}) = \frac{\Delta T_{\text{lim}}}{R_{\text{die}}}. \quad (2.2)$$

q_{wall} depends only on the die itself and the temperature budget. It is the hard wall that conventional cooling *cannot cross*, no matter how aggressively the convective loop is driven. Note carefully: q_{wall} is *not* the typical thermal wall reported in industrial cooling reports — those are $q_{\text{passive}}(\dot{V}_{\text{rated}})$ at a fixed-flow rated operating point. The ultimate wall q_{wall} is approached in the limit of unbounded fan flow at the cost of vanishing COP.

COP of the convective loop

Model the fan / pump work as $W_{\text{f}}(\dot{V}) = \alpha \dot{V}^{\beta}$ with $\beta \in [1, 3]$ covering laminar through turbulent regimes; the external resistance scales inversely, $R_{\text{ext}}(\dot{V}) = K/\dot{V}$. The convective COP is

$$\text{COP}_{\text{conv}}(q, \dot{V}) = \frac{q}{W_{\text{f}}(\dot{V})}, \quad (2.3)$$

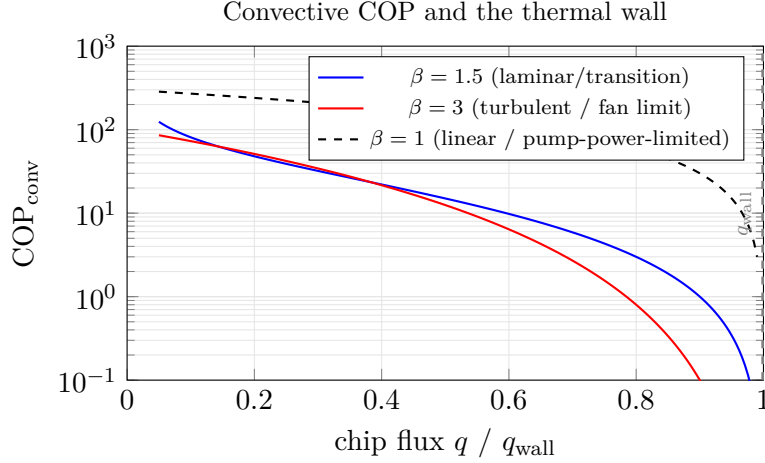


Figure 2.1: Convective $\text{COP}_{\text{conv}}(q)$ from (2.4), normalized so that $q = q_{\text{wall}}$ corresponds to $R_{\text{ext}} = 0$ and infinite flow. The curve crashes to zero at the wall regardless of fan-curve exponent β ; what differs between regimes is the rate of approach.

subject to the constraint that the operating point must satisfy $q \leq q_{\text{passive}}(\dot{V})$, i.e. $\dot{V} \geq K q / (\Delta T_{\text{lim}} - R_{\text{die}} q)$. Solving the equality (just-meeting the thermal limit) and substituting,

$$\boxed{\text{COP}_{\text{conv}}(q) = \frac{q}{\alpha \left[\frac{K q}{\Delta T_{\text{lim}} - R_{\text{die}} q} \right]^{\beta}}}. \quad (2.4)$$

Several features:

- For $q \ll q_{\text{wall}}$, the bracket is small, so $\text{COP}_{\text{conv}} \propto q^{1-\beta} / (\Delta T_{\text{lim}})^{-\beta}$ — a polynomial decay set by the fan curve.
- As $q \rightarrow q_{\text{wall}}$, the denominator $\Delta T_{\text{lim}} - R_{\text{die}} q \rightarrow 0$, so $\dot{V} \rightarrow \infty$ and $\text{COP}_{\text{conv}} \rightarrow 0$. The wall is a soft asymptote in flow rate but a hard *efficiency* crash in COP.

This is the picture in Figure 2.1: the convective COP is finite for $q < q_{\text{wall}}$ and falls catastrophically as $q \rightarrow q_{\text{wall}}^-$. The chip simply cannot operate at $q > q_{\text{wall}}$ under any classical convective scheme.

Cold-plate sub-model

The single-knob $R_{\text{ext}}(\dot{V}) = K/\dot{V}$ used above is the right cartoon, but the actual flow-to-resistance map of a real cold plate has more structure. Following the design-deck conventions, we split the source-to-coolant-inlet path into a cold-plate internal piece R_{cp} and a bulk coolant-warming piece R_{bulk} :

$$R_{\text{ext}}(\dot{V}) = R_{\text{cp}}(\dot{V}) + R_{\text{bulk}}(\dot{V}), \quad (2.5)$$

with

$$R_{\text{cp}}(\dot{V}) = R_{\text{series}} + \frac{R_{\text{stag}}}{1 + (\dot{V}/\dot{V}_c)^m}, \quad (2.6)$$

$$R_{\text{bulk}}(\dot{V}) = R_{\text{plumb}} + \frac{\beta}{\rho C_p \dot{V}^m}, \quad (2.7)$$

where R_{series} is the flow-independent contact / spreading / wall conduction baseline, R_{stag} is the zero-flow / stagnation residual that smoothly turns off as the forced-convection number $\dot{V}/\dot{V}_c$ crosses

unity ($m \in [0.3, 0.8]$ for laminar / turbulent regimes), and the $\beta/(\rho C_p \dot{V}^m)$ term captures coolant warming ($\beta = 1$ for outlet, $\beta \sim \frac{1}{2}$ for mean rise; zero for an air cold plate). The pump or fan electrical work is

$$W_f(\dot{V}) = W_{\text{static}} + K \dot{V}^n, \quad n \in [2, 3], \quad (2.8)$$

with W_{static} the controller / idle baseline (motor efficiency and electronics losses) and $K \dot{V}^n$ the hydraulic component; $n \rightarrow 2$ for laminar-dominated pressure drop, $n \rightarrow 3$ for fully turbulent regimes.

Three operating regimes of the bare convective system. The constraint $R_{\text{die}} + R_{\text{ext}}(\dot{V}) \leq \Delta T_{\text{lim}}/q$ partitions the $(q, \dot{V})$ plane into three pieces, and the COP (2.3) takes a different form in each:

$$\text{COP}_{\text{conv}}(q) = \begin{cases} \frac{\Delta T_{\text{lim}}}{R_{\text{ext}}(0) W_{\text{static}}}, & 0 \leq q < q_{\text{passive}}^{(0)}, \\ \frac{q}{W_{\text{static}} + K \dot{V}_c^n \left[\frac{R_{\text{stag}}}{\Delta T_{\text{lim}}/q - R_{\text{series}}} - 1 \right]^{n/m}}, & q_{\text{passive}}^{(0)} \leq q \leq q_{\text{real-wall}}, \\ 0, & q > q_{\text{real-wall}}, \end{cases} \quad (2.9)$$

with the three reference fluxes

$$q_{\text{passive}}^{(0)} = \frac{\Delta T_{\text{lim}}}{R_{\text{ext}}(0)}, \quad (\text{passive-to-forced-flow turn-on}) \quad (2.10)$$

$$q_{\text{real-wall}} = \frac{\Delta T_{\text{lim}}}{R_{\text{die}} + R_{\text{series}}}, \quad (\text{pump-limited wall, } \dot{V} \rightarrow \infty) \quad (2.11)$$

$$q_{\text{wall}} = \frac{\Delta T_{\text{lim}}}{R_{\text{die}}}, \quad (\text{idealised wall, all } R_{\text{ext}} \rightarrow 0). \quad (2.12)$$

Three points worth stating explicitly — they decide everything that follows:

1. *The widely quoted “thermal wall” is $q_{\text{real-wall}}$, not q_{wall} .* Real cold plates have a non-zero series resistance R_{series} (TIM, spreading, contact) that does not scale away with flow; the ultimate $q_{\text{wall}} = \Delta T_{\text{lim}}/R_{\text{die}}$ is only approached asymptotically.
2. *COP must be defined relative to the useful extracted cooling power.* Writing $\text{COP}_{\text{conv}} = q/W_f$ with a fixed W_f hides the wall; the moment we let $\dot{V}$ be the free parameter required to satisfy the temperature constraint, (2.9) crashes to zero at $q \rightarrow q_{\text{real-wall}}^-$. This is the feature microrefrigeration is supposed to attack.
3. *Below $q_{\text{passive}}^{(0)}$, the convective loop is in passive mode:* the cold plate’s stagnation resistance R_{stag} is already sufficient at zero forced flow, and only the controller idle power W_{static} appears in the denominator. COP is constant and large in this regime; below $q_{\text{passive}}^{(0)}$ microrefrigeration is unhelpful because conventional cooling has not yet engaged its expensive degrees of freedom.

Figure 2.2 shows the three regimes side by side; the curve we labelled “classical thermal wall” in Figure 2.1 is the cliff at $q \rightarrow q_{\text{real-wall}}^-$, not the idealised q_{wall} itself.

2.5 1D thermal model with microrefrigeration

Place a microrefrigerator at the active layer. Its effect, per the “thermal shunt” picture of Section 2.3, is to extract a fraction of the chip heat *before* it has to flow through R_{die} and R_{ext} : the photonic

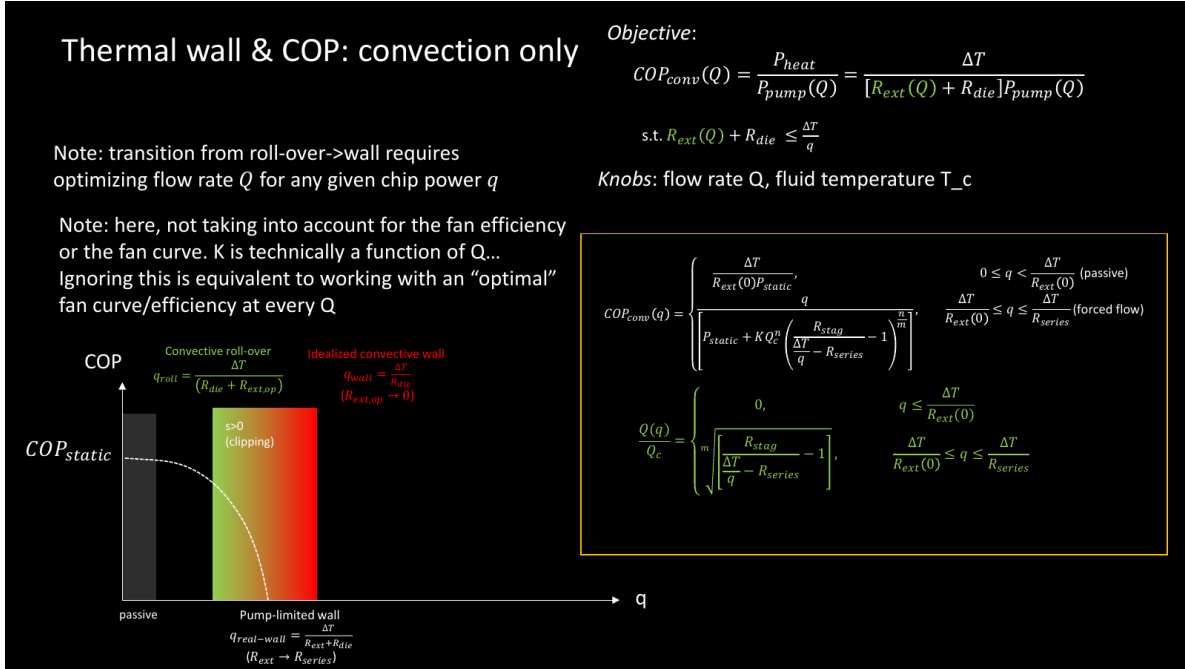


Figure 2.2: Cold-plate sub-model and the three convective COP regimes (reproduced from the design deck). Below $q_{passive}^{(0)}$ (gray, passive) the controller idle power W_{static} sets a constant COP_{static} . Between $q_{passive}^{(0)}$ and $q_{real-wall}$ (green/red, forced flow) the pump engages and the COP obeys the explicit closed form (2.9). Approaching the idealised $q_{wall} = \Delta T_{lim}/R_{die}$ requires $\dot{V} \rightarrow \infty$, where the COP crashes to zero.

cold plate absorbs phonon energy locally and exits via the photonic network at the speed of light. Define the dimensionless *microrefrigeration loading*

$$s \equiv \frac{Q_{MR}}{q}, \quad (2.13)$$

the fraction of chip heat lifted optically. The remaining $(1 - s)q$ flows through the conventional path; the laser-extracted fraction bypasses both R_{die} and R_{ext} and exits through the photonic stack.

Three regimes of MR operation

Per the design slides, three qualitatively distinct regimes appear as s is increased:

(I) Thermal clipping, $0 < s \ll 1$. The MR provides a small thermal lift relative to the bulk ΔT ; the local extractor temperature T_m stays close to ambient. In this regime the MR’s role is to clip the upper tail of transient hot spots, leaving the bulk system in its high-COP regime. The convective side still does most of the work; the MR contributes only a correction.

(II) Hot-spot interception, intermediate s . The MR provides a meaningful ΔT_{lift} , keeping the hot spot at the thermal-limit threshold even when bulk convection alone would fail. This is the regime where MR *turns on* as the convective COP collapses ($q \rightarrow q_{wall}$). The bulk system can stay in its efficient operating point because the MR handles the peak demand.

(III) Resistance nullification, $s > 1$. Here the MR goes beyond cancelling the downstream resistance and actually drives T_m below ambient, pulling heat from the bulk *as well as* the hot spot. This effectively reduces the die thermal resistance (the hot spot becomes a thermal sink to the bulk), at the cost of additional MR work. It is also the regime where ASF operating below ambient would consume exergy (Case B of Section 1.13); positive net generation is no longer possible without drawing on additional free energy.

Hybrid COP

Each MR mechanism has its own intrinsic $\text{COP}_{\text{MR}}(s, T_m, T_h)$ governed by the thermophotonic loop of Chapter 1. In the cooling regime the relevant COP_{MR} is positive; near the self-powered threshold of (1.18) it can be very large or even diverge. The hybrid system COP is

$$\boxed{\text{COP}_{\text{hyb}}(q; s, \dot{V}) = \frac{q}{W_f(\dot{V}) + Q_{\text{MR}}/\text{COP}_{\text{MR}}} = \frac{q}{W_f(\dot{V}) + s q/\text{COP}_{\text{MR}}}.} \quad (2.14)$$

The constraint becomes *the chip-with-MR thermal limit*: only the fraction $(1-s)q$ flows through $R_{\text{die}} + R_{\text{ext}}$, so the hybrid passive limit is

$$q_{\text{passive}}^{(\text{hyb})}(\dot{V}, s) = \frac{\Delta T_{\text{lim}}}{(1-s)(R_{\text{die}} + R_{\text{ext}}(\dot{V}))}. \quad (2.15)$$

Equation (2.15) is the formal expression of the “zero-resistance die”: as $s \rightarrow 1$, $q_{\text{passive}}^{(\text{hyb})} \rightarrow \infty$ at *fixed* $\dot{V}$, so the bulk system never has to enter the inefficient high-flow regime that crashed COP_{conv} in Figure 2.1.

Wall comparison

Comparing (2.2) and (2.15): the classical wall sits at $q_{\text{wall}} = \Delta T_{\text{lim}}/R_{\text{die}}$, fixed by the die alone. With MR at fraction s , the wall moves to $q_{\text{wall}}^{(\text{hyb})} = \Delta T_{\text{lim}}/[(1-s)R_{\text{die}}]$, which diverges as $s \rightarrow 1$. The MR has effectively *cancelled* the die thermal resistance for the heat it intercepts. The price is $s q$ of MR work (or, per Chapter 1, possibly negative electrical input if the loop is in the self-powered regime).

Normalised λ -form and closed-form optimal MR strength

The clearest way to compare bulk convection against MR — and to pick the optimal MR strength s^* for a given chip flux q — is to normalise both COPs to the bulk system’s design operating point, denoted $\text{COP}_{b,0} \equiv \text{COP}_{\text{conv}}(q_{b,0})$ at some reference flux $q_{b,0}$. Define the dimensionless ratios

$$\boxed{\lambda_b \equiv \frac{\text{COP}_{\text{conv}}(q; \dot{V})}{\text{COP}_{b,0}}, \quad \lambda_m \equiv \frac{\text{COP}_{\text{MR}}(s, T_m, T_h)}{\text{COP}_{b,0}}.} \quad (2.16)$$

λ_b measures how far the bulk loop has wandered from its sweet spot — it is ~ 1 in the high-COP plateau region below $q_{\text{passive}}^{(0)}$ and collapses to zero as $q \rightarrow q_{\text{real-wall}}^-$. λ_m is the technology signature of the microrefrigerator; it is constant or slowly varying along s . For thin-film TECs $\lambda_m \sim 0.3$; for laser cooling the *gross* value is smaller still ($\text{COP} \simeq \eta_P \eta_{\text{ASF}} \simeq 0.02\text{--}0.07$, since η_{ASF} is bounded by the quantum defect), and it is LPC recycling of the fluorescence that lifts the *net* λ_m toward 0.1–0.3, approaching ~ 1 only in the limit $\eta_P \eta_{\text{LPC}} \eta_{\text{cpl}} \rightarrow 1$ (Chapter 7).

s range	$R_{\text{eff}} \equiv R_{\text{die}} + (1 - s)R_{\text{ext}}$	Physical meaning
$s = 0$	$R_{\text{die}} + R_{\text{ext}}$	passive bulk, no MR
$0 < s < 1$	$R_{\text{die}} + (1 - s)R_{\text{ext}}$	thermal clipping; fans slow down
$s = 1$	R_{die}	hot-spot interception; $T_{\text{m}} = T_{\text{a}}$
$1 < s < s_0$	$0 < R_{\text{eff}} < R_{\text{die}}$	resistance nullification; $T_{\text{m}} < T_{\text{a}}$
$s = s_0 = 1 + R_{\text{ext}}/R_{\text{die}}$	0	zero-resistance die; $T_{\text{j}} = T_{\text{a}}$
$s > s_0$	$R_{\text{eff}} < 0$	over-cooling; $T_{\text{j}} < T_{\text{a}}$, wasteful

Table 2.1: The six regimes of the dimensionless MR strength s , and the apparent effective resistance $R_{\text{eff}}(s) = R_{\text{die}} + (1 - s)R_{\text{ext}}$ the bulk system sees. Optimal control (2.19) always picks the smallest feasible s — over-cooling is never optimal.

In the λ -form the per-flux work decomposes additively:

$$\frac{W_{\text{f}} + W_{\text{MR}}}{q \text{COP}_{b,0}^{-1}} = \frac{1}{\lambda_b} (1 - s) + \frac{s}{\lambda_m}, \quad (2.17)$$

where the first term is the bulk contribution (only the $(1 - s)q$ of heat that still flows downstream pays for fan work) and the second is the MR contribution. The hybrid COP from (2.14) becomes

$$\text{COP}_{\text{hyb}}(q; s) = \text{COP}_{b,0} \left[\frac{1-s}{\lambda_b} + \frac{s}{\lambda_m} \right]^{-1}. \quad (2.18)$$

Optimal MR strength. The architectural objective is to minimise total work subject to the junction-temperature constraint $q[(1 - s)R_{\text{ext}}(\dot{V}) + R_{\text{die}}] \leq \Delta T_{\text{lim}}$. The objective (2.17) is linear and increasing in s whenever $\lambda_m < \lambda_b$ (the typical regime: bulk in its sweet spot, MR less efficient). The optimiser therefore picks the *smallest* s that satisfies the constraint, giving the closed-form rule

$$s^*(q) = \max \left(0, 1 - \frac{\Delta T_{\text{lim}}/q - R_{\text{die}}}{R_{\text{ext}}(\dot{V})} \right). \quad (2.19)$$

The structure is intuitive. If $q \leq q_{\text{passive}}^{(\text{hyb})} = \Delta T_{\text{lim}}/(R_{\text{die}} + R_{\text{ext}})$, then $s^* = 0$ — the bulk system has enough headroom on its own and the MR stays off. Above that flux, s^* ramps linearly with q until $s^* = 1$ at $q = \Delta T_{\text{lim}}/R_{\text{die}} = q_{\text{wall}}$ (downstream fully cancelled, junction node pinned to ambient); beyond that, MR begins to cancel the die resistance itself, eventually reaching $s^* = s_0 = 1 + R_{\text{ext}}/R_{\text{die}}$ at the *zero-resistance die* point where $T_{\text{j}} = T_{\text{a}}$ exactly. Above s_0 the MR overcools ($T_{\text{j}} < T_{\text{a}}$) — thermodynamically allowed, but a waste of work (per Equation (2.17)) since the temperature constraint is already over-satisfied.

The four design-relevant regimes packaged in s are summarised in Table 2.1.

2.6 2D thermal model: hot spots and the geometric crisis

The 1D model above treats the chip as a single thermal node. In a real chip the heat is concentrated at functional-unit hot spots — ALU clusters, AVX units, tensor cores, HBM stacks — on top of a much cooler bulk substrate. The transition from a 1D “average” model to a 2D / 3D thermal landscape evolves the conduction bottleneck from a material limitation into a geometric crisis: bulk cooling cannot reach internal hot spots effectively, and the global COP is dragged down by a small number of localized peaks.

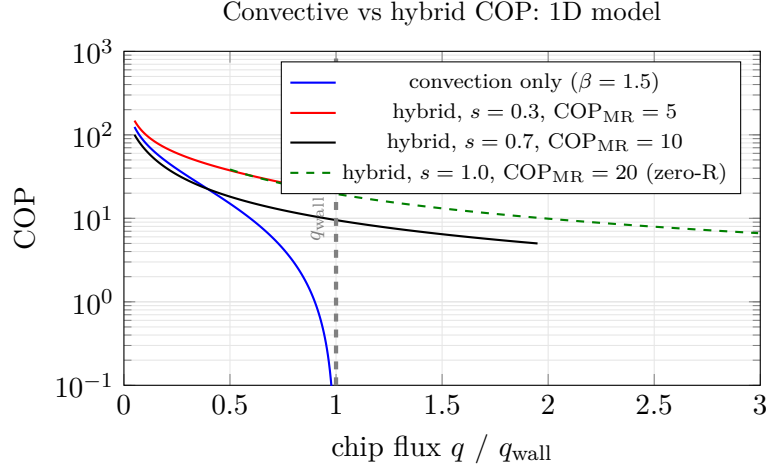


Figure 2.3: Convective and hybrid (convection + MR) COP from (2.4) and (2.14). Increasing the MR loading s pushes the operable flux past q_{wall} and replaces the catastrophic convective collapse with a smooth roll-off set by COP_{MR} . Numerical values are illustrative; the physical message is that the wall moves rightward and softens with increasing s .

Multi-node decomposition

Replace the single thermal node by a network: each functional unit i carries a local heat flux q_i , with a local thermal resistance $R_{\text{die},i}$ to the package back-side and a shared external resistance $R_{\text{ext}}(\dot{V})$ to the coolant. The shared coolant must satisfy the worst-case thermal limit:

$$T_c + R_{\text{ext}}(\dot{V}) \bar{q} + R_{\text{die},i} q_i \leq T_{\text{lim}} \quad \forall i, \quad (2.20)$$

where $\bar{q} = \sum_i A_i q_i / \sum_i A_i$ is the area-weighted bulk flux into the cold plate. The hottest q_i sets the required $\dot{V}$; since $W_f(\dot{V})$ is monotonic and convex, the worst-case node forces the entire convective system into a high-flow regime even when the bulk $\bar{q}$ is moderate.

The ratio $\alpha \equiv \max_i q_i / \bar{q}$ (peak-to-bulk power density) is the geometric crisis parameter. $\alpha = 1$ recovers the 1D model; modern CPU / GPU values are $\alpha \sim 5\text{--}20$ (short-lived bursts can be much higher). In multi-node form,

$$\text{COP}_{\text{conv}}^{(2D)}(\bar{q}, \alpha) = \frac{\bar{q} A_{\text{tot}}}{W_f(\dot{V}_\star(\alpha \bar{q}))}, \quad (2.21)$$

where $\dot{V}_\star$ is set by the worst-case node. For fixed $\bar{q}$, increasing α raises $\dot{V}_\star$ (more flow needed to handle the peak) and lowers $\text{COP}_{\text{conv}}^{(2D)}$. The bulk cooler is paying for cooling the entire die just to satisfy one hot spot — the energy waste that microrefrigeration is designed to eliminate.

Spot cooling and System COP

Place an MR locally at each hot spot and let it lift only the peak heat flux. Then the convective side sees the bulk $\bar{q}$ without the geometric inflation: the worst-case constraint (2.20) is satisfied at the hot spot by the MR (which carries $(q_i - \bar{q}) A_i$ of the local excess), and the convective $\dot{V}$ drops to the bulk requirement. Schematically

$$\text{COP}_{\text{hyb}}^{(2D)} = \frac{\bar{q} A_{\text{tot}}}{W_f(\dot{V}(\bar{q})) + \sum_i (q_i - \bar{q}) A_i / \text{COP}_{\text{MR}}}. \quad (2.22)$$

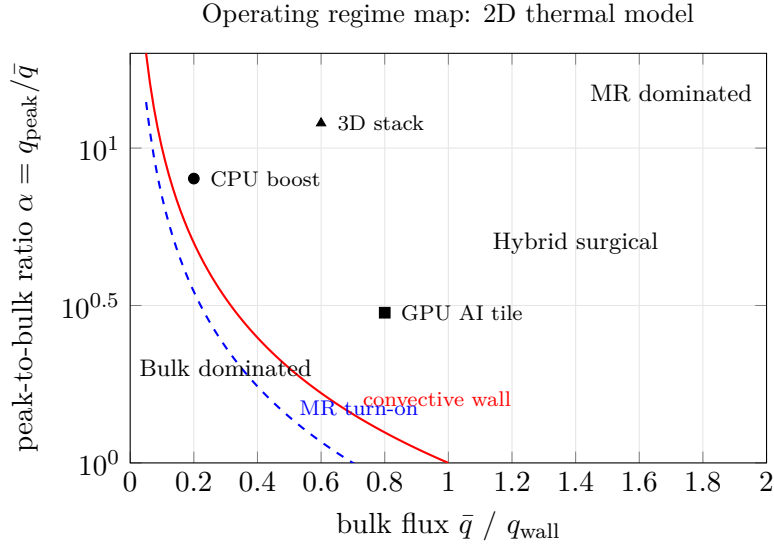


Figure 2.4: Operating regime map in the bulk-flux $\bar{q}$ vs peak-to-bulk ratio α plane. The convective wall (red) is $\alpha\bar{q} = q_{\text{wall}}$; below it, classical convection alone is sufficient. Above the MR turn-on contour (blue, dashed), microrefrigeration is required to keep the bulk system in its high-COP regime. Markers indicate representative trajectories of three chip architectures, after the architecture table on slide 49 of the design deck.

The MR pays only for the *excess* flux above the bulk; the convective system stays in its high-COP regime; and the system COP recovers to the 1D-bulk value plus a small MR-loop correction. This is what slide 28 of the design deck calls “managing the System COP by avoiding the energy waste of cooling the entire die to satisfy a single hot spot.”

Regime map in $(\bar{q}, \alpha)$

Figure 2.4 plots the operating regimes in the $(\bar{q}, \alpha)$ plane. Three regions emerge:

- **Bulk-dominated** (low α , low $\bar{q}$): convection alone is sufficient; MR provides marginal benefit.
- **Hybrid surgical relief** (intermediate α , moderate $\bar{q}$): convection handles the bulk, MR intercepts hot spots. This is where the design wins.
- **MR-dominated** (high α or high $\bar{q}$ beyond q_{wall}): convection cannot reach internal hot spots; MR carries most of the load.

Different chip architectures occupy different paths through this plane: CPU-boost workloads sit at low $\bar{q}$ but high α (the “edge crowding” regime); GPU / AI tile densification tracks toward high $\bar{q}$ with moderate α ; 3D-stacked logic combines high $\bar{q}$ and very high α (the “catastrophic regime” of slide 52).

Final COP-vs- q comparison

Figure 2.5 replots the convective COP and the hybrid COP in the 2D model, with the convective wall now set by the worst-case node. The convective curve crashes earlier than in 1D because the geometric concentration α effectively shrinks the operable bulk flux to q_{wall}/α ; the hybrid curve, by contrast, removes that penalty completely if the MR can intercept the peak excess. Two qualitative wins emerge:

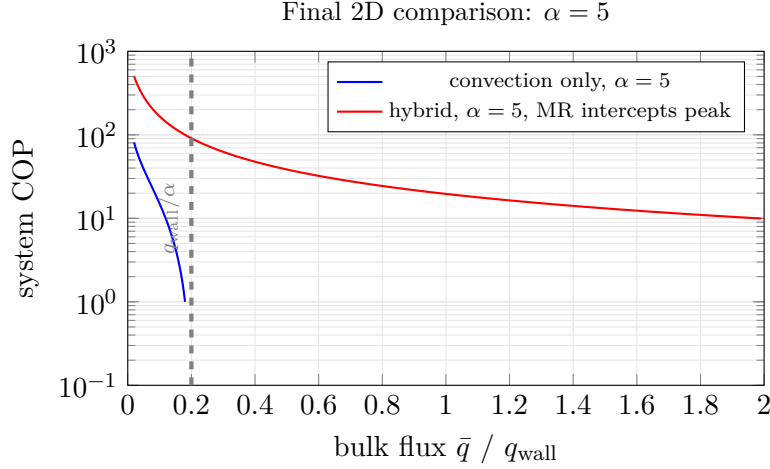


Figure 2.5: 2D-model COP vs bulk flux for $\alpha = 5$. The convective-only curve crashes at $\bar{q} = q_{\text{wall}}/\alpha$ (the geometric wall, much earlier than the 1D wall); the hybrid system replaces this crash with a smooth roll-off because the MR removes the geometric penalty. Both curves use illustrative numerics; the hard message is the $5\times$ rightward shift of the operable region.

- the operable bulk flux extends well past q_{wall}/α without the COP crash;
- at any given bulk flux, the hybrid COP is significantly higher because the convective system stays in its efficient regime.

2.7 Temporal targeting: matching the workload

The 2D analysis above assumes *steady-state* $\bar{q}$ and α . Real chip workloads are temporally bursty: AVX bursts, matrix-multiply phases, memory-bandwidth-limited stalls all produce $q_i(t)$ profiles with sub-microsecond fluctuations (Figure 1.8). The convective system has no temporal agility — pumps and fans run on millisecond-to-second timescales, so they must be sized for the *worst-case-over-window* flux. Photonic cooling is fundamentally different: the laser pump, the photonic network, and the LPC all respond on ns– μ s timescales, so the MR can be gated to follow the workload directly.

From peak-sized to RMS-sized cooling

Let the chip workload have temporal moments $\bar{q}_t = \langle q(t) \rangle_t$ and $q_{\text{peak}}^{(t)} = \max_t q(t)$, with $\alpha_t \equiv q_{\text{peak}}^{(t)} / \bar{q}_t$. Convective cooling, sized to satisfy the thermal limit at the temporal peak, delivers

$$W_f^{(\text{conv})} = W_f(\dot{V}_*(\alpha_t \bar{q}_t)), \quad (2.23)$$

i.e. pays for cooling sized at $q_{\text{peak}}^{(t)}$, even when the time-averaged $\bar{q}_t$ is much smaller. Photonic microrefrigeration with sub-microsecond gating instead pays only

$$W_f^{(\text{conv})}(\bar{q}_t) + \langle Q_{\text{MR}}(t) \rangle_t / \text{COP}_{\text{MR}}, \quad (2.24)$$

where the MR is engaged only when $q(t) > \bar{q}_t$. The temporal work saved is roughly

$$\Delta W_f^{(\text{temporal})} = W_f(\dot{V}_*(\alpha_t \bar{q}_t)) - W_f(\dot{V}_*(\bar{q}_t)), \quad (2.25)$$

which for a fan curve $W_f \propto \dot{V}^\beta$ and $R_{\text{ext}} \propto 1/\dot{V}$ translates to a factor of order α_t^β reduction in convective work — the same factor that distinguishes peak-sized from RMS-sized cooling. The MR cost is proportional to the *temporal excess* $\langle \max(q(t) - \bar{q}_t, 0) \rangle_t$, which is the duty cycle times the burst amplitude. For a typical CPU workload with $\sim 10\%$ duty cycle on AVX bursts, this is roughly $0.1 \times (\alpha_t - 1)\bar{q}_t$.

Spatial + temporal targeting: the full model

Combining the 2D spatial decomposition with sub-microsecond temporal gating, the operative figure of merit becomes

$$\text{COP}_{\text{system}} = \frac{\bar{q}_t A_{\text{tot}}}{W_f(\dot{V}_*(\bar{q}_t)) + \sum_i \langle \max(q_i(t) - \bar{q}_t, 0) \rangle_t A_i / \text{COP}_{\text{MR}}}. \quad (2.26)$$

The convective denominator term is now sized at the time-averaged *bulk* flux; the MR term picks up only the spatial-temporal excess. Both savings (the α factor from spatial targeting and the duty-cycle factor from temporal targeting) compound.

Why photonic cooling can do this and convection cannot. The photonic cold plate has three timescales available to its control loop:

- *Laser pulse rate* (MHz–GHz): the laser can be driven in CW, Q-switched, or pulsed mode (Figure 1.8). Pulse repetition rate is matched to the workload’s burst frequency.
- *Photonic-network reconfiguration* (μs –ms): metasurface or switchable-coupler elements can route pump light to specific tiles within $\sim 1 \mu\text{s}$, tracking which functional units are presently active.
- *Thermal control loop* (ms): the sensor layer of Section 4.18 closes a feedback loop on T_h at each tile.

Convective systems lack the first two of these. A pump or fan has inertia and cannot reroute coolant to a specific hot spot at μs timescales; the only tool is to oversize the flow. A photonic cold plate gates the cooling at the speed of light and reroutes it at the speed of the photonic network — both faster than the thermal time constants of the chip — so the time-averaged work is bounded below by the bulk requirement, not the peak.

2.8 Architecture matrix: who needs microrefrigeration?

Table 2.2 consolidates the regime map of Figure 2.4 with the convection-wall analysis and the hotspot-amplification ratio α/β (Equation (2.20)) into a chip-class ranking. The opportunity for microrefrigeration scales with α/β and the absolute peak flux; for 3D-stacked compute, RF / GaN / SiC power dies, and chiplet packages with HBM, MR is no longer optional but a survival factor.

2.9 Why laser cooling: contrast with thin-film TECs

The microrefrigeration mechanism is not specific to laser cooling; thin-film thermoelectric coolers (μTECs) implement the same *s*-knob via the Peltier effect with established Bi_2Te_3 or SiGe materials. Two recent demonstrations [4, 5] have shown μTEC hotspot clipping with measurable performance recovery on real silicon. However, μTECs face four physics constraints that laser cooling does not:

Rank	Chip type	α/β	Peak flux (W/mm ²)	MR opportunity
1	NAND / low-power memory	0.3–3	0.01–0.1	low
2	DRAM / commodity memory	0.5–5	0.05–0.3	low–medium
3	HBM stack	0.8–12	0.2–1	moderate–high
4	Mobile SoC	1.3–25	0.5–3	high (bursty)
5	Server CPU	2–25	1–8	high
6	GPU / AI accelerator	1.5–14	1–10	very high
7	Chiplet / 2.5D HPC package	2–60	2–15	very high
8	3D logic-on-logic stack	4–120	5–30	extremely high (mandatory)
9	RF / GaN / SiC power IC	7–800	20–300+	extreme (no alternative)

Table 2.2: Architecture opportunity matrix for microrefrigeration. α is the hot-region power fraction; β is the hot-region area fraction; the ratio α/β is the local heat-flux amplification relative to chip average. Bulk convection sees $\bar{q}$; silicon performance sees the peak. Rows 4–9 sit above the MR-turn-on contour of Figure 2.4 during normal operation.

- **Parasitic Joule heating.** TEC cooling power is $\dot{Q}_c = ST_c I - K\Delta T - \frac{1}{2}I^2 R$; the $I^2 R$ term grows quadratically with current and eventually dominates. Practical Bi₂Te₃ thin-film TECs sit at COP_{MR} $\sim$ 0.2–0.5 at $\Delta T_{\text{lift}} \sim$ 10 K with peak cooling density $\dot{Q}_c/A \sim$ 1–10 W/mm².
- **Electrical contact resistance.** Direct contact with the silicon active layer is required, and contact resistance scales adversely below $\sim$ 100 μm tile pitch; sub-10 μm spatial targeting is hard.
- **Bulk system absorbs MR waste.** Both $\dot{Q}_c$ and the electrical input $\dot{Q}_c/\text{COP}_{\text{MR}}$ eventually exit through the same cold plate, so the bulk loop must size for the increased load. This eats into the effective hybrid COP.
- **Limited lift.** $\Delta T_{\text{lift}} \lesssim$ 10 K per stage; multi-stage cascades multiply parasitics.

Laser cooling sidesteps all four:

- No parasitic Joule heating; the optical pump is coherent / work-like (Chapter 1), and the heat exits as fluorescence photons.
- No electrical contact; spatial targeting is set by the photonic-network resolution ($\sim \mu\text{m}$ achievable, $\sim$ ns switching).
- Bulk system does *not* reject MR waste; the fluorescence couples directly to the LPC and feeds back into the electrical loop (Chapter 7).
- ΔT_{lift} is set by phonon relaxation and material purity, not by Joule heating; $\sim$ 100 K lifts have been demonstrated with rare-earth-doped crystals [6].

The implication for Equation (2.18) is more nuanced than a simple COP comparison. The gross laser-cooling COP is *lower* than a μTEC 's ($\eta_P \eta_{\text{ASF}} \simeq$ 0.02–0.07), because η_{ASF} is bounded by the anti-Stokes quantum defect; what the photonic loop offers instead is (i) recovery of most of that pump energy through the LPC, so that the *net* λ_m reaches 0.1–0.3 and approaches $\sim$ 1 only as $\eta_P \eta_{\text{LPC}} \eta_{\text{cpl}} \rightarrow$ 1 (Chapter 7); (ii) no waste-heat deposition on the die — the unrecovered pump energy leaves as fluorescence to the LPC, not as Joule heat next to the hot spot; and (iii) spatial and temporal targeting two to three orders of magnitude finer than a μTEC . At the level of the hybrid system, (ii) and (iii) are what make laser cooling a hotspot razor whose ROI does not depend on beating the μTEC 's COP (Section 10.6.6); (i) is what would let it improve system perf/W uniformly across the chip flux, and it is contingent on the LPC. Table 2.3 summarises the comparison.

Metric	Thin-film TEC (μTEC)	Laser / photonic cold plate
Mechanism	Peltier (electrical)	Anti-Stokes fluorescence (optical)
Intrinsic COP_{MR}	0.2–0.5	$\eta_{\text{P}}\eta_{\text{ASF}} \simeq 0.02\text{--}0.07$ gross (bulk); 0.1–0.3 with LPC recycling; $\rightarrow 1$ and beyond at the Tier II/III design point (Section 8.3.4)
$\lambda_m = \text{COP}_{\text{MR}}/\text{COP}_{b,0}$	0.1–0.3	0.05–0.3 (bulk, recycling-limited); $\gtrsim 1$ at the design point
Spatial resolution	$\gtrsim 100\ \mu\text{m}$	$\sim 1\ \mu\text{m}$
Temporal resolution	$\sim \text{ms}$	$\sim \text{ns--}\mu\text{s}$
Cooling density	1–10 W/mm ²	0.1–10 W/mm ² (bulk extractors, Table 8.4); $\gtrsim 10^3$ W/mm ² at the photonic design point (Table 8.2)
ΔT_{lift}	$\lesssim 10\ \text{K}$	$\lesssim 100\ \text{K}$
Bulk-loop waste-heat penalty	yes (MR waste $\rightarrow$ cold plate)	no (recycled via LPC)
Electrical contact to active layer	required	none (free-space / waveguide pump)

Table 2.3: Side-by-side comparison of thin-film thermoelectric (μTEC) and laser / photonic cold-plate microrefrigeration. Both implement the same s -knob of (2.13), but laser cooling’s spatial / temporal targeting is two to three orders of magnitude finer and the bulk loop is not burdened by MR waste heat — the pump energy leaves as fluorescence and is recycled via the LPC of Chapter 7. Its gross COP is lower than a μTEC ’s; its net COP depends on $\eta_{\text{P}}\eta_{\text{LPC}}$.

2.10 Chapter summary

The microrefrigeration story collapses to a small set of design relations:

- *Three thermal walls*, not one: $q_{\text{passive}}^{(0)}$ (passive turn-on), $q_{\text{real-wall}}$ (pump-limited), q_{wall} (idealised die-only) — (2.12).
- *Cold-plate sub-model* with explicit $R_{\text{series}}/R_{\text{stag}}/R_{\text{plumb}}$ and pump law $W_{\text{static}} + K\dot{V}^n$, giving the piecewise COP (2.9).
- *Normalised λ -form* (2.18): a single dimensionless comparison of bulk loop versus MR technology.
- *Closed-form optimal MR strength* (2.19): $s^*(q)$ ramps linearly from zero at $q_{\text{passive}}^{(\text{hyb})}$ through $s = 1$ at q_{wall} to s_0 at the zero-resistance-die point.
- *Six regimes of s* (Table 2.1): clipping $\rightarrow$ interception $\rightarrow$ nullification $\rightarrow$ zero-resistance $\rightarrow$ over-cooling.
- *2D + temporal targeting*: spatial selectivity (α/β amplification) and temporal selectivity (workload-aligned duty cycle) together yield the system COP of (2.26).
- *Architecture matrix* (Table 2.2): the MR opportunity rises monotonically with α/β and peak flux; 3D stacks, GPUs / AI accelerators, and power ICs are the binding-constraint applications.
- *Laser cooling vs. μTEC* (Table 2.3): laser cooling gives sub-micron spatial / sub-microsecond temporal resolution and zero bulk-loop waste-heat penalty at a gross COP below a μTEC ’s; its net λ_m is set by LPC recycling. These are why the rest of the book is about photonic, not thermoelectric, microrefrigeration.

Chapter 3

Thermodynamic Considerations for Lasers, Fibers & Laser Power Converters

3.1 Scope

Chapter 1 introduced four loop-level efficiencies — $\eta_P, \eta_{\text{cpl}}, \eta_{\text{ASF}}, \eta_{\text{LPC}}$ — and bundled the photonic and electro-optical infrastructure (laser, fiber, LPC) into single numerical multipliers. Chapter 2 compared the cooling architecture against classical convection. This chapter unpacks the three component efficiencies that the master expression (1.17) folds together:

- **Laser** — wall-plug efficiency η_P including wavelength regime and commercial-technology constraints (edge-emitting diodes vs. VCSELs);
- **Fiber / pump delivery** — the coupling factor η_{cpl} including standard SMF vs. hollow-core fiber physics, coupling losses, and damage / power limits;
- **LPC** — the optical-to-electrical efficiency η_{LPC} , revisited from the wavelength-constraint and radiation-entropy point of view of Chapter 7.

A photonic cooling loop with poor η_P , η_{cpl} , or η_{LPC} fails the feasibility test (1.17) regardless of how good the ASF physics is; conversely, perfecting the ASF physics pays no dividend if these three components throw away the gain. We then close the chapter with an *information-theoretic* view of the same machine: a chip is a Landauer-bounded computing device whose irreversible bit operations are the deepest origin of the heat that ASF cooling extracts. Photonic cooling exports that computational entropy through the fluorescence channel and the LPC at higher exergy than convective cooling can, with material consequences for the system-level entropy budget.

The information-theory portion is built directly on Gershenfeld’s “Signal entropy and the thermodynamics of computation” [7] in IBM Systems Journal Vol. 35 (1996), and its parent Landauer / Bennett / Shannon / Fredkin–Toffoli chain [8–11].

3.2 Laser thermodynamics

Wall-plug efficiency

The wall-plug efficiency $\eta_P \equiv P_L/P_{in}$ of a semiconductor laser is the product of several internal efficiency factors:

$$\eta_P = \eta_{inj} \eta_{rad} \eta_{ext} \eta_V, \quad (3.1)$$

with

- η_{inj} — carrier injection efficiency (how many of the electrons that cross the junction reach the active region without thermalising in cladding / contact layers);
- η_{rad} — radiative recombination efficiency of the active region (radiative vs. non-radiative recombination);
- η_{ext} — external photon-extraction efficiency through the front facet / aperture;
- $\eta_V \equiv \hbar\omega/(qV_{op})$ — the voltage efficiency, the ratio of the photon energy to the applied junction voltage at the operating point.

η_V is bounded above by unity and is reached only at ideal operating points; series resistance, contact drops, and thermal voltage rise above $qV > \hbar\omega$ at finite current.

The energy lost from $1 - \eta_P$ is dissipated as heat at the laser junction, which must itself be removed by a (typically conventional) thermal management scheme. This heat is part of the $Q_{rej,L} = (1 - \eta_P)P_{in}$ flow into the ambient sink in the exergy diagram of Figure 1.5.

Wavelength regimes

The laser must absorb resonantly into the extractor material, and the extractor in turn determines the available cooling regime. Common high-WPE laser-emitter combinations:

Wavelength	Active region	Pump for ASF in	Typical WPE
808 nm–980 nm	GaAs / InGaAs QW	Yb-doped crystals / glasses [6]	0.50–0.70
1020 nm–1080 nm	InGaAs / GaInP	Yb:YLF (Section 8.1.1)	0.55–0.65
1450 nm–1530 nm	InGaAsP / InP	Er-doped, Tm-doped [6]	0.30–0.50
1900 nm–2050 nm	GaSb / AlGaInAsSb	Tm:YLF, Ho:YAG [6]	0.20–0.40

Two physical regularities sit behind the table. First, WPE peaks near 980 nm for InGaAs because the QW band-edge sits near the GaAs substrate’s transparency window and the voltage drop qV_{op} is well-matched to $\hbar\omega$; this is also where Yb-doped extractors have their absorption peak, which is why Yb:YLF is the dominant ASF platform. Second, longer-wavelength GaSb-class lasers suffer from larger non-radiative recombination (Auger), so the loop’s η_P is intrinsically lower for the 2 μ m regime; this argues against placing the loop on the longer-wavelength side unless the available extractor physics demands it.

Edge-emitting diodes vs. VCSELs

Two commercial laser geometries dominate practical pump sources for photonic cooling.

Edge-emitting diode lasers (EELs). Stripe-geometry lasers in which the optical mode propagates along the substrate; the emission is from a polished facet at one edge. Single-emitter devices reach $\sim 60\%$ – 70% WPE at 1 W-class output near 980 nm; bar / stack arrays scale to multi-kW

with somewhat lower per-emitter WPE ($\sim 50\%$ – 60%). The output beam is asymmetric (highly divergent in the fast axis, less so in the slow axis) and requires aspheric collimation before efficient fiber coupling.

Vertical-cavity surface-emitting lasers (VCSELs). The optical mode propagates vertical to the wafer, with DBRs forming the cavity and a small (μm -scale) aperture defining the mode. Per-emitter powers are smaller ($\sim 1\text{ mW}$ – 1 W) but VCSELs scale beautifully into 2D arrays — thousands of emitters on a single chip — and exhibit a circular, symmetric beam that couples efficiently into multimode fibers. Per-emitter WPE at 980 nm is typically $\sim 40\%$ – 55% ; the array geometry and high modulation bandwidth make VCSELs the natural choice when either many spatially distributed pump beams (one per cold-plate tile, Figure 1.14) or fast pulse modulation (Section 2.7) is required.

The choice is a trade between WPE (favour EELs) and beam quality / spatial multiplexing / modulation bandwidth (favour VCSELs). For the device of this book, where the cold plate is a tiled array driven by many independent pump channels with sub- μs temporal gating, VCSEL arrays are the operating-point baseline; the loop-level effect of their lower WPE is captured directly in (1.17).

3.3 Fiber delivery and the coupling efficiency η_{cpl}

The pump optical power produced by the laser must be delivered to the extractor. In the rack-scale architecture of Figure 1.12, this is via fiber: a top-of-rack pump source distributes light through a fiber bundle to many cold plates. The fiber thermodynamics matter because every loss between the laser and the extractor is an additive contribution to $1 - \eta_{\text{cpl}}$, which enters the loop bound (1.17) linearly.

Standard silica fiber

A single-mode silica fiber (SMF) at 980 nm – $1.5\mu\text{m}$ has propagation loss of $\sim 0.5\text{ dB/km}$ and tolerates $\sim 1\text{ W}$ – 10 W CW before nonlinear damage (stimulated Raman scattering, stimulated Brillouin scattering, or material breakdown at high local intensities at the launch facet). Multimode silica fiber tolerates higher absolute power but at lower brightness. For practical pump-delivery applications:

- Per-emitter VCSEL-to-fiber coupling: $\sim 50\%$ – 80% (Fresnel + mode mismatch);
- EEL-to-fiber coupling with aspheric optics: $\sim 70\%$ – 90% ;
- Fiber-to-cold-plate coupling (free-space focus through a coupler): another $\sim 80\%$ – 95% .

Stack these and the laser-to-extractor coupling efficiency η_{cpl} is typically in the range 40% – 75% .

Hollow-core fibers

Hollow-core fibers (HCFs) — anti-resonant tube lattices, photonic bandgap fibers, nested anti-resonant nodeless fibers (NANFs) — guide light through air rather than glass, which collapses the nonlinearity by a factor of 10^3 – 10^4 and raises the damage threshold to the kW-class. For the photonic-cooling application, this matters because the per-tile pump-flux requirement ($\sim 100\text{ W}$ at 1 mm pitch in Figure 1.11) is precisely the regime where SMF is marginal and HCF is comfortable. HCF additionally:

- carries less thermal load (the fiber itself absorbs much less, so it does not heat in the way a high-power-loaded SMF does);

- supports broadband transmission (the photonic-bandgap regime can be designed for visible, near-IR, or mid-IR operation, so a single fiber type can deliver pumps at any of the wavelength regimes of Section 3.2);
- has slightly higher propagation loss (~ 1 dB/km typical, lab record ~ 0.1 dB/km at $1.55 \mu\text{m}$), which is irrelevant for the m-scale fiber lengths at the rack level but worth noting.

Coupling efficiency into HCF is comparable to SMF ($\sim 70\%$ – 90% from a free-space beam) once the launch optics are matched to the hollow-core mode. The per-fiber damage threshold is ~ 1 kW– 10 kW CW. Hollow-core fiber is the operating-point baseline for the rack-scale architecture of this book.

What η_{cpl} actually folds into

In the loop equation (1.4), $P_{\text{abs}} = \eta_{\text{cpl}} P_{\text{L}}$, so η_{cpl} is the product of every loss between the laser-output facet and the extractor’s absorption volume:

$$\eta_{\text{cpl}} = \eta_{\text{laser} \rightarrow \text{fiber}} \eta_{\text{fiber prop}} \eta_{\text{fiber} \rightarrow \text{coupler}} \eta_{\text{coupler} \rightarrow \text{extractor}}. \quad (3.2)$$

The first three are conventional fiber-optics numbers; the last is the photonic-network in-coupling of Chapter 4, which is set by the impedance match between the photonic-network mode and the extractor’s electronic dipole. Improvements to any factor enter (1.17) multiplicatively; the loop is most sensitive to whichever is currently smallest. For the present operating-point assumptions ($\eta_{\text{laser} \rightarrow \text{fiber}} \sim 0.8$, HCF propagation ~ 0.99 at m-scale lengths, $\eta_{\text{fiber} \rightarrow \text{coupler}} \sim 0.9$, $\eta_{\text{coupler} \rightarrow \text{extractor}} \sim 0.9$), the bottleneck is at the laser-to-fiber stage — the same stage where high-WPE EEL bars fight a brightness disadvantage that VCSEL arrays handle with grace.

3.4 LPC thermodynamics revisited

Chapter 7 developed the LPC bound from radiation exergy (7.1) and the entropy-management view of multi-junction architectures (7.5). We collect two component-level points that sit alongside the laser and fiber analysis above.

Wavelength matching to the fluorescence

The LPC’s bandgap must match the fluorescence centre frequency $\bar{\omega}_{\text{f}}$; the per-junction efficiency drops sharply when $\hbar\bar{\omega}_{\text{f}} \not\approx E_{\text{g,LPC}}$ for either of two reasons. Excess photon energy ($\hbar\omega - E_{\text{g}} > 0$) thermalises in the junction — the standard Shockley–Queisser thermalisation loss — and contributes to $Q_{\text{rej,P}}$ in Figure 1.5. Photons with $\hbar\omega < E_{\text{g}}$ are not absorbed, so they leak past the LPC and either dissipate elsewhere or escape the package. Both penalties grow with the spectral width of the fluorescence relative to the bandgap, which is the engineering case for the multi-junction architectures of Section 7.4.

Where the LPC’s entropy rejection physically goes

The recovered electrical power $P_{\text{rec}} = \eta_{\text{LPC}} P_{\text{f}}$ is exergy; the rejected $Q_{\text{rej,P}} = P_{\text{f}} - P_{\text{rec}}$ is heat at T_0 dumped from the LPC substrate / heat-sink to the same ambient that absorbs the laser’s rejected heat. The bound $\eta_{\text{LPC}} \leq \eta_{\text{LPC,max}}$ of (1.15) says only the heat-derived portion of P_{f} is Carnot-limited; the pump-derived portion is in principle fully recoverable. At the component level, this distinction shows up as a difference between the LPC’s open-circuit voltage V_{oc} at low fluorescence flux (where the heat-derived contribution is a small fraction of P_{f}) and at high flux

(where the cooling-cycle is loaded so that $\eta_{\text{ASF}}P_{\text{abs}}$ is comparable to P_{abs}). Calibrating V_{oc} as a function of incident fluorescence flux is therefore the natural laboratory test of the exergy ceiling (1.15) — a useful experimental marker for next-generation LPCs.

Temperature dependence of η_{ASF} and the per-zone operating optimum

The exergy bound (1.15) carries an explicit T_{h} through the Carnot factor $(1 - T_0/T_{\text{h}})$; less obviously, η_{ASF} itself has a T_{h} -dependence that couples the two through the LPC. Two microscopic origins are worth flagging because they determine the material choice for each thermal zone of Section 10.9:

- *Sub-quenching regime*, η_{ASF} rises with T_{h} . The anti-Stokes shift per absorbed pump photon is $\hbar(\omega_{\text{f}} - \omega_{\text{p}}) \sim k_{\text{B}}T_{\text{h}}\bar{n}_{\text{ph}}$, so the cooling energy per absorbed photon grows with T_{h} ; each pump photon in the extractor carries away a larger phonon-derived fraction of its emitted energy. In this regime the Carnot factor $(1 - T_0/T_{\text{h}})$ and the numerator η_{ASF} both widen with T_{h} , and the LPC exergy ceiling $\eta_{\text{LPC,max}} = [1 + \eta_{\text{ASF}}(1 - T_0/T_{\text{h}})]/(1 + \eta_{\text{ASF}})$ improves with the product.
- *Above the quenching temperature*, η_{ASF} rolls over. Concentration quenching, thermal population of upper-manifold phonon sidebands, and thermally-activated non-radiative recombination ($w_{\text{nr}}(T)$) reduce η_{EQE} . For Yb:YLF the rolloff sets in above ~ 400 K [6]; for Ho- and Tm-doped hosts the useful window extends further, and for wide-bandgap-hosted rare-earth systems (e.g. SiC:Er) the ceiling has been reported into the 600–800 K range.

The consequence is a *material-dependent optimum temperature* T_{h}^* per tile, determined by the crossover of the rising Carnot factor and the falling $\eta_{\text{EQE}}(T)$. Below T_{h}^* the loop trades exergy recovery for cooler operation; above it, quenching wins and net recoverable work falls. The heterogeneous-thermal template of Section 10.9 exploits this in two directions: compute-zone tiles are matched to extractor materials whose T_{h}^* sits in the 500–600 K band where the Carnot factor is wide, while storage / control zone tiles use low-quenching Yb-based hosts whose T_{h}^* is nominal but whose sub-ambient operation collapses subthreshold leakage. The LPC exergy bound is therefore not a single number for the whole chip; it is a per-zone number tied to each tile’s operating temperature and extractor material.

3.5 Information-theoretic considerations

The chapters so far have treated the chip as a thermal source: it dissipates power q , which must leave through the package. The chip is also *a computation*: it consumes information at its inputs, transforms it through irreversible logic operations, and emits a structured information stream at its outputs. The two views are linked by Landauer’s principle and the broader signal-entropy framework of Gershenfeld [7]. Following that paper: “computers are unambiguously thermodynamic engines that do work and generate waste heat . . . developing low-power computing is going to require as much attention to entropy as to energy.”

3.5.1 Three energy scales: $k_{\text{B}}T$, $CV^2/2$, and the workload-entropy bound

The chip-scale information thermodynamics has three energy scales that differ by orders of magnitude [7].

(i) **Landauer floor.** The erasure of one bit of information at temperature T_{chip} costs at least

$$E_{\text{Landauer}} = k_{\text{B}} T_{\text{chip}} \ln 2 \quad (3.3)$$

of dissipated heat [8]. Each irreversible logic operation reduces the logical state space of the system; if the binary value of a bit is unknown, erasing it shrinks the phase space available to the system by $k_B \ln 2$, and the corresponding heat $T dS$ is dumped to the environment. At $T_{\text{chip}} = 350 \text{ K}$, $k_B T_{\text{chip}} \ln 2 \approx 3.3 \times 10^{-21} \text{ J per bit}$.

(ii) CMOS electrostatic erasure. In conventional CMOS, each transistor’s gate / line capacitance C is charged to supply voltage V and dumped to ground on each switching event, dissipating $\frac{1}{2}CV^2$ per transition. Following [7], for a “hot” chip with $\sim 10^7$ transistors switching at $\sim 100 \text{ MHz}$, $\frac{1}{2}CV^2$ per bit is ~ 7 orders of magnitude above $k_B T$:

$$\frac{\frac{1}{2}CV^2}{k_B T_{\text{chip}} \ln 2} \sim 10^6 - 10^7. \quad (3.4)$$

The Landauer bound is therefore far below the operating-point per-bit energy, but the same arguments *about erasing and moving bits* apply at the much larger CMOS scale: charge-recovery (adiabatic) logic exists precisely to recover some of $\frac{1}{2}CV^2$ by treating it as free energy that can be returned to the supply, exactly as Landauer’s bit-erasure argument applies at $k_B T$ [7, 9]. The two-orders-of-magnitude Sommerfeld-electronic-entropy term sits between $k_B T$ and $CV^2/2$ and is also worth a thought, but neither dominates the dissipation.

(iii) Workload signal entropy. Gershenfeld’s central contribution is to make the bridge between the workload’s *measurable* signal entropy at the chip’s I/O boundary and the thermodynamic limit on dissipation. Define for a sequential computation with input symbol stream $\{x_n\}$ and output stream $\{y_n\}$ the per-symbol *reverse* information

$$I_{\text{reverse}} = H(x_n; x_{n+1}, \dots, x_{n+d}; y_n, y_{n+1}, \dots, y_{n+d}; n) - H(x_{n+1}, \dots, x_{n+d}; y_n, \dots, y_{n+d}; n), \quad (3.5)$$

which measures the bits of x_n that cannot be predicted from the future of the system, and analogously the per-output *forward* information I_{forward} . The difference $\Delta I = I_{\text{reverse}} - I_{\text{forward}}$ is the net information *consumed* per cycle by the system, and bounds the informational heating / cooling load that the underlying physics must absorb [7]:

$$\boxed{\dot{S}_{\text{comp}} \geq k_B \ln 2 \dot{N}_{\text{ops}} \Delta I,} \quad (3.6)$$

where $\dot{N}_{\text{ops}}$ is the per-second operation rate and ΔI is in bits/op. The workload itself sets a data-driven, technology-independent floor on the entropy production of any implementation. No measurement of the chip’s actual temperature can tell you whether π ’s digits are random [7]; equivalently, no clever engineering can reduce dissipation below the workload-entropy floor ΔI .

3.5.2 Functional units as thermalised information buffers

A functional unit (an ALU, an AVX execution port, a tensor-core matrix-multiply lane) holds a working state in flip-flops, SRAM caches, and register files. Each clocked operation:

1. reads a (low-entropy) input state from registers,
2. performs an irreversible logic operation,
3. writes a (low-entropy) output state back,
4. dissipates $\sim \frac{1}{2}CV^2$ of charge-erasure energy plus, in principle, no less than $k_B T_{\text{chip}} \ln 2 \Delta I$ of bit-erasure heat (3.6) into the local lattice.

The local lattice is the *entropy buffer*: it absorbs the information-erasure entropy generated by the operation and stores it as phonon energy at T_{chip} until it can be carried away. At hot-spot rates of TFLOPS per mm^2 , with bit-flip counts of $\sim 10^2$ per FLOP, and the $\sim 10^6$ - 10^7 above-Landauer CMOS overhead of (3.4), local heat-flux densities reach the kW/mm^2 regime — exactly the geometric crisis flagged in Section 2.6. In Gershenfeld’s framing, the hot spot *is* an entropy buffer; in our framing, ASF cooling empties that buffer through the optical channel before the phonon-conduction pathway has to do all the work.

3.5.3 Information content vs. phonon entropy of the lattice

A computation’s information content (the bits in its inputs, outputs, and internal state) is *small* compared to the physical-state entropy of the chip’s lattice. A modern processor core might hold $\sim 10^6$ bits of architectural state, while the same chip’s lattice at 350 K carries phonon entropy of order $N_{\text{atoms}} k_B \sim 10^{23} k_B$ — $\sim 10^{17}$ times more. This is the “many-orders-of-magnitude gap between thermodynamic and information-bearing degrees of freedom” explicitly named in [7]. Removing the chip’s heat is removing phonon entropy; the architectural state (the information being computed) flows through the chip’s I/O on a separate electrical channel. The two pathways must not be conflated — ASF cooling does not “cool the information,” it removes the phonon entropy generated *by* the information processing, while the chip’s information output runs unimpeded. The point of photonic cooling specifically is to engineer the phonon-entropy pathway through the optical channel so that the chip’s compute throughput is not throttled by the package’s thermal resistance.

3.5.4 Reversible logic and the limits of clever engineering

[7] stresses an awkward fact: a reversible primitive (e.g. the Fredkin gate of [11]) used in an *irreversible* computation does *not* avoid dissipation. Once a circuit allocates degrees of freedom (extra wires, ancilla bits) that the workload does not actually need, erasure must pay the full penalty of eliminating all the available configurations when those wires are reset. Conversely, an irreversible primitive used in a partially reversible task wastes information that the workload would not have required. The general lesson:

There is an entropic cost to allocating degrees of freedom, whether or not they are used. Thermodynamically optimal computing should add degrees of freedom only as needed by the calculation [7].

This is the deepest content of the workload-entropy bound: the floor on dissipation is set by the *coding* of the computation, not by the choice of primitive. Photonic cooling operates above this floor — like all real chips at present, the device sits in the CMOS regime (3.4) where $\frac{1}{2}CV^2$ dominates both the Landauer floor and the workload-coded entropy. But the exergy-recovery framing of the photonic loop turns out to be a direct refinement of the same argument, applied at a different energy scale.

3.5.5 Entropy balance: convective vs. photonic cooling

The same Landauer-and-CMOS-derived entropy can leave the chip through two quite different pathways, and they have different downstream exergy budgets.

Convective cooling. Computational entropy $\dot{S}_{\text{comp}} = \dot{Q}_{\text{die}}/T_{\text{chip}}$ flows out through R_{die} and R_{ext} as heat at progressively lower temperatures, ending up in the ambient at T_0 . The exergy $\dot{Q}_{\text{die}}(1 - T_0/T_{\text{chip}})$ is destroyed at every conduction step — the irreversibility shows up as entropy production $\dot{S}_{\text{gen}} = \dot{Q}_{\text{die}}(1/T_0 - 1/T_{\text{chip}})$. The destroyed exergy is just heat at ambient; it cannot be recovered.

Photonic (ASF) cooling. The same $\dot{Q}_{\text{die}}$ is converted to fluorescence at the active layer; the fluorescence has higher exergy than the heat would have had on its conduction path because the entropy is exported through the optical channel into a smaller modal phase space than the phonon bath occupied (Chapter 6). The LPC converts a fraction ζ_f of the fluorescence energy back to electrical work (6.16). The exergy that survives this round trip is bounded by B_f of (1.14).

Information-theoretic interpretation. The chip’s irreversible bit operations produce, per (3.6), a workload-determined entropy flux. Convection turns that flux into “classical” phonon disorder at T_0 — thermodynamically useless. Photonic cooling re-encodes the same flux into the entropy of a structured radiation field with engineered low $\dot{S}_f$ per unit P_f ; the LPC then extracts the work content through quasi-Fermi-level splitting (Section 3.4) and returns a fraction of the computational free energy as recovered electricity. In Gershenfeld’s framing, the photonic loop is a *partial reversibilisation* of the chip’s irreversible-computation entropy production: it does not violate the workload-entropy bound (3.6), but it captures more of the available exergy than convection ever can.

3.5.6 Temporal targeting as Maxwell-demon-like sorting

Temporal targeting (Section 2.7) compounds in the information picture: when photonic cooling is engaged only during a hot-spot burst, the entropy being exported through the optical channel is the entropy of a *specific* computational phase (an AVX matmul, an HBM access burst). That phase has a characteristic spectral / spatial signature that the photonic network can match; the resulting B_f/P_f is higher than for a time-averaged steady-state load because the engineered output mode is matched to the active functional unit. In Gershenfeld’s language, the photonic cooling becomes a kind of Maxwell demon acting on the entropy stream — sorting bits by the functional unit that produced them, and routing each substream through a matched optical channel with its own $\dot{B}_{f,i}$. This is the information-theoretic counterpart of the multi-junction LPC story of Section 7.4: spectral sorting at the LPC and temporal / spatial sorting at the cold plate are two facets of the same “decompose entropy into low-entropy substreams” principle.

3.5.7 What the photonic loop *is* thermodynamically

Bringing all four threads together: a chip is a computation [7]; its dissipation is set by the workload’s signal entropy times $k_B T_{\text{chip}} \ln 2$, sitting many orders of magnitude above the Landauer floor at the CMOS scale (3.4) but obeying the same free-energy / erasure framework. ASF photonic cooling does not reduce $\dot{S}_{\text{comp}}$; the workload sets that floor. What it *does* is choose the channel through which $\dot{S}_{\text{comp}}$ is exported: rather than dumping it as phonon disorder into ambient (convection), it re-encodes it into a structured radiation field whose entropy fraction $\dot{S}_f/\dot{E}_f$ can be made small, and then recovers a fraction of the resulting free energy through the LPC. The chapters that follow (Chapters 4 to 7) develop the photonic engineering that minimises that entropy fraction; this chapter has shown why doing so is a direct extension of Gershenfeld’s information–entropy argument applied to the electrical / thermal / optical / electrical loop of the device.

3.6 Component WPE summary table

A consolidated summary, with order-of-magnitude numerical guidance. Specific commercial-device numbers should be filled in from the to-be-built tables flagged in `transcripts_v2/LaserPowerConverters_and_Thermodynamics.txt` and `LaserWallPlugEfficiency_and_Thermodynamics.txt`; here we quote the regime that the loop-level analysis assumes.

Component	Quantity	Value	Notes
Diode laser (EEL, 980 nm)	η_P	0.50–0.70	best for high-power single emitter
VCSEL array (980 nm)	η_P	0.40–0.55	spatial multiplexing, fast modulation
Laser $\rightarrow$ fiber (HCF)	coupling efficiency	0.80–0.90	per-stage
HCF propagation, m-scale	loss	0.99	negligible at rack lengths
Fiber $\rightarrow$ coupler	coupling efficiency	0.80–0.95	launch optics
Coupler $\rightarrow$ extractor	critical coupling (4.11)	0.80–0.95	EO-engineered, Chapter 4
Net η_{cpl}		0.40–0.75	product of above
LPC (single junction, 980 nm)	η_{LPC}	0.55–0.70	[12]
LPC (multi-junction, 808 nm)	η_{LPC}	0.65–0.75	[13]
LPC (photon-recycling, 850 nm)	η_{LPC}	0.69	[14]
ASF, Yb:YLF (room T)	η_{ASF}	0.02–0.05	[6], before Purcell
ASF, Yb:YLF (Purcell-enhanced)	η_{ASF}	0.20–0.30	target, [15]

Plugging the operating-point row of each component into the feasibility test (1.17), with T_h in the hot-spot regime, gives the design ranges of Chapter 1. The chapter has thus closed the loop on the four loop-level efficiencies that drive every later chapter’s optimisation target.

3.7 Chapter summary

- Laser WPE is the product of injection, radiative-recombination, extraction, and voltage-efficiency factors (3.1); near 980 nm, EELs reach $\sim 70\%$, VCSEL arrays $\sim 50\%$, and the choice trades WPE for spatial multiplexing and modulation bandwidth.
- Standard SMF is power-limited at ~ 10 W by nonlinearity / damage; hollow-core fibers raise the damage-threshold to the kW class while keeping coupling efficiency comparable, so HCF is the operating-point baseline for the rack-scale architecture.
- η_{cpl} is the product of laser-to-fiber, fiber-propagation, fiber-to-coupler, and coupler-to-extractor coupling efficiencies (3.2); for the operating-point assumptions, $\eta_{cpl} \sim 0.5$ –0.75.
- LPC efficiency is wavelength-matched to the fluorescence and constrained by both the bandgap-thermalisation penalty and the radiation-entropy bound (7.1); only the heat-derived portion of the fluorescence is Carnot-limited (1.15).
- Information-theoretically, the chip is a Landauer-bounded device whose irreversible-computation entropy must leave as heat (3.3); CMOS technology presently sits $\sim 10^6$ – 10^7 times above that floor (3.4). Gershenfeld’s signal-entropy bound (3.6) [7] sets a workload-coded floor on dissipation that no engineering can cross. Convection turns that entropy into useless ambient heat; photonic ASF cooling re-encodes it into a structured radiation field whose exergy is partially recovered by the LPC. The photonic loop is a partial reversibilisation of the chip’s computational entropy production — a direct extension of Gershenfeld’s framework to the electrical / thermal / optical / electrical loop.

- Temporal targeting (Section 2.7) raises the recovery ceiling by matching the engineered optical channel to the spectral / spatial signature of specific computational phases, acting as a Maxwell-demon-like sorter on the entropy stream.
- The component-WPE table (Section 3.6) feeds (1.17) with operating-point values and is the final input to the device-level feasibility analysis.

What still needs to be filled in. The commercial-laser and demonstrated-LPC tables flagged in Section 7.11 feed directly into the value column of Section 3.6 above and are the next concrete deliverable for this chapter. Specific WPE numbers for laser-diode bars, VCSEL arrays at the device wavelengths of interest, and HCF power-handling data for matched mode launches should come from manufacturer datasheets and the recent literature; the **References.xlsx** laser sheet is the natural source.

Part II

Electro-Optic Foundations

Chapter 4

Electro-Optics, the Single-Mode Picture, and Its Thermodynamic Translation

4.1 Scope

This chapter does two related things. In Section 4.2–Section 4.18 we collect the standard electromagnetic and photonic machinery needed to describe the passive optical behaviour of the layered stack: Maxwell’s equations in stratified media, the transfer-matrix method, leaky resonances, the local density of optical states (LDOS) and the Purcell factor, and input–output coupling. Then in Section 4.8–Section 4.17 we use those results, under the single-mode approximation, to write the cooling power, the COP, the net-power criterion, and the second-law / exergy limit on power generation in a form that connects directly to the thermophotonic master equations of Chapter 1. This second part is the bridge: it turns “the cavity has a Purcell factor” into a quantitative statement about how much of the loop’s $\eta_{\text{conv}} \dot{B}_{\text{f}}$ survives the second law. The active-medium population dynamics sit on top of both halves and are taken up in Chapter 5.

4.2 Maxwell’s equations in stratified media

For a planar stack with z -dependent permittivity $\varepsilon(\omega, z) = [n(\omega, z) + i\kappa(\omega, z)]^2$, non-magnetic $\mu = \mu_0$, and time-harmonic fields $\mathbf{E}(\mathbf{r}, t) = \mathbf{E}(\mathbf{r}) e^{-i\omega t}$, the source-free wave equation in each slab of uniform ε reduces to

$$\left[\nabla^2 + \varepsilon(\omega) \frac{\omega^2}{c^2} \right] \mathbf{E}(\mathbf{r}) = 0. \quad (4.1)$$

Plane-wave solutions of fixed in-plane wavevector $\mathbf{k}_{\parallel}$ give the slab-wise vertical wavenumber

$$k_z(\omega) = \sqrt{\varepsilon(\omega) \frac{\omega^2}{c^2} - k_{\parallel}^2}, \quad (4.2)$$

with the branch chosen so that evanescent waves decay. Each slab supports upward/downward travelling TE and TM solutions; interface continuity of tangential $\mathbf{E}$ and $\mathbf{H}$ connects them.

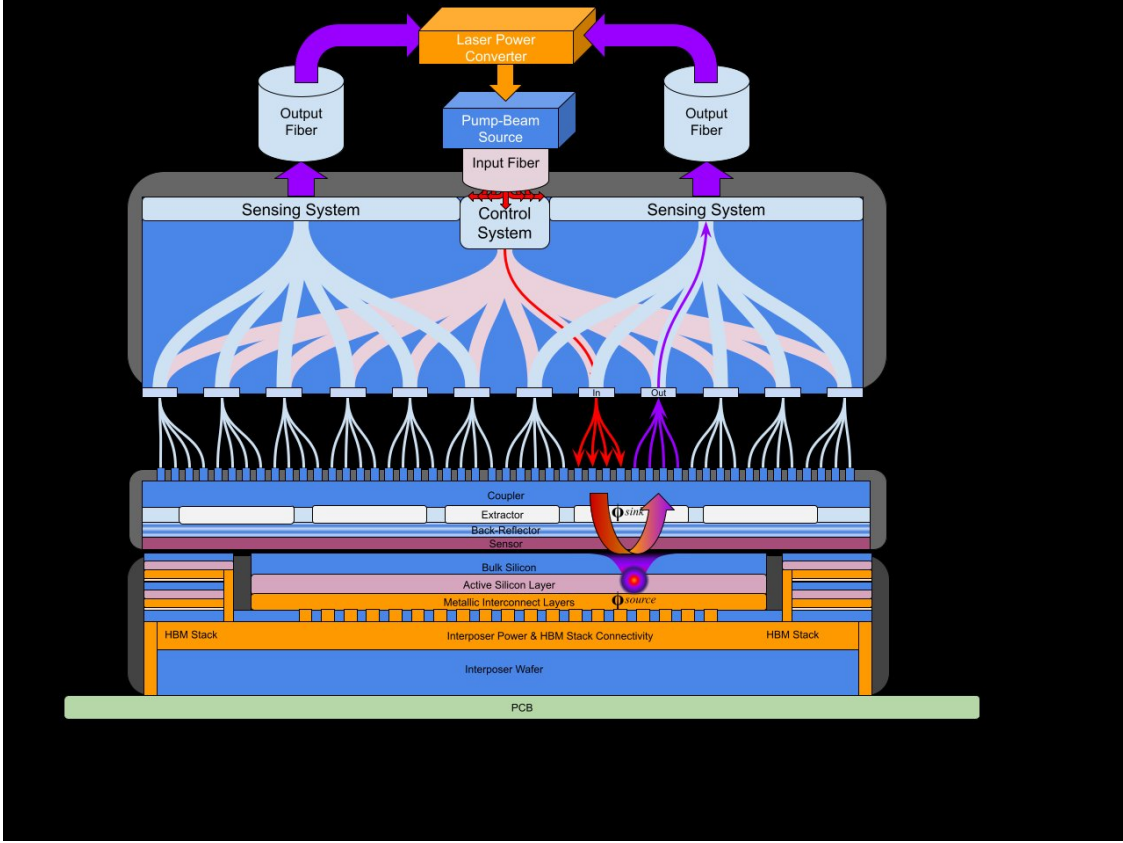


Figure 4.1: Full-stack view of the photonic cold plate and its optical interface. From top: pump-beam source and output fibers, sensing and control system, an active photonic network that fans pump power into individual in-couplers and merges fluorescence into outgoing fibers, the four functional layers of the cold plate itself (coupler / extractor / back-reflector / sensor), and finally the processor die (bulk silicon / active layer / metallic interconnect) with HBM stacks flanking it on an interposer wafer.

4.3 Transfer-matrix method

The field in slab j is the pair $(E_j^+ e^{ik_{z,j}z}, E_j^- e^{-ik_{z,j}z})$. The *interface matrix* $I_{j \rightarrow j+1}$ enforces Fresnel continuity at $z_{j,j+1}$, and the *propagation matrix* $P_j = \text{diag}(e^{+ik_{z,j}d_j}, e^{-ik_{z,j}d_j})$ advances the field through slab thickness d_j . The total transfer matrix

$$M = I_{0 \rightarrow 1} P_1 I_{1 \rightarrow 2} P_2 \cdots P_N I_{N \rightarrow N+1} \quad (4.3)$$

relates the amplitudes on the two sides of the stack. Reflectance, transmittance, and the in-stack field profile follow directly from M . For the cold-plate stack this allows us to (a) compute pump absorption versus angle / polarization, (b) design a wavelength-selective back reflector (a distributed Bragg reflector DBR or high-contrast grating) that is low-loss at ω_p and high-loss at ω_f , or vice versa, and (c) design fluorescence out-coupling via a frequency- and angle-selective top layer.

4.4 Leaky resonances: Q and V

A resonance in the photonic network is characterized by two numbers.

Quality factor. If the stored electromagnetic energy decays as $U(t) = U_0 e^{-\omega_0 t/Q}$, the line shape of the resonance is a Lorentzian of half-width at half-maximum (HWHM) $\Gamma/2 = \omega_0/(2Q)$. We routinely decompose Q into partial contributions from independent loss channels (Matthiessen's rule):

$$\frac{1}{Q} = \frac{1}{Q_e} + \frac{1}{Q_i} + \frac{1}{Q_{\text{abs}}}, \quad (4.4)$$

with Q_e external (desired output coupling), Q_i intrinsic (scattering / radiation leakage into unwanted channels), and Q_{abs} due to medium absorption. The absorption-limited ceiling

$$Q_{\text{abs}} \approx \frac{1}{\Gamma_{\text{conf}}} \frac{\text{Re } \varepsilon}{\text{Im } \varepsilon} = \frac{n}{2\kappa \Gamma_{\text{conf}}}, \quad (4.5)$$

with Γ_{conf} the fraction of the mode energy stored in the absorbing medium ($\kappa = \text{Im } n$ its extinction coefficient; the energy decay rate is $\omega \Gamma_{\text{conf}} \text{Im } \varepsilon / \text{Re } \varepsilon$, so Q_{abs} is dimensionless and *falls* as more of the mode sits in the absorber), is the ceiling that recurs throughout Section 4.12 and Chapter 6. No design can usefully push Q past Q_{abs} : beyond the ceiling, additional cavity confinement is swallowed by medium absorption and reabsorption.

Mode volume. For a cavity mode with electric field $\mathbf{E}(\mathbf{r})$,

$$V = \frac{\int \varepsilon(\mathbf{r}) |\mathbf{E}(\mathbf{r})|^2 d^3r}{\max_{\mathbf{r}} \varepsilon(\mathbf{r}) |\mathbf{E}(\mathbf{r})|^2}. \quad (4.6)$$

Engineered subwavelength structures (photonic-crystal defect cavities, plasmonic antennas, bound states in the continuum) can achieve $V \ll (\lambda/n)^3$.

4.5 Local density of states and Purcell enhancement

A dipole $\mathbf{d}$ at $\mathbf{r}_0$ radiates into the full electromagnetic bath at a rate proportional to the projected local density of states

$$\rho_{\mathbf{d}}(\omega; \mathbf{r}_0) = \frac{2\omega}{\pi c^2} \hat{\mathbf{d}} \cdot \text{Im } \mathbf{G}(\mathbf{r}_0, \mathbf{r}_0; \omega) \cdot \hat{\mathbf{d}}, \quad (4.7)$$

with $\mathbf{G}$ the electromagnetic Green dyadic. A single-mode Lorentzian cavity of frequency ω_0 , width ω_0/Q , and mode volume V gives the Purcell enhancement of the rate into that mode,

$$F_P = \frac{3}{4\pi^2} \left(\frac{\lambda_0}{n} \right)^3 \frac{Q}{V} \eta_{\text{orient}} \eta_{\text{spec}}, \quad (4.8)$$

where $\eta_{\text{orient}} = |\hat{\mathbf{d}} \cdot \hat{\mathbf{e}}(\mathbf{r}_0)|^2$ and η_{spec} is the spectral overlap of the emitter line with the cavity resonance. Purcell enhancement does two distinct jobs for the cooling cycle:

1. It routes a larger fraction of the spontaneous-emission rate into the designed output channel, increasing η_{ASF} .
2. It can reshape the emission into a narrower, more directional mode, which — as Section 4.9 and Chapter 6 will show — *reduces the entropy per unit energy* of the fluorescence and raises the allowable LPC conversion efficiency toward unity.

4.6 What Purcell enhancement does — and does not — do

Purcell enhancement is often quoted as a single number, $F_P = (3/4\pi^2)(\lambda_0/n)^3(Q/V)$, that “boosts the spontaneous-emission rate into the mode.” The interpretation matters for design, because not every Q/V helps.

What Purcell buys: branching into the engineered mode. The emitter sees *all* available optical modes of the structure. The spontaneous-emission rate into the free-space continuum is unchanged by the cavity — the cavity only redistributes the projection onto its mode basis. What changes is the mode-resolved rate $\gamma_{\text{cav}} = F_{\text{P}} \gamma_{\text{free-space}}$ into the engineered mode; whenever $F_{\text{P}} \gg 1$ the branching ratio into that mode approaches unity.

What Purcell does *not* do: create photon energy. F_{P} says nothing about the energy per photon; that is set by the anti-Stokes shift $\hbar(\omega_{\text{f}} - \omega_{\text{p}})$. A very high F_{P} with bad spectral tuning will rate-enhance an emission line that does *not* help cooling. Spectral overlap (η_{spec} in (4.8)) is the gatekeeper.

What Purcell does *not* do: reduce entropy for free. Narrowing the mode reduces the *modal* entropy of the emitted field at fixed photon number. But if the emitter’s upper manifold is thermally redistributed and each cycle re-thermalizes in-manifold, the radiated field can still carry the full Lorentzian line shape of the bulk transition unless the cavity is narrow enough to *filter* re-emitted photons into the mode-selected channel.

What Purcell does *not* do: beat the absorption ceiling. Pushing Q past Q_{abs} only moves optical energy from the desired mode into medium absorption. Design target: $Q \lesssim Q_{\text{abs}}$, plus enough V^{-1} headroom to still produce $F_{\text{P}} \gg 1$. In practice that means shrinking V (subwavelength photonic-crystal defect cavities, guided-mode resonances, or high-contrast gratings) rather than chasing Q .

The relevant figure of merit for extractor-layer design is therefore not Q , not Q/V , but $F_{\text{P}} \cdot \eta_{\text{spec}} \cdot \beta_{\text{sel}}$, where β_{sel} is the branching ratio into the engineered mode and η_{spec} is the spectral overlap. This is the triplet that appears in the master equation (5.20) of the next chapter.

4.7 Coupled-mode theory: input–output for a single resonance

Temporal coupled-mode theory (CMT) describes a single resonance amplitude $a(t)$ driven by input port s_+ and radiating to output s_- :

$$\dot{a} = \left[-i\omega_0 - \frac{\kappa_{\text{i}} + \kappa_{\text{e}} + \kappa_{\text{abs}}}{2} \right] a + \sqrt{\kappa_{\text{e}}} s_+, \quad (4.9)$$

$$s_- = -s_+ + \sqrt{\kappa_{\text{e}}} a, \quad (4.10)$$

with $\kappa_{\text{e,i,abs}} = \omega_0/Q_{\text{e,i,abs}}$. The steady-state absorption on resonance into the medium is

$$\eta_{\text{abs}} = \frac{4 \kappa_{\text{e}} \kappa_{\text{abs}}}{(\kappa_{\text{e}} + \kappa_{\text{i}} + \kappa_{\text{abs}})^2}, \quad (4.11)$$

which is maximized (equal to unity in the lossless-intrinsic limit) at *critical coupling* $\kappa_{\text{e}} = \kappa_{\text{i}} + \kappa_{\text{abs}}$. The pump channel in Chapter 5 is a direct instance of (4.9); the fluorescence channel adds a stimulated-emission gain term and requires the fully coupled treatment of that chapter.

4.8 Single-mode electro-optic results as power flows

We now translate the single-fluorescence-mode coupled-mode model into the thermodynamic / exergy form of Chapter 1. The cleanest route is to take the steady-state photon fluxes the cavity model produces and reinterpret each one first as an *energy* flow and then as an *exergy* flow.

Under the single-fluorescence-mode approximation, the coupled-mode model produces two key steady-state rates:

- a pump absorption rate into the active medium, R_p ,
- a fluorescence emission rate into the selected cavity mode, R_f .

The corresponding powers are

$$P_{\text{abs}} = \hbar\omega_p R_p, \quad P_f = \hbar\omega_f R_f. \quad (4.12)$$

Schematically, the cavity rate is

$$R_f = \eta_{\text{esc}} \Gamma_f^{\text{cav}} N_2, \quad \Gamma_f^{\text{cav}} \propto \frac{Q_f}{V_f}, \quad (4.13)$$

with N_2 the excited-state population (set by the pump and the total decay channels) and η_{esc} the fraction of the intracavity decay that escapes to the useful output port. *All of the cavity electrodynamics enters thermodynamics through the single-mode fluorescence power P_f , which is itself controlled by the Purcell-enhanced rate into the selected mode.*

The first-law statement for the active medium is

$$P_f = P_{\text{abs}} + P_{\text{cool}} - P_{\text{parasitic}}, \quad (4.14)$$

or equivalently

$$\boxed{P_{\text{cool}} = P_f - P_{\text{abs}} - P_{\text{parasitic}}} \quad (4.15)$$

where $P_{\text{parasitic}}$ collects everything that heats rather than cools the medium: nonradiative decay, background absorption, fluorescence escaping outside the useful channel, imperfect extraction. The electro-optic model thus translates into thermodynamics through one central identity:

$$\boxed{P_{\text{cool}} = \hbar\omega_f R_f - \hbar\omega_p R_p - P_{\text{parasitic}}} \quad (4.16)$$

In the idealized single-mode, unity-escape, unity-radiative-efficiency limit this reduces to the familiar anti-Stokes balance

$$P_{\text{cool}} \approx P_{\text{abs}} \left(\frac{\omega_f}{\omega_p} - 1 \right). \quad (4.17)$$

4.9 Exergy form: pump, fluorescence, extracted heat

Energy alone is not enough if we want to connect to COP and net power generation; we need exergy. For any stream, $\dot{B} = \dot{E} - T_0 \dot{S}$, where T_0 is the ambient reference temperature, $\dot{E}$ is the energy flow, and $\dot{S}$ is the entropy flow. We evaluate this for each of the three streams crossing the active-medium boundary.

Pump stream. A laser pump is very nearly a work-like input; its entropy per unit energy is negligible, so to a very good approximation

$$\dot{B}_p \approx P_p. \quad (4.18)$$

Fluorescence stream. The selected fluorescence mode carries energy P_f , but not all of it is work-like because it can carry entropy:

$$\dot{B}_f = P_f - T_0 \dot{S}_f. \quad (4.19)$$

In the ideal narrowband single-mode limit $\dot{S}_f$ is minimized and the fluorescence stream becomes increasingly work-like: $\dot{B}_f \rightarrow P_f$. This is exactly why the single-mode cavity picture is thermodynamically powerful: the cavity not only increases P_f , it can also concentrate fluorescence *exergy* by reducing the spectral / spatial entropy of the emitted field.

Heat extracted from the cooled body. If the medium extracts heat P_{cool} from a body at T_c (the local extractor temperature, which is equal to the chip temperature T_h in the absence of thermal-boundary resistance),

$$\dot{B}_Q = \left(1 - \frac{T_0}{T_c}\right) P_{\text{cool}}. \quad (4.20)$$

The sign of the prefactor $1 - T_0/T_c$ separates the two operating regimes: $T_c > T_0$ means extracted heat carries positive exergy (power generation while cooling is allowed); $T_c < T_0$ means refrigeration below ambient, which *consumes* exergy.

Reversible exergy balance for the cooling stage. The fluorescence-mode exergy is bounded by the sum of the pump exergy and the heat-derived exergy:

$$\dot{B}_f \leq \dot{B}_p + \left(1 - \frac{T_0}{T_c}\right) P_{\text{cool}}, \quad (4.21)$$

or equivalently

$$P_f - T_0 \dot{S}_f \leq P_p + \left(1 - \frac{T_0}{T_c}\right) P_{\text{cool}}. \quad (4.22)$$

This is the first cleanly Carnot-like statement of the chapter: the coupled electro-optic model produces P_f from the cavity physics, while thermodynamics determines how much of that fluorescence power is convertible to useful work. Combining the bound with a downstream converter of efficiency η_{conv} (the LPC),

$$P_{e,\text{out}} \leq \eta_{\text{conv}} \dot{B}_f, \quad (4.23)$$

and a pump generated by an upstream laser of wall-plug efficiency η_P (so that $P_{e,\text{in}} = P_p/\eta_P$), the net electrical power of the closed loop satisfies

$$P_{\text{net}} \leq \eta_{\text{conv}} \left[P_p + \left(1 - \frac{T_0}{T_c}\right) P_{\text{cool}} \right] - \frac{P_p}{\eta_P}. \quad (4.24)$$

This is the cleanest “Carnot-like” statement of the single-mode system at this level of approximation, and it is the form in which Section 1.11 would write the exergy ceiling on the LPC.

The COP that follows. Defining net work input $W_{\text{net,in}} = P_{e,\text{in}} - P_{e,\text{out}}$,

$$\text{COP} = \frac{P_{\text{cool}}}{W_{\text{net,in}}} = \frac{P_{\text{cool}}}{P_p/\eta_P - P_{e,\text{out}}}. \quad (4.25)$$

For refrigeration below ambient ($T_c < T_0$), the reversible-limit minimum work input is $W_{\min} = P_{\text{cool}}(T_0/T_c - 1)$, so

$$\boxed{\text{COP} \leq \frac{T_c}{T_0 - T_c}.} \quad (4.26)$$

The single-mode model is therefore consistent with the standard Carnot refrigerator bound, but it produces P_{cool} and the net-electrical bookkeeping directly from the electromagnetic parameters ($Q_f/V_f, \eta_{\text{esc}}, N_t, \dots$).

4.10 The trap of treating fluorescence as Carnot-bounded

Equation (4.21) as written treats the fluorescence stream as if it were just a thermal heat flow at T_c with a Carnot-fraction exergy. That is too pessimistic. The fluorescence emerging from a single cavity mode is not an incoherent thermal reservoir at T_c ; it is a *nonequilibrium photon reservoir* whose work potential is set by its non-equilibrium free energy — by its energy *and* its entropy, not by a Carnot factor attached to a notional reservoir temperature. The right object is therefore the exergy of the fluorescence mode relative to the ambient electromagnetic environment.

This is the conceptual move that takes us from (4.21) to a sharper bound that respects the single-mode structure of the cavity. The next two sections build it out.

4.11 Single-mode fluorescence as a nonequilibrium bosonic reservoir

Let the fluorescence cavity mode be a single bosonic mode b at frequency ω_f , linewidth κ_f , with total mean occupation $n_f \equiv \langle b^\dagger b \rangle$. To keep coherent and incoherent parts of the state distinct, write the mode as a displaced noisy state,

$$b = \beta + \delta b, \quad \beta \equiv \langle b \rangle, \quad n_f = |\beta|^2 + n_{\text{inc}}, \quad (4.27)$$

with $n_{\text{inc}} \equiv \langle \delta b^\dagger \delta b \rangle$ the phase-insensitive incoherent population. This is the key decomposition: $|\beta|^2$ is the coherent, work-like population, and n_{inc} is the incoherent / entropic population.

Single-mode entropy. For a single bosonic mode the maximum-entropy state at fixed mean occupation is a thermal (geometric-distribution) state with number probabilities $p_m = n^m / (1+n)^{m+1}$ and von Neumann entropy $S(n) = k_B g(n)$, where

$$\boxed{g(n) = (n+1) \ln(n+1) - n \ln n.} \quad (4.28)$$

This is not a fitting function: it is the entropy of one bosonic degree of freedom carrying mean incoherent occupation n . Useful properties: $g(0) = 0$; for $n \ll 1$, $g(n) \approx n(1 - \ln n)$; for $n \gg 1$, $g(n) \approx \ln n + 1$; and $g'(n) = \ln(1 + 1/n) > 0$ so the entropy grows monotonically with incoherent occupation.

For a displaced thermal (or Gaussian, phase-insensitive) state, the entropy depends only on n_{inc} , not on the displacement:

$$\boxed{S_f = k_B g(n_{\text{inc}}).} \quad (4.29)$$

Coherent energy raises exergy without raising entropy; incoherent population raises both.

Mode exergy. At ambient temperature T_0 , the appropriate exergy of the mode is its non-equilibrium free energy relative to the ambient equilibrium photon state at the same frequency. Let $n_0 = (e^{\hbar\omega_f/k_B T_0} - 1)^{-1}$ be the ambient thermal occupancy. Then

$$B_f = (E_f - E_0) - T_0(S_f - S_0), \quad (4.30)$$

with $E_f = \hbar\omega_f n_f$, $E_0 = \hbar\omega_f n_0$, $S_0 = k_B g(n_0)$; explicitly

$$B_f = \hbar\omega_f (|\beta|^2 + n_{\text{inc}} - n_0) - k_B T_0 [g(n_{\text{inc}}) - g(n_0)]. \quad (4.31)$$

Equivalently, $B_f = k_B T_0 D(\rho_f \| \rho_0)$, the quantum relative entropy of the cavity state with respect to the ambient thermal state — the cleanest thermodynamic definition.

Three useful limits.

- **Pure coherent state.** $n_{\text{inc}} = 0$, $S_f = 0$, and at optical frequencies $n_0 \approx 0$, so $B_f \approx \hbar\omega_f |\beta|^2 = E_f$. A purely coherent fluorescence mode is essentially all exergy.
- **Thermal single-mode radiation.** $\beta = 0$, so $B_f = \hbar\omega_f (n_{\text{inc}} - n_0) - k_B T_0 [g(n_{\text{inc}}) - g(n_0)]$ — smaller than the energy because of entropy.
- **Optical-frequency regime.** For visible / near-IR fluorescence at room temperature $n_0 \ll 1$, so

$$B_f \simeq \hbar\omega_f n_f - k_B T_0 g(n_{\text{inc}}). \quad (4.32)$$

This is the working formula throughout the rest of the chapter.

Mode flux. If the useful outcoupling rate is $\kappa_{f,\text{out}}$, then $P_f = \hbar\omega_f \kappa_{f,\text{out}} n_f$, $\dot{S}_f = k_B \kappa_{f,\text{out}} g(n_{\text{inc}})$, and the exergy flux is

$$\dot{B}_f = \kappa_{f,\text{out}} B_f \simeq P_f - k_B T_0 \kappa_{f,\text{out}} g(n_{\text{inc}}) \quad (4.33)$$

in the $n_0 \approx 0$ optical-frequency limit.

The bridge. The cavity / electronic physics feeds thermodynamics through the pair

$$(P_f, \dot{S}_f) \quad \text{or equivalently} \quad (n_f, n_{\text{inc}}). \quad (4.34)$$

P_f is set by the rate / coupled-mode model; the entropy side requires the additional information of how much of the mode population is in the coherent vs incoherent component.

Exergy efficiency of the fluorescence mode. Define the fluorescence exergy fraction

$$\eta_{X,f} \equiv \frac{\dot{B}_f}{P_f} \simeq 1 - \frac{k_B T_0 g(n_{\text{inc}})}{\hbar\omega_f n_f}. \quad (4.35)$$

Three design levers feed $\eta_{X,f}$: a stronger coherent component $|\beta|^2$ raises it; a larger incoherent occupation n_{inc} lowers it (entropy penalty); a higher photon energy $\hbar\omega_f$ raises it. Narrowing into a high- Q , low- V single-mode cavity therefore does two thermodynamic things: (i) it raises the useful mode population via Purcell-enhanced emission, (ii) it can reduce the entropy per emitted joule by concentrating emission into a low-entropy nonequilibrium mode.

Net-power criterion in the mode picture. Combining the exergy-bounded converter $P_{e,\text{out}} \leq \eta_{\text{conv}} \dot{B}_{\text{f}}$ with the laser input $P_{e,\text{in}} = P_{\text{abs}}/\eta_{\text{P}}$ gives

$$P_{\text{net}} \leq \eta_{\text{conv}} \dot{B}_{\text{f}} - \frac{P_{\text{abs}}}{\eta_{\text{P}}}, \quad (4.36)$$

and the condition for net power generation becomes

$$\eta_{\text{conv}} \dot{B}_{\text{f}} > \frac{P_{\text{abs}}}{\eta_{\text{P}}}. \quad (4.37)$$

Not “the extracted heat has positive Carnot exergy,” but rather: the nonequilibrium free energy of the fluorescence mode exceeds the input work, divided by the LPC efficiency. That is the correct single-mode statement, and it is the form Chapter 7 uses.

The COP with fluorescence-exergy recovery is

$$\text{COP} \leq \frac{P_{\text{cool}}}{P_{\text{abs}}/\eta_{\text{P}} - \eta_{\text{conv}} \dot{B}_{\text{f}}}. \quad (4.38)$$

This sharpens (4.25): the fluorescence is recovered as non-equilibrium free energy, not as Carnot-fraction-of-heat.

4.12 Single-mode steady state: the cavity FOMs enter explicitly

We now push the exergy expression all the way back to the cavity electrodynamics so the absorption-limited Q_{f} and the Purcell-enhanced rate appear explicitly. Take a single pump channel at ω_{p} and a single fluorescence cavity mode at $\omega_{\text{f}} > \omega_{\text{p}}$, with intracavity field b . Let

- N_{t} — total dopant population, N_2 excited, $N_1 = N_{\text{t}} - N_2$,
- $\kappa_{\text{f}} = \kappa_{\text{f},\text{out}} + \kappa_{\text{f},\text{int}}$ — total cavity decay rate,
- $\eta_{\text{esc}} = \kappa_{\text{f},\text{out}}/\kappa_{\text{f}}$ — escape efficiency,
- W_{p} — pump-induced excitation rate per available lower-state ion,
- Γ_{nr} — nonradiative decay rate,
- Γ_{other} — all non-useful radiative channels,
- $\Gamma_{\text{f}}^{\text{cav}}$ — selected fluorescence-mode channel,
- $\Gamma_{2,\text{tot}} = \Gamma_{\text{nr}} + \Gamma_{\text{other}} + \Gamma_{\text{f}}^{\text{cav}}$ — total upper-state decay.

Cavity-enhanced rate. The Purcell-type emission rate into the selected mode is

$$\Gamma_{\text{f}}^{\text{cav}} = F_{\text{P}} \Gamma_{\text{f}}^{(0)} = A_{\text{f}} \frac{Q_{\text{f}}}{V_{\text{f}}}, \quad (4.39)$$

with F_{P} given by (4.8) and A_{f} absorbing all geometric / spectral overlap factors.

Excited-state population. Steady state of $\dot{N}_2 = W_{\text{p}}(N_{\text{t}} - N_2) - \Gamma_{2,\text{tot}} N_2$ gives

$$N_2 = \frac{W_{\text{p}}}{W_{\text{p}} + \Gamma_{\text{nr}} + \Gamma_{\text{other}} + A_{\text{f}} Q_{\text{f}}/V_{\text{f}}} N_{\text{t}}. \quad (4.40)$$

This is the first place where N_{t} and $Q_{\text{f}}/V_{\text{f}}$ enter explicitly.

Useful fluorescence power. Each upper-state ion emits into the selected mode at rate Γ_f^{cav} , escaping through the useful port at fraction η_{esc} :

$$P_f = \hbar\omega_f \eta_{\text{esc}} \Gamma_f^{\text{cav}} N_2 = \hbar\omega_f \eta_{\text{esc}} \left(A_f \frac{Q_f}{V_f} \right) \frac{W_p N_t}{W_p + \Gamma_{\text{nr}} + \Gamma_{\text{other}} + A_f Q_f / V_f}. \quad (4.41)$$

Pump absorption power. $P_{\text{abs}} = \hbar\omega_p W_p (N_t - N_2)$, so

$$P_{\text{abs}} = \hbar\omega_p W_p \frac{\Gamma_{\text{nr}} + \Gamma_{\text{other}} + A_f Q_f / V_f}{W_p + \Gamma_{\text{nr}} + \Gamma_{\text{other}} + A_f Q_f / V_f} N_t. \quad (4.42)$$

Cooling power. Combining (4.41) and (4.42) into (4.15),

$$P_{\text{cool}} = \hbar N_t \frac{W_p}{W_p + \Gamma_{2,\text{tot}}} \left[\omega_f \eta_{\text{esc}} \left(A_f \frac{Q_f}{V_f} \right) - \omega_p (\Gamma_{\text{nr}} + \Gamma_{\text{other}} + A_f Q_f / V_f) \right] - P_{\text{parasitic}}^{(\text{extra})}. \quad (4.43)$$

This is the direct single-mode expression for cooling power in terms of the bare parameters (N_t , Q_f/V_f , η_{esc} , W_p , Γ_{nr} , Γ_{other}). Cooling requires $\omega_f \eta_{\text{esc}} \Gamma_f^{\text{cav}} > \omega_p \Gamma_{2,\text{tot}}$ plus the extra-parasitic margin.

Mode entropy and exergy with cavity FOMs. With P_f given by (4.41) and the entropy flux (4.29),

$$\dot{B}_f = P_f - k_B T_0 \kappa_{f,\text{out}} g(n_{\text{inc}}) = \hbar\omega_f \eta_{\text{esc}} \Gamma_f^{\text{cav}} N_2 - k_B T_0 \kappa_{f,\text{out}} g(n_{\text{inc}}). \quad (4.44)$$

Each ingredient on the right is now an explicit electromagnetic / material parameter — subject to a closure model for n_{inc} (Section 4.14).

Absorption-limited Q_f . The cavity Q_f is not free: because the mode overlaps absorbing material at ω_f , (4.5) gives $Q_f \leq Q_{\text{abs}}(\omega_f)$, so the realisable Purcell-enhanced rate is bounded by

$$\Gamma_f^{\text{cav}} \leq A_f \frac{Q_{\text{abs}}}{V_f}, \quad (4.45)$$

and the useful fluorescence power and exergy flux are bounded correspondingly. Beyond this ceiling, additional Q_f buys nothing: any extra confinement is absorbed, with the absorbed photons either heating the medium parasitically or recycling back into the population dynamics of Chapter 5.

Optimality tradeoff in Q_f . Increasing Q_f does three competing things. *Helps:* $\Gamma_f^{\text{cav}} \propto Q_f/V_f$ loads more power into the selected mode. *Also helps:* a tighter single-mode cavity can reduce the entropic fraction n_{inc}/n_f , lowering the entropy penalty. *Hurts / saturates:* as $Q_f \rightarrow Q_{\text{abs}}$, absorption-induced loss prevents arbitrary increase. The true optimization problem is therefore not “maximise P_f ” but

$$\max_{Q_f, V_f} \dot{B}_f \quad \text{subject to} \quad Q_f \leq Q_{\text{abs}}(\omega_f). \quad (4.46)$$

That is the exergy-based design objective for the photonic network.

4.13 Why n_{inc} is not a primitive: cooperativity, branching, output etendue

n_{inc} is *not* an independent thermodynamic nuisance parameter. It should be understood as an emergent quantity set by the optical-electronic dynamics, and therefore depends on:

1. the degree of collective / single-mode coherence the emitter ensemble develops;
2. the cooperativity between emitters and the selected cavity mode;
3. the existence and nature of a well-defined output channel;
4. the branching between the selected mode and all other radiative and reabsorbing channels.

Sources of incoherent population. The fluctuations that contribute to n_{inc} come from spontaneous emission from uncorrelated emitters into the selected mode, dephasing of the electronic polarization, random re-injection from non-resonant or leaky channels, reabsorption followed by incoherent re-emission, imperfect outcoupling that destroys phase information, and internal cavity loss that mixes the mode with uncontrolled bath channels. n_{inc} is the part of the selected-mode population fed by entropy-producing reservoirs rather than by phase-organized transfer into the useful channel.

Cooperativity / coherence. A natural effective coherence parameter is

$$C_{\text{coh}} \sim \frac{\Gamma_{\text{sel}}}{\gamma_{\phi} + \Gamma_{\text{other}} + \Gamma_{\text{nr}} + \kappa_{\text{f,mix}}}, \quad (4.47)$$

where γ_{ϕ} is electronic dephasing, $\Gamma_{\text{sel}} \equiv \Gamma_{\text{f}}^{\text{cav}}$, and $\kappa_{\text{f,mix}}$ is mixing into uncontrolled bath channels. In a more cavity-QED-like form, $C_{\text{coh}} \sim N_{\text{eff}} g_{\text{f}}^2 / (\kappa_{\text{f}} \gamma_{\perp})$, but the rate form above is closer to the operating regime here. As C_{coh} increases, $n_{\text{inc}}/n_{\text{f}}$ decreases.

Output-channel phase-space preservation. Even if the intracavity field is concentrated into one mode, the emitted field can lose exergy if it is dumped into a large external manifold. The right language is *etendue* or phase-space compression. Introduce a factor

$$\boxed{\eta_{\text{ps}}} \quad (4.48)$$

that measures how well the selected internal mode maps onto a low-entropy external output manifold: $\eta_{\text{ps}} \approx 1$ means emission into a small-etendue, well-defined external channel; $\eta_{\text{ps}} \ll 1$ means emission spreading over many external states. This is not the same as “collection efficiency”: two systems can have the same collected power but different exergy if one preserves a tighter angular / spatial / mode manifold. A simple closure is $\eta_{\text{ps}} \sim 1/(1 + \ln M_{\text{out}})$ with M_{out} the effective number of external modes occupied.

Branching ratio. Decompose $\Gamma_{2,\text{tot}} = \Gamma_{\text{nr}} + \Gamma_{\text{sel}} + \Gamma_{\text{other}}$, where $\Gamma_{\text{other}} = \Gamma_{\text{guided}} + \Gamma_{\text{leaky}} + \Gamma_{\text{free}} + \Gamma_{\text{reabs,loss}}$ collects all non-selected radiative channels. Define the selected-mode branching ratio (full and radiative-only):

$$\boxed{\beta_{\text{sel}} = \frac{\Gamma_{\text{sel}}}{\Gamma_{2,\text{tot}}}, \quad \beta_{\text{sel}}^{(\text{rad})} = \frac{\Gamma_{\text{sel}}}{\Gamma_{\text{sel}} + \Gamma_{\text{other}}}.} \quad (4.49)$$

Better selected-mode branching not only increases n_{f} ; it also lowers $n_{\text{inc}}/n_{\text{f}}$, because the mode is fed by phase-organized transfer rather than by spontaneous loading from a sea of other channels.

Reabsorption: helpful or hurtful. Reabsorption is not automatically bad. Let $\mathcal{R}_{\text{reabs}}$ measure the strength of recycling of off-manifold fluorescence back into the active medium. If recycled photons are funnelled back into the selected high- Q , low-entropy manifold (“beneficial recycling”), they raise mean photon energy and can lower $n_{\text{inc}}/n_{\text{f}}$. If recycled photons randomize phase / direction / mode assignment before re-emission (“entropic recycling”), they raise entropy production and lower $\dot{B}_{\text{f}}/P_{\text{f}}$. The right question is therefore not “is reabsorption present?” but “does reabsorption renormalize Γ_{sel} upward, or does it renormalize $n_{\text{inc}}/n_{\text{f}}$ upward?” An auxiliary *recycling-quality* factor η_{rec} captures the distinction.

4.14 A two-tier framework: branching β_{sel} + entropy quality χ

It is convenient to separate the two distinct optimization questions the cavity poses:

- **(A) Selected-mode branching** — how much of the excited-state radiative decay goes into the chosen fluorescence mode / manifold? Captured by β_{sel} and η_{esc} .
- **(B) Selected-mode entropy quality** — given that photons go into that mode, how ordered is the output field? Captured by a single *mode-purity* / *coherent-fraction* parameter

$$\chi \equiv \frac{n_{\text{inc}}}{n_{\text{f}}}, \quad 0 \leq \chi \leq 1. \quad (4.50)$$

These two are not the same: a mode can have high branching but still large entropy if loaded incoherently, or modest branching but very low entropy if the output is highly phase-space compressed.

Closure model for χ . A practical phenomenological closure that respects the physics is

$$\chi \approx \frac{\chi_0}{1 + \alpha_1 \frac{\Gamma_{\text{sel}}}{\gamma_{\phi} + \Gamma_{\text{nr}} + \Gamma_{\text{other}} + \kappa_{\text{f,mix}}} + \alpha_2 \beta_{\text{sel}}^{(\text{rad})} + \alpha_3 \eta_{\text{ps}} + \alpha_4 \eta_{\text{rec}} \mathcal{R}_{\text{reabs}}} \quad (4.51)$$

with $\chi_0 \leq 1$ a baseline incoherent fraction and $\alpha_{1,2,3,4} \geq 0$ phenomenological weights. This has the right qualitative behaviour: χ decreases with larger selected-mode coupling, smaller Γ_{other} and γ_{ϕ} , better phase-space compression, and beneficial recycling.

Resulting n_{inc} . With $n_{\text{f}} = \eta_{\text{esc}} \Gamma_{\text{sel}} N_2 / \kappa_{\text{f,out}}$,

$$n_{\text{inc}} = \chi n_{\text{f}} = \chi \frac{\eta_{\text{esc}} \Gamma_{\text{sel}}}{\kappa_{\text{f,out}}} \frac{W_{\text{p}} N_t}{W_{\text{p}} + \Gamma_{\text{nr}} + \Gamma_{\text{sel}} + \Gamma_{\text{other}}}. \quad (4.52)$$

n_{inc} is no longer free; it is determined by pump loading, the competition among radiative and nonradiative rates, the selected-mode branching, the output coupling, and the coherence / phase-space closure embedded in χ .

Entropy and exergy with the closure. The single-mode entropy flux is $\dot{S}_{\text{f}} = k_B \kappa_{\text{f,out}} g(\chi n_{\text{f}})$, and the exergy flux is

$$\dot{B}_{\text{f}} = \hbar \omega_{\text{f}} \eta_{\text{esc}} \Gamma_{\text{sel}} \frac{W_{\text{p}} N_t}{W_{\text{p}} + \Gamma_{2,\text{tot}}} - k_B T_0 \kappa_{\text{f,out}} g \left(\chi \frac{\eta_{\text{esc}} \Gamma_{\text{sel}}}{\kappa_{\text{f,out}}} \frac{W_{\text{p}} N_t}{W_{\text{p}} + \Gamma_{2,\text{tot}}} \right). \quad (4.53)$$

Both the energy term and the entropy term are now explicit functions of $(N_t, Q_{\text{f}}/V_{\text{f}}, \eta_{\text{esc}}, W_{\text{p}}, \Gamma_{\text{nr}}, \Gamma_{\text{other}}, \chi)$, with χ supplied by the closure (4.51).

4.15 Concrete output-channel scenarios

How η_{ps} (and therefore χ) depends on the specific output architecture is best illustrated by two examples.

Scenario (i): metasurface narrow-cone radiator into a multimode fiber. A leaky metasurface resonance emits predominantly into a narrow angular cone near the surface normal; collection is by a multimode fiber matched in numerical aperture. The radiation is concentrated into a small angular range, the external field has low angular entropy, and collection is power-efficient. The subtlety is that a multimode fiber accepts many transverse fiber modes, so the launched mode set need not be maximally low-entropy. Still, compared with broad Lambertian fluorescence this is a much lower-entendue channel; η_{ps} can be high even with a multimode fiber provided the emitted cone is narrow and the launched mode set is limited. In this case Γ_{sel} is the resonant radiative coupling, η_{esc} is the radiative outcoupling to free space, η_{ps} is boosted by narrow-cone directionality, and χ is substantially smaller than for isotropic fluorescence.

Scenario (ii): coupling to a multimode waveguide. The selected manifold is now a guided set of modes rather than a free-space cone. Escape can be efficient, reabsorption in bulk free space is strongly suppressed, and routing is easy. But if the waveguide is truly multimode the output field populates many guided spatial / polarization modes: energy capture is excellent but mode entropy is higher than for single-mode or few-mode collection. Power capture may be higher than in scenario (i) and reabsorption losses may be lower, but η_{ps} is generally worse unless the waveguide is few-mode or coupling is biased toward a small subset of modes. Multimode waveguides are good *energetically*, but not automatically optimal exergy-wise.

The clean separation *between energy collection and exergy collection* is the design lesson: chasing high P_{f} is not the same as chasing high $\dot{B}_{\text{f}}$.

4.16 Compact final formulation

Collecting the core results of Section 4.8–Section 4.15 in one place:

$$N_2 = \frac{W_{\text{p}}}{W_{\text{p}} + \Gamma_{\text{nr}} + \Gamma_{\text{sel}} + \Gamma_{\text{other}}} N_t, \quad (4.54)$$

$$P_{\text{f}} = \hbar\omega_{\text{f}} \eta_{\text{esc}} \Gamma_{\text{sel}} N_2, \quad (4.55)$$

$$P_{\text{abs}} = \hbar\omega_{\text{p}} W_{\text{p}} (N_t - N_2), \quad (4.56)$$

$$P_{\text{cool}} = P_{\text{f}} - P_{\text{abs}} - P_{\text{parasitic}}, \quad (4.57)$$

$$S_{\text{f}} = k_B g(n_{\text{inc}}), \quad g(n) = (n+1) \ln(n+1) - n \ln n, \quad (4.58)$$

$$n_{\text{f}} = \frac{\eta_{\text{esc}} \Gamma_{\text{sel}}}{\kappa_{\text{f,out}}} N_2, \quad n_{\text{inc}} = \chi n_{\text{f}}, \quad (4.59)$$

$$\dot{B}_{\text{f}} = P_{\text{f}} - k_B T_0 \kappa_{\text{f,out}} g(\chi n_{\text{f}}), \quad (4.60)$$

$$\text{COP} \leq \frac{P_{\text{cool}}}{P_{\text{abs}}/\eta_{\text{p}} - \eta_{\text{conv}} \dot{B}_{\text{f}}}, \quad (4.61)$$

$$\text{net-gen criterion: } \eta_{\text{conv}} \dot{B}_{\text{f}} > P_{\text{abs}}/\eta_{\text{p}}, \quad (4.62)$$

$$Q_{\text{f}} \leq Q_{\text{abs}}(\omega_{\text{f}}), \quad \Gamma_{\text{sel}} \leq A_{\text{f}} Q_{\text{abs}}/V_{\text{f}}. \quad (4.63)$$

The two design objectives are now cleanly separated:

- **Energy / loading:** $\Gamma_{\text{sel}} \sim Q_{\text{f}}/V_{\text{f}}$ subject to $Q_{\text{f}} \leq Q_{\text{abs}}$.
- **Entropy quality:** $\chi = \chi(\gamma_{\phi}, \Gamma_{\text{sel}}, \Gamma_{\text{other}}, \eta_{\text{ps}}, \mathcal{R}_{\text{reabs}}, \dots)$, with closure (4.51).

Maximising performance means maximising not just P_{f} but $\dot{B}_{\text{f}}/P_{\text{abs}}$, subject to the absorption-limited Q_{f} and the output-channel entropy constraints.

4.17 Asymptotic regimes

The compact set above admits a number of useful asymptotic regimes that organize the design space.

1. Weak-pump limit, $W_{\text{p}} \ll \Gamma_{2,\text{tot}}$. Then $N_2 \approx W_{\text{p}}N_t/\Gamma_{2,\text{tot}}$, and

$$P_{\text{f}} \approx \hbar\omega_{\text{f}} \eta_{\text{esc}} \beta_{\text{sel}} W_{\text{p}}N_t, \quad (4.64)$$

$$P_{\text{abs}} \approx \hbar\omega_{\text{p}} W_{\text{p}}N_t, \quad (4.65)$$

$$P_{\text{cool}} \approx \hbar W_{\text{p}}N_t(\eta_{\text{esc}} \beta_{\text{sel}} \omega_{\text{f}} - \omega_{\text{p}}) - P_{\text{parasitic}}. \quad (4.66)$$

Both cooling and fluorescence are linear in W_{p} , and the binding parameter is the effective useful branching $\eta_{\text{esc}} \beta_{\text{sel}}$. If $n_{\text{inc}} \ll 1$, then $g(n_{\text{inc}}) \approx n_{\text{inc}}(1 - \ln n_{\text{inc}})$ and the entropy penalty is sublinear, so the exergy fraction can be high even at small total power. This is the regime in which *single-mode purity matters most*: low total power, but potentially very high exergy per photon.

2. Strong-pump (saturation) limit, $W_{\text{p}} \gg \Gamma_{2,\text{tot}}$. $N_2 \rightarrow N_t$, $P_{\text{f}} \rightarrow \hbar\omega_{\text{f}} \eta_{\text{esc}} \Gamma_{\text{sel}}N_t$, $P_{\text{abs}} \rightarrow \hbar\omega_{\text{p}} \Gamma_{2,\text{tot}}N_t$, and

$$P_{\text{cool}} \approx \hbar N_t \Gamma_{2,\text{tot}} (\eta_{\text{esc}} \beta_{\text{sel}} \omega_{\text{f}} - \omega_{\text{p}}) - P_{\text{parasitic}}. \quad (4.67)$$

Power saturates because the excited-state population saturates. For exergy this regime is not automatically best: P_{f} increases, but the mode can become increasingly incoherent with pump loading, raising χ and the entropy penalty.

3. High-branching selected-mode limit, $\Gamma_{\text{sel}} \gg \Gamma_{\text{nr}} + \Gamma_{\text{other}}$. $\beta_{\text{sel}} \rightarrow 1$, $\Gamma_{2,\text{tot}} \approx \Gamma_{\text{sel}}$, and $P_{\text{cool}} \approx \hbar N_t \Gamma_{\text{sel}} W_{\text{p}} / (W_{\text{p}} + \Gamma_{\text{sel}}) (\eta_{\text{esc}} \omega_{\text{f}} - \omega_{\text{p}})$. If also $\eta_{\text{esc}} \approx 1$, the cooling condition reduces essentially to $\omega_{\text{f}} > \omega_{\text{p}}$ — ideal anti-Stokes cooling. If high selected-mode branching also lowers χ , then $\eta_{X,\text{f}} \rightarrow 1$, $\dot{B}_{\text{f}} \rightarrow P_{\text{f}}$, and the fluorescence is both energetically dominant and low-entropy.

4. Weak-branching regime, $\Gamma_{\text{sel}} \ll \Gamma_{\text{nr}} + \Gamma_{\text{other}}$. $\beta_{\text{sel}} \ll 1$; generically $P_{\text{cool}} < 0$ unless parasitics are extraordinarily small and $\omega_{\text{f}} \eta_{\text{esc}} \beta_{\text{sel}}$ still beats ω_{p} (hard when $\beta_{\text{sel}} \ll 1$). Exergy is poor for both reasons: little power into the useful mode, and χ typically large because the selected mode is fed from a mostly incoherent radiative background.

5. Nonradiative-loss-dominated regime, $\Gamma_{\text{nr}} \gg \Gamma_{\text{sel}}, \Gamma_{\text{other}}$. $\Gamma_{2,\text{tot}} \approx \Gamma_{\text{nr}}$, $\beta_{\text{sel}} \approx \Gamma_{\text{sel}}/\Gamma_{\text{nr}} \ll 1$. Pump absorption produces heating through the high non-radiative channel; P_{cool} is strongly negative and $\dot{B}_{\text{f}}$ is small. This is essentially fatal for both refrigeration and power recovery, and explains why low background absorption and low non-radiative decay are the principal material requirements of Chapter 8.

6. Low-entropy selected-mode limit, $\chi n_f \ll 1$. $g(\chi n_f) \approx \chi n_f(1 - \ln(\chi n_f))$, so

$$\eta_{X,f} \approx 1 - \frac{k_B T_0}{\hbar \omega_f} \chi (1 - \ln(\chi n_f)). \quad (4.68)$$

At optical frequencies $\hbar \omega_f \gg k_B T_0$, so this is close to 1 unless χ is large. In this regime $\dot{B}_f \approx P_f$ and $P_{\text{net}} \lesssim \eta_{\text{conv}} P_f - P_{\text{abs}}/\eta_P$ — the fluorescence behaves almost as a work reservoir. This is the correct replacement for the naive Carnot-like fluorescence picture.

7. Highly incoherent selected-mode limit, $\chi \rightarrow 1$. The selected mode is essentially thermalised within its own bosonic degree of freedom; the entropy term $k_B T_0 \kappa_{f,\text{out}} g(n_f)$ can become substantial, so $\eta_{X,f} \ll 1$ even when P_f is large. Maximising fluorescence *power* alone can therefore mislead: a large P_f with $\chi \approx 1$ may carry much less recoverable work than a smaller but more ordered output.

8. Absorption-limited Q regime, $Q_f \rightarrow Q_{\text{abs}}$. $\Gamma_{\text{sel}}^{\text{max}} = A_f Q_{\text{abs}}/V_f$. Beyond this, energetic optimisation saturates — pushing Q_f higher is impossible. Further improvement must come from lowering V_f , improving η_{esc} , lowering Γ_{other} and Γ_{nr} , and lowering χ via better output-mode purity and phase-space compression. Beyond the absorption ceiling, exergy optimization shifts from “more Purcell” to “better entropy quality.”

9. Strong-reabsorption regime. As discussed above, reabsorption can help or hurt. If recycling feeds back into the selected manifold with high quality, $\Gamma_{\text{sel}}^{\text{eff}} > \Gamma_{\text{sel}}$ and $\chi^{\text{eff}} < \chi$, and both P_{cool} and $\dot{B}_f$ improve. If recycling mostly randomises phase / direction, it adds to Γ_{other} and / or χ and hurts exergy even if the average fluorescence frequency is blue-shifted. The right question is the effective renormalisation, not the bare reabsorption strength.

10. Narrow-cone / low-etendue output regime. High η_{ps} , low χ , large $\eta_{X,f}$. $P_{\text{net}} \leq \eta_{\text{conv}} \dot{B}_f - P_{\text{abs}}/\eta_P$ can become positive much sooner than in a broad-etendue output because $\dot{B}_f/P_f$ is high.

11. Multimode-output regime. Decent P_f and P_{cool} , but poorer $\dot{B}_f$ and therefore poorer P_{net} and COP enhancement. This is the clean example of decoupling between energy collection and exergy collection.

Best-case full regime. The most favourable regime combines several of the above:

$$\boxed{\Gamma_{\text{sel}} \gg \Gamma_{\text{nr}} + \Gamma_{\text{other}}, \quad Q_f \approx Q_{\text{abs}}, \quad \eta_{\text{esc}} \approx 1, \quad \chi \ll 1.} \quad (4.69)$$

Under these conditions

$$P_{\text{cool}} \approx \hbar N_t \Gamma_{\text{sel}} \frac{W_p}{W_p + \Gamma_{\text{sel}}} (\omega_f - \omega_p) - P_{\text{parasitic}}, \quad (4.70)$$

$$\dot{B}_f \approx P_f, \quad (4.71)$$

i.e. the fluorescence has nearly all its energy as exergy, and the loop is at most a Carnot factor away from saturating (1.17). All the device-level engineering of Chapters 5 and 6 is the program of pushing each of these inequalities into the favourable regime.

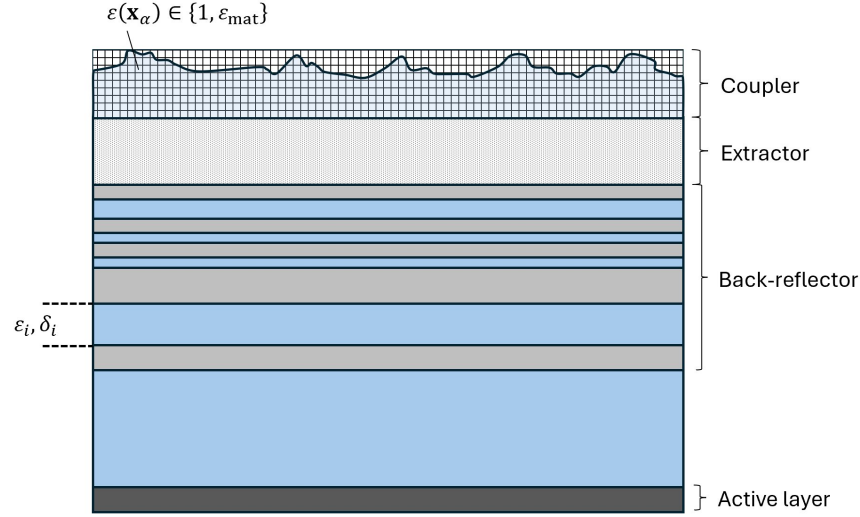


Figure 4.2: Cross-section of the stack as an inverse-design domain. Each voxel of the coupler layer carries a binary permittivity $\varepsilon(\mathbf{x}_\alpha) \in \{1, \varepsilon_{\text{mat}}\}$; the extractor, back-reflector, and active layer are parameterized by their per-layer $(\varepsilon_i, \delta_i)$. The figure of merit that the optimizer maximizes is the closed-loop COP($\delta, g; \Lambda, r$) of Chapter 6 (or equivalently $\hat{B}_f/P_{\text{abs}}$ of Equation (4.46)), subject to the absorption ceiling (4.5).

4.18 Thin-film stack design for the cold plate

The five functional layers translate to the following EO requirements, each tied to the asymptotic-regime analysis above:

1. **Photonic network.** Two nearly decoupled waveguide modes, one resonant at ω_p (pump delivery), one resonant at ω_f (fluorescence collection), with crosstalk < -30 dB. Practical realizations: patterned guided-mode resonances (GMR), photonic-crystal slabs, thin-film DBRs with engineered defects. Independent tuning of Q_p and Q_f is what enables critical pump coupling without compromising the fluorescence-mode purity.
2. **Coupler.** Index-matched spacer that mode-matches the two network resonances to the electronic dipoles of the extractor while maintaining the field overlap Γ_{conf} of (4.5).
3. **Extractor.** Active laser-cooled medium. Its dielectric response enters (4.5) and Chapter 5. Material choice drives Chapter 8.
4. **Back reflector.** Broadband, high-reflectivity at ω_p to force a second pump pass; wavelength-selective at ω_f (low reflectivity, high transmission) so that ASF exits toward the LPC rather than back into the extractor. Metal, DBR, or high-contrast-grating realizations all work; Section 4.3 supplies the design tools.
5. **Sensor.** A thermo-optic or photovoltaic micro-sensor embedded below the reflector, used for a local feedback loop on T_h ; its EO footprint must not perturb the reflector passband.

4.19 Chapter summary

Every quantity appearing in the thermophotonic feasibility condition (1.17) has an electromagnetic pre-image, and this chapter has supplied them all:

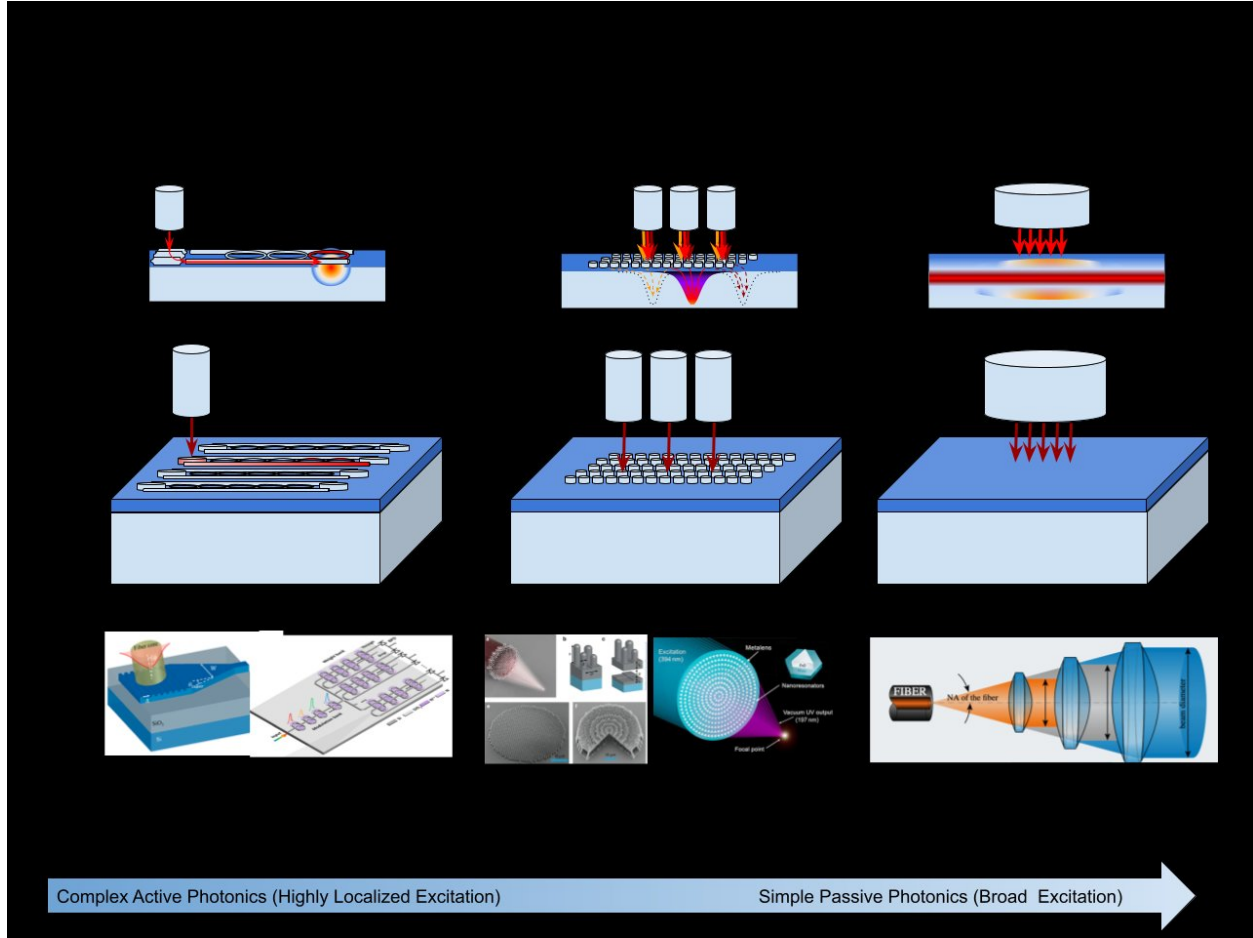


Figure 4.3: Design ladder from complex active photonics (left) to simple passive photonics (right). The cold-plate coupler can be realised along the entire span: photonic-crystal slab with embedded gain (left), metasurface / meta-lens arrays (centre), or a broadband dome / index-matching layer (right). Trade-off: left increases spectral / angular selectivity (raises η_{ps} , lowers χ); right increases tolerance and pump acceptance angle.

thermo quantity	EO pre-image
$\eta_{ASF} = Q_c/P_{abs}$	Purcell-enhanced rate into engineered mode + reabsorption vs. $\Gamma_{nr} + \Gamma_{other}$
pump absorption η_{cpl}	critical coupling of pump resonance, (4.11)
fluorescence exergy $\dot{B}_f$	$P_f - T_0 \dot{S}_f$, (4.33)
fluorescence exergy fraction	$\eta_{X,f} = 1 - k_B T_0 g(\chi n_f)/(\hbar \omega_f n_f)$
parasitic P_{nr}	Q_i , background absorption, imperfect escape
cavity Q ceiling	absorption-limited Q_{abs} , (4.5)
branching ratio	$\beta_{sel} = \Gamma_{sel}/\Gamma_{2,tot}$, (4.49)
entropy quality	$\chi = n_{inc}/n_f$, closure (4.51)
output etendue	η_{ps} , narrow cone vs. multimode

The two-tier structure is the key conceptual export: design the photonic network for *both* branching (raise β_{sel} , push Q_f toward Q_{abs}) *and* entropy quality (lower χ by suppressing dephasing, leaky channels, and multimode output spread). Optimising for P_f alone is not enough; the design

objective is $\dot{B}_f/P_{\text{abs}}$ at fixed $Q_f \leq Q_{\text{abs}}$.

Chapter 5 assembles the rate-equation half (excited-state population, reabsorption, escape efficiency) into the coupled emitter-resonance dynamics; Chapter 6 pushes the spectral / angular entropy bookkeeping to its formal limit; and Chapter 7 inherits the $\dot{B}_f$ bound as the ceiling on what the LPC can do.

Chapter 5

Coupled Electro-Optic Model of the Extractor

5.1 Objective

Chapter 4 treated the photonic stack passively and showed how single-mode results bridge to the thermodynamic / exergy picture of Chapter 1. Here we couple the cavity to the electronic populations of the extractor medium and derive $P_{\text{abs}}, P_{\text{f}}, P_{\text{cool}}$ self-consistently. The chapter is organized as three nested levels of the same coupled cavity–electro-optic theory:

1. the *rate-equation level* (photon number and population), which is what the cooling-power, threshold, and ASF-efficiency expressions are built on;
2. the *field–polarization level* (cavity amplitude and medium polarization), which produces the fluorescence spectrum and gives the dressed cavity pole and the absorption-limited Q_{f} self-consistently rather than by assumption;
3. the *design / optimization level*, where everything is recast in dimensionless form and the optimal external coupling, quality factor, fluorescence detuning, and pump frequency are derived.

The treatment follows the Q&A–style construction in `transcripts_v2/MXL-CoupledElectroOptics.md`; we keep the notation aligned with Chapters 1 and 4 so that $P_{\text{abs}}, P_{\text{f}}, \eta_{\text{esc}}, \eta_{\text{ASF}}, \dot{B}_{\text{f}}, \beta_{\text{sel}}$ all carry their previous meanings.

5.2 Minimal physical picture

Take a cooling medium with total active-ion number N_t , ground population N_1 and excited population N_2 , with $N_1 + N_2 = N_t$. Assume fast intramanifold thermalization so the medium can be represented by an effective two-manifold model with

- bulk radiative decay rate Γ_{r}^0 ,
- bulk nonradiative decay rate Γ_{nr}^0 ,
- absorption / emission cross-sections $\sigma_{\text{a}}(\omega), \sigma_{\text{e}}(\omega)$,
- background absorption coefficient $\alpha_{\text{b}}(\omega)$ capturing impurities, host disorder, and any frequency-dependent passive loss.

Couple the medium to two cavity modes:

- a *pump mode* b at ω_{p} with intracavity photon number $n_{\text{p}} = |b|^2$, responsible for injecting

excitations into the cooling medium;

- a *fluorescence mode* a at $\omega_f > \omega_p$ with photon number $n_f = |a|^2$, where Purcell enhancement and useful reabsorption happen.

The fluorescence mode is blue-detuned relative to the pump (anti-Stokes geometry) so it can both enhance emission and be reabsorbed. Bulk background absorption α_b enters as a parallel photon-removal channel at every frequency and shows up in the cavity linewidths of Sections 5.3 and 5.4.

5.3 Pump mode

For the pump mode driven by external source s_{in} at detuning Δ_p ,

$$\dot{b} = \left(i\Delta_p - \frac{\kappa_p}{2} \right) b + \sqrt{\kappa_{\text{ex},p}} s_{\text{in}}, \quad |s_{\text{in}}|^2 = \frac{P_{\text{in}}}{\hbar\omega_p}, \quad (5.1)$$

with linewidth budget

$$\kappa_p = \kappa_{\text{ex},p} + \kappa_{\text{i},p} + \kappa_{\text{bg},p} + \kappa_{\text{act},p}, \quad (5.2)$$

where the four contributions are the external coupling, passive intrinsic loss, background absorption ($\kappa_{\text{bg},p} = v_{g,p} \xi_p \alpha_b(\omega_p)$), and active-medium loading ($\kappa_{\text{act},p} = 2c \sigma_a(\omega_p)(N_1 - N_2) \Gamma_{\text{conf}}$). On pump resonance ($\Delta_p = 0$),

$$n_p = \frac{4 \kappa_{\text{ex},p}}{\kappa_p^2} \frac{P_{\text{in}}}{\hbar\omega_p}. \quad (5.3)$$

The pump-induced upward and downward stimulated rates are

$$W_{12}^{(p)} = \mathcal{G}_p \sigma_a(\omega_p) n_p, \quad W_{21}^{(p)} = \mathcal{G}_p \sigma_e(\omega_p) n_p, \quad (5.4)$$

with $\mathcal{G}_p$ the usual overlap / mode-volume / group-velocity factor. The net pump excitation rate is

$$R_p = W_{12}^{(p)} N_1 - W_{21}^{(p)} N_2, \quad (5.5)$$

which automatically captures the fact that as the pump approaches transparency, stimulated emission at ω_p suppresses useful excitation. Pump optimization therefore enters through both ω_p and κ_p .

Two consequences of (5.4)–(5.5) are used repeatedly in Chapters 8 and 9 and are worth stating here. First, with $\mathcal{G}_p = v_{g,p} \xi_p / V_p$ (the pump analogue of (5.8)) and the local intensity $I_p = \hbar\omega_p v_{g,p} n_p \xi_p / V_p$, the per-ion pump rate is the familiar

$$W_{12}^{(p)} = \frac{\sigma_a(\omega_p) I_p}{\hbar\omega_p}, \quad (5.6)$$

which is the link between a material's cross-section and the pump power the photonic network must deliver. Second, since $R_p \rightarrow 0$ when $N_2/N_1 = \sigma_a(\omega_p)/\sigma_e(\omega_p)$, the steady-state excited fraction is capped, at *any* pump intensity, by the *pump-transparency limit*

$$x_{\text{max}} \equiv \left(\frac{N_2}{N_1} \right)_{I_p \rightarrow \infty} = \frac{\sigma_a(\omega_p)}{\sigma_a(\omega_p) + \sigma_e(\omega_p)} = \left[1 + e^{(\hbar\omega_{00} - \hbar\omega_p)/k_B T} \right]^{-1}, \quad (5.7)$$

where the last form uses the McCumber / Kennard–Stepanov relation (5.21) with $\hbar\omega_{00} \equiv \hbar\omega_{\text{mc}}$ the zero-line (absorption–emission crossing) energy. Because anti-Stokes pumping places $\hbar\omega_p$ several

$k_B T$ below $\hbar\omega_{00}$, $x_{\max}$ is small — $\sim 10^{-3}$ – 10^{-4} for a rhodamine pumped 6–9 $k_B T$ into its red tail, but ~ 0.3 for Yb:YLF pumped one $k_B T$ below its zero line. The maximum cycling rate per ion is thus not $1/\tau$ but $x_{\max}/\tau$, and the volumetric cooling-power ceiling of a red-tail-pumped extractor is set by $x_{\max}$, not by its radiative lifetime (Section 8.3.3). The associated saturation intensity is $I_{\text{sat}} = \hbar\omega_p/[(\sigma_a + \sigma_e)\tau]$, which for a rhodamine is of order 10^6 W/cm^2 — reachable inside a critically coupled resonator, but *not* at the level of incident tile flux.

5.4 Fluorescence mode with reabsorption

Couple the cavity-selected fluorescence mode at ω_f to the medium with effective Einstein-like coefficients

$$G_f = \frac{v_{g,f} \xi_f}{V_f} \sigma_e(\omega_f), \quad A_f = \frac{v_{g,f} \xi_f}{V_f} \sigma_a(\omega_f), \quad (5.8)$$

representing stimulated emission and reabsorption into the selected mode, plus a spontaneous-emission seed coefficient $S_f \propto F_P(\omega_f) \Gamma_r^0$. The coupled electronic-photonic equations are

$$\dot{N}_2 = R_p - (\Gamma_r^0 + \Gamma_{nr}^0) N_2 - G_f N_2 (n_f + 1) + A_f N_1 n_f, \quad (5.9)$$

$$\dot{n}_f = -\kappa_f n_f + S_f N_2 + G_f N_2 n_f - A_f N_1 n_f, \quad (5.10)$$

$$\kappa_f = \kappa_{\text{ex},f} + \kappa_{i,f} + \kappa_{\text{bg},f}, \quad \kappa_{\text{bg},f} = v_{g,f} \xi_f \alpha_b(\omega_f). \quad (5.11)$$

Interpretation of each term:

- $S_f N_2$: spontaneous emission into the selected mode (the +1 seed in (5.9)’s $n_f + 1$ factor),
- $G_f N_2 n_f$: stimulated emission,
- $A_f N_1 n_f$: reabsorption of the blue-detuned fluorescence photon,
- $\kappa_f n_f$: cavity loss (escape + intrinsic + background).

This structure already captures the device’s defining feature: reabsorption is not just a loss because it re-excites the medium and recycles energy through a blue-detuned radiative channel.

5.5 Steady state, threshold, and figures of merit

Setting $\dot{n}_f = 0$ in (5.10) gives the central result

$$n_f = \frac{S_f N_2}{\kappa_f + A_f N_1 - G_f N_2}. \quad (5.12)$$

The denominator exposes the competition between desired buildup ($G_f N_2$), ordinary cavity loss (κ_f), and reabsorption loading ($A_f N_1$).

Below-threshold stability. Stability requires

$$\kappa_f + A_f N_1 - G_f N_2 > 0, \quad (5.13)$$

which is the modal transparency / self-oscillation threshold. Solving for the excited-state population using $N_1 = N_t - N_2$,

$$N_{2,\text{th}} = \frac{\kappa_f + A_f N_t}{A_f + G_f}. \quad (5.14)$$

This is the maximum excited population for stable below-threshold cooling operation. As $N_2 \rightarrow N_{2,\text{th}}$, the cavity-mode occupation diverges and the system crosses out of the cooling regime into transparency or self-oscillation.

Population balance and useful simplification. Setting $\dot{N}_2 = 0$ in (5.9) and using $\dot{n}_f = 0$ at steady state to substitute $S_f N_2 + G_f N_2 n_f - A_f N_1 n_f = \kappa_f n_f$, the population balance collapses to

$$R_p = (\Gamma_r^0 + \Gamma_{nr}^0) N_2 + \kappa_f n_f. \quad (5.15)$$

Pump-created excitations leave either through ordinary bulk decay or through the cavity-fluorescence channel. This is the bridge from the dynamical equations to the compact cooling-power expressions of Section 5.6.

Weak-inversion form. In the cooling regime $N_2 \ll N_t$, $N_1 \simeq N_t$, so (5.12) reduces to

$$n_f \approx \frac{S_f N_2}{\kappa_f + A_f N_t}. \quad (5.16)$$

5.6 Cooling power and the parasitic budget

A robust definition of the net cooling power is $P_{\text{cool}} = P_{\text{opt,out}} - P_{\text{opt,abs}}$, where the useful escaped power has two contributions:

- *uncoupled bulk fluorescence* that escapes from the medium directly, $P_{\text{u,esc}} = \eta_u \hbar \bar{\omega}_u \Gamma_{r,u}^0 N_2$, with η_u the bulk-escape fraction and $\bar{\omega}_u$ the mean uncoupled-fluorescence frequency;
- *cavity-coupled fluorescence* routed through the engineered mode, $P_{\text{c,esc}} = \hbar \omega_f \kappa_{\text{ex},f} n_f$.

The deposited (parasitic) optical power has three components: absorbed pump $\hbar \omega_p R_p$, pump-frequency background absorption $\hbar \omega_p \kappa_{\text{bg},p} n_p$, and fluorescence-frequency background absorption $\hbar \omega_f \kappa_{\text{bg},f} n_f$. Combining,

$$P_{\text{cool}} = \eta_u \hbar \bar{\omega}_u \Gamma_{r,u}^0 N_2 + \hbar \omega_f \kappa_{\text{ex},f} n_f - \hbar \omega_p R_p - \hbar \omega_p \kappa_{\text{bg},p} n_p - \hbar \omega_f \kappa_{\text{bg},f} n_f. \quad (5.17)$$

Non-radiative decay enters implicitly: it reduces N_2 at fixed R_p and therefore the conversion of absorbed pump power to escaped optical power.

A more revealing rearrangement

Use (5.15) to eliminate R_p from (5.17). Grouping the cavity contributions,

$$P_{\text{cool}} = N_2 [\Gamma_{r,u}^0 (\eta_u \hbar \bar{\omega}_u - \hbar \omega_p) - \Gamma_{nr}^0 \hbar \omega_p] - \hbar \omega_p \kappa_{\text{bg},p} n_p + n_f [\hbar (\omega_f - \omega_p) \kappa_{\text{ex},f} - \hbar \omega_p \kappa_{i,f} - \hbar (\omega_f + \omega_p) \kappa_{\text{bg},f}]. \quad (5.18)$$

Reading the cavity-contribution bracket: the term $+\hbar(\omega_f - \omega_p)\kappa_{\text{ex},f}n_f$ is the anti-Stokes lift from extracted photons; the $-\hbar\omega_p\kappa_{i,f}n_f$ is the passive cavity penalty; and the $-\hbar(\omega_f + \omega_p)\kappa_{\text{bg},f}n_f$ fluorescence-background absorption penalty is especially dangerous when recycling boosts n_f (the photon resides in the medium long enough to be absorbed). This last asymmetry — a short-residence photon rejects much less heat than a long-residence photon absorbs — is exactly what caps the benefit of arbitrarily high Q_f .

ASF efficiency in this language

A useful single-pass measure is the *fluorescence yield ratio* $\rho_f \equiv P_{c,esc}/(\hbar\omega_p R_p) = P_f/P_{abs}$; the canonical ASF cooling efficiency of Table 1.2 is then $\eta_{ASF} = \rho_f - 1$ (the “−1” is the absorbed pump that must be paid back before any heat is lifted). Cooling requires

$$\rho_f > 1 + \frac{\kappa_{bg,p} n_p}{R_p} + \frac{\omega_f}{\omega_p} \frac{\kappa_{bg,f} n_f}{R_p}. \quad (5.19)$$

Using the steady-state n_f of (5.16),

$$\rho_f \equiv \frac{P_f}{P_{abs}} = \frac{\omega_f}{\omega_p} \frac{A_{eff} \eta_{esc}}{\gamma_r + \gamma_{nr} + A_{eff}(1 + \xi_{rec})}, \quad \eta_{ASF} = \rho_f - 1, \quad (5.20)$$

where A_{eff} is the spontaneous rate into the engineered mode ($\propto F_p$), η_{esc} is the useful-port escape fraction, γ_r, γ_{nr} are the bulk radiative and non-radiative rates, and ξ_{rec} is the dimensionless reabsorption factor determined by A_f/G_f and the population ratio. Equation (5.20) is the electro-optic companion of the thermodynamic master expression (1.11): the numerator is the engineered channel, the denominator is the competition between it and every bulk loss path.

5.7 Why reabsorption is a feature, not a bug

A standard simplification of ASF rate equations drops reabsorption entirely, treating the upper-manifold photon as emitted and gone. For the device of this book that simplification removes one of the dominant efficiency-boost mechanisms. It is worth laying out what $A_f N_1 n_f$ actually does.

(i) It blue-shifts the mean emitted frequency. The bare fluorescence spectrum is centred at $\bar{\omega}_f$ and extends several $k_B T_h/\hbar$ below and above it. If the photon-recycling cavity is spectrally on the blue side — frequencies $\omega_f > \bar{\omega}_f$, favouring the anti-Stokes tail — low-frequency photons leave easily, while any high-frequency photon that is emitted into the cavity has a high probability of being reabsorbed (A_f is large near the peak of σ_a) and re-emitted via the engineered S_f/G_f into the blue-weighted cavity mode. After many such recycles, the emitted mean frequency $\langle \omega_f \rangle$ climbs toward ω_f — each bounce shifts mean energy up by roughly $k_B T_h$, so the net cooling power $\propto \langle \omega_f \rangle - \omega_p$ grows.

(ii) Entropy stays bounded if the recycling channel is narrow. A naive concern is that reabsorption must export entropy somewhere and that entropy should end up in the mode we care about. It does not, as long as the intermediate excitation relaxes by intra-manifold phonon thermalization *before* re-emission: each cycle resets the photon back to the top of the upper manifold and the next emission step is a fresh, cavity-selected spontaneous process. The entropy of the engineered output mode is controlled by n_{inc} of Section 4.11, not by the number of recycles; the entropy cost of the recycling is paid in the phonon bath — which is exactly the channel we *want* to pay it, since it is the bath providing the cooling.

(iii) It amplifies cooling power, not P_f . The reabsorption factor ξ_{rec} in (5.20) increases the effective decay rate out of the upper manifold, so the steady-state N_2^{ss} decreases for fixed R_p . One might read this as “reabsorption *reduces* the fluorescence output.” It does — per pump photon — but every recycled photon that eventually escapes carries more energy away (point (i) above), so the cooling power grows even as P_f stays roughly flat.

(iv) **It is the mechanism through which Q_{abs} of Section 4.4 enters the design.** Making the cavity narrower than Q_{abs} yields diminishing returns: the excess Q is absorbed inside the medium and returned via S_{f} , so raising Q stops helping once ξ_{rec} saturates. This is the quantitative version of Chao–Rodriguez’s observation [15] that “no need to push any single mode far beyond Q of the atomic transition ~ 100 , in which case cavity radiation dominates over reabsorption anyway.” The “reabsorption dominates” regime is what the recycling benefit lives in.

These four points are why reabsorption appears as a *gain* term $+A_{\text{f}}|a_{\text{f}}|^2 N_1$ in (5.9), not as a pure loss, and why ξ_{rec} in (5.20) can be made large without collapsing the cooling cycle.

5.8 Why absorption is blue-shifted relative to fluorescence

For transitions between thermalized manifolds, the absorption and emission cross-sections satisfy a McCumber / Kennard–Stepanov relation,

$$\sigma_{\text{e}}(\omega, T) = \sigma_{\text{a}}(\omega, T) \exp\left(\frac{\hbar\omega_{\text{mc}} - \hbar\omega}{k_{\text{B}}T}\right), \quad (5.21)$$

where ω_{mc} is the McCumber frequency (the zero-phonon-line position). The exponential factor $\sigma_{\text{e}}/\sigma_{\text{a}}$ *decreases* with ω — it exceeds unity below the zero line and falls below it above — so absorption is weighted more strongly on the *blue* side and emission more strongly on the *red* side. Therefore $\bar{\omega}_{\text{a}} > \bar{\omega}_{\text{e}}$ automatically: the familiar absorption-blue / fluorescence-red offset is not added by hand, it is built into the thermalized manifold structure.

This relation is what $\xi_{\text{e}}(\omega)$ and $\xi_{\text{a}}(\omega)$ enforce in the field-amplitude formulation of Section 5.9: they are the spectrally weighted emission and absorption envelopes connected by (5.21).

5.9 Field–polarization formulation: where the Lorentzian comes from

The rate-equation level above is enough for cooling-power and threshold bookkeeping, but it does *not* produce the fluorescence spectrum on its own — only the time-domain field-amplitude equation does. The Lorentzian single-mode kernel of Section 5.11 should not be postulated; it is the leading approximation to the pole structure of the self-consistent coupled system. The aim of this section is to make that explicit.

Field, polarization, population

Replace the photon-number equation (5.10) by an amplitude equation, plus a polarization equation:

$$\dot{a} = \left(i\omega_{\text{c}} - \frac{\kappa_{\text{c}}}{2}\right) a + ig P + F_{\text{a}}(t), \quad (5.22)$$

$$\dot{P} = (i\omega_0 - \gamma_{\perp}) P + ig \Xi(N_1, N_2; \omega) a + F_{\text{P}}(t), \quad (5.23)$$

$$\dot{N}_2 = R_{\text{p}} - \Gamma_{\text{nr}}^0 N_2 - \Gamma_{\text{r}}^0 N_2 - \mathcal{R}_{\text{cav}}[a, P], \quad (5.24)$$

with κ_{c} the bare cavity linewidth, g the cavity–medium coupling, ω_0 the electronic-transition frequency, $\gamma_{\perp}$ the transverse decoherence rate, and $F_{\text{a}}, F_{\text{P}}$ the vacuum / Langevin noise sources. The cavity-mediated exchange term is $\mathcal{R}_{\text{cav}} = 2g \text{Im}(a^* P)$. The population-weighted optical response is

$$\Xi(N_1, N_2; \omega) = N_2 \xi_{\text{e}}(\omega) - N_1 \xi_{\text{a}}(\omega), \quad (5.25)$$

with ξ_e, ξ_a tied by the McCumber relation (5.21). This is the microscopic origin of “absorption bluer than fluorescence” inside the single-mode theory.

Frequency domain and the dressed pole

Treating N_1, N_2 as quasi-static on the optical timescale and Fourier transforming, (5.22)–(5.23) give

$$\chi_{\text{eff}}^{-1}(\omega) = -i(\omega - \omega_c) + \frac{\kappa_c}{2} + \Sigma(\omega), \quad \Sigma(\omega) = \frac{g^2 \Xi(\omega)}{\gamma_{\perp} - i(\omega - \omega_0)}, \quad (5.26)$$

with the medium susceptibility $\chi_m(\omega) = [\gamma_{\perp} - i(\omega - \omega_0)]^{-1}$. Near $\omega \approx \omega_c$ the dressed cavity is approximately $\chi_{\text{eff}}^{-1} \approx -i(\omega - \omega_{\text{eff}}) + \kappa_{\text{eff}}/2$ with

$$\boxed{\omega_{\text{eff}} = \omega_c + \text{Im } \Sigma(\omega_c), \quad \kappa_{\text{eff}} = \kappa_c + 2 \text{Re } \Sigma(\omega_c).} \quad (5.27)$$

The medium therefore renormalizes both the cavity frequency (Lamb-like shift) and the cavity linewidth. Since $\Xi(\omega_c) = N_2 \xi_e(\omega_c) - N_1 \xi_a(\omega_c)$, absorption-dominated regimes ($N_1 \xi_a > N_2 \xi_e$) *broaden* the mode, while strong emission-dominance narrows it toward threshold.

Modified fluorescence spectrum

The output spectrum, ignoring cross-correlations,

$$S_{\text{out}}(\omega) \propto \kappa_{\text{ex},f} |\chi_{\text{eff}}(\omega)|^2 g^2 |\chi_m(\omega)|^2 D_P(\omega, N_2, T), \quad (5.28)$$

with D_P the polarization-noise spectral density. Near a single isolated dressed pole this is approximately Lorentzian, but with parameters renormalized by (5.27). The Lorentzian is emergent, not assumed.

Reduction to the rate coefficients

Decomposing $\Sigma(\omega)$ into real and imaginary parts and identifying emission-driven, absorption-driven, and noise-driven pieces, the rate-equation coefficients of (5.9)–(5.10) become

$$G_f \sim 2g^2 \frac{\gamma_{\perp} \xi_e(\omega_f)}{\gamma_{\perp}^2 + (\omega_f - \omega_0)^2}, \quad (5.29)$$

$$A_f \sim 2g^2 \frac{\gamma_{\perp} \xi_a(\omega_f)}{\gamma_{\perp}^2 + (\omega_f - \omega_0)^2}, \quad (5.30)$$

$$S_f \sim \Gamma_{r,0} \xi_e(\omega_f) F_P^{(\text{cav})}(\omega_f), \quad (5.31)$$

with the cavity Purcell factor

$$F_P^{(\text{cav})}(\omega_f) = \frac{2 \zeta_f \mu^2}{\hbar \epsilon_0 n^2} \frac{Q_f}{V_f} \frac{\gamma_{\perp}}{\gamma_{\perp}^2 + (\omega_f - \omega_0)^2}, \quad (5.32)$$

where ζ_f is the spatial / orientation overlap factor, μ is the dipole moment, and the single-photon coupling is $g_f^2 = \zeta_f \mu^2 \omega_f / (2\hbar \epsilon_0 n^2 V_f)$. Equations (5.29)–(5.32) are the explicit microscopic forms of the reduced coefficients used at the rate-equation level.

5.10 Where $Q_f \leq Q_{\text{abs}}$ comes from

The dressed-pole expression (5.27) for κ_{eff} is also the cleanest derivation of the absorption-limited Q_f . Splitting the dressed linewidth,

$$\kappa_f = \kappa_{\text{c,bare}} + \kappa_{\text{act}}(\omega_f) + \kappa_{\text{bg}}(\omega_f), \quad (5.33)$$

where $\kappa_{\text{c,bare}}$ collects external coupling and passive cavity loss, while

$$\kappa_{\text{act}}(\omega_f) \approx 2g^2 \gamma_{\perp} \frac{N_1 \xi_a(\omega_f) - N_2 \xi_e(\omega_f)}{\gamma_{\perp}^2 + (\omega_f - \omega_0)^2}, \quad \kappa_{\text{bg}}(\omega_f) \sim v_g \xi_f \alpha_b(\omega_f). \quad (5.34)$$

In the cooling regime $N_1 \gg N_2$, absorption dominates and $\kappa_{\text{act}} > 0$: the medium contributes directly to the linewidth. Once the bare cavity linewidth becomes smaller than the absorption-induced linewidth, the dressed linewidth saturates at $\kappa_f \gtrsim \kappa_{\text{abs}} \equiv \kappa_{\text{act}} + \kappa_{\text{bg}}$, i.e.

$$Q_f = \frac{\omega_f}{\kappa_f} \lesssim \frac{\omega_f}{\kappa_{\text{abs}}} \equiv Q_{\text{abs}}(\omega_f). \quad (5.35)$$

Q_{abs} is not an independent constraint imported from elsewhere; it is the part of the dressed linewidth generated by the medium itself. Since $g^2 \propto 1/V_f$, the medium-induced linewidth does not disappear as the cavity is made nominally narrower — the same coupling that produces useful Purcell enhancement also produces the absorption-induced loss.

Effective Q_f for design. The design ceiling is therefore the harmonic combination

$$Q_f^{\text{eff}} = \left(\frac{1}{Q_{\text{design}}} + \frac{1}{Q_{\text{abs}}(\omega_f)} \right)^{-1}, \quad (5.36)$$

which interpolates between two regimes: when $Q_{\text{design}} \ll Q_{\text{abs}}$, improving fabrication helps; when $Q_{\text{design}} \gg Q_{\text{abs}}$, the system is absorption-limited and further fabrication improvements do not. In practice the optimum is $Q_{\text{design}}^{\text{opt}} \sim Q_{\text{abs}}(\omega_f)$, slightly below it so the cavity is not dominated by excess internal absorption.

Detuning-dependence. Pushing ω_f bluer increases the per-photon energy lift but typically increases $\xi_a(\omega_f)$ relative to $\xi_e(\omega_f)$, which raises κ_{act} and lowers $Q_{\text{abs}}(\omega_f)$. This is the principal reason the optimum ω_f does not sit at arbitrarily large blue detuning.

5.11 Reduction to a Lorentzian single-mode model

For design purposes it is convenient to collapse the cavity filter onto a single Lorentzian of centre ω_0 , HWHM $\Gamma/2$, spectrally overlapped with a Lorentzian bare-fluorescence line of centre $\bar{\omega}_{\text{fl}}$ and HWHM $\gamma_{\text{fl}}/2$. The overlap integral

$$\mathcal{O}(\Delta, \Gamma) = \int \frac{(\Gamma/2) (\gamma_{\text{fl}}/2)}{[(\omega - \omega_0)^2 + (\Gamma/2)^2][(\omega - \bar{\omega}_{\text{fl}})^2 + (\gamma_{\text{fl}}/2)^2]} d\omega \quad (5.37)$$

is itself a Lorentzian in the detuning $\Delta = \omega_0 - \bar{\omega}_{\text{fl}}$ with sum-width $(\Gamma + \gamma_{\text{fl}})/2$. Closed-form overlap with bookkeeping of the shifted mean frequency and routed fluorescence power carries through to Chapter 6, which differentiates $\text{COP}(\Delta, \Gamma)$ to produce the design rules Δ^*, Γ^* .

5.12 Design knobs in dimensionless form

Collect the optimization quantities into a small set of dimensionless numbers:

$$B \equiv \frac{\omega_f}{\omega_p} > 1 \quad (\text{blue-detuning factor}) \quad (5.38)$$

$$\eta_{\text{out},f} \equiv \frac{\kappa_{\text{ex},f}}{\kappa_f} \quad (\text{fluorescence-mode extraction fraction}) \quad (5.39)$$

$$R_f \equiv \frac{A_f N_t}{\kappa_f} \quad (\text{reabsorption / photon-recycling loading}) \quad (5.40)$$

$$C_f \equiv \frac{S_f}{\kappa_f} \quad (\text{cavity-capture / Purcell parameter}) \quad (5.41)$$

$$\mathcal{G}_f \equiv \frac{N_t G_f}{\kappa_f} \quad (\text{stimulated-emission loading}) \quad (5.42)$$

$$\phi_f \equiv \frac{\kappa_{\text{bg},f}}{\kappa_f} \quad (\text{fluorescence parasitic fraction}). \quad (5.43)$$

In the cooling regime $N_2 \ll N_t$ the photon occupation reduces to

$$n_f \approx \frac{C_f N_2}{1 + R_f}. \quad (5.44)$$

The cavity contribution to the fluorescence yield ratio (the escaped-photon power per unit absorbed pump power; note the factor κ_f , without which the expression is not dimensionless) becomes

$$\rho_f^{(f)} \approx \frac{\omega_f}{\omega_p} \eta_{\text{out},f} \frac{\kappa_f C_f N_2 / R_p}{1 + R_f}, \quad \eta_{\text{ASF}}^{(f)} = \rho_f^{(f)} - 1, \quad (5.45)$$

Spectral asymmetry packed into a single coupling strength. Combining (5.29)–(5.32) with (5.43), all of the spectral asymmetry collapses into one dimensionless coupling

$$\Lambda_f \equiv N_t \zeta_f \frac{\mu^2}{\hbar \epsilon_0 n^2} \frac{Q_f^2}{\omega_f V_f} \frac{\gamma_{\perp}}{\gamma_{\perp}^2 + \Delta_f^2}, \quad (5.46)$$

with $\Delta_f = \omega_f - \omega_0$, in terms of which $R_f = \Lambda_f \xi_a(\omega_f)$, $\mathcal{G}_f = \Lambda_f \xi_e(\omega_f)$, and the threshold excited fraction $x_{\text{th}} \equiv N_{2,\text{th}}/N_t$ is

$$x_{\text{th}} = \frac{1 + \Lambda_f \xi_a(\omega_f)}{\Lambda_f [\xi_a(\omega_f) + \xi_e(\omega_f)]}. \quad (5.47)$$

Note the asymptotic behavior: as $\Lambda_f \rightarrow \infty$, $x_{\text{th}} \rightarrow \xi_a/(\xi_a + \xi_e)$ — the threshold saturates to a spectral-asymmetry limit set by the McCumber relation, not zero. Pushing Q^2/V does not give arbitrary benefit even before fabrication or absorption ceilings are imposed.

5.13 Largest stable extracted cooling power

The threshold pump excitation rate at which N_2 reaches $N_{2,\text{th}}$ is approximately

$$R_{p,\text{th}} \approx \Gamma_{\text{tot,eff}} N_{2,\text{th}} = (\Gamma_{\text{nr}}^0 + \Gamma_{r,u}^0 + S_f) \frac{\kappa_f + A_f N_t}{A_f + G_f}. \quad (5.48)$$

Two qualitatively different things happen as the photonic design varies.

Purcell enhancement raises the tolerable pump. In the weak-photon regime $\dot{N}_2 \approx R_p - \Gamma_{\text{tot,eff}} N_2$, so for fixed R_p , increasing S_f lowers N_2 . Since the instability threshold is set by $N_2 < N_{2,\text{th}}$, a larger Purcell-enhanced radiative drain lets the system tolerate a larger pump rate before hitting transparency / self-oscillation. This is the mechanism by which Purcell enhancement raises the largest stable extracted cooling power.

Reabsorption and recycling are double-edged. Larger A_f raises the numerator of (5.14) (more loss to balance) but it also recycles photons through the medium, which in the favourable regime increases the mean escaped photon energy. Whether (5.48) grows or shrinks depends on which of these dominates — captured by whether $\partial R_{p,\text{th}}/\partial Q_f$ at fixed ω_f is positive or negative.

Four physical regimes. The reduced model thus predicts

- **(i) Low- Q , weak cavity influence:** C_f small, selected-mode channel unimportant; bulk ASF dominates.
- **(ii) Useful cavity enhancement:** C_f increases, the blue-detuned mode becomes a significant escape channel, and cooling improves.
- **(iii) Reabsorption-assisted spectral sharpening:** R_f is non-negligible but parasitic loss is still controlled. Recycling raises the mean escaped photon energy and P_{cool} grows even as P_f plateaus.
- **(iv) Over-coupled / high-residence-time regime:** as Q approaches Q_{abs} , photons dwell too long, R_f grows, and useful extraction saturates or falls.

There is therefore an interior optimum, not a monotonic trend.

5.14 Optimization conditions

The reduced cavity contribution to cooling is $\Phi_f \propto (\kappa_{\text{ex},f} - \kappa_{\text{bg},f}) C_f / (1 + R_f)$. Optimizing it with respect to each design knob in turn:

Optimal external coupling: generalized critical coupling

Write $\kappa_f = \kappa_{\text{ex},f} + \kappa_{\text{par},f}$ with $\kappa_{\text{par},f} = \kappa_{i,f} + \kappa_{\text{bg},f} + \kappa_{\text{act},f}$. In the weak-recycling limit $R_f \Lambda_f \ll 1$, $\Phi_f \propto \kappa_{\text{ex},f} / (\kappa_{\text{ex},f} + \kappa_{\text{par},f})^2$, maximised at

$$\kappa_{\text{ex},f}^{\text{opt}} \approx \kappa_{\text{par},f} = \kappa_{i,f} + \kappa_{\text{bg},f} + \kappa_{\text{act},f}, \quad (5.49)$$

or equivalently $Q_{\text{ex},f}^{-1} \approx Q_{i,f}^{-1} + Q_{\text{bg},f}^{-1} + Q_{\text{act},f}^{-1}$. The conventional critical-coupling criterion reappears, but with the parasitic channel enlarged to include active reabsorption loading. With strong recycling ($R_f \gtrsim 1$), the optimum shifts slightly toward over-coupling: $\kappa_{\text{ex},f}^{\text{opt}} \sim \kappa_{\text{par},f}(1 + R_f)$, because shorter residence times beat repeated reabsorption.

Optimal design Q_f

Combining (5.36) with the saturation behaviour of (5.47),

$$Q_{\text{design}}^{\text{opt}} \sim Q_{\text{abs}}(\omega_f), \quad (5.50)$$

more precisely just below it. Pushing Q_{design} above Q_{abs} does not help because medium absorption caps the realisable Q_f ; pushing it well below leaves Purcell benefit on the table.

Optimal fluorescence detuning ω_f

Differentiating $\ln \Phi_f$ with respect to ω_f ,

$$\boxed{\frac{1}{\omega_f} + 2 \frac{d \ln Q_f^{\text{eff}}}{d\omega_f} + \frac{d \ln \xi_e}{d\omega_f} - \frac{2\Delta_f}{\gamma_{\perp}^2 + \Delta_f^2} - \frac{d \ln(1 + R_f)}{d\omega_f} \approx 0.} \quad (5.51)$$

The five terms are: energetic incentive to go bluer ($1/\omega_f$); emission-spectrum penalty ($d \ln \xi_e/d\omega < 0$ because ξ_e drops on the blue side); optical detuning penalty (the medium susceptibility falls off the electronic resonance); the medium-induced Q penalty (absorption strengthens blueward); and reabsorption-loading penalty. The optimum sits between the emission peak ω_e and the absorption peak ω_a ,

$$\boxed{\omega_e < \omega_f^{\text{opt}} < \omega_a,} \quad (5.52)$$

typically closer to the blue shoulder of fluorescence than to the absorption peak.

Optimal pump detuning ω_p

The pump excitation rate scales as

$$R_p \propto \sigma_a(\omega_p) \frac{\kappa_{\text{ex,p}}}{\Delta_p^2 + (\kappa_p/2)^2} P_{\text{in}}/(\hbar\omega_p), \quad (5.53)$$

and the cooling lift per absorbed pump photon scales like $(\omega_f^{\text{opt}}/\omega_p - 1)\sigma_a(\omega_p) \cdots$ minus the pump-background-absorption penalty. The optimum is *interior*: too blue gives strong absorption but small anti-Stokes lift; too red gives strong lift but $\sigma_a(\omega_p)$ collapses and background heating dominates. Concretely,

$$\boxed{\text{maximize } \sigma_a(\omega_p) \frac{\kappa_{\text{ex,p}}}{\Delta_p^2 + (\kappa_p/2)^2} \left(\frac{\omega_f^{\text{opt}}}{\omega_p} - 1 \right) - \text{pump-bg penalty.}} \quad (5.54)$$

Threshold extracted power at the optimum

At the optimum design, combining (5.48) with (5.47),

$$\boxed{R_{\text{p,th}}^{\text{opt}} \sim N_t \left[\Gamma_b + \kappa_f^{\text{opt}} \frac{C_f^{\text{opt}}}{1 + R_f^{\text{opt}}} \right] \frac{1 + R_f^{\text{opt}}}{R_f^{\text{opt}} + \mathcal{G}_f^{\text{opt}}},} \quad (5.55)$$

with $\Gamma_b = \Gamma_{\text{nr}}^0 + \Gamma_{r,u}^0$. This is the largest stable extracted cooling power in single-mode form: it sets the absolute scale that the thermophotonic master equation (1.17) must respect.

Four physical conclusions

1. The apparent Q_f^2/V_f gain from Purcell enhancement is always moderated by the identical Q_f^2/V_f growth of reabsorption loading, and by the absorption-imposed ceiling $Q_f \leq Q_{\text{abs}}$.
2. The best cavity is *not* the highest- Q cavity, but one designed so that the usable Q is near Q_{abs} and the external coupling is near generalized critical coupling.
3. The best fluorescence mode is blue of the pump and usually blue of the fluorescence peak, but not so blue that ξ_e collapses while ξ_a and Q_{abs}^{-1} explode.
4. The optimal pump is red of the absorption peak but not too far red, because the anti-Stokes lift must be balanced against the ability to actually absorb pump photons.

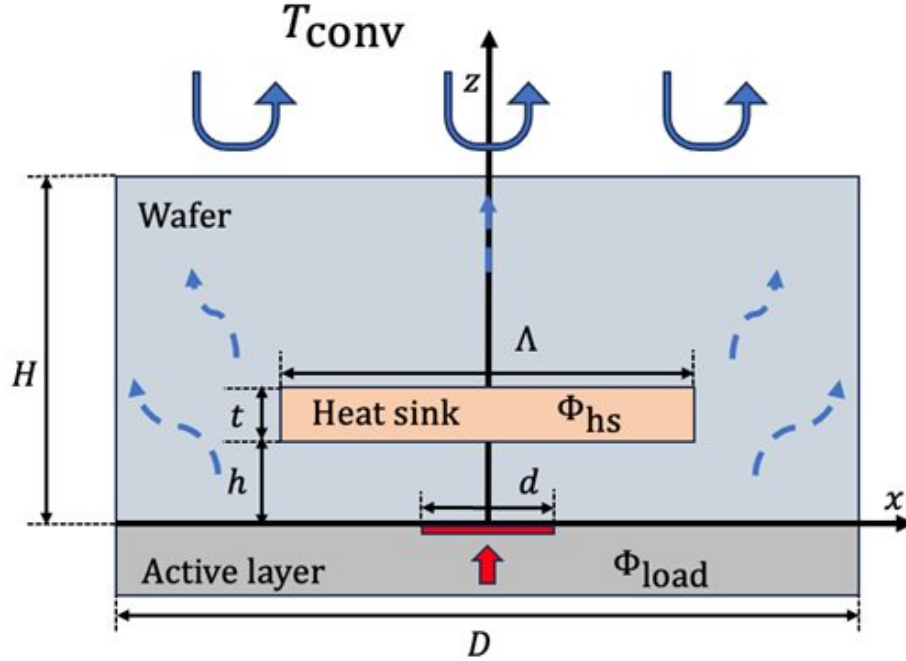


Figure 5.1: Thermal boundary condition under which the coupled EO equations are solved. The active layer (bottom) carries the hotspot heat flux Φ_{load} ; a secondary heat sink of pitch Λ sits at height h within the wafer and receives Φ_{hs} . The extractor temperature T_h entering (5.9),(5.20) is the temperature of the active layer in the immediate vicinity of the pump spot of width d ; $\gamma_{\text{nr}}(T_h)$ and the reabsorption factor $\xi_{\text{rec}}(T_h)$ are temperature-dependent and must be recomputed self-consistently with the thermal profile.

5.15 Thermal boundary condition on the extractor

The coupled-mode equations above assume a fixed lattice temperature T_h . In the actual device T_h is set self-consistently by the interplay of (a) the non-radiative heating channel $\Gamma_{\text{nr}}^0 N_2$ from (5.9), (b) the ASF cooling flux $P_{\text{cool}} = P_f - P_{\text{abs}} - P_{\text{nr}}$ of (1.6), and (c) the external thermal resistance to the heat sink (Figure 5.1). In steady state the extractor temperature balances laser-delivered cooling power against the chip heat load Φ_{load} and the secondary heat-sink flux Φ_{hs} on a characteristic pitch Λ .

5.16 Implications for inverse design of the stack

Every symbol in the master equation (5.20) — equivalently in (5.45) or (5.46) — is an inverse-design target for the layered stack of Chapter 4:

- **Photonic network.** Sets $\kappa_{\text{ex,p}}, \kappa_{\text{ex,f}}$ and the Q, V entering F_P , hence A_{eff} and Λ_f . Independent tuning of the two channels (pump vs. fluorescence) is what enables critical coupling at ω_p without sacrificing narrowness at ω_f .
- **Coupling layer.** Controls Γ_{conf} (field overlap) and therefore the absorption-limited ceiling (4.5); sets the $\zeta_f \mu^2 Q_f^2 / (\omega_f V_f)$ product entering Λ_f .
- **Extractor.** Provides $\sigma_a, \sigma_e, \Gamma_r^0, \Gamma_{\text{nr}}^0, \alpha_b, \xi_e, \xi_a$, and the McCumber asymmetry; choice of platform (crystal, semiconductor, dispersion) partitions these quantities as summarized in Chapter 8.
- **Back reflector.** Enforces a second pump pass (raising $\kappa_{\text{p,act}}/\kappa_p$ toward critical coupling) and sets the angular / spectral boundary on the fluorescence mode that controls η_{esc} .

- **Sensor layer.** Does not enter (5.20) at steady state, but closes the thermal control loop. It must be placed beneath a reflector node of the fluorescence mode to avoid parasitic absorption contributing to $\kappa_{\text{bg},f}$.

5.17 Chapter summary

- The extractor + photonic network is a two-resonance emitter–cavity system whose steady-state photon occupation is given by (5.12), with threshold (5.14) and population balance (5.15).
- Cooling power has the explicit form (5.18); the cavity contribution carries the $\hbar(\omega_f - \omega_p)\kappa_{\text{ex},f}n_f$ anti-Stokes lift but pays the $\hbar(\omega_f + \omega_p)\kappa_{\text{bg},f}n_f$ background-absorption penalty.
- Reabsorption is a feature, not a bug (Section 5.7): four mechanisms through which it raises P_{cool} without collapsing the cooling cycle.
- Absorption-blue / fluorescence-red asymmetry comes from the McCumber relation (5.21); in the field-amplitude formulation it is encoded by ξ_e/ξ_a and the medium susceptibility.
- The Lorentzian single-mode model is *emergent*: it is the dressed-pole approximation (5.27) to the self-consistent field–polarization theory, with the medium contributing $\Sigma(\omega)$ to the cavity self-energy.
- $Q_f \leq Q_{\text{abs}}(\omega_f)$ is not an external constraint — it follows directly from $\kappa_f = \kappa_c + 2 \text{Re } \Sigma$ once medium absorption is included (Section 5.10).
- The reduction $\eta_{\text{out},f}, R_f, C_f, \mathcal{G}_f, \Lambda_f$ of Section 5.12 reduces the optimization to a small set of dimensionless numbers; the threshold fraction x_{th} saturates at the spectral asymmetry $\xi_a/(\xi_a + \xi_e)$ as $\Lambda_f \rightarrow \infty$.
- Optimal design sits at generalized critical coupling (5.49) with $Q_{\text{design}} \sim Q_{\text{abs}}$ (5.50), fluorescence detuning between ω_e and ω_a (5.52), and pump detuning red of the absorption peak (5.54).

The Lorentzian reduction (Section 5.11) is the hand-off to Chapter 6, where the detailed design rules for the resonance shape emerge in fully nondimensional form.

Part III

Limits and Conversion

Chapter 6

Energy and Fluorescence Limits

6.1 Why the Carnot bound is not about bandwidth

A recurring confusion is that, in a closed loop pumped by a coherent laser and read out by an LPC, the laser pump is treated as essentially pure exergy while the anti-Stokes fluorescence — which is nearly narrowband compared to a thermal source — is treated as Carnot-bounded. Why does the LPC see the laser at $\sim 100\%$ recovery efficiency, but the fluorescence at a Carnot factor?

The Carnot bound is not about bandwidth. It is about whether the radiation carries entropy — equivalently, whether it is in thermal equilibrium with a temperature.

Why the laser pump is treated as work-like. A laser beam, even though it has finite bandwidth, is highly coherent, directional, and far from thermal equilibrium. Its entropy per unit energy is extremely low; in the thermodynamic sense it is very close to a work reservoir. An LPC reaching $\sim 100\%$ on a laser input is not violating Carnot, because it is not converting heat into work — it is converting work-like radiation into electrical work.

Why fluorescence is treated as Carnot-limited (in the many-mode limit). Spontaneous fluorescence, even if narrowband, is incoherent, isotropic, and carries entropy tied to the lattice temperature. Treating it as a thermal radiation reservoir at T_h gives the familiar Carnot-fraction exergy $(1 - T_0/T_h)$. But this is the *many-mode, thermally occupied* limit; it is not a fundamental statement about narrowband fluorescence. When the same energy is engineered into a small number of low-entropy modes, the bound relaxes. The discriminator that Section 6.5 formalizes is *entropy per unit energy*, equivalently phase-space density / étendue / brightness — not bandwidth per se.

Result 6.1 (Exergy of a radiation stream). For a stream carrying power $\dot{H}$ with accompanying entropy flux $\dot{S}$, exergy evaluated against a reference reservoir at T_0 is $\mathcal{E} = \dot{H} - T_0 \dot{S}$. The entropy flux depends on the full photon distribution function $f(\omega, \hat{\Omega}, \lambda)$ — frequency, solid angle, polarization — via $\dot{S} \propto \int [(1 + f) \ln(1 + f) - f \ln f] dN$. Work-like limits ($\dot{S} \rightarrow 0$) are approached by single-mode coherent fields regardless of bandwidth.

Result 6.1 is the hinge of the entire chapter. The task is to evaluate $\dot{S}$ in the canonical regimes of interest — diffuse fluorescence (the “Carnot limit”) and a single engineered leaky resonance (the photonic-network limit) — and then to make the connection back to P_{cool} , η_{ASF} , and the closed-loop COP of Chapters 1 and 5.

6.2 Photonic engineering and the exergy ladder

If you engineer a photonic resonant system that does photon recycling and converts the fluorescence into narrowband, more spectrally coherent, highly directional emission — though not a coherent laser — how should the exergy be treated?

Stop thinking of the output as “just thermal fluorescence.” Treat it as a nonequilibrium photon reservoir whose exergy depends on its full optical state: spectrum, angular distribution, polarization, coherence properties, and (most importantly) photon occupation / chemical potential.

What resonant engineering actually changes.

- **Spectral narrowing** reduces the number of occupied frequency modes; alone it does not automatically raise exergy, but it helps when accompanied by controlled modal occupation.
- **Directionality** is decisive. Ordinary fluorescence spreads into many angular modes, which is a major entropy source. Funnelling the output into a narrow angular cone drops the entropy substantially.
- **Photon recycling** helps if it preferentially re-emits into low-entropy modes; if it merely traps photons until they thermalize, it does *not* raise exergy.
- **Purcell enhancement** reshapes the modal entropy of the output, but does not, on its own, create work from heat; it only helps if it channels emission into a lower-entropy optical state.

An exergy ladder. The emitted radiation can be placed on a spectrum:

Regime	Optical character	Exergy fraction $\mathcal{E}/\dot{H}$
high-entropy	broad, incoherent, isotropic, many modes	$\sim 1 - T_0/T_h$ (Carnot)
single-mode	narrowband, directional, mode-selected	$\rightarrow 1$ up to entropy correction
coherent laser	phase-coherent, highly occupied	≈ 1

Resonant engineering is the work of moving the ASF output from the top row toward the middle row. This does not evade the second law: if a passive structure reduces the entropy of the extracted optical field, the entropy reduction has to be paid somewhere else in the full-system accounting (rejected power, parasitic absorption, or drawing on the nonequilibrium electronic free energy of the active medium).

6.3 Mode-counting toy model

Before the formal spectral–angular functional, work out a mode-counting toy model that makes the entropy compression benefit of photonic engineering quantitatively explicit. Treat the emitted fluorescence as a set of bosonic modes of energy $\varepsilon = \hbar\omega_{\text{fl}}$ each, with total photon number N_{ph} distributed uniformly over $\mathcal{N}_{\text{mode}}$ accessible modes, mean occupation $\bar{n} = N_{\text{ph}}/\mathcal{N}_{\text{mode}}$. Total energy $E = N_{\text{ph}} \varepsilon$; single-mode bosonic entropy $s(\bar{n}) = k_B[(1 + \bar{n}) \ln(1 + \bar{n}) - \bar{n} \ln \bar{n}]$; total

$$S = \mathcal{N}_{\text{mode}} k_B [(1 + \bar{n}) \ln(1 + \bar{n}) - \bar{n} \ln \bar{n}], \quad \mathcal{E} = E - T_0 S. \quad (6.1)$$

Dilute limit, $\bar{n} \ll 1$ (many modes per photon). $S \simeq N_{\text{ph}} k_B [1 - \ln \bar{n}]$, so

$$\mathcal{E}_{\text{dilute}} \simeq E - N_{\text{ph}} k_B T_0 [1 + \ln(\mathcal{N}_{\text{mode}}/N_{\text{ph}})]. \quad (6.2)$$

Entropy is logarithmic in the mode count; halving $\mathcal{N}_{\text{mode}}$ buys a constant exergy boost $k_B T_0 \ln 2$ per photon, which at $T_0 = 300 \text{ K}$ is $\sim 18 \text{ meV}$ ($k_B T_0 \ln 10 \simeq 60 \text{ meV}$ per decade). Over a 1-eV-photon stream this is a $\sim 2\%$ exergy improvement per *halving*, i.e. $\sim 6\%$ per decade of mode-count reduction.

Concentrated limit, $\bar{n} \gg 1$ (few modes, heavily occupied). $\ln(1 + \bar{n}) \simeq \ln \bar{n}$ and $S \simeq \mathcal{N}_{\text{mode}} k_B [\ln(N_{\text{ph}}/\mathcal{N}_{\text{mode}}) + 1]$, so as $\mathcal{N}_{\text{mode}}/N_{\text{ph}} \rightarrow 0$, $\mathcal{E} \rightarrow E$ and the fluorescence becomes work-like. This is the formal justification for $\mathcal{E} \rightarrow P_{\text{f}}$ in the single-mode limit.

What “mode count” means physically. Factor $\mathcal{N}_{\text{mode}} \simeq \mathcal{N}_{\omega} \cdot \mathcal{N}_{\Omega} \cdot \mathcal{N}_{\text{spatial}} \cdot \mathcal{N}_{\text{pol}}$: each of (spectral width) $\times$ (angular spread) $\times$ (spatial modes) $\times$ (polarizations) multiplies entropy. This is why “bandwidth alone” is the wrong axis: a narrowband but isotropic field can have enormous $\mathcal{N}_{\Omega}$ and large entropy, whereas a somewhat broader but single-spatial-mode field can have far less. Purcell enhancement reduces $\mathcal{N}_{\Omega} \cdot \mathcal{N}_{\text{spatial}} \cdot \mathcal{N}_{\text{pol}}$ by raising the branching ratio into the engineered mode; a mode-selective β -factor near unity collapses several decades of entropy at once.

Why this does not violate the second law. A passive photonic structure cannot create exergy. If the extracted optical field has smaller S than a diffuse reference, one of three things happens elsewhere: (i) total extracted optical power drops (“cleaner light, but less of it”); (ii) entropy is exported into non-radiative or rejected channels; (iii) the nonequilibrium free energy stored in the electronic population of the extractor is being drawn down. The second law holds at the full-system level; the toy model bounds only the particular stream we care about.

6.4 Master framework

Let the emitted optical field occupy modes m with mean occupation n_m and frequency ω_m , so $E_{\text{out}} = \sum_m \hbar \omega_m n_m$ and $S_{\text{out}} = k_B \sum_m [(1 + n_m) \ln(1 + n_m) - n_m \ln n_m]$. At optical frequencies the ambient thermal photon occupation is negligible, so

$$\boxed{B_{\text{out}} = E_{\text{out}} - T_0 S_{\text{out}}.} \quad (6.3)$$

Every result in the chapter is a consequence of this object.

ASF + cavity + LPC loop bound

For the closed-loop loop of Chapter 1: pump optical power P_L , pump coupling η_{cpl} , pump-derived absorbed power $P_{\text{abs}} = \eta_{\text{cpl}} P_L$, chip-derived heat $Q_c = \eta_{\text{ASF}} P_{\text{abs}}$, emitted fluorescence P_{f} , photonic-structure-shaped delivery η_{del} , LPC efficiency η_{LPC} . The recovered electrical power is $P_{\text{rec}} = \eta_{\text{LPC}} \eta_{\text{del}} P_{\text{f}}$. The second law does not bound η_{LPC} directly by Carnot; it bounds the recoverable electrical power by the exergy of the delivered optical field:

$$\boxed{P_{\text{rec}} \leq B_{\text{del}} = E_{\text{del}} - T_0 S_{\text{del}},} \quad (6.4)$$

or equivalently $\eta_{\text{LPC}} \leq B_{\text{del}}/E_{\text{del}} = 1 - T_0 S_{\text{del}}/E_{\text{del}}$ with equality only for a reversible converter. In closed-loop form, using $P_{\text{f}} = P_{\text{abs}} + Q_c$,

$$P_{\text{rec}} \leq B_{\text{f}} \leq P_{\text{abs}} + Q_c(1 - T_0/T_{\text{h}}) = \eta_{\text{cpl}} P_L [1 + \eta_{\text{ASF}}(1 - T_0/T_{\text{h}})]. \quad (6.5)$$

This is the same exergy ceiling that Section 1.11 extracted at the bookkeeping level; the cavity / metagrating matters because it changes B_{del} (the optical exergy quality), not the energy E_{del} itself.

Étendue / brightness theorem

The number of accessible radiation microstates is essentially phase-space volume: area $\times$ solid angle $\times$ bandwidth $\times$ polarization, i.e. étendue. In the toy model this is $\mathcal{N}_{\text{mode}} \propto (\Delta\omega/2\pi)(A\Omega/\lambda^2)\mathcal{N}_{\text{pol}}\mathcal{N}_{\text{spatial}}$. The brightness theorem says a passive optical system cannot raise radiance beyond the source value; thermodynamically, a passive system cannot reduce the phase-space entropy of radiation without paying elsewhere. The cavity / metagrating moves fluorescence from large $\mathcal{N}_{\text{mode}}$ (low brightness, high entropy per unit energy) toward small $\mathcal{N}_{\text{mode}}$ (high brightness, low entropy per unit energy) — but only by rejecting power, relying on source nonequilibrium, or generating entropy elsewhere. This is the optical version of the second law.

Carnot factor as the high-entropy thermal limit

Specialize (6.1) to a thermalized photon reservoir at effective hot temperature T_h with optical-frequency dilute occupation $\bar{n} \approx e^{-\hbar\omega/k_B T_h}$. Then $\ln(\mathcal{N}_{\text{mode}}/N_{\text{ph}}) \approx \hbar\omega/k_B T_h$, and (6.2) gives

$$\frac{B}{E} \approx 1 - \frac{T_0}{T_h} - \frac{k_B T_0}{\hbar\omega}. \quad (6.6)$$

For optical photons $\hbar\omega \gg k_B T_0$, so the last term is a small correction:

$$\boxed{\frac{B}{E} \approx 1 - \frac{T_0}{T_h} \quad (\text{thermal narrowband optical, high-entropy limit}).} \quad (6.7)$$

The Carnot factor emerges naturally as the high-entropy thermal limit of the toy model. Exact thermal-radiation exergy (Petela / Landsberg-type) refines the constant, but the Carnot $(1 - T_0/T_h)$ is the leading scaling.

Stronger bound for a metagrating / single leaky mode

Now the low-entropy structured-emission case. Q and V do not directly set exergy by themselves; they enter through mode selectivity, Purcell factor, β -factor into the desired mode, linewidth, angular channel count, outcoupling, photon lifetime, and reabsorption. Split the output into a structured channel (fraction β) and a diffuse background (fraction $1 - \beta$): $E_{\text{out}} = E_1 + E_{\text{bg}}$, $S_{\text{out}} = S_1 + S_{\text{bg}}$. For the dominant single mode of occupation n_1 , $S_1 = k_B[(1 + n_1)\ln(1 + n_1) - n_1 \ln n_1]$. In the heavily-occupied limit $n_1 \gg 1$,

$$\boxed{\frac{B_1}{E_1} \approx 1 - \frac{k_B T_0}{\hbar\omega_f n_1} (1 + \ln n_1),} \quad (6.8)$$

which approaches unity as $n_1 \rightarrow \infty$. This is the cleanest “stronger-than-Carnot” bound for low-entropy emission. The role of Q and V is via the effective mode count $\mathcal{N}_{\text{mode,eff}} \sim (\omega_0\tau/2\pi Q)\mathcal{N}_\Omega\mathcal{N}_{\text{pol}}$ and the β -factor $\beta \sim F_P/(F_P + F_{\text{other}} + \Gamma_{\text{nr}}/\Gamma_0)$ with $F_P \propto Q/V$. Larger Q/V raises $\beta \rightarrow 1$, reduces $\mathcal{N}_{\text{mode,eff}}$, and pushes B/E toward unity.

6.5 Spectral-angular exergy functional

For a steady photon field with occupation $\bar{n}(\omega, \hat{\Omega}, \lambda)$ propagating through surface A , the energy flux is

$$\dot{H} = \sum_{\lambda} \int d\omega d\hat{\Omega} \frac{\omega^2}{4\pi^3 c^2} \hbar\omega \bar{n} \cos\theta, \quad (6.9)$$

and the entropy flux is

$$\dot{S} = \sum_{\lambda} \int d\omega d\hat{\Omega} \frac{\omega^2}{4\pi^3 c^2} k_B [(1 + \bar{n}) \ln(1 + \bar{n}) - \bar{n} \ln \bar{n}] \cos\theta. \quad (6.10)$$

Exergy is $\mathcal{E} = \dot{H} - T_0 \dot{S}$. Two limits follow immediately.

6.5.1 Many-mode, thermally occupied: Carnot

If the emitter radiates into $\mathcal{N}_{\text{mode}} \gg 1$ modes at lattice brightness temperature T_h , with $\bar{n} = (e^{\hbar\omega/k_B T_h} - 1)^{-1}$, direct evaluation gives

$$\mathcal{E} \leq \dot{H} \left(1 - \frac{T_0}{T_h}\right), \quad (6.11)$$

reproducing the diffuse-fluorescence Carnot bound. This is *where the Carnot factor comes from*: from the entropy of a thermally occupied many-mode radiation bath, not from any assumption about bandwidth.

6.5.2 Single leaky resonance: cavity-limited exergy

Specialize to a single resonance at ω_0 , HWHM $\Gamma/2$, mode volume V , occupation $\bar{n}_0$. The radiated power into that mode is

$$P_{\text{fl}} = \hbar\omega_0 \frac{\Gamma}{2} \bar{n}_0 \eta_{\text{esc}}. \quad (6.12)$$

Because the mode is a single mode (angular, polarization, spectral), its entropy flux is controlled by $\bar{n}_0$ alone: the entropy per photon is $k_B[(1 + \bar{n}_0) \ln(1 + \bar{n}_0) - \bar{n}_0 \ln \bar{n}_0]/\bar{n}_0 \simeq k_B(1 - \ln \bar{n}_0)$ for $\bar{n}_0 \ll 1$, which *diverges* logarithmically as the mode empties. The work-like limit is therefore the *bright* one, $\bar{n}_0 \gg 1$ (cf. (6.2) and Section 7.3),

$$\mathcal{E} \rightarrow P_{\text{fl}} \quad (\text{work-like, } \bar{n}_0 \gg 1), \quad \frac{\mathcal{E}}{P_{\text{fl}}} \simeq 1 - \frac{k_B T_0}{\hbar\omega_0} \left[1 + \ln \frac{1}{\bar{n}_0}\right] \quad (\bar{n}_0 \ll 1), \quad (6.13)$$

so that even a dim single mode at $\hbar\omega_0 = 2 \text{ eV}$, $\bar{n}_0 = 10^{-3}$ carries $\simeq 0.90$ of its energy as exergy — the “ ≈ 1 ” owes more to $\hbar\omega_0 \gg k_B T_0$ than to single-mode-ness. Purcell enhancement helps by concentrating the emission into few modes at high occupation. The Carnot bound (6.11) is an artefact of summing over many modes at equal brightness temperature; engineering the photonic stack to radiate into one low-entropy mode *beats* (6.11) without violating the second law — it violates the many-mode premise.

6.6 Single-mode exergy as nonequilibrium free energy

The Carnot / many-mode bound and the single-mode work-like limit are the two asymptotes of a cleaner bookkeeping: the non-equilibrium free energy of the fluorescence mode *relative to the ambient photon environment*. This is the formulation Chapters 4 and 5 already adopted; we collect the key results here.

Displaced-thermal decomposition. Represent the fluorescence mode's state as a displaced thermal state with coherent amplitude α and incoherent occupation $\bar{n}_{\text{th}}$, $\bar{n} = |\alpha|^2 + \bar{n}_{\text{th}}$. The von Neumann entropy depends only on the incoherent part, $s_{\text{m}}(\bar{n}_{\text{th}}) = k_{\text{B}} g(\bar{n}_{\text{th}})$ with $g(n) = (n+1) \ln(n+1) - n \ln n$. The coherent displacement shifts the mode energy by $\hbar\omega_0 |\alpha|^2$ without paying any entropy penalty.

Mode exergy. At ambient T_0 with negligible thermal $\bar{n}_{\text{a}}$ at optical frequencies,

$$\mathcal{E}_{\text{mode}} \approx \hbar\omega_0 (|\alpha|^2 + \bar{n}_{\text{th}}) - T_0 k_{\text{B}} g(\bar{n}_{\text{th}}). \quad (6.14)$$

The coherent part is all exergy; the incoherent part is partially recoverable as a nonequilibrium free energy — not a Carnot factor applied to a heat flow.

Exergy flux. If the mode escapes at rate $\kappa_{\text{f,e}}$,

$$\dot{\mathcal{E}}_{\text{f}} = \kappa_{\text{f,e}} [\hbar\omega_0 \bar{n} - T_0 k_{\text{B}} g(\bar{n}_{\text{th}})]. \quad (6.15)$$

This is the quantity that caps the recoverable electrical power of the LPC in Chapter 7, replacing any Carnot-times- P_{f} estimate.

Exergy fraction.

$$\zeta_{\text{f}} \equiv \frac{\dot{\mathcal{E}}_{\text{f}}}{P_{\text{fl}}} = 1 - \frac{T_0 k_{\text{B}} g(\bar{n}_{\text{th}})}{\hbar\omega_0 \bar{n}}. \quad (6.16)$$

Two design levers raise ζ_{f} : increasing the coherent fraction $|\alpha|^2/\bar{n}$ through stimulated / recycling processes, and increasing $\hbar\omega_0$ relative to $k_{\text{B}}T_0$ through anti-Stokes blue shift.

Power-generation condition. Combining (6.15) with the closed-loop bookkeeping of Chapter 1,

$$\eta_{\text{LPC}} \zeta_{\text{f}} P_{\text{fl}} > \frac{P_{\text{pump}}}{\eta_{\text{P}}} \quad (6.17)$$

is the single-mode refinement of the Carnot condition of Section 1.13. The relevant thermodynamic potential is the fluorescence mode's non-equilibrium free energy, not any externally imposed reservoir temperature.

6.7 Resonance impacts on extracted power and ASF efficiency

A photonic resonance does not only improve the *quality* of emitted light. It also changes the *quantity*, the *mean emission frequency*, and (sometimes) the *pump absorption* and *excited-state population dynamics*. The device problem is therefore not just “maximize exergy fraction,” but jointly optimize cooling power, optical recovery, COP, and possibly net electrical generation.

Standard ASF cooling power. An idealized expression is

$$\dot{Q}_{\text{c}} = P_{\text{abs}} \left(\eta_{\text{ext}} \frac{\bar{\omega}_{\text{f}}}{\omega_{\text{p}}} - 1 \right), \quad (6.18)$$

with η_{ext} the external fluorescence quantum efficiency and $\bar{\omega}_{\text{f}}$ the mean emitted frequency. The resonance can modify all of (i) P_{abs} , (ii) η_{ext} , (iii) $\bar{\omega}_{\text{f}}$, and (iv) the exergy quality of the fluorescence.

Resonance-modified mean frequency. Let the resonance spectral weighting be $W(\omega)$ and the bare emission spectrum $S_0(\omega)$. The resonance-modified spectrum is $S_r(\omega) \propto S_0(\omega) F_P(\omega) \eta_{\text{esc}}(\omega)$, and

$$\bar{\omega}_f^{(r)} = \frac{\int d\omega \omega S_r(\omega)}{\int d\omega S_r(\omega)}. \quad (6.19)$$

If the resonance preferentially enhances the high-frequency side of the fluorescence, $\bar{\omega}_f^{(r)} > \bar{\omega}_f^{(0)}$, which directly raises $\eta_{\text{ASF}}^{(\text{ideal})} \sim \bar{\omega}_f/\omega_p - 1$. This is the thermodynamic value of photon-recycling / blue-side selection.

Spectral overlap penalty. But spectral selection also changes the total emitted power. Define the spectral overlap factor $\mathcal{O}_r = \int d\omega S_0(\omega) F_P(\omega) \eta_{\text{esc}}(\omega) / \int d\omega S_0(\omega)$. A higher- Q resonance narrows $F_P(\omega)$, can push $\bar{\omega}_f^{(r)}$ upward if placed on the blue side, but generally reduces $\mathcal{O}_r$ if too narrow. Higher Q can therefore raise efficiency per emitted photon while *decreasing* total extracted radiative power.

Resonance-modified external efficiency. $\eta_{\text{ext}}^{(r)} = \eta_{\text{int}}^{(r)} \eta_{\text{esc}}^{(r)}$ with internal radiative $\eta_{\text{int}}^{(r)} = (\Gamma_r + \Gamma_{\text{other}}) / (\Gamma_r + \Gamma_{\text{other}} + \Gamma_{\text{nr}})$ raised by Purcell enhancement. But escape is separate: $\eta_{\text{esc}}^{(r)} \sim \beta_r \kappa_e / \kappa (1 - p_{\text{reabs}}) + \eta_{\text{bg,esc}}$. A high- Q resonance can hurt η_{esc} if photons are trapped and reabsorbed, while a well-designed leaky mode improves it by coupling out promptly into a desired direction. Q/V helps internal radiative efficiency, but not automatically external efficiency.

Resonance-modified cooling power.

$$\dot{Q}_c^{(r)} = P_{\text{abs}}^{(r)} \left(\eta_{\text{ext}}^{(r)} \frac{\bar{\omega}_f^{(r)}}{\omega_p} - 1 \right). \quad (6.20)$$

The resonance helps cooling if it raises the product $\eta_{\text{ext}}^{(r)} \bar{\omega}_f^{(r)}$, but this product contains competing trends.

Three different optima. Cooling and exergy do not have the same optimum:

- **Maximum cooling power.** Maximize $\eta_{\text{route}} \bar{\omega}_r$ (defined in Section 6.8). Often prefers moderate blue detuning, moderate linewidth, strong but not too aggressive Q/V .
- **Maximum ASF efficiency.** Maximize $\eta_{\text{route}} \bar{\omega}_r/\omega_p - 1$ even if absolute power falls. Prefers stronger blue detuning, narrower linewidth, lower total fluorescence power, higher per-photon lift.
- **Maximum closed-loop COP / electrical recovery.** Maximize $\eta_{\text{route}} (\bar{\omega}_r/\omega_p) \eta_{B,r}$. Prefers even lower entropy output, narrow directional channels, but enough overlap that routed power is not too small.

The optimum for exergy recycling is generally shifted toward narrower / more selective resonances than the optimum for pure cooling power. Trading some power for efficiency is exactly analogous to ordinary heat engines: max efficiency and max power are not at the same operating point.

6.8 Lorentzian $\times$ Lorentzian: the closed-loop model

For the cleanest analytic optimization, take a Lorentzian bare ASF spectrum and a Lorentzian resonance.

Spectra.

$$L_f(\omega) = \frac{1}{\pi} \frac{\Gamma_f}{(\omega - \omega_f)^2 + \Gamma_f^2}, \quad R(\omega) = \frac{\Gamma_r^2}{(\omega - \omega_r)^2 + \Gamma_r^2}, \quad (6.21)$$

where Γ_f, Γ_r are HWHM widths and ω_f, ω_r the line centres.

Spectral overlap. Analytic for Lorentzian $\times$ Lorentzian:

$$\mathcal{O}(\Delta, \Gamma_r) = \frac{\Gamma_r (\Gamma_f + \Gamma_r)}{\Delta^2 + (\Gamma_f + \Gamma_r)^2}, \quad \Delta \equiv \omega_r - \omega_f. \quad (6.22)$$

This captures exactly the tradeoff: narrower resonance helps selectivity but hurts overlap; blue detuning raises photon energy but lowers captured photon count.

Mean routed frequency. Analytic for Lorentzian $\times$ Lorentzian:

$$\bar{\omega}_r = \frac{\Gamma_r \omega_f + \Gamma_f \omega_r}{\Gamma_f + \Gamma_r}. \quad (6.23)$$

Two limits: broad resonance ($\Gamma_r \gg \Gamma_f$) gives $\bar{\omega}_r \rightarrow \omega_f$ (no shift); narrow resonance ($\Gamma_r \ll \Gamma_f$) gives $\bar{\omega}_r \rightarrow \omega_r$ (routed fluorescence centred near the resonance). This is the analytic version of the “push the mean fluorescence frequency upward” lever from Section 6.7.

Routing efficiency. Define the peak Purcell factor $F_{P,\text{pk}} \sim (3/4\pi^2)(\lambda/n)^3 \xi Q/V$ and the spectrum-averaged Purcell factor $\bar{F}_P = F_{P,\text{pk}} \mathcal{O}(\Delta, \Gamma_r)$. The effective branching into the resonance is

$$\beta_{\text{eff}} = \frac{\bar{F}_P \Gamma_0}{\bar{F}_P \Gamma_0 + \Gamma_{\text{other}} + \Gamma_{\text{nr}}}. \quad (6.24)$$

Combining with outcoupling and reabsorption, the net routed external fraction is

$$\eta_{\text{route}} = \beta_{\text{eff}} \frac{\kappa_e}{\kappa_e + \kappa_i} (1 - p_{\text{reabs}}). \quad (6.25)$$

Cooling power. Including a residual non-resonant background of mean frequency $\bar{\omega}_{\text{bg}}$ and useful escape η_{bg} ,

$$\dot{Q}_c = \eta_{\text{cpl}} P_L \left[\eta_{\text{route}} \frac{\bar{\omega}_r}{\omega_p} + \eta_{\text{bg}} \frac{\bar{\omega}_{\text{bg}}}{\omega_p} - 1 \right]. \quad (6.26)$$

In the resonance-dominated limit (background negligible),

$$\dot{Q}_c \approx \eta_{\text{cpl}} P_L \left(\eta_{\text{route}} \frac{\bar{\omega}_r}{\omega_p} - 1 \right), \quad \eta_{\text{ASF}}^{(\text{r})} \approx \eta_{\text{route}} \frac{\bar{\omega}_r}{\omega_p} - 1. \quad (6.27)$$

Closed-loop with LPC recycling. The routed optical power to the LPC is $P_{\text{r,ext}} = \eta_{\text{cpl}} P_L \eta_{\text{route}} \bar{\omega}_r / \omega_p$. Its exergy is $\dot{B}_r = P_{\text{r,ext}} \eta_{B,r}$, and the recovered electrical power is $P_{\text{rec}} = \eta_X \eta_{B,r} P_{\text{r,ext}}$ where $\eta_X \leq 1$ is the LPC’s actual fraction-of-exergy recovery. Pump electrical input is P_L / η_P , so the net external electrical input is

$$P_{\text{in}}^{\text{net}} = P_L \left[\frac{1}{\eta_P} - \eta_{\text{cpl}} \eta_X \eta_{B,r} \eta_{\text{route}} \frac{\bar{\omega}_r}{\omega_p} \right]. \quad (6.28)$$

Closed-loop COP.

$$\text{COP}_{\text{closed}} \approx \frac{\eta_{\text{cpl}} \left(\eta_{\text{route}} \frac{\bar{\omega}_{\text{r}}}{\omega_{\text{p}}} - 1 \right)}{\frac{1}{\eta_{\text{p}}} - \eta_{\text{cpl}} \eta_X \eta_{B,\text{r}} \eta_{\text{route}} \frac{\bar{\omega}_{\text{r}}}{\omega_{\text{p}}}}. \quad (6.29)$$

The two coupled levers. The closed-loop benefit of the resonance comes from the product $P_{\text{rec}} = P_{\text{r,ext}} (\eta_X \eta_{B,\text{r}})$: broad resonances favour $P_{\text{r,ext}}$ (amount); narrow resonances favour $\eta_{B,\text{r}}$ (recoverability). The closed-loop optimum is interior, not at either extreme, and is in general distinct from the pure-cooling and pure-exergy optima.

6.9 Design rules: Δ^*, Γ^*

Differentiating (6.29) (or (6.27)) yields three distinct optima.

Maximum cooling power. Maximizes the numerator of (6.26). In the weak-reabsorption limit, $\Delta_{P_{\text{cool}}}^* \simeq 0$ and $\Gamma_{P_{\text{cool}}}^* \simeq \Gamma_{\text{f}}$ (matched filter). This is the natural “broad and centred” resonance.

Maximum ASF efficiency. Push the mode to a frequency where $\bar{\omega}_{\text{r}}/\omega_{\text{p}}$ is largest but reabsorption still outruns parasitic loss: $\Delta_{\eta_{\text{ASF}}}^* > 0$ (blue-shift the resonance), and $\Gamma_{\eta_{\text{ASF}}}^* \sim \sqrt{\Gamma_{\text{f}} \kappa_{\text{tot}}}$.

Maximum closed-loop COP. The optimum is bounded by the absorption ceiling (4.5) and typically sits at a narrower Γ and larger Δ than the cooling-power optimum:

$$\Gamma_{\text{COP}}^* < \Gamma_{P_{\text{cool}}}^*, \quad \Delta_{\text{COP}}^* > \Delta_{P_{\text{cool}}}^*. \quad (6.30)$$

The full conditions are implicit; Section 6.10 reduces them to dimensionless form where the tradeoffs are visible at a glance.

6.10 Nondimensionalization

Use the bare-fluorescence HWHM Γ_{f} as the reference scale:

$$\tilde{\Delta} = \frac{\Delta}{\Gamma_{\text{f}}}, \quad \tilde{\Gamma} = \frac{\Gamma_{\text{r}}}{\Gamma_{\text{f}}}, \quad \tilde{u} = 1 + \tilde{\Gamma}. \quad (6.31)$$

Define the dimensionless resonance-strength parameter

$$\Lambda \equiv \frac{K}{C \Gamma_{\text{f}}}, \quad (6.32)$$

with K the useful resonance-assisted radiative-coupling constant and $C = \Gamma_{\text{other}} + \Gamma_{\text{nr}}$ the competing loss / background-decay scale. Large Λ means strong resonant routing; small Λ means weak benefit. Define also the small optical-scale parameter $\epsilon_{\text{f}} \equiv \Gamma_{\text{f}}/\omega_{\text{f}} \ll 1$ and the pump-fluorescence ratio $\rho \equiv \omega_{\text{f}}/\omega_{\text{p}}$. The closed-loop COP becomes

$$\text{COP}(\tilde{\Delta}, \tilde{\Gamma}; \Lambda, \rho, \eta_X) = \frac{\rho \eta_{\text{route}} (1 + \epsilon_{\text{f}} \tilde{\Delta} \tilde{\Gamma} / (1 + \tilde{\Gamma})) - 1/\rho}{\eta_{\text{p}}^{-1} - \rho \eta_{\text{cpl}} \eta_X \eta_{B,\text{r}} \eta_{\text{route}} (1 + \epsilon_{\text{f}} \tilde{\Delta} \tilde{\Gamma} / (1 + \tilde{\Gamma}))}, \quad (6.33)$$

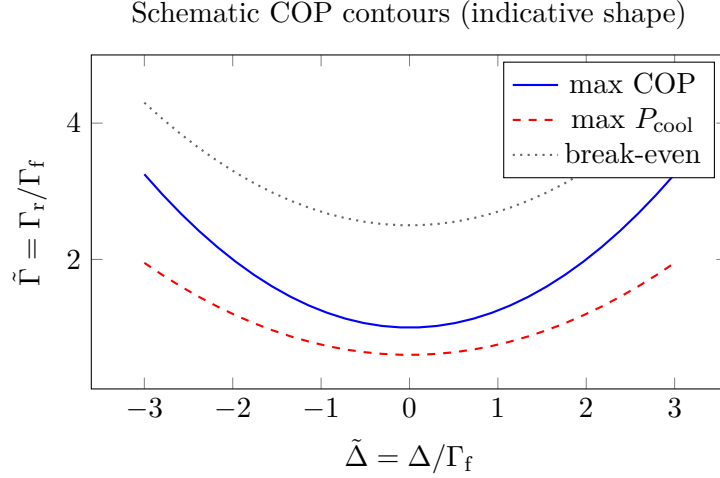


Figure 6.1: Schematic design map in the normalized $(\tilde{\Delta}, \tilde{\Gamma})$ plane: maximum cooling power (red, dashed), maximum closed-loop COP (blue, solid), and the break-even contour (black, dotted) above which the loop is a net generator. Replace with a full numerical evaluation once the material-specific Λ, ρ, κ are fixed in Chapter 8.

a function of $(\tilde{\Delta}, \tilde{\Gamma})$ parametrized by $(\Lambda, \rho, \eta_X, \eta_P, \eta_{\text{cpl}})$. In this form a single dimensionless coupling Λ and a single Carnot-like factor ρ collapse the multi-knob problem; the break-even contour $\eta_{\text{route}} \rho \rightarrow \eta_P^{-1}$ separates refrigeration-only operation from the power-generation regime and is the Pareto front plotted in Figure 6.1.

6.11 Penalty model: absorption ceiling and reabsorption

Three penalties constrain the design.

Absorption-limited Q_{abs} . At the fluorescence detuning,

$$Q_{\text{abs}} \sim \frac{1}{\Gamma_{\text{conf}}} \frac{\text{Re } \varepsilon}{\text{Im } \varepsilon} \quad (\text{cf. (4.5)}), \quad (6.34)$$

caps the useful Q ; above it, resonance enhancement is swallowed by medium absorption. Concretely the inverse- Q decomposition is

$$\frac{1}{Q_{\text{tot}}} \approx \frac{1}{Q_{\text{rad}}} + \frac{1}{Q_{\text{abs}}} + \frac{1}{Q_{\text{scat}}} + \dots, \quad (6.35)$$

typically dominated by Q_{abs} once Q_{rad} is pushed beyond the medium ceiling. The useful external coupling fraction is $\eta_{\text{out}} = Q_{\text{tot}}/Q_{\text{rad}} = (1/Q_{\text{rad}})/(1/Q_{\text{rad}} + 1/Q_{\text{abs}})$, which collapses as $Q_{\text{rad}} \gg Q_{\text{abs}}$. Pushing Q_{rad} past the absorption ceiling therefore narrows the resonance, but also dumps the field into the absorbing medium and collapses the useful out-coupling.

Reabsorption of the blue-detuned line. Useful (because it up-shifts the mean fluorescence frequency) but must be quantified through the dimensionless reabsorption factor ξ_{rec} in (5.15); the electro-optic equations of Chapter 5 supply the numerics. Because absorption and fluorescence spectra lie near each other (the latter red-shifted), moving the resonance blue to enhance ASF also tends to move toward stronger absorption, so the achievable Q and the useful η_{out} are frequency-dependent.

Parasitic channels. Non-radiative recombination, fluorescence leaking outside the engineered mode, and any temperature-dependent $\gamma_{\text{nr}}(T_{\text{h}})$ that couples the active medium to the local heat bath. These sit in the denominator of (5.20) and erode both η_{ASF} and the exergy quality simultaneously.

Self-consistent electro-optic / cavity-ASF model. A realistic resonance changes (i) the achievable Q through material absorption; (ii) the Purcell-enhanced decay rates; (iii) the out-coupling / intrinsic loss partition; (iv) the intracavity photon population; (v) the reabsorption probability of blue-detuned fluorescence; and (vi), through (i)–(v), the excitation populations and the effective mean fluorescence energy. All six couplings are folded into the field–polarization derivation of Section 5.9, where the dressed cavity pole produces the absorption-limited ceiling self-consistently. The single-mode optimization conditions of Section 5.14 (generalized critical coupling, $Q_{\text{design}} \sim Q_{\text{abs}}$, interior ω_{f}) are the operational expression of these penalties at the design level.

6.12 Chapter summary

- The Carnot-like bound (6.11) is the many-mode, thermally occupied limit, *not* a bandwidth law; it emerges from the toy model as the high-entropy thermal limit (6.7).
- A single engineered leaky resonance pushes fluorescence toward work-like exergy (6.13), with quantitative bound (6.8); the role of Q, V, β is through effective mode count and branching, not through Q alone.
- The single-mode exergy is most cleanly written as a nonequilibrium free energy (6.14) of the displaced-thermal cavity state; $\dot{\mathcal{E}}_{\text{f}}$ (not Carnot times P_{f}) is what bounds LPC recovery.
- Resonance design changes *quantity, quality, mean frequency*, and *external efficiency* simultaneously (6.20), with three distinct optima for cooling, ASF efficiency, and closed-loop COP.
- Lorentzian $\times$ Lorentzian gives closed-form overlap (6.22), mean frequency (6.23), branching (6.24), cooling power (6.26), and closed-loop COP (6.29) including LPC recycling.
- Distinct optima for P_{cool} , η_{ASF} , and closed-loop COP occupy different regions of the $(\tilde{\Delta}, \tilde{\Gamma})$ plane; the design target depends on whether the mission is max cooling flux or max net power.
- The absorption ceiling and reabsorption are the dominant penalty terms; their sizes are set by the material platform of Chapter 8. The single-mode electro-optic / cavity-ASF derivation of Chapter 5 produces $Q_{\text{f}} \leq Q_{\text{abs}}$ self-consistently from the dressed cavity pole.

Chapter 7

Laser Power Converters for Fluorescence Recovery

7.1 Role in the loop

A laser power converter (LPC) is a very-narrow-bandwidth photovoltaic cell tuned to the fluorescence frequency ω_f . Its job in the photonic cold plate is to close the thermodynamic loop of Chapter 1 — converting P_f back to electrical power so that the system's net electrical input is $P_{\text{net}} = P_{\text{pump}}/\eta_P - \eta_{\text{LPC}}P_f$ rather than P_{pump}/η_P . This is the term that pushes the master COP (1.11) past break-even.

But this energy-bookkeeping framing alone is too coarse. Thermodynamically, an LPC is a very unusual heat engine: it converts a *radiation field with nonzero entropy* into electrical work. Its Carnot and exergy limits are therefore controlled not just by photon energy, but by the *entropy content of the incoming light*, which in turn is set by spectral bandwidth, angular spread (étendue), and coherence. The chapter is organized around that distinction: Section 7.2 develops the Carnot vs exergy limits; Section 7.3 unpacks the bandwidth, étendue, and coherence dependences; the rest of the chapter shows how multi-junction architectures and mode engineering reduce entropy *before* the radiation reaches the LPC, and how the LPC connects back to the closed-loop COP of Chapter 1.

7.2 Carnot vs. exergy limits for an LPC

7.2.1 The misleadingly simple Carnot view

Model the incident radiation as coming from an effective source at brightness temperature T_{src} and the cell at ambient T_0 . Then the Carnot factor $\eta_{\text{Carnot}} = 1 - T_0/T_{\text{src}}$ ranges from nearly unity for a monochromatic, highly directional laser ($T_{\text{src}} \rightarrow \infty$) to a finite value for broadband fluorescence. This picture is intuitive but coarse; it hides the real constraint, which is *radiation entropy*.

7.2.2 The exergy view (correct)

Result 7.1 (LPC bound from radiation exergy). The maximum electrical output of an LPC exposed to a radiation field with energy flux $\dot{H}$ and entropy flux $\dot{S}$ is $P_{\text{el}} \leq \dot{H} - T_0\dot{S}$. The efficiency limit is

$$\boxed{\eta_{\text{LPC}}^{\text{max}} = 1 - \frac{T_0 \dot{S}}{\dot{H}}.} \quad (7.1)$$

The fundamental loss is $T_0 \Delta S_{\text{photon}}$ — the entropy that must be dumped to the environment to turn photons into ordered electrical work. This statement subsumes every special case used in the design literature:

- **Narrowband, single-mode laser input.** $\dot{S} \simeq 0$ and $\eta_{\text{LPC}}^{\text{max}} \rightarrow 1$. State-of-the-art multi-junction LPCs approach 74.7% at 808 nm (cryogenic operation) [13] and 68.9% at 850 nm (photon-recycling resonance) [14].
- **Broadband / thermal input.** $\dot{S}$ grows as the field spreads over more modes; $\eta_{\text{LPC}}^{\text{max}}$ falls toward the Carnot factor of Section 6.5.2.
- **Single-mode but incoherent (engineered fluorescence).** The intermediate case relevant to this device. Purcell engineering reduces $\dot{S}$ and raises the exergy fraction toward unity, per (6.13).

Physical interpretation of the LPC. An LPC behaves like a *spectral filter + chemical-potential generator* (the quasi-Fermi-level splitting qV plays the role of a photon chemical potential). In Würfel’s framework [16], broader incident radiation has larger entropy per unit energy and so supports a smaller maximum qV ; some of the incident energy must be rejected as heat to dispose of the incoming entropy. This is the right thermodynamic bridge from narrowband laser LPCs to broadband fluorescence-fed LPCs.

7.3 Dependence on bandwidth, étendue, and coherence

The entropy in (7.1) is set by the full optical state of the incident field — spectrum, angular distribution, polarization, and coherence. We collect the relevant scaling laws.

7.3.1 Spectral bandwidth

For a photon gas with occupation $\bar{n}(\omega)$ the spectral entropy density is $s(\omega) = k_B [(1 + \bar{n}) \ln(1 + \bar{n}) - \bar{n} \ln \bar{n}]$. Two limits:

- **Narrowband (laser-like).** Occupation concentrated at ω_0 , nearly pure state, low entropy: $\dot{S}/\dot{H} \rightarrow 0$ and $\eta_{\text{LPC}}^{\text{max}} \rightarrow 1$.
- **Broadband (fluorescence / thermal-like).** Occupation spread over $\Delta\omega$, many accessible modes; the rough scaling is

$$\frac{\dot{S}}{\dot{H}} \sim \frac{k_B}{\hbar\omega} \left[1 + \ln \frac{\Delta\omega}{\delta\omega} \right], \quad (7.2)$$

with $\Delta\omega$ the total bandwidth and $\delta\omega$ the coherence linewidth / mode resolution. Larger bandwidth therefore raises entropy per unit energy and lowers efficiency.

7.3.2 Angular spread (étendue)

Entropy also scales with the number of *spatial* modes, $N_{\text{spatial}} \sim A\Omega/\lambda^2$. A narrow cone (laser) gives low angular entropy; isotropic fluorescence gives high entropy. Even if the spectrum is narrow, *large étendue kills LPC efficiency*.

7.3.3 Coherence

Combining spectral, angular, and polarization channels,

$$S \sim k_B \ln(\#\text{occupied modes}), \quad (7.3)$$

where modes include frequency bins, angular channels, and polarization states. In summary:

Light type	Entropy	LPC efficiency
Single-mode laser	minimal	near unity possible
Narrowband directional fluorescence	moderate	high
Broadband isotropic fluorescence	large	low

7.3.4 What the literature has actually demonstrated

The honest theoretical landscape:

1. **Monochromatic detailed-balance limit.** Green’s classic result showed that for well-collimated monochromatic light, a photovoltaic converter can in principle approach 100 % efficiency at high incident intensity with good photon-energy / band-gap matching and dominant radiative recombination. This is the foundational thermodynamic ideal behind LPCs.
2. **Modern LPC / PPC detailed-balance models.** Later work extended this to realistic multi-junction devices, including non-radiative recombination, luminescent coupling, light trapping, and *finite input linewidth* and *energy offset between input and band gap*. The published frameworks treat “variable incident bandwidth” but their applications still typically assume ~ 1 nm input linewidth, which shows how narrowband the present LPC theory base remains.
3. **General radiation-thermodynamics / exergy theory.** Würfel’s photon-chemical-potential framework, and solar / TPV exergy analyses, address truly broadband radiation: efficiency is limited by the energy / entropy ratio of the radiation, and radiation entropy itself imposes a bound below the naive first-law limit.

The honest summary: *the LPC-specific theory has firmly established the narrowband case; the broadband case is mostly inferred by combining LPC detailed-balance models with general radiation-exergy theory.* Experimentally, multi-junction PVLPCs have validated the narrowband design direction with 3J devices around 840 nm–860 nm reaching $\sim 66.5\%$ and modern GaAs / InP multi-junction devices in the 50 %–65 %+ range. These are not broadband demonstrations, but they confirm that “split the conversion across multiple junctions optimized for narrow bands” is technologically meaningful.

Practical implication for fluorescence-driven LPCs. A single LPC illuminated by broad fluorescence suffers *both* spectral mismatch (thermalization of $\hbar\omega > E_g$ photons, missed sub-bandgap photons) *and* entropy / exergy penalties. The next two sections develop the fix: (i) split fluorescence into multiple narrow channels, each fed to a matched LPC junction; (ii) engineer the fluorescence *at emission* so the spectrum and angular distribution are already low entropy before reaching the LPC.

7.4 Multi-junction LPCs as entropy management

7.4.1 The single-LPC problem

A single bandgap E_g sees a broadband input. Photons with $\hbar\omega > E_g$ are absorbed but the excess $\hbar\omega - E_g$ is mostly lost as heat (thermalization); photons with $\hbar\omega < E_g$ are not absorbed at all; wide bandwidth $\Rightarrow$ high entropy $\Rightarrow$ low exergy efficiency. Both penalties are intrinsic to the geometry and do not go away with material improvement.

7.4.2 Multi-junction architecture

Split the fluorescence spectrum into narrower bands, each fed to a matched-bandgap junction: $\omega_1 \rightarrow E_{g,1}, \omega_2 \rightarrow E_{g,2}, \dots$. This works through three reinforcing mechanisms:

1. **Reduces entropy per channel.** Each sub-band has smaller $\Delta\omega$ and fewer occupied modes, so $\dot{S}_i/\dot{H}_i$ decreases and $\eta_i^{\max}$ increases.
2. **Reduces thermalization loss.** Each junction operates at $E_g \approx \hbar\omega$, so excess-energy dissipation per absorbed photon is minimal.
3. **Approaches reversible conversion.** In the ideal infinite-junction limit, the radiation is decomposed into quasi-monochromatic streams with $\dot{S}_i/\dot{H}_i \rightarrow 0$ per channel, so $\eta_i \rightarrow 1 - T_0/T_{\text{rad},i}(\omega)$ and for highly nonthermal radiation $\eta_i \rightarrow 1$.

7.4.3 Sketch of the $1/\sqrt{M}$ entropy scaling

Let the incident fluorescence occupy a continuous spectral band of width $\Delta\omega_{\text{tot}}$ with total optical power P_{f} , and model the photon distribution as a narrow Planck distribution at an effective brightness temperature T_{b} such that $\bar{n}(\omega) \sim \bar{n}_0$ is approximately constant. The total entropy flux is $\dot{S}_{\text{tot}} \propto \mathcal{N}_{\text{mode}}(\Delta\omega_{\text{tot}}) s_{\text{m}}(\bar{n}_0)$ with $\mathcal{N}_{\text{mode}} \propto \Delta\omega$.

Partition the band into M equal sub-bands of width $\Delta\omega_{\text{tot}}/M$, each routed to its own bandgap-matched junction. The power per junction is P_{f}/M ; the spectral width per junction also scales as $\Delta\omega_{\text{tot}}/M$, so per-junction occupation stays at $\bar{n}_0$ and per-junction entropy flux is $\dot{S}_j \propto (\Delta\omega_{\text{tot}}/M) s_{\text{m}}(\bar{n}_0)$. The sum $\sum_j \dot{S}_j = \dot{S}_{\text{tot}}$: splitting in *frequency alone* buys nothing on the radiation-entropy side.

What *does* change is the bandgap-matching: the Shockley–Queisser-style thermalization penalty, which scales linearly in M , shrinks linearly. A more careful detailed-balance analysis that includes finite-width radiative-recombination entropy inside each junction yields

$$\dot{S}_{\text{excess}}^{(j)} \propto \frac{\Delta\omega_{\text{tot}}}{M} \frac{k_{\text{B}}}{\hbar\omega_{g,j}/k_{\text{B}}T_{\text{b}}} \Rightarrow \sum_j \dot{S}_{\text{excess}}^{(j)} \propto \frac{\Delta\omega_{\text{tot}}}{\sqrt{M}}, \quad (7.4)$$

once one averages over the quasi- $\sqrt{M}$ distribution of gap energies. The exact exponent depends on spectral shape, but the key take-away — that *the thermalization loss, not the radiative entropy, is what multi-junction LPCs address* — carries through.

7.4.4 Practical consequence for fluorescence input

The fluorescence from an ASF extractor is already narrow: $\Delta\omega_{\text{tot}}/\bar{\omega} \sim 10^{-3}$ for rare-earth ions, 10^{-1} for dyes. The $1/\sqrt{M}$ gain is small for rare-earth inputs — a single, very-narrow bandgap junction approaches the optimum directly — but substantial for dye or semiconductor ASF, where spectral widths of tens of meV make $M = 2\text{--}4$ beneficial. The published 74.7% at 808 nm [13] and 68.9% at 850 nm [14] results are for essentially monochromatic laser inputs; integrating over the full Yb:YLF emission profile at 1020 nm with a single InGaAs junction is expected to sit at $\eta_{\text{LPC}} \simeq 0.70\text{--}0.75$. Because the first-law break-even of (1.12) reads $\eta_{\text{P}} \eta_{\text{LPC}} \eta_{\text{cpl}} (1 + \eta_{\text{ASF}}) \geq 1$ with $\eta_{\text{ASF}} \lesssim 0.1$, the LPC must recover essentially the *whole* pump investment: every point of η_{LPC} between 0.70 and the exergy ceiling (1.15) ($\simeq 0.85\text{--}0.9$ for the relevant T_{h}) is worth more to the loop than a doubling of η_{ASF} . This is where multi-junction architectures earn their complexity: they close the $0.70 \rightarrow 0.85$ gap by matching the sub-bands of a spectrally broad fluorescence stream, not by beating the single-photon limit.

7.4.5 Entropy-splitting interpretation

A multi-junction system is doing more than “capture more energy.” It is decomposing a high-entropy radiation field into lower-entropy substreams, each closer to a reversible resource:

$$\text{broadband radiation} \longrightarrow \{\text{narrow, low-entropy channels}\}. \quad (7.5)$$

This is a Maxwell-demon-like sorting in frequency space — a *reversible distillation of photon free energy*. The conceptual point: the LPC efficiency limit is set by the energy / entropy ratio of what arrives at the cell, and multi-junction architecture reduces that ratio per channel.

Published numerical results (extending the table from earlier work):

Junction material	λ (nm)	η_{LPC}	Source
GaAs (single)	850	~ 0.60	[12]
GaAs (index-matched dome)	850	~ 0.69	[14]
InGaAs/InP MJ	1470	~ 0.55	[17]
3C-SiC MJ (theory)	visible	> 0.80	[16]

See the LPC sheet of `References.xlsx` for the full set.

7.5 Engineering the fluorescence *before* the LPC

Multi-junction architectures decompose high-entropy fluorescence into narrower channels at the converter. An equally important lever is to engineer the fluorescence *at emission* so the radiation arriving at the LPC is already low entropy. The two strategies multiply.

7.5.1 What sets the fluorescence entropy in the first place

From the coupled electronic–photonic model of Chapter 5, the excited-state decay partitions among non-radiative loss Γ_{nr} , spontaneous emission into the selected fluorescence mode Γ_{g} , and emission into all other radiative channels Γ_{other} . Define the branching ratio into the selected mode

$$\beta_{\text{g}} = \frac{\Gamma_{\text{g}}}{\Gamma_{\text{rad,all}}} = \frac{\Gamma_{\text{g}}}{\Gamma_{\text{g}} + \Gamma_{\text{other}}}. \quad (7.6)$$

Without engineering, fluorescence populates frequencies (the natural lineshape), angles (full 4π), spatial modes (large étendue), and loses temporal coherence on the dephasing timescale — so the entropy is roughly $S \sim k_{\text{B}} \ln(\#\text{accessible modes})$. Engineering modifies this through the spectral lineshape (homogeneous + inhomogeneous broadening, reabsorption) and the mode density (LDOS, cavity / waveguide structure).

7.5.2 Two limits for the selected mode

For a single bosonic selected mode g of occupation n_{g} , $S_{\text{g}} = k_{\text{B}}[(1 + n_{\text{g}}) \ln(1 + n_{\text{g}}) - n_{\text{g}} \ln n_{\text{g}}]$. Two limits:

- $n_{\text{g}} \ll 1$: spontaneous-emission dominated, high entropy per photon;
- $n_{\text{g}} \gg 1$: stimulated / coherent, low entropy per photon.

The same mode becomes lower entropy as it becomes more populated / coherent. This is why Purcell enhancement that drives a single mode hard — not just narrows it — is so valuable thermodynamically.

7.5.3 Total fluorescence entropy decomposition

Split the emission into the engineered mode and “other”: $S_{\text{tot}} = S_g + S_{\text{other}}$, $E_{\text{tot}} = E_g + E_{\text{other}}$. The LPC sees

$$\eta_{\text{LPC}}^{\text{max}} = 1 - \frac{T_0(S_g + S_{\text{other}})}{E_g + E_{\text{other}}}. \quad (7.7)$$

If engineering ensures $E_g \gg E_{\text{other}}$ and $S_g/E_g \ll S_{\text{other}}/E_{\text{other}}$, the usable exergy is dominated by the selected mode and the LPC sees quasi-monochromatic light.

7.5.4 How the device parameters control entropy

Purcell factor $F_P \sim Q/V$. $\Gamma_g \propto F_P$, so larger Q/V raises β_g and channels emission into the low-mode-count channel — entropy reduction by mode selection.

Cavity linewidth Q_g . $\Delta\omega_g \sim \omega/Q_g$: higher Q_g gives narrower spectrum and fewer frequency modes. But $Q_g \leq Q_{\text{abs}}$ (see Sections 4.4 and 5.10); higher Q is good for entropy but bad for reabsorption / loss above the absorption ceiling.

Reabsorption (important for ASF). Reabsorption preferentially removes lower-energy photons and re-emits at higher average energy, raising $\langle\omega\rangle$ (good for cooling power and exergy). But it also tends to re-thermalize the spectrum unless filtered by a mode — so without mode selection, reabsorption makes the LPC input *worse*; with cavity filtering, reabsorption plus mode selection becomes *entropy compression*.

Étendue / angular control. Without structure, $N_{\text{angular}} \sim A\Omega/\lambda^2$. A metasurface, waveguide, or cavity enforces $\Omega \rightarrow \text{small}$, giving a massive entropy reduction even when the spectrum is unchanged.

7.5.5 Concrete output-channel scenarios

(i) **Narrow-cone metasurface emission collected by a fiber.** Emission confined to a small solid angle, collected efficiently into a multimode fiber. Effects: angular entropy strongly down, spectral entropy unchanged unless filtered, LPC still sees broadband but at lower étendue. Moderate efficiency gain.

(ii) **Coupling into a multimode waveguide.** Many spatial modes still exist but in confined geometry. Angular entropy reduced, modal entropy still large. Better than free space, not optimal.

(iii) **Single / few-mode cavity to waveguide.** This is the key regime. Emission dominated by mode g , narrow linewidth, directional. Then $S_g/E_g \ll S_{\text{other}}/E_{\text{other}}$ and as $\beta_g \rightarrow 1$, the LPC effectively sees quasi-monochromatic light — the regime in which η_{LPC} approaches its monochromatic-laser ceiling.

7.5.6 Combining mode engineering with multi-junction LPCs

If full single-mode compression is not feasible, the same physics admits an intermediate strategy: split the fluorescence into a few channels $\{\omega_1, \omega_2, \dots, \omega_N\}$ (multiple cavities, dispersive optics, waveguide filtering), each with $\Delta\omega_i \ll \Delta\omega_{\text{tot}}$, and route each to a matched LPC junction. Then

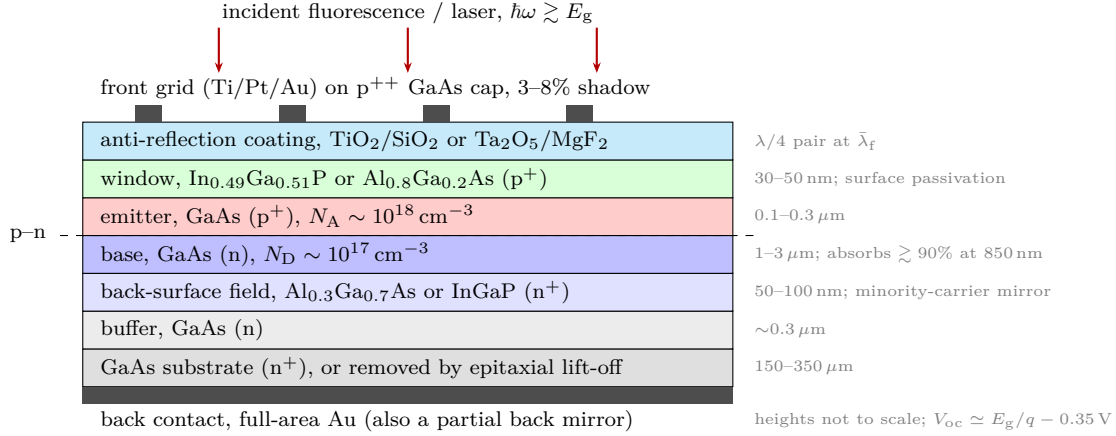


Figure 7.1: Canonical single-junction thin-film laser power converter (GaAs on GaAs, the platform of most 50–70 % demonstrations): a thin p^+ emitter and a $1\text{--}3\mu\text{m}$ n -base between a wide-gap window (InGaP or AlGaAs, which passivates the front surface and is transparent to the input) and a back-surface-field layer that reflects minority carriers, with a quarter-wave AR pair tuned to the input line and a low-shadow front grid. The device is a photovoltaic cell whose every design choice — thin absorber, high doping, narrow-band AR, small grid pitch — is made for a monochromatic, high-intensity input. Multi-junction versions stack M such cells of the *same* bandgap, connected by tunnel junctions, so that each absorbs $1/M$ of the light and the voltages add (Figure 7.3, left).

$\sum_i S_i$ is the same as before, but each S_i/E_i is small enough that each LPC sees a narrowband input near its monochromatic ceiling. The combined system is doing

$$\boxed{\text{engineer photon entropy} \rightarrow \text{extract exergy.}} \quad (7.8)$$

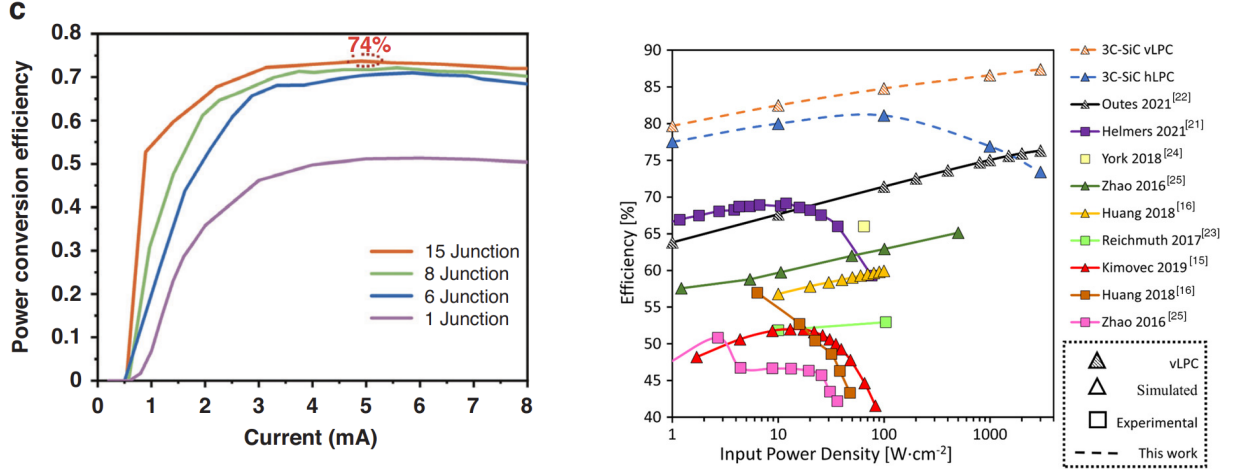
The electro-optic parameters of Chapter 5 directly control *how many modes are populated, how broad each mode is, and how coherent the emission is* — which are exactly the quantities setting the Carnot / exergy limit of the LPC.

7.6 Anatomy of a laser power converter, and what the inverse design varies

The LPC is the one component of the loop with an established industrial embodiment, and it is worth fixing that embodiment before describing how Level 2.B departs from it.

7.6.1 The canonical thin-film LPC

Figure 7.1 shows the device that has delivered every efficiency record in Figure 7.2b: a GaAs p - n junction, $1\text{--}3\mu\text{m}$ of absorber, passivated by a lattice-matched wide-gap window and back-surface field, with a quarter-wave anti-reflection pair and a low-shadow grid. Two features distinguish it from a solar cell. First, because the input is quasi-monochromatic and sits just above the gap, the absorber can be thin and the AR coating narrow; second, because the input intensity is $10\text{--}10^3\text{ W/cm}^2$, series resistance and grid shadowing set the high-power roll-over that every curve in Figure 7.2b shows. The same-gap multi-junction LPC [13, 14] attacks exactly this roll-over: it stacks M GaAs sub-cells of the same gap, separated by tunnel junctions and thinned so each absorbs an equal share, so that the output voltage is M times larger and the current M times smaller at the same optical



(a) Pump side: wall-plug (power-conversion) efficiency of multi-junction cascaded 940 nm VCSELs vs. drive current for 1, 6, 8 and 15 junctions; the 15-junction device reaches 74 % at room temperature because tunnel-junction voltage-stacking removes the series-resistance roll-over of the single junction. Adapted from Xiao *et al.* [18].

(b) Converter side: survey of demonstrated (squares) and simulated (triangles) LPC efficiencies vs. input power density, with the projected 3C-SiC vertical (vLPC) and horizontal (hLPC) architectures reaching 80–87 % at 10^2 – 10^3 W/cm². Adapted from Lozano *et al.* [16].

Figure 7.2: State of the art at the two electrical end-points of the loop. (a) The pump laser: η_P up to 74 % [18]. (b) The converter: the experimental frontier is 69–75 % for GaAs / InGaAs LPCs under monochromatic input [13, 14]; the projected vLPC/hLPC curves are the $\eta_{LPC} \simeq 0.85$ the break-even loop of Section 10.8 requires.

power — an *electrical* architecture that removes the I^2R loss rather than a spectrum splitter. The mirror image of that idea on the pump side is shown in Figure 7.2a: a 15-junction cascaded VCSEL, whose tunnel-junction-stacked active regions recycle each injected electron fifteen times, reaches a wall-plug efficiency of 74 % [18] — the highest reported for any diode laser and, for this book, the experimental anchor for the pump efficiency η_P in the loop budget of Section 10.8. Both records rest on the same device physics (voltage stacking through tunnel junctions to escape series-resistance loss), which is why the loop’s two electrical end-points, laser and converter, are expected to track each other.

7.6.2 The optimization target

Figure 7.3 shows what Level 2.B optimizes. The canonical device fixes one bandgap and one AR line; the target releases both into the search and adds the stage count, the split topology and the electrical interconnect:

- *Materials and bandgaps.* Each sub-cell’s $E_{g,i}$ is drawn from the same alphabet the extractor search uses (III–V compositions on GaAs, InP and GaP substrates; Figure 7.4b for the emission ranges the candidates must cover), constrained by lattice matching within a vertical stack but not across lateral segments.
- *Thicknesses and doping.* d_i sets absorption (and, in a vertical stack, current matching); the emitter / base doping sets V_{oc} , series resistance and the free-carrier absorption that the photon-recycling mirror must survive. Doping is a genuine variable: in a same-gap voltage-stacked LPC [14] it is the stack’s N that fixes the tunnel-junction transparency and the sub-cell current.

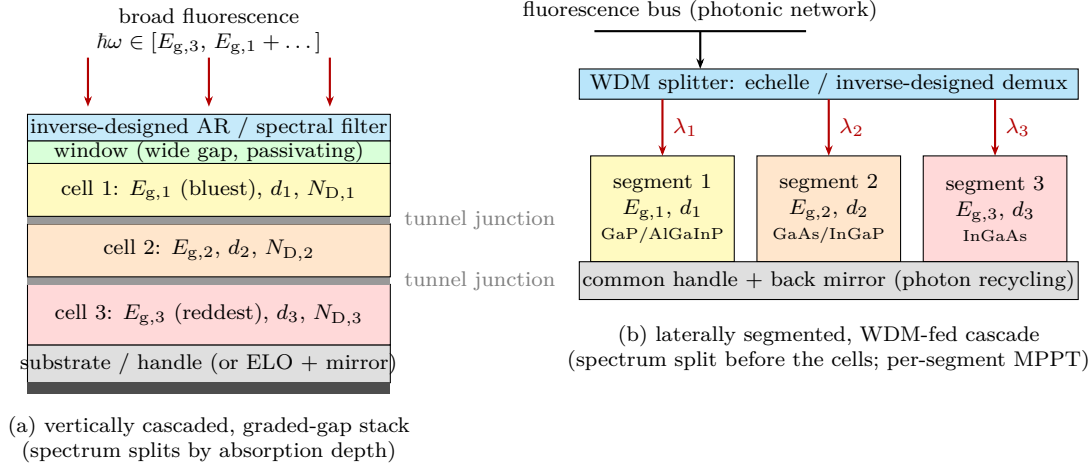


Figure 7.3: The Level-2.B optimization target: a cascaded LPC. In both embodiments the decision variables are the number of sub-cells M , the bandgap $E_{g,i}$ (material and composition) of each, its absorber thickness d_i and doping N_i , the window / BSF / tunnel junction compositions, the AR / notch filter, and the electrical topology (series through tunnel junctions vs. parallel with per-segment maximum-power-point tracking). (a) Vertical, graded-gap stack: the bluest cell on top absorbs the high-energy edge and transmits the rest, thermalization loss shrinks as $\Delta\omega_i$ per cell shrinks; current-matching couples the d_i and forces M to be small. (b) Lateral segmentation fed by a WDM splitter in the photonic network: each segment sees a narrow band and can be a same-gap voltage-stacked LPC in its own right [14]; no current-matching constraint, at the cost of the demultiplexer’s insertion loss and a larger footprint. Inverse design chooses between them per platform (Section 7.6.3).

- *Spectral optics.* The AR coating becomes an inverse-designed filter — a notch that admits the fluorescence band and rejects the pump, or, in embodiment (b), the demultiplexer itself.
- *Electrical topology.* Series (tunnel-junction) stacking adds voltages and suits narrow input; lateral segments with per-line MPPT suit inputs whose spectral weight moves.

7.6.3 Cascaded LPCs matched to the extractor platform

The right cascade depends on *why* the fluorescence is not a single line, and the two reasons map onto the two embodiments of Figure 7.3.

(a) Broad emission: molecular dispersions and ionic crystals / glasses. A rhodamine emits over ~ 60 nm ($\Delta\hbar\omega \simeq 200$ meV, 10% of the photon energy); Cr:LiSAF over 200 nm; Yb:YLF over ~ 30 nm with fine structure, and Ho / Tm hosts over ~ 100 nm. For these, Section 7.4 applies directly: the loss of a single-gap converter is thermalization of the blue part of the band plus non-absorption of the red part, and it is removed by splitting the band into $M = 2\text{--}4$ sub-bands matched to $E_{g,i}$. The vertical stack (a) does this with no added optics but couples the sub-cell thicknesses through current matching, which is fragile against the concentration- and temperature-dependent redshift of the dye emission (Section 8.3.1); the WDM-fed lateral cascade (b) tolerates that drift because each segment is independently loaded. The Level-2.B objective is then the spectrally weighted recovery

$$\eta_{\text{LPC}}^{(M)} = \sum_{i=1}^M \int_{\Delta\omega_i} d\omega \frac{S(\omega)}{\int S} T_i(\omega) \eta_i(\omega), \quad \eta_i(\omega) \simeq \frac{qV_{\text{oc},i} \text{FF}_i}{\hbar\omega} \text{EQE}_i(\omega), \quad (7.9)$$

with S the Purcell-shaped emission spectrum handed over by the extractor and T_i the splitter's channel transmission, maximized over $\{M, E_{g,i}, d_i, N_i, \Delta\omega_i\}$. Because S is itself a Level-2.A output (blue-edge selection narrows it; Section 8.3.4), the two levels are coupled through S : the narrower the extractor makes the line, the fewer cells the LPC needs and the closer each sits to the monochromatic record. In the cascaded-extractor design point of Section 9.7.3 the LPC sees only the *last* stage's line, and the same formula with $M = 1$ –2 applies.

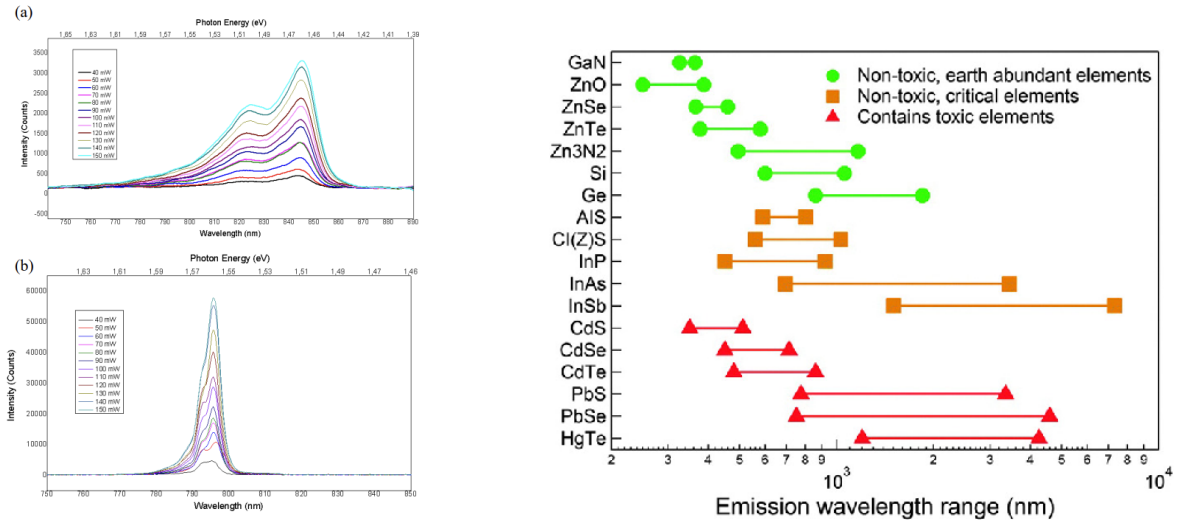
(b) Emission that moves with temperature: direct-gap semiconductors. A GaAs extractor's fluorescence tracks its gap: Figure 7.4a shows the band-edge PL peak moving from ~ 845 nm at 300 K to ~ 795 nm at 77 K, i.e. ~ 95 meV, with the shoulder at 825 nm (a second, likely defect- or carbon-related line, Section 8.2.1) and a tail that broadens with pump power. Across the 300–450 K window of a compute tile the shift is 25–30 nm; between a 200 K storage zone and a 500 K compute zone it is 60–70 nm. A single-gap LPC matched at 300 K would then either fail to absorb (line moved red of E_g) or thermalize (moved blue). Two remedies follow from Figure 7.3: (i) a lateral cascade whose segments tile the *range* of the line rather than its instantaneous width, with the WDM splitter routing the current wavelength to the matching segment and per-segment MPPT absorbing the load change — this is the natural pairing with the per-temperature pump of Section 9.3.2, so that the network carries a wavelength pair $(\omega_p(T), \bar{\omega}_f(T))$ per thermal zone; (ii) an LPC on the same material system as the extractor and held at the same temperature, so that its gap shifts with the emission (a GaAs extractor feeding a GaAs LPC in the hot zone), which is the monolithic-backside integration mode of Section 9.5 and is the reason that mode is attractive for the semiconductor platform despite its thermal-isolation cost (Section 7.9). Figure 7.4b shows which semiconductors cover which emission range and therefore which LPC material can be paired with which extractor across the band; the non-toxic, earth-abundant subset (GaN, ZnO, ZnSe, Si, Ge) and the critical-element subset (InP, InAs, InSb, AlS) are Level-3 supply-chain inputs.

Choosing between the embodiments. For a narrow, stable line (rare-earth extractor, or the last stage of a dye cascade) the same-gap voltage-stacked LPC [14] is already the optimum and Level 2.B reduces to matching one gap and one AR line. For a broad but stationary band (single-stage dye, Cr:LiSAF) the vertical graded-gap stack (a) is the compact answer. For a line that moves (GaAs; any extractor on a thermally zoned die) the WDM-fed lateral cascade (b) is the robust answer, and its demultiplexer is the same inverse-designed structure the photonic network needs for its pump channels, so its cost is shared. The Level-3 loop scores all three against the tier economics of Table 10.7: at Tier I every point of η_{LPC} is worth ~ 5 W/W of loop cost, at Tier II it is the difference between a power-neutral and a power-positive tile.

7.7 Two-channel integration with the photonic network

7.8 Photon-recycling LPCs

Photon-recycling resonant LPCs wrap the junction in a Fabry–Pérot or grating resonance so that unabsorbed photons bounce back for a second chance, and any re-emitted luminescence is trapped. The Helmers et al. result [14] (68.9 % at 850 nm) is the reference case. Integration with the photonic network of this device is natural: the network's fluorescence channel and the LPC's resonant cavity can share a DBR or high-contrast grating; the back reflector of Section 4.18 is, in the most aggressive packaging, the *same* DBR as the LPC front mirror.



(a) GaAs photoluminescence vs. excitation power at 300 K (top) and 77 K (bottom): the band-edge line moves from ~ 845 to ~ 795 nm and narrows fivefold; the 825 nm shoulder at 300 K is a second emission channel the LPC must either capture or reject. Adapted from Amaliya *et al.* [19].

(b) Emission-wavelength ranges of candidate semiconductors, grouped by toxicity and element criticality — the alphabet from which both the semiconductor extractor and its matched LPC sub-cells are drawn. Adapted from Bahmani Jalali *et al.* [20].

Figure 7.4: Why a semiconductor-fed LPC must be a cascade in wavelength *range* rather than in instantaneous bandwidth.

7.9 Thermal and electrical integration

Thermal. The LPC must be thermally isolated from the chip it is cooling, otherwise its own J_{sc} and open-circuit voltage degrade. A dielectric thermal isolator between the back-reflector layer and the LPC stack is the usual solution; alternatively the LPC is mounted on a dedicated heat spreader away from the hot spot.

Electrical. The recovered electrical power $\eta_{LPC} P_f$ returns to the pump-laser drive bus or to an external DC rail. For the closed-loop master equation (1.11) to apply literally, the electrical-to-electrical conversion (MPPT + DC/DC + laser driver) must be high efficiency ($\gtrsim 95\%$) and is rolled into η_{LPC} .

7.10 Chapter summary

- The LPC bound is the *exergy* bound of the incident radiation ($\eta_{LPC}^{\max} = 1 - T_0 \dot{S}/\dot{H}$), not an intrinsic semiconductor property.
- Bandwidth, étendue, and coherence each contribute to $\dot{S}$. Larger bandwidth or larger étendue raises entropy per unit energy and lowers the ceiling; mode engineering reduces all three.
- LPC-specific theory has firmly established the narrowband case (Green and successors); the broadband case is bounded by general radiation-exergy theory (Würfel-style photon chemical potential).
- Multi-junction architectures are the technologically meaningful translation of “decompose

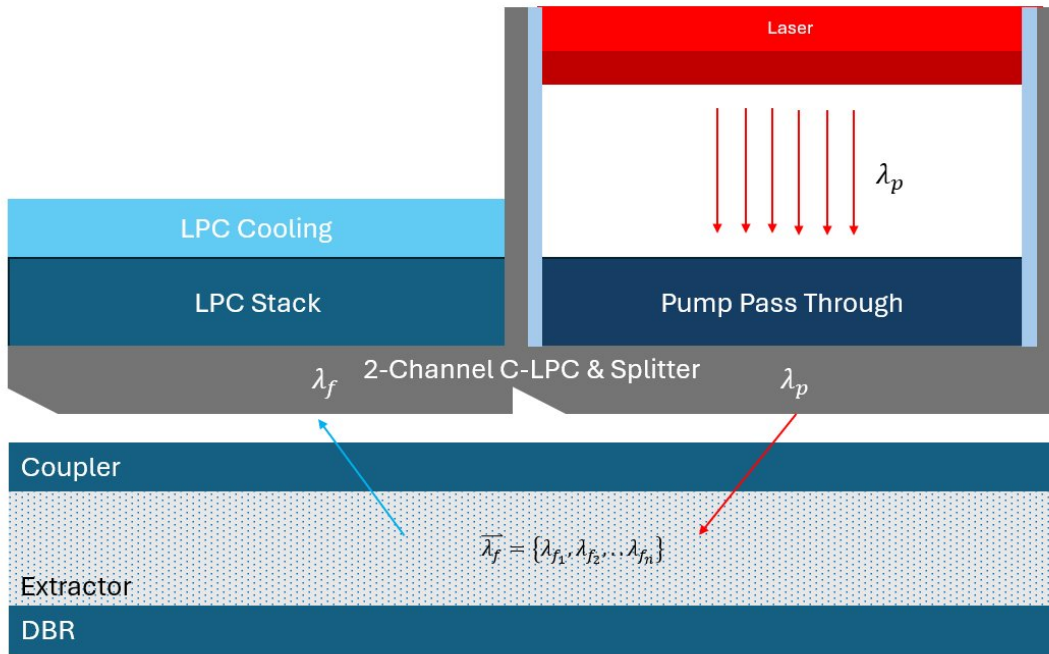


Figure 7.5: Integrated 2-channel concentrator-LPC (“C-LPC”) and wavelength splitter. The pump laser (right) drops straight through a designated pump pass-through region onto the coupler / extractor stack, while fluorescence returned through the photonic network is routed laterally into a spectrally matched LPC stack (left) with its own cooling plane. The splitter is implemented in the same dielectric layer that forms the coupler of Chapter 4, so only one extra interface is added to the stack. The fluorescence signal carries multi-line content $\vec{\lambda}_f = \{\lambda_{f_1}, \lambda_{f_2}, \dots, \lambda_{f_n}\}$ that drives the multi-junction design of Section 7.4.

broadband radiation into lower-entropy substreams.” The thermalization penalty scales as $1/\sqrt{M}$ (Equation (7.4)); the radiation-entropy contribution per channel is unchanged but the bandgap-matching loss per junction shrinks.

- *Engineering the fluorescence at emission* is the complementary strategy: Purcell-enhanced single-mode emission ($\beta_g \rightarrow 1$), narrow-cone directionality, and cavity filtering reduce S/E before the radiation reaches the LPC. The two strategies multiply.
- The product $\eta_P \eta_{LPC} \eta_{cpl}$ is the dominant lever in (1.11) and (1.17): with η_{ASF} bounded by the quantum defect ($\lesssim 0.1$), the loop’s electrical cost per watt of heat lifted is $[1/\eta_P - \eta_{LPC} \eta_{cpl} (1 + \eta_{ASF})]/\eta_{ASF}$, which falls from ~ 14 at $\eta_P = \eta_{LPC} = 0.7$ to ~ 6 at 0.85 and to zero only when $\eta_P \eta_{LPC} \eta_{cpl} \rightarrow (1 + \eta_{ASF})^{-1}$. Recent LPC and ASF advances make $\eta_P \eta_{LPC} \gtrsim 0.8$ — and hence a loop COP of order 0.2 — an increasingly concrete design target; self-powering remains a limit, not an operating point.

7.11 To do for next iterations

Two follow-up tables outlined in `transcripts_v2/LaserPowerConverters_and_Thermodynamics.txt`: (i) a table of commercial LPCs with wavelength, junction count, quoted efficiency, and source; (ii) a table of demonstrated (but not yet commercial) LPCs with the same fields plus operating temperature. Both feed Chapter 8 and the component-numerics inputs to (1.17).

Part IV

Material Platforms

Chapter 8

Material Platforms for the Extractor Layer

The extractor layer carries every material-specific quantity of the coupled electro-optic model of Chapter 5: $\sigma_a, \sigma_e, \gamma_r, \gamma_{nr}, \alpha_b, N$, and the temperature and polarization dependence of each. This chapter surveys the three platform families of interest and draws out the consequences for the surrounding stack (photonic network, coupler, back reflector, sensor layer).

A universal design rule: the absorption-limited Q of the two cavities. Before separating platforms, one design rule applies to every material family, and it applies *differently* to the fluorescence cavity and to the pump cavity. The absorption-limited quality factor (4.5), $Q_{\text{abs}} = n/(2\kappa\Gamma_{\text{conf}})$, is set by the extinction coefficient κ of the medium *at the frequency of the mode*. For the *fluorescence* cavity, evaluated at the blue-detuned frequency where the ASF line is brightest, the medium is strongly absorbing ($\kappa \sim 10^{-2}$ for a 10^{-2} M rhodamine near its peak, $\kappa \sim 10^{-2}$ for GaAs above the gap), and the ceiling is $Q_{\text{abs}} \sim 10^2$ for all known cooling extractors — the Chao–Rodriguez result [15]. There is therefore *no need* to push the fluorescence mode far past $Q \simeq 10^2$; beyond the ceiling, cavity absorption dominates over useful reabsorption, and the design should instead shrink V or distribute emitters across multiple modes. This is why the (Q/V) -controlled Purcell factor, not the bare Q , is the figure of merit for the fluorescence channel.

For the *pump* cavity the situation is reversed: anti-Stokes pumping sits in the red tail of the absorption band where κ is 10^2 – 10^4 times smaller ($\kappa \sim 10^{-4}$ for 10^{-2} M R101 at 617–640 nm, Section 9.7.2), so $Q_{\text{abs}} \sim n/2\kappa \sim 10^3$ – 10^4 and critical coupling of the pump *requires* a mode of that quality. The inverse-designed pump resonators of Section 9.7.2 ($Q \simeq 3.5 \times 10^3$ – 1.3×10^4) sit exactly at this second ceiling. The two cavities are thus designed against opposite ends of the same rule: a low- Q , small- V fluorescence mode and a high- Q pump mode whose linewidth is matched to the red-tail absorption it must exhaust.

8.1 Doped crystals and glasses

Rare-earth- and transition-metal-doped insulators are the historical workhorse of ASF cooling. We partition them by the nature of the dipole transition.

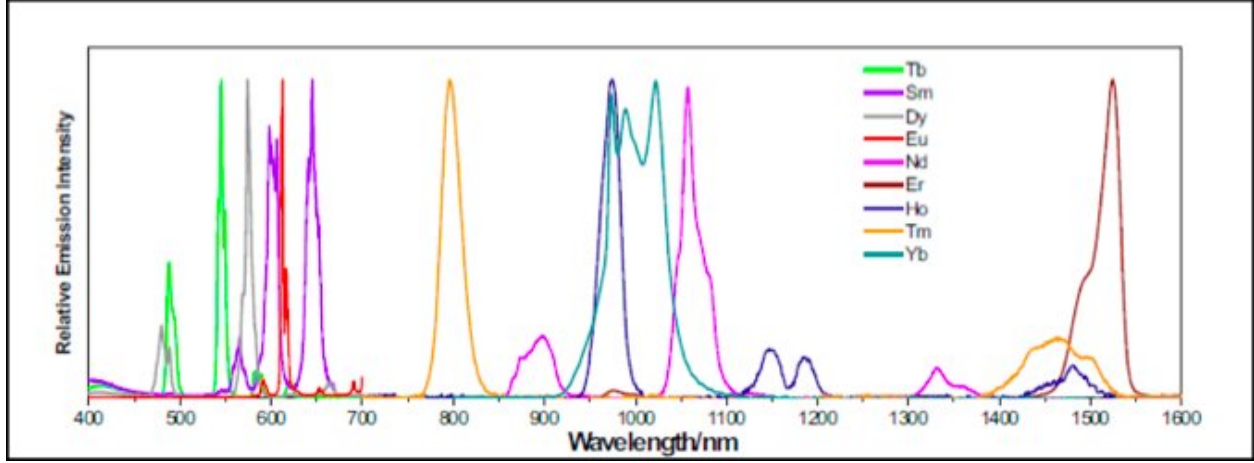


Figure 8.1: Representative room-temperature emission spectra of the principal rare-earth cooling dopants — Tb, Sm, Dy, Eu, Nd, Er, Ho, Tm, Yb — spanning the visible to mid-infrared. The Yb line near 1000 nm, the Er line near 1530 nm, and the Tm line near 1480 nm are the three most developed cooling platforms; Ho and Tm emission in the 1.5 – 2.0 μm range mates with the InGaAs / InP multi-junction LPCs of Section 7.4.

8.1.1 Forbidden-transition ion dopants

Platforms. $\text{Yb}^{3+}:\text{YLF}$ and $\text{Yb}^{3+}:\text{KYW}$ crystals, Yb-doped ZBLAN and silica glasses, and $\text{Ho}^{3+} / \text{Tm}^{3+}$ -doped hosts [21–25]. Transitions are parity- or spin-forbidden (weak electric dipole / magnetic dipole or their vibronic sidebands), giving small absorption cross-sections ($\sigma_a \sim 10^{-20}$ – 10^{-21} cm^2) and long upper-state lifetimes ($\gtrsim \text{ms}$).

Consequences for the stack.

- Small σ_a demands either high dopant density (to keep the extractor thin) or strong resonant enhancement of the pump channel — or both. The photonic network’s pump Q_p can be pushed near the absorption-limited ceiling (4.5) because the host is optically clean.
- Long $\tau_r = 1/\gamma_r$ means the Purcell enhancement F_p has lots of room to grow before the cavity rate saturates into the bulk rate; single-mode routing of fluorescence is particularly effective.
- Low background absorption α_b — this is the platform’s principal advantage — makes $\eta_{\text{external}} \rightarrow 1$ achievable and parasitic heating minimal. Reported external quantum efficiencies exceed 97 % in optimized Yb:YLF [6, 26].

Representative numbers ($\text{Yb}^{3+}:\text{YLF}$). The workhorse platform of solid-state ASF cooling. Typical operating values at the ${}^2F_{7/2} \rightarrow {}^2F_{5/2}$ pump transition near 1020 nm are $\sigma_a \simeq 8 \times 10^{-21} \text{ cm}^2$, $\sigma_e(\omega_f) \sim 10^{-20} \text{ cm}^2$, $\tau_r \simeq 1.5$ – 2.0 ms , $N_{\text{Yb}} \simeq 5 \times 10^{25} \text{ m}^{-3}$ ($\sim 5\%$ doping) below the concentration-quenching threshold [15, 26]. Mean fluorescence wavelength sits at $\sim 1000 \text{ nm}$, giving an anti-Stokes shift of $\sim 25 \text{ meV}$ per photon — one $k_B T_h$ at room temperature — and peak cooling has been demonstrated to 91 K in bulk crystals [6]. In the 100 μm -pitch device geometry of Figure 8.2, the pump absorption coefficient is only $\alpha_p = \sigma_a N_{\text{Yb}} \simeq 0.4 \text{ cm}^{-1}$, so a 10 μm film absorbs $\sim 4 \times 10^{-4}$ of the pump in a single pass; a 40 % single-pass absorption would need $\sim 1.3 \text{ cm}$ of crystal. The thin-film form factor therefore stands or falls on the pump cavity: a critically coupled resonator with $Q_p \sim n/(2\kappa\Gamma_{\text{conf}}) \sim 10^4$ – 10^5 ($\kappa \simeq \alpha_p \lambda / 4\pi \simeq 3 \times 10^{-6}$) is what brings the absorbed fraction toward unity, exactly as for the red-tail-pumped dye of Section 9.7.2. Yb-doped ZBLAN and silica glasses are an amorphous-host alternative with slightly lower σ_a but much easier thin-film integration [15].

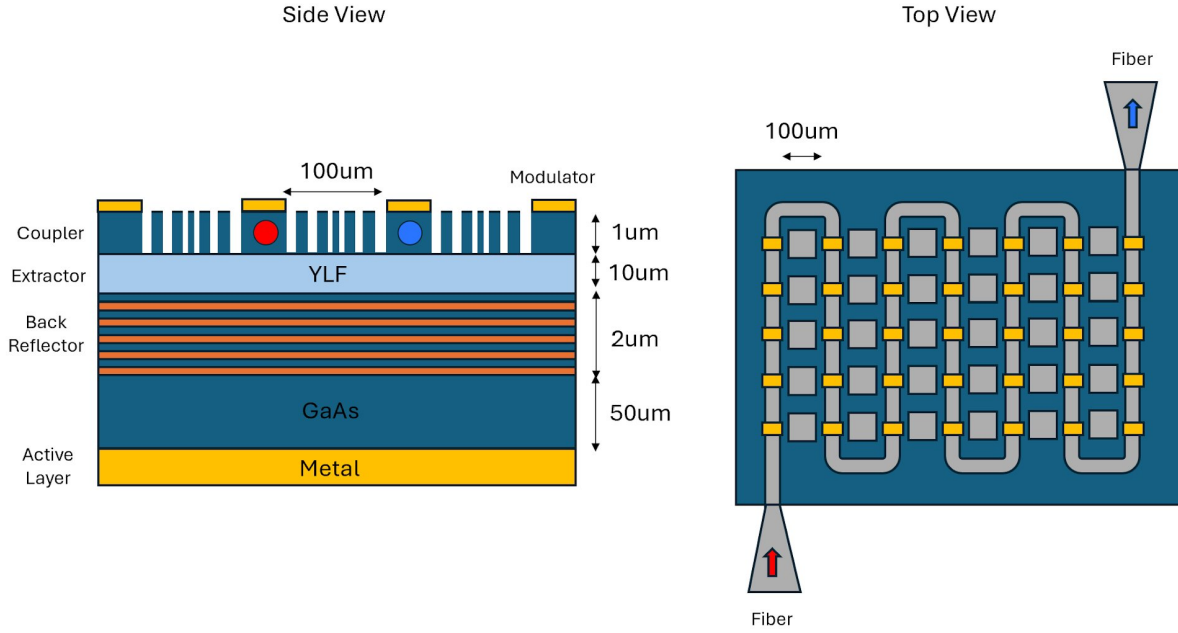


Figure 8.2: Yb:YLF-based cold-plate realization (side + top view). A 10 μm -thick Yb:YLF extractor sits between a 1 μm structured coupler and a 2 μm DBR back-reflector; the active GaAs layer is at the bottom. The top-view shows the serpentine microfluidic channel that delivers coolant to a 100 μm -pitch array of modulator/sensor pads. This is the “very narrow bandwidth, low α_b ” reference design of Section 8.1.1; pump / fluorescence fibers connect to the left/right ends of the serpentine.

8.1.2 Allowed-transition ion dopants

Platforms. $\text{Cr}^{3+}:\text{LiSAF}$, $\text{Cr}^{3+}:\text{colquiriites}$, $\text{Ti}^{3+}:\text{sapphire}$ [27–31]. Transitions are electric-dipole allowed, giving much larger σ_a ($\sim 10^{-19}\text{cm}^2$) and nanosecond-class lifetimes. Pumping is typically on a phonon sideband of the allowed line — absorption on the red side, fluorescence on the full line up to and beyond the zero-phonon frequency.

Consequences for the stack.

- Larger σ_a eases critical coupling of the pump channel: $\kappa_{p,\text{act}} \gg \kappa_{p,\text{b}}$ can be reached at lower Γ_{conf} , relaxing the photonic network design.
- Fast γ_r means the Purcell ceiling is lower; *benefit*: much higher cooling-power density because $R_p N_2^{\text{ss}}$ does not saturate as early.
- Broader emission band means the Lorentzian $\times$ Lorentzian overlap of Section 6.8 uses a larger Γ^* , favouring a wider fluorescence resonance and a less aggressive LPC bandgap match; multi-junction LPC architectures of Section 7.4 are especially relevant.
- Nonradiative decay of ${}^2E \rightarrow {}^4A_2$ and the phonon sideband details dominate the thermal penalty, so the sensor layer has to resolve T_h transients on the μs scale rather than the ms scale of the forbidden-transition case.

Representative numbers ($\text{Cr}^{3+}:\text{LiSAF}$). At the *heavily-doped* 7.4%-Cr concentration that the Chao-Rodriguez analysis identifies as the platform sweet spot [15], $\sigma_a \simeq 5 \times 10^{-20}\text{cm}^2$ on the

red side of the ${}^4A_2 \rightarrow {}^4T_2$ band, $\tau_r \simeq 67 \mu\text{s}$, $N_{\text{Cr}} \simeq 10^{27} \text{ m}^{-3}$, and the internal radiative efficiency $\eta_r = \gamma_r/(\gamma_r + \gamma_{\text{nr}}) \simeq 0.85\text{--}0.95$ in well-grown single crystals [30, 31]. The phonon-sideband emission extends from 780 nm to 1000 nm, which makes the single-pass fluorescence bandwidth much broader than any rare-earth platform; the effective anti-Stokes shift scales with the red-edge detuning of the pump rather than with a fixed line.

The Cr:LiSAF red tail. The quantity that decides whether the broad band can be pumped anti-Stokes is the absorption tail beyond the emission mean ($\sim 850 \text{ nm}$). Digitizing the room-temperature absorption of 4% Cr^{3+} :LiSAF from Silva *et al.* [32] over 700–1075 nm gives α falling from 9.0 cm^{-1} to $1 \times 10^{-3} \text{ cm}^{-1}$ ($E \parallel c$) and from 3.2 to $4 \times 10^{-4} \text{ cm}^{-1}$ ($E \perp c$) — three to four orders of magnitude, the signature of an exponential tail. A linear fit of $\ln \alpha$ against photon energy over 750–1050 nm gives $E_u = 97 \text{ meV}$ (π) and 92 meV (σ), i.e. an Urbach steepness $\sigma = k_B T/E_u \simeq 0.26\text{--}0.27$ at 290 K, with $R^2 \simeq 0.98$ (the fit-window sensitivity is 0.23–0.34). Two consequences follow. First, the tail is four times shallower than the Boltzmann limit $\sigma = 1$ of a thermalized vibronic manifold (Section 9.3.2): it is a strong-coupling / disorder tail, so absorption at 900 nm is $2 \times 10^{-2} \text{ cm}^{-1}$ rather than the $\sim 10^{-5} \text{ cm}^{-1}$ a $\sigma = 1$ tail anchored at 750 nm would predict, and the pump can sit $\sim 50 \text{ nm}$ beyond the emission mean at usable absorption. Second, the shallow tail does not by itself relax the transparency cap (5.7): with $\sigma_a(900 \text{ nm}) \simeq \alpha/n \simeq 4 \times 10^{-23} \text{ cm}^2$ at 4% doping and an emission cross-section of order 10^{-20} cm^2 in the same band, $x_{\text{max}} \simeq 4 \times 10^{-3}$ — the McCumber crossing implied by the measured emission is near 815 nm, not at the 720 nm the absorption/emission peaks would suggest. With $\tau = 67 \mu\text{s}$, $n = 10^{21} \text{ cm}^{-3}$ and $\eta_q = 6\%$ (900 nm on 850 nm) the volumetric ceiling is $\sim 0.8 \text{ W/mm}^3$ bulk and $20\text{--}80 \text{ W/mm}^3$ with $F_P = 30\text{--}100$ (Table 1.1), which places Cr:LiSAF a decade above Yb:YLF and two below the dye at the same Purcell factor. The binding constraint remains η_{EQE} : $\eta_q = 6\%$ requires $\eta_{\text{EQE}} > 0.94$, and the best single-crystal internal efficiency is 0.85–0.95; Cr:LiSAF is therefore a Level-1 candidate whose fate is decided by Purcell-enhanced radiative rate (which raises $\eta_{\text{IQE}} = \gamma_r F_P/(\gamma_r F_P + \gamma_{\text{nr}})$) rather than by absorption. Titanium:sapphire is the analogous "early transition-metal" platform with similar allowed-transition kinetics; both are the natural extractors whenever very high cooling-power density is the binding constraint.

8.2 Semiconductors

8.2.1 Direct-bandgap semiconductors

Platforms. GaAs / $\text{Ga}_{0.5}\text{In}_{0.5}\text{P}$ double heterostructures; InP-based InGaAs/InGaAsP; GaN-based emitters; SiC MJ stacks [33–35]. $\sigma_a \sim 10^{-16} \text{ cm}^2$, nanosecond lifetimes, external radiative quantum efficiencies up to 96 % in index-matched GaAs/GaInP at 850 nm [35] and 99 % at cryogenic temperatures [36].

Consequences for the stack.

- Very strong absorption allows sub-micron extractor layers; the photonic network and the active medium can share the same epitaxial stack. This is the route closest to monolithic integration with the LPC.
- Interface recombination and surface passivation dominate γ_{nr} — an external quantum efficiency of 99 % is only obtained with epitaxial-lift-off-free interfaces [36]. The coupler layer must serve as a passivation dielectric.
- Light extraction is limited by total internal reflection; solutions include high-index domes [35], micropillars [37], and photonic-crystal gratings [38]. All of these co-design with the fluorescence

channel of the photonic network: they *are* the fluorescence channel when done right.

- Back-reflector design has to handle the large index contrast with air/solvent above and must reflect the (slightly) Stokes-shifted fluorescence differently from the anti-Stokes pump — the DBR must be wavelength-selective (Section 4.18).

GaAs cooling equations and the cubic in carrier density. Unlike rare-earth platforms whose two-manifold description reduces the extractor to an effective two-level system, semiconductors have to track the carrier density N directly. The steady-state balance between pump-generated carriers and recombination gives, following Chao-Rodriguez [15] and Sheik-Bahae–Epstein [39],

$$G_p = A_{\text{SRH}} N + B N^2 + C N^3, \quad (8.1)$$

with A_{SRH} the Shockley-Read-Hall nonradiative rate, B the radiative recombination coefficient, and C the Auger rate. The volumetric cooling power density follows directly from (8.1): every escaped photon removes $\hbar\bar{\omega}_f = (1 + \eta_q)\hbar\omega_p$ while every generated carrier cost $\hbar\omega_p$, so

$$p_{\text{cool}}(N) = \hbar\omega_p [\eta_q \eta_e B N^2 - A_{\text{SRH}} N - C' N^3], \quad \eta_q \equiv \frac{\bar{\omega}_f - \omega_p}{\omega_p}, \quad (8.2)$$

with η_e the photon-extraction efficiency and C' the Auger coefficient augmented by free-carrier absorption (Section 9.3.3). The cooling term is *quadratic* in N ; the Auger loss is cubic, which is what produces an interior optimum,

$$N^* = \frac{\eta_q \eta_e B}{3C'} \left[1 + \sqrt{1 - 3A_{\text{SRH}}/(4A_0)} \right] \xrightarrow{A_{\text{SRH}} \rightarrow 0} \frac{2\eta_q \eta_e B}{3C'}, \quad (8.3)$$

(A_0 is the break-even SRH rate of (9.7)), which for bulk GaAs at 300 K with $\eta_e = 0.25$ is $N^* \simeq 5.7 \times 10^{17} \text{ cm}^{-3}$ (Table 9.2); the 10^{18} – 10^{19} cm^{-3} range is reached only once photon extraction and Purcell enhancement raise $\eta_e B$ by an order of magnitude. At the optimum $p_{\text{cool}}^* \simeq (4/27) \eta_q^3 (\eta_e B)^3 \hbar\omega_p / C'^2 \simeq 80 \text{ W/mm}^3$ for bulk GaAs — i.e. $\sim 100 \text{ W/cm}^2$ for a $10 \mu\text{m}$ layer, in line with the Sheik-Bahae–Epstein estimate [39] and two to three orders of magnitude above any rare-earth platform at the same thickness. Note the price: the absorbed pump density at N^* is $\hbar\omega_p (AN + BN^2 + CN^3) \simeq 3 \times 10^4 \text{ W/mm}^3$, so the bulk optimum lifts only $\sim 0.3\%$ of the pump power as heat ($\eta_{\text{ASF}} \simeq 0.003$); the figure rises toward $\eta_q \simeq 3.5\%$ only as $\eta_e \rightarrow 1$. This is the reason to keep semiconductors in the stack *and* the reason their loop economics hinge on power recovery (Chapter 7).

Representative numbers (GaAs/GaInP double heterostructure). $\sigma_a \simeq 10^{-16} \text{ cm}^2$ at band-edge, $B \simeq 2 \times 10^{-10} \text{ cm}^3/\text{s}$, $C \simeq 7 \times 10^{-30} \text{ cm}^6/\text{s}$, $A_{\text{SRH}} \lesssim 10^5 \text{ s}^{-1}$ in best-of-class samples, external radiative efficiency 96 % at room temperature in index-matched geometries [35] and 99 % at cryogenic conditions [36]. Natural photonic realisation: the GaInP passivation layer of the extractor doubles as the coupler-layer dielectric of Section 4.18, making this the closest-to-monolithic option for on-chip integration.

8.2.2 Indirect-bandgap semiconductors

Platforms. Germanium nanocrystals [40], Si (proposed), and SiC variants on the allowed end. Absorption and emission are phonon-assisted; the zero-phonon line is weak and the vibrational sideband carries most of the oscillator strength.

Consequences for the stack.

- Low oscillator strength forces extremely high confinement ($\Gamma_{\text{conf}} \rightarrow 1$) or nanostructuring; the coupler layer becomes very demanding.
- Broad, structured spectra favour multi-junction LPCs of Section 7.4.
- Attractive for CMOS-compatible integration; indirect-bandgap platforms are the long-horizon entry for monolithic on-chip placement.

8.3 Molecular dispersions: dyes and nanoparticles in solvents**8.3.1 Fluorescent dyes: solvent-based dispersions and their wall**

The solvent-based baseline. Rhodamine 101 (R101) in acidic ethanol (99% EtOH / 1% HCl) remains the canonical demonstration of liquid-phase laser cooling [41, 42]; Rhodamine 6G (R6G), Rhodamine B, Rhodamine 3B, and the Alexa Fluor family [43, 44] extend the palette. Rhodamines are the dominant candidate class because they alone satisfy the joint requirement of high internal quantum efficiency (IQE), sub-5 ns radiative lifetime, good photostability, narrow Stokes shift, and large molar extinction coefficient — the combined chemistry constraints that eliminate coumarins (large Stokes shift) and, as isolated monomers, cyanines (poor QY, poor photostability) and NIR dyes (QY down, poorer stability) — a verdict that the aggregate and solid-state emitters of Section 8.3.6 reopen. Red-edge pumping in the long-wavelength tail of σ_a produces anti-Stokes shifts of tens of meV per absorbed photon.

The experimental record. The record on which the platform rests is thinner than its citation count suggests, and the loop of Chapter 9 should know exactly what has been shown. The earliest controlled result is Zander and Drexhage’s [45]: R6G in deuterated ethanol, with the *sign* of the photothermal lens used as the thermometer — a converging lens beyond 570 nm is a direct optical signature of net heat extraction, immune to the slow drifts of a bulk thermometer — giving an absolute cooling efficiency $P_{\text{cool}}/P_{\text{laser}}$ of 3.2×10^{-5} at 575 nm (~ 1 mW of cooling at a relative efficiency $\eta_{\text{cool}} \simeq 1.1\%$). Clark and Rumbles’ bulk temperature drop in R101/acidified ethanol [41] (0.7 K/h under 350 mW, ~ 0.6 mW net) drew a formal Comment [46] on open-cell thermal drift and noise floor that the spectroscopic follow-up [42] did not fully answer; Mungan and Gosnell’s review [47] rates the Zander–Drexhage result “unequivocal” and the Clark–Rumbles one plausible but inconclusive, and no independent group has reproduced a bulk temperature drop in a rhodamine solution — in contrast to the rapid multi-laboratory reproduction of Yb:ZBLAN in the same years. The only post-2005 primary work is the Kyiv tricarbo-cyanine series [48, 49]: NIR dyes in DMF under 980 nm CW pumping with anti-Stokes shifts up to 147 nm (225 meV, i.e. η_q up to 18% per photon, three to four times the rhodamine figure) and again $\sim 1\%$ net cooling — large η_q bought with a bare IQE of ~ 0.2 . That handicap is the one the photonic levers of Section 8.3.4 remove most cheaply: a Purcell factor acts on the radiative rate alone, so

$$\eta_{\text{IQE}}(F_P) = \frac{F_P \eta_{\text{IQE},0}}{F_P \eta_{\text{IQE},0} + 1 - \eta_{\text{IQE},0}}, \quad (8.4)$$

which takes 0.2 to 0.71, 0.88 and 0.96 at $F_P = 10, 30$ and 100 . Evaluated on the same assumptions as the R640 row of Table 1.1 (10^{-1} M isolated, 400 K, $F_P = 100, 50 \mu\text{m}$), a 980 nm-pumped tricarbo-cyanine reaches $1.7 \times 10^5 \text{ W/mm}^3$ at $\eta_{\text{cool}} = 13\%$ — the R640 density at a third of the circulated pump, at the pump wavelength where η_P is highest and with fluorescence on a GaAs LPC — and two or three such stages reach the 30 and 60% tiers. Photostability and the aggregation

behaviour of cyanines at 10^{-1} M are the open questions, and Section 8.3.6 is where the second is answered. Three lessons carry into the design. First, the methodological bar set by the 1996 Comment — a thermometer that cannot be fooled by convection, and a signal above the drift floor — is the acceptance criterion for any tile-level demonstration, and the thermal-lens sign and the differential (pump-on/off, tail-vs-band) methods of Section 9.7.4 are how a thin film meets it. Second, the $\sim 1\%$ that every liquid experiment reached is exactly rung 0 of Table 8.2: the bulk record neither contradicts nor supports the design point, which lives on rungs 3–7. Third, the photophysics itself has since been validated at scale by a community that never set out to cool anything: hot-band-absorption anti-Stokes imaging uses the same thermally populated vibrational levels in rhodamines, cyanines, BODIPYs and ATTO dyes — indocyanine green under 915 nm [50], Alexa Fluor 568 measured side by side with R101 [51] — and its systematic spectral mapping of anti-Stokes shift, linewidth and yield across dye families is a ready-made Level-0 database (Section 9.3.2) for the extractor alphabet.

What integrating-sphere measurements show. Recent measurements of high-purity R101 in acidic ethanol using the Nature-protocol integrating-sphere method establish the ceiling of the solvent-based platform (see Figure 8.4; the acid-free R640 series discussed below is overlaid):

- *IQE stays remarkably high across four decades of concentration.* With aggressive cleaning, R101 IQE remains above 95% from 10^{-6} M to $\sim 10^{-3}$ M, drops to $\sim 90\%$ at 3×10^{-3} M, then crosses a knee and falls to $\sim 63\%$ by 10^{-2} M. The molecule itself is nearly ideal well past the traditional 1-mM “aggregation” rule of thumb.
- *EQE crashes far earlier because escape efficiency collapses.* The photon-escape probability drops from $\sim 95\%$ at 10^{-6} M to $\sim 14\%$ at 10^{-2} M; EQE follows, dropping from $\sim 95\%$ to $\sim 19\%$ over the same range. Cooling scales with EQE, not IQE, so the practical wall of the solvent-based system sits around 10^{-3} M, not the 10^{-2} M where aggregation quenching finally hits the dye itself.
- *Mean fluorescence wavelength red-shifts ~ 60 nm* over the concentration range, from ~ 605 nm at 10^{-6} M to ~ 662 nm at 10^{-2} M, driven by iterative self-absorption. This shifts the mean emitted-photon energy toward the pump band and directly reduces the per-photon cooling shift ($\omega_f - \omega_p$).

Both effects are intrinsic to a solvent environment: dye monomers diffuse freely and eventually form H- and J-aggregates that quench, and photons emitted anywhere in the cuvette re-encounter the absorption band of every other monomer along the escape path. R6G shows the same shape at slightly higher concentrations (IQE $\geq 95\%$ up to $\sim 10^{-3}$ M, escape efficiency from 90% down to $\sim 40\%$ over the tested range) but the same qualitative wall.

The acid-free R640 series: the IQE knee is a preparation artefact, not a molecular limit. A second integrating-sphere series (August 2026) replaces the acidified R101 inner salt with high-purity Rhodamine 640 — the perchlorate salt of the same chromophore (Section 8.3.2) — dissolved directly in ethanol with no HClO_4 . The two preparations are indistinguishable up to 10^{-3} M, but diverge sharply above it (Figure 8.4, solid vs. dashed):

- *IQE:* R640 holds $\simeq 98\%$ to 10^{-3} M, $\simeq 93\%$ at 4.6×10^{-3} M, and $\simeq 81\%$ at 10^{-2} M — against $\simeq 62\%$ for the acidified R101 at the same concentration. Roughly half of the non-radiative loss previously attributed to “aggregation quenching” at 10^{-2} M is therefore introduced by the acidification step (counter-ion / impurity chemistry), not by the chromophore density itself.
- *Escape efficiency and EQE* are unchanged by the preparation ($\eta_e \simeq 15\%$, $\eta_{\text{EQE}} \simeq 39\%$ vs. 19%)

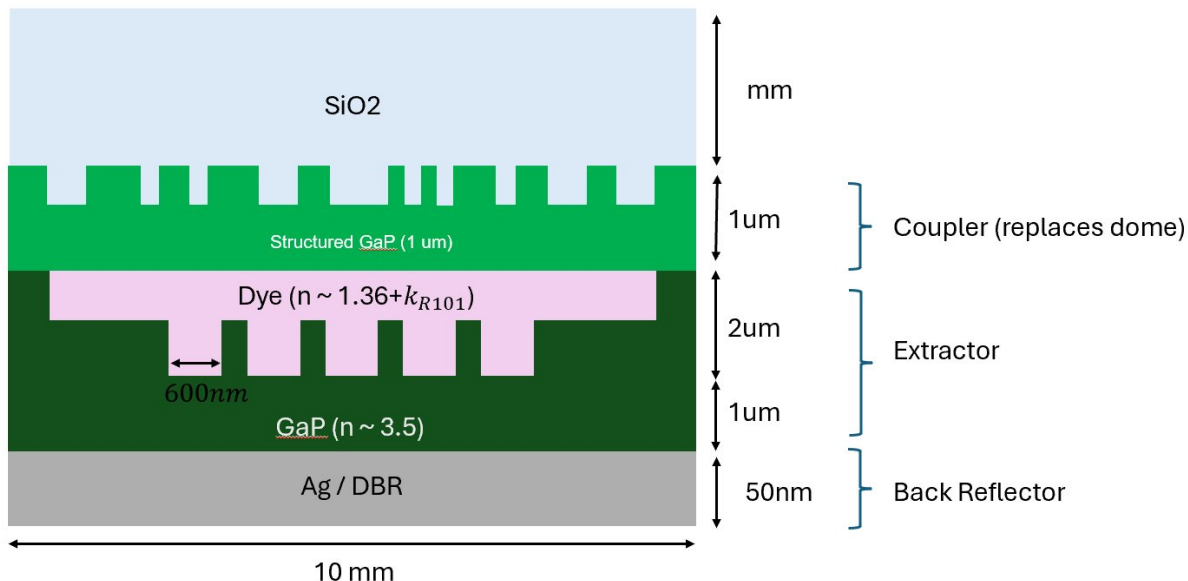


Figure 8.3: Legacy cold-plate realization with a solvent-phase dye-loaded extractor (Rhodamine 101 in acidic ethanol, $n \simeq 1.36 + ik_{R101}$) infiltrated into a structured GaP matrix. The GaP coupler ($n \simeq 3.5$, $1 \mu\text{m}$ thick) provides sub-wavelength index contrast that routes the pump while leaving the dye at the field maximum; the extractor is a $2 \mu\text{m}$ GaP/dye composite over a $1 \mu\text{m}$ plain-GaP coupling layer, with a 50 nm Ag / DBR back-reflector. The whole tile is $10 \times 10 \text{ mm}$ and mounts under a 1 mm -thick SiO_2 thermal window. This microfluidic form factor is intrinsically capped at $\sim 10^{-3} \text{ M}$ by the escape-efficiency and aggregation-quenching walls of Figure 8.4; the SMILES-in-polymer thin-film form factor of Section 8.3.2 replaces it and lifts the concentration ceiling by two orders of magnitude.

at 10^{-2} M); the EQE gain is entirely the IQE gain propagated through the recycling identity $\eta_{\text{EQE}} = \eta_{\text{IQE}}\eta_e/[1 - (1 - \eta_e)\eta_{\text{IQE}}]$, which reproduces the measured 39% from (0.81, 0.15) exactly. Escape is a cuvette-geometry property; it is what the thin-film form factor is meant to change.

- *Mean fluorescence wavelength* red-shifts from 619 nm at 10^{-5} M to 641 nm (10^{-4}), 652 nm (10^{-3}) and 659 nm (10^{-2} M) in the 1 cm cuvette, the same reabsorption-driven shift as R101 (662 nm). The *intrinsic* (short-path) mean emission of the monomer is $\bar{\lambda}_f \simeq 615\text{--}620 \text{ nm}$, not the 605 nm sometimes quoted from the emission peak; this is the value a thin film with a sub-extinction-length escape path should be designed to.
- *Stability:* the acid-free samples re-measured after 24 h are unchanged within error at all four concentrations.

The design consequence is quantitative: with the acidification step removed, the concentration at which IQE crosses 90% moves from $\sim 3 \times 10^{-3}$ to $\sim 6 \times 10^{-3} \text{ M}$, and the residual 19% loss at 10^{-2} M is the target that SMILES steric isolation (Section 8.3.2) must remove. Since the cyanostar complex binds the R640 perchlorate directly, the SMILES film inherits the acid-free chemistry; an IQE above 90% at 10^{-2} M in the polymer is therefore the expected, not the optimistic, outcome — but it remains a projection until the integrating-sphere series is repeated on films.

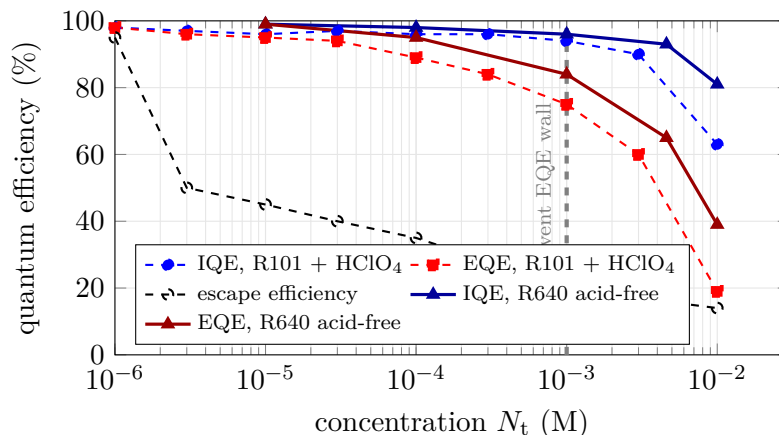


Figure 8.4: Solvent-phase concentration wall (integrating-sphere measurements, Nature-protocol method). Dashed: acidified R101 inner salt in ethanol; solid triangles: acid-free R640 in ethanol (August 2026). IQE stays above 90% up to $\sim 3 \times 10^{-3}$ M (R101) or $\sim 6 \times 10^{-3}$ M (R640); at 10^{-2} M the acid-free preparation retains 81% IQE against 63%. EQE tracks IQE only at low concentration and collapses above $\sim 10^{-3}$ M because the cuvette escape efficiency (identical for both preparations) falls to $\sim 15\%$ via inner-filter reabsorption. The R101 EQE point at 3×10^{-3} M has been re-derived from the measured IQE and escape values through the recycling identity (60%, replacing an inconsistent 45%). Cooling power scales with EQE, so the solvent-phase platform is practically limited to $\lesssim 10^{-3}$ M; the thin-film form factor attacks the escape term, the acid-free / SMILES chemistry the IQE term.

Consequences for the legacy stack. The solvent-based form factor drives the design of the legacy extractor (Figure 8.3):

- A liquid extractor means the “layer” is a microfluidic channel with dielectric walls. The coupler, network, back reflector, and sensor must all be dielectric or passivated surfaces in contact with the solvent.
- Solvent choice is part of the design: dielectric constant sets Γ_{conf} , polarity sets dye aggregation (and hence γ_{nr} via self-quenching).
- Photobleaching is a failure mode absent in solids: the sensor layer must include a fluorescence-intensity lock to detect degradation. In exchange, the extractor is refreshable without opening the photonic stack — a meaningful reliability advantage for production use.
- The Lorentzian $\times$ Lorentzian overlap of Section 6.8 translates directly to dye-solvent systems; resonance-tuned extraction can spectrally isolate a bright sub-peak of the dye emission, raising effective exergy.

But the concentration wall of Figure 8.4 caps cooling power density. With $N_t \lesssim 10^{-3}$ M and typical $\sigma_a \sim 10^{-17}$ cm² in the red tail, even a 100 μm path length only reaches $\sigma_a N_t L \sim 0.06$, so a substantial single-pass absorption requires geometrically large cuvettes — exactly the form factor incompatible with chip-scale integration. A different material state is needed to break this wall.

8.3.2 SMILES-in-polymer thin films: breaking the concentration wall

The concentration wall of Figure 8.4 is not a property of the rhodamine molecule; it is a property of the molecule’s *environment*. If neighbouring monomers can be mechanically prevented from forming aggregates and if the emitted photon can be given an escape path that does not require traversing many extinction lengths of same-band dye, both walls disappear. The SMILES (Small-Molecule

Ionic Isolation Lattices) approach¹ delivers exactly this: each rhodamine monomer is complexed with a pair of *cyanostar* macrocycles that shield it from neighbouring monomers, and the resulting complexes are dispersed in a transparent polymer host that provides a solid, structurable matrix.

Mechanism. The cyanostar macrocycle binds tightly to the fluorophore’s counter-ion and physically encapsulates the ionic pair, holding each dye monomer at a fixed separation from its neighbours regardless of loading fraction. In practice this suppresses two aggregation pathways simultaneously:

- *H- and J-aggregates*, which drive concentration quenching in solvents by opening non-radiative decay channels;
- *ion-pair-driven self-quenching*, which is the dominant loss channel in salt-forming solvents.

The suppression is a structural (*steric*) rather than electronic effect, and it holds for essentially any polymer host tested: polyurethane, polypropylene, polystyrene, PMMA, polyvinylpyrrolidone, polycarbonate, PEG ethers, polyacrylic acid, and photopolymer resins. Solution-processable transparent hosts (PMMA, PS, optical polycarbonate) are the natural fit for chip-scale integration; polyurethane has been demonstrated as a lasing medium in the SMILES form, which is the strongest experimental evidence that the extractor’s radiative rate is preserved.

R640 as the practical experimental system. Rhodamine 640 (R640) is the perchlorate salt of Rhodamine 101 — the same xanthene chromophore, permanently locked in its fluorescent zwitterionic tautomer without requiring free acid in the extractor environment. Because R640 already ships as the salt, it is directly compatible with SMILES encapsulation without adding HCl or HClO₄ to the polymer precursor, which is the current experimental route at Maxwell. The chromophore, and hence the absorption / emission line shapes, radiative lifetime and low-concentration IQE, are those of R101; “R640” names the counter-ion (ClO₄[−]) and the resulting dominance of the fluorescent form. The two are *not* interchangeable at high concentration, however: the acid-free R640 series of Section 8.3.1 retains 81% IQE at 10^{−2} M where the acidified R101 falls to 62%, because the perchloric-acid step needed to lock R101 into its fluorescent tautomer introduces a non-radiative channel that R640 avoids. Where the discussion below refers to R101 spectra the physics is unchanged; the practical extractor material, and the one whose high-concentration IQE should be used in any performance estimate, is R640-SMILES-in-polymer.

The design regime the wall shift opens. With SMILES encapsulation expected to preserve IQE at concentrations $\gtrsim 10^{-2}$ M and the escape-efficiency problem attacked geometrically by a 1–10 μm thin-film form factor (in place of the mm-scale cuvette), the extractor becomes a solid, structurable layer with the following characteristics:

- *IQE target* $\gtrsim 90\%$ at 10^{−2} M, projected from the acid-free R640 solution value of 81% (Section 8.3.1) plus the steric suppression of the residual aggregation channel; the intrinsic monomer value is $\simeq 98\%$. This is a projection to be confirmed on films, and it is the single most leveraged number in the platform: the required EQE for net cooling is $\geq 0.92\text{--}0.95$ (Section 8.3.5).
- *Concentration* $N_t \gtrsim 10^{-2}$ M, two orders of magnitude beyond the solvent wall. Loading fractions in the 1–10% w/w range in polycarbonate ($\rho \simeq 1.2\text{ g/cm}^3$, $M_w(\text{R640}) = 591\text{ g/mol}$) correspond to $\sim 20\text{--}200\text{ mM}$ before the cyanostar mass is counted.
- *Thin-film form factor*: a 1–10 μm structured layer replaces the microfluidic cuvette. This is directly integrable with photonic-network waveguides, DBR back-reflectors, and Purcell-

¹Not to be confused with the SMILES chemical-notation string format — unrelated abbreviations that share three letters.

enhancing resonators of Chapters 4 and 5 — the geometry the doped crystal and semiconductor platforms already exploit becomes available to molecular-dispersion chemistry.

- *Escape efficiency recovered by geometry:* at 1–10 μm film thickness with Purcell-enhanced single-mode extraction, the path-length problem that crashed escape efficiency in the mm-scale solvent case is bypassed — fluorescence exits the film into the photonic network before it can be reabsorbed. The film thickness is nevertheless bounded from below by the transparency ceiling derived next.

8.3.3 From concentration to cooling-power density: the volumetric route

The first-law bookkeeping of (4.15) does not by itself connect a material's constitutive parameters to a cooling power; that link is the per-ion pump rate (5.6) and the transparency cap (5.7). Writing them per unit volume gives a route that goes directly from a concentration to a *volumetric* cooling power density p_{cool} (W/mm^3), and only then to an areal density by integrating through the film thickness d . This is the form the chip-level thermal model of Chapter 10 consumes, and it applies verbatim to molecular dispersions (with the molecular number density $n_t = N_A C$) and to semiconductors (with the carrier density, (8.2)).

Molecular dispersions. Let n_t be the dye number density, $\sigma_a(\lambda_p, T)$ and $\sigma_e(\lambda_p, T)$ the cross-sections at the pump wavelength (related by McCumber, (5.21)), τ the excited-state lifetime, and I_p the local pump intensity. The steady-state excited fraction $x = N_2/n_t$ follows from (5.5),

$$x = \frac{I_p/I_{\text{sat}}}{1 + (I_p/I_{\text{sat}})(1 + \sigma_e/\sigma_a)}, \quad I_{\text{sat}} \equiv \frac{\hbar\omega_p}{\sigma_a\tau}, \quad (8.5)$$

so that $x \rightarrow x_{\text{max}} = \sigma_a/(\sigma_a + \sigma_e)$ as $I_p \rightarrow \infty$. The absorbed pump power per unit volume is $\hbar\omega_p n_t x/\tau$, and the net cooling power per unit volume is

$$\boxed{p_{\text{cool}} = \frac{n_t x \hbar\omega_p}{\tau} \eta_{\text{cool}}, \quad \eta_{\text{cool}}(\lambda_p) = \eta_{\text{abs}}(\lambda_p) \eta_{\text{EQE}} \frac{\lambda_p}{\lambda_f} - 1,} \quad (8.6)$$

where $\eta_{\text{abs}} = \alpha_r/(\alpha_r + \alpha_b)$ is the fraction of absorbed pump taken by the dye rather than the host ($\alpha_r = \ln 10 C \varepsilon(\lambda_p) = n_t \sigma_a$). Two limits matter:

$$\text{unsaturated } (I_p \ll I_{\text{sat}}) : \quad p_{\text{cool}} \simeq \alpha_r(\lambda_p, T) I_p \eta_{\text{cool}}, \quad (8.7)$$

$$\text{transparency ceiling } (I_p \gg I_{\text{sat}}) : \quad p_{\text{cool}}^{\text{max}} = \frac{n_t \hbar\omega_p}{\tau} \frac{\sigma_a}{\sigma_a + \sigma_e} \eta_{\text{cool}} \simeq \frac{n_t \hbar\omega_p \eta_{\text{cool}}}{\tau} e^{-(\hbar\omega_0 - \hbar\omega_p)/k_B T}. \quad (8.8)$$

The areal cooling density is then $P_{\text{cool}}/A = \int_0^d p_{\text{cool}} dz$, which for a film thin compared with the (cavity-enhanced) absorption length is simply $p_{\text{cool}} d$; with a critically coupled pump resonator that absorbs the incident flux completely, the unsaturated branch becomes $P_{\text{cool}}/A \simeq I_{\text{inc}} \eta_{\text{cool}}$ *provided* the film is thick enough that the absorbed power per unit volume stays below the ceiling:

$$d \geq d_{\text{min}} = \frac{I_{\text{inc}} \tau}{n_t x_{\text{max}} \hbar\omega_p}. \quad (8.9)$$

Where τ comes from: the Strickler–Berg relation. The lifetime in (8.6)–(8.9) is not an independent material parameter. For a strongly allowed molecular transition the radiative rate is fixed by the *absorption* spectrum through the Strickler–Berg relation [52], the molecular form of Einstein’s A/B ratio corrected for the absorption–emission frequency difference and the host index:

$$\Gamma_r \equiv \frac{1}{\tau_0} = 2.880 \times 10^{-9} n^2 \langle \tilde{\nu}_f^{-3} \rangle_{\text{av}}^{-1} \frac{g_l}{g_u} \int \frac{\varepsilon(\tilde{\nu})}{\tilde{\nu}} d\tilde{\nu}, \quad (8.10)$$

with ε in $\text{M}^{-1} \text{cm}^{-1}$, $\tilde{\nu}$ in cm^{-1} , $\langle \tilde{\nu}_f^{-3} \rangle_{\text{av}}$ the fluorescence-weighted mean of $\tilde{\nu}^{-3}$, and $g_l/g_u = 1$ for singlet–singlet transitions. For R101 in ethanol ($\varepsilon_{\text{max}} \simeq 9.5 \times 10^4 \text{M}^{-1} \text{cm}^{-1}$ at 565 nm, $\int \varepsilon d\tilde{\nu}/\tilde{\nu} \simeq 7.9 \times 10^3$, $n = 1.36$, $\bar{\lambda}_f = 605 \text{nm}$) it gives $\tau_0 \simeq 5.2 \text{ns}$ against the measured 4.2–4.6 ns at near-unity quantum yield — the $\sim 20\%$ agreement the original paper reports for rhodamines. Four consequences follow for the ladder. (i) *The same integral sets both ends of the cycle:* the oscillator strength that makes the dye a strong absorber (large σ_a , short d_{min}) makes it a fast emitter (large Γ_r , large ceiling), so (8.11) cannot be improved by choosing a molecule with a longer τ at the same ε — only the non-radiative rate is free, and it only hurts. (ii) *IQE is measurable without an integrating sphere:* $\eta_{\text{QE}} = \tau/\tau_0$, with τ from a lifetime measurement and τ_0 from the absorption spectrum, which is the cross-check the R640 series of Figure 8.4 should carry, and which separates a true IQE loss (τ shortens) from an escape loss (τ lengthens through reabsorption–re-emission). (iii) *Host index is a lever:* $\Gamma_r \propto n^2$ is the homogeneous-medium Purcell factor, so moving the chromophore from ethanol (1.36) into a polymer (1.5) or a high-index nanocomposite (2.0) raises the ceiling by $1.2\times$ and $2.2\times$ before any cavity is added — the first, structureless rung of the Γ_{tot} lever. (iv) *The $\langle \tilde{\nu}_f^{-3} \rangle$ factor is the emission-side lever in a different guise:* blue-shifting the emission stream (rung 3) raises Γ_r as $\bar{\nu}_f^3$, which is why a cavity that selects the blue edge of the band raises both η_q and the rate. The relation holds only where its premises hold — strongly allowed transitions ($\varepsilon_{\text{max}} \gtrsim 10^4$), mirror-image bands, and emission from the thermalized S_1 — so it applies to rhodamines, cyanines and AIE/J-aggregate emitters but *not* to the forbidden-transition ions of Section 8.1.1, whose rates come from Judd–Ofelt or F  chtbauer–Ladenburg analysis, nor to the vibronic d – d band of Cr:LiSAF, where the 10^2 – 10^3 times smaller ε is exactly why its lifetime is μs rather than ns.

The thickness window, and why the film is a photon-recycling device. Equation (8.9) bounds d from below; reabsorption bounds it from above. For an emission line-shape tuned to the measured low-concentration mean of 619 nm and the same absorption spectrum, the single-pass escape probability of an isotropically emitting slab (no index trapping, i.e. assuming photonic extraction) is shown in Figure 8.5: $\eta_e \geq 0.95$ requires $d \leq 2.3 \mu\text{m}$ at 10^{-2}M and $d \leq 0.25 \mu\text{m}$ at 10^{-1}M , while $\eta_e \geq 0.99$ requires 0.36 and $< 0.1 \mu\text{m}$. Against these, d_{min} at $10 \text{W}/\text{mm}^2$ incident is 18 and $1.8 \mu\text{m}$ (300 K), 3.4 and $0.3 \mu\text{m}$ (400 K). The single-pass window is therefore *closed* by roughly an order of magnitude at every concentration, and it closes further when the emission is photonicallly blue-shifted toward the zero line (escape at $5 \mu\text{m}$, 10^{-2}M , falls from 0.92 to 0.71 when $\bar{\lambda}_f$ moves from 616 to 590 nm, because the blue edge sits where the absorption is larger). It opens only for incident fluxes below ~ 0.5 – $1 \text{W}/\text{mm}^2$, i.e. the near-term hot-spot razor. The design point of Section 8.3.4 is thus necessarily a *photon-recycling* device: photons are reabsorbed several times before leaving, and the film’s external efficiency is $\eta_{\text{EQE}} = \eta_{\text{IQE}}\eta_e/[1 - (1 - \eta_e)\eta_{\text{IQE}}]$ with per-pass $\eta_e \simeq 0.4$ – 0.9 , so $\eta_{\text{EQE}} \geq 0.95$ is available only with $\eta_{\text{IQE}} \geq 0.99$ or with Purcell-directed extraction that couples emission into the network mode within one absorption length. This is the precise sense in which “escape efficiency $\simeq 1$ ” is a photonic requirement on the film rather than a thickness one, and why the emission-side objective of Section 9.7.2 is as important as the pump side. The same figure shows the reward: moving $\bar{\lambda}_f$ from 618 to 590 nm at fixed λ_p raises the peak η_{cool} from 4% to 9% (10^{-2}M , 300 K) and lowers the EQE required for net cooling from 0.95 to 0.90.

C (M)	T (K)	λ_p (nm)	ε (L mol ⁻¹ cm ⁻¹)	α_r (cm ⁻¹)	η_{cool}	x_{max}	$p_{\text{cool}}^{\text{max}}$ (W/mm ³)	P_{cool}/A , 5 μm (W/mm ²)
10 ⁻²	300	640	29	0.67	3.2%	1.3×10^{-3}	18	0.09
10 ⁻²	300	651	8.2	0.19	4.0%	3.7×10^{-4}	6.3	0.03
10 ⁻²	400	651	39	0.90	5.1%	2.7×10^{-3}	58	0.29
10 ⁻²	400	680	3.7	0.085	7.0%	2.5×10^{-4}	7.3	0.04
10 ⁻¹	300	651	8.2	1.9	3.2%	3.7×10^{-4}	51	0.26
10 ⁻¹	400	651	39	9.0	3.3%	2.7×10^{-3}	380	1.9
10 ⁻¹	400	680	3.7	0.85	7.6%	2.5×10^{-4}	79	0.40

Table 8.1: Operating points from (8.8) for R101/R640 ($\eta_{\text{EQE}} = 1$, $\tau = 4.3$ ns, $\lambda_{00} = 588$ nm; $\bar{\lambda}_f = 618$ nm at 10⁻² M and 630 nm at 10⁻¹ M to reflect the concentration-dependent reabsorption shift of Section 8.3.1). Every entry in the last two columns scales linearly with the actual η_{EQE} -dependent η_{cool} , which is positive only for $\eta_{\text{EQE}} \gtrsim 0.92$ –0.95. The 10⁻¹ M, 400 K row is the design point at which a 5 μm film first reaches the W/mm² class needed for hot-spot service.

Figure 8.6 evaluates (8.7)–(8.8) for R101/R640 using the measured molar extinction anchored at $\varepsilon(600 \text{ nm}) = 4290 \text{ L mol}^{-1} \text{ cm}^{-1}$ and the exponential red tail $\varepsilon(E) = \varepsilon(E_0) \exp[\sigma(E - E_0)/k_B T]$, $\sigma = 1$, of Section 9.3.2 (with $\lambda_{00} = 588$ nm, $\tau = 4.3$ ns, Kedenburg ethanol background, $\eta_{\text{EQE}} = 1$); Table 8.1 lists the numbers at representative operating points. Three conclusions follow.

1. *The ceiling is exponentially small in the anti-Stokes shift.* Because the pump must sit 6 – $9 k_B T$ below the zero line to be below the *mean* emission, $x_{\text{max}} \sim 10^{-3}$ – 10^{-4} and every molecule can cycle only a few times per μs regardless of pump power. A 10⁻² M, 5 μm film at 300 K tops out near 0.03–0.1 W/mm²; the 10^3 – 10^4 W/mm² obtained by letting every molecule cycle at 10^9 s^{-1} is not reachable *in a bulk medium* at any intensity, because stimulated emission at ω_p shuts the pump off long before the fluorescence lifetime is the limit. It is, however, the book's design target, and Section 8.3.4 shows which photonic levers lift the ceiling to it: the cap is on $x_{\text{max}} \Gamma_{\text{tot}}$, and both factors are engineerable.
2. *Temperature and concentration are the levers, and they multiply.* x_{max} rises as $e^{-\Delta E/k_B T}$ and the tail absorption as $e^{\sigma(E-E_0)/k_B T}$, so a 400 K die gives 5–10 $\times$ the ceiling of a 300 K one at the same pump; 10⁻¹ M gives another 10 $\times$. At 10⁻¹ M and 400 K a 5 μm film reaches $\sim 2 \text{ W/mm}^2$ at $\lambda_p \simeq 650 \text{ nm}$ — the hot-spot regime of Chapter 10 — with an unsaturated slope $p_{\text{cool}}/I_p \simeq 0.9 \text{ mm}^{-1} \times \eta_{\text{cool}}$. The dye platform is therefore *intrinsically a hot-die platform*: the same physics that makes it marginal at room temperature makes it competitive on a 400 K compute tile.
3. *The pump-cavity Q must be designed at the operating excitation.* As $x \rightarrow x_{\text{max}}$ the medium's effective extinction falls as $\kappa_{\text{eff}} = \kappa [1 - x(1 + \sigma_e/\sigma_a)]$, so a resonator critically coupled at low power over-couples at high power. Operating at $I_p \lesssim I_{\text{sat}}/3$ keeps κ_{eff} within 30% of its small-signal value; equivalently, (8.9) with $I_{\text{inc}} \simeq 10 \text{ W/mm}^2$ gives $d_{\text{min}} \simeq 18 \mu\text{m}$ at 10⁻² M/300 K, 3.4 μm at 10⁻² M/400 K, and 0.3 μm at 10⁻¹ M/400 K. The sub-micron liquid layer of the first inverse designs (Section 9.7.2) is a small-signal demonstration; its authors' own recommendation to thicken the dye layer and lower Q is exactly (8.9).

Semiconductors. The same route for a direct-gap semiconductor is (8.2) with N set by the pump through $\alpha(\lambda_p, T) I_p / \hbar \omega_p = AN + BN^2 + C'N^3$; $P_{\text{cool}}/A = p_{\text{cool}}(N) d$ for a layer thinner than the diffusion length. The transparency cap is replaced by the Auger roll-over at N^* , and the table-of-record is Table 9.2: 80 W/mm³ for bulk GaAs at 300 K, i.e. 0.8 W/mm² for a 10 μm layer at $\eta_{\text{ASF}} \simeq 0.3\%$. Raising η_e from 0.25 to 0.9 (photonic extraction) lifts this to $\sim 4 \times 10^3 \text{ W/mm}^3$ (40 W/mm² per 10 μm) at $\eta_{\text{ASF}} \simeq 1\%$, because $p_{\text{cool}}^* \propto (\eta_e B)^3$; the absorbed pump density at that

point is $\sim 4 \times 10^5 \text{ W/mm}^3$, so the practical limit is not the extractor but the pump the network can deliver and the LPC can recover. Because η_q for GaAs is only $\simeq 3.5\%$ (890 vs. 860 nm), the semiconductor platform buys its high p_{cool} with a pump power 30–300 $\times$ the heat lifted — the opposite trade from the dye, and the reason the two are complementary rather than competing (Section 8.4).

The cooling-density comparison across platforms is therefore made in Table 8.4 at a common thickness, from the volumetric ceiling of each material: the solution-processable SMILES-R640 film does *not* reach the direct-bandgap semiconductor's W/mm^3 class at room temperature, but at $\gtrsim 400 \text{ K}$ and $\gtrsim 10^{-1} \text{ M}$ it enters the same W/mm^2 -per- $5 \mu\text{m}$ regime with a 5–10 $\times$ better η_{ASF} , no epitaxy, and compatibility with essentially any substrate.

8.3.4 The photonic ladder: from bulk limits to the design point

The book's design point is not the cuvette: it is a hypothetical tile in which photonic engineering, inverse design, Purcell enhancement and nanostructuring have been used to lift each of the efficiencies that Section 8.3.3 shows to be limiting in bulk. The purpose of the bulk analysis is therefore not to bound the device but to name its *requirements*: every factor in

$$p_{\text{cool}}^{\text{max}} = n_t x_{\text{max}}(\omega_p, \bar{\omega}_f; T) \Gamma_{\text{tot}} \hbar \omega_p \eta_{\text{cool}}, \quad \frac{P_{\text{cool}}}{A} = p_{\text{cool}}^{\text{max}} d, \quad (8.11)$$

is a lever that some level of the optimization loop of Chapter 9 owns. Reading (8.11) from left to right:

1. n_t — *chemistry (Level 0/1)*. SMILES isolation lifts the concentration wall from 10^{-3} to $\gtrsim 10^{-1} \text{ M}$ at preserved IQE (Section 8.3.2): $\times 100$.
2. T — *architecture (Level 3 / Chapter 10)*. The tail absorption and x_{max} both scale as $e^{-\Delta E/k_B T}$; a 400 K compute tile gives $\times 5$ –10, a 600 K wide-bandgap tile another $\times 3$ –5 and, more importantly, doubles the anti-Stokes energy obtainable per Boltzmann factor.
3. x_{max} at fixed η_q — *emission-side photonics (Level 2.A)*. x_{max} depends on how far the pump sits below the *zero line*, whereas η_q depends on how far it sits below the *mean emission*. In bulk the two differ by the Stokes shift ($\sim 100 \text{ meV}$ for R101, half the Boltzmann budget wasted). A fluorescence cavity that Purcell-selects the blue edge of the emission band — the mechanism by which photonic engineering blue-shifts and narrows the ASF stream in Chapter 6 — moves λ_f from 618 nm toward the zero line (588 nm), so that at 651 nm η_q rises from 5% to 10% at unchanged x_{max} , or equivalently the same η_q is had at 625 nm with $\times 10$ –40 the x_{max} . Because Purcell enhancement multiplies σ_a and σ_e equally *at the same frequency*, it cannot change the McCumber ratio at ω_p ; it changes where the emission goes, which is the lever that matters.
4. Γ_{tot} — *broadband Purcell (Level 2.A)*. The cap is on the excited-state *population*; the cycling rate through it is $\Gamma_{\text{tot}} = F_P/\tau$. A band-averaged Purcell factor $\bar{F}_P = 30$ –100 — plasmonic or hyperbolic-metamaterial LDOS, or dense nanocavity arrays, over the $\sim 60 \text{ nm}$ emission band — multiplies the ceiling by the same factor, at the cost of raising I_{sat} and hence the pump flux the resonator must sustain. This is the single largest rung, and the one the joint absorption–emission inverse design of Section 9.7.2 is now being extended to. The same factor can be bought from the materials side — n^2 of the host through (8.10), and N_c -fold superradiance in a J-aggregate (Section 8.3.6) — and the two multiply.
5. η_{cool} — *extraction (Level 2.A/2.B)*. $\eta_{\text{EQE}} \geq 0.92$ –0.99 through Purcell-enhanced single-mode extraction, and $\eta_{\text{abs}} \rightarrow 1$ through low- α_b hosts; without these the other rungs multiply zero.

Rung (cumulative)	η_{cool}	x_{max}	$p_{\text{cool}}^{\text{max}}$ (W/mm ³)	P_{cool}/A (W/mm ²)	pump flux (W/m ²)
0. Bulk: 10^{-2} M, 300 K, 651 nm, $\bar{\lambda}_f = 618$ nm, 5 μm	3.9%	3.7×10^{-4}	6.3	0.03	0.8
1. + SMILES, 10^{-1} M	3.2%	3.7×10^{-4}	51	0.26	8
2. + 400 K tile	3.3%	2.7×10^{-3}	380	1.9	57
3. + blue-edge emission, $\bar{\lambda}_f \rightarrow 590$ nm	10.3%	2.7×10^{-3}	1.2×10^3	5.9	57
4. + broadband Purcell $\bar{F}_P = 30$	10.3%	2.7×10^{-3}	3.5×10^4	180	1.7×10^3
5. + 20 μm , resonator-absorbed	10.3%	2.7×10^{-3}	3.5×10^4	7×10^2	6.9×10^3
6. + $\bar{F}_P = 100$, 50 μm	10.3%	2.7×10^{-3}	1.2×10^5	5.9×10^3	5.7×10^4
7. 600 K tile, pump 700 nm	18.5%	1.5×10^{-3}	1.1×10^5	5.4×10^3	2.9×10^4

Table 8.2: The photonic ladder for an R640-class molecular extractor, evaluated with (8.11) and the tail model of Section 9.3.2 ($\eta_{\text{EQE}} = 1$; every row scales with the realised η_{EQE} -dependent η_{cool}). Rungs 0–2 are the experimental programme of Sections 8.3.1 and 8.3.2; rungs 3–6 are the photonic-engineering requirements that the inverse-design loop of Chapter 9 must deliver to reach the book’s Tier-I design point ($\gtrsim 10^3$ W/mm² at $\eta_{\text{cool}} \geq 10\%$). The last column is the pump flux the photonic network must deliver and the LPC must recover, $P_{\text{cool}}/(A\eta_{\text{cool}})$: at the design point it is of order 1 MW/cm², which is why the design point is a *pulsed, targeted* tile (Section 10.8) and why $\eta_P\eta_{\text{LPC}}$ enters every rung.

6. *d* — *pump resonator designed at the operating excitation (Level 2.A)*. 20–50 μm of extractor, fully absorbing through a $Q \sim 10^3$ – 10^4 mode whose critical coupling is set at the bleached κ_{eff} , not the small-signal one (8.9).

Table 8.2 evaluates the rungs cumulatively. Rungs 0–2 are where the experimental programme stands; rungs 3–6 are the requirements flow-down for the design point, and they are the objective functions of the joint absorption–emission inverse design that Chapter 9 describes. Two further rungs take the device to the higher-efficiency tiers:

Tier II, $\eta_{\text{cool}} \geq 30\%$: cascaded anti-Stokes stages. A single thermalized transition cannot deliver an anti-Stokes shift of 0.3–0.6 eV (12 – $24 k_B T$ at 300 K) at useful rate, because x_{max} carries the full Boltzmann factor. N stages each lifting a *small* shift do not: if the fluorescence of stage i is routed by the photonic network into the red tail of stage $i + 1$ (a bluer emitter), each stage works near $x_{\text{max}} \sim 10^{-2}$ and the shifts compound, $\omega_N/\omega_0 = \prod_i (1 + \eta_{q,i})$, with net efficiency $\eta_{\text{cool}} = \prod_i [(1 + \eta_{q,i})\eta_{\text{EQE},i}] - 1$. Ten stages of 5% at $\eta_{\text{EQE}} = 0.99$ give 47%; six of 10% at 0.99 give 67%; ten of 5% at 0.97 give 20% — the cascade converts the efficiency problem into a per-stage EQE problem, which is precisely the problem Purcell-enhanced single-mode extraction solves. The extractor becomes a multi-material, graded-zero-line stack (a multi-junction LPC in reverse), and “all materials $\times$ all pump, emission and LPC wavelengths” is the search space the loop must cover.

Tier III, $\eta_{\text{cool}} \geq 60\%$: the self-powered threshold. With $\eta_P\eta_{\text{cpl}} \simeq 0.75$ (1.17) requires $\eta_{\text{ASF}} \geq 0.33/\phi$: 0.67 on a 600 K tile, 2.4 (impossible) on a 350 K one. The 60% tier is therefore not an arbitrary milestone; it is the efficiency at which a hot compute tile crosses into net electrical generation with an ideal LPC, and Tier III is reached only by the cascade on a wide-bandgap, 600 K zone (Section 10.9).

An eighth rung, still open: asymmetric pump-side engineering. Every rung above respects the McCumber ratio $\sigma_e(\omega_p)/\sigma_a(\omega_p)$ of the emitter: the photonic environment is used to move emission, not to change the material’s detailed balance at the pump frequency. That is a consequence of Lorentz reciprocity and time-invariance of the environment, not a law of nature, and it is the assumption behind the x_{max} cap and its $e^{-\Delta E/k_B T}$ temperature dependence. Two experimentally established classes of structure break exactly these symmetries. *Magneto-optical nonreciprocity* makes a structure’s spectral-directional absorptivity and emissivity unequal —

Kirchhoff's law violated directly — first in resonantly coupled InAs at 1 T [53, 54], then broadband in gradient-ENZ InAs [55], and most recently with an emissivity–absorptivity contrast of 0.43 in graded InGaAs at 5 T [56] (theory: [57]; review: [58]). *Time and space–time modulation* achieves the same without a magnet: a modulated antenna absorbs at one frequency and emits at another [59], indirect photonic transitions give complete isolation and electrically driven nonreciprocity on a silicon chip [60–62], space–time gratings violate Kirchhoff's law [63], a GHz-modulated graphene metasurface shows nonreciprocal frequency conversion in the mid-infrared [64], and ENZ films translate optical frequencies by tens of nanometres in a single sub-cycle event [65, 66]. Most directly relevant, time-modulated thermal emission has been shown to constitute a refrigeration cycle in its own right, with modulation work replacing the luminescence efficiency as the resource, approaching the Carnot limit in principle [67, 68].

What these results establish is that the two ingredients exist: structures whose absorption and emission channels are decoupled, and structures that change a photon's frequency after it has entered the environment. What they do *not* yet establish is the specific combination the ladder needs — an environment in which the emitter absorbs from the pump channel at ω_p while its stimulated emission at ω_p is removed from that channel (frequency-translated, or routed out non-reciprocally) before it can return energy to the pump — so that the excited-state cap is set by the modulation rather than by $e^{-(\hbar\omega_{00}-\hbar\omega_p)/k_B T}$. If it can be built, it decouples the volumetric ceiling from the die temperature and collapses the exponential cost of a large anti-Stokes shift, i.e. it attacks Tiers II–III directly rather than through the cascade. It is an active research direction at Maxwell and is carried in the loop of Chapter 9 as a Level-2.A search dimension (modulation frequency, depth and phase; magneto-optical layer) whose fitness is scored, like every other rung, by (8.11).

The rest of this chapter and Chapters 9 and 10 are written against these tiers: Tier I is the hot-spot razor ($\gtrsim 10^3$ W/mm², $\eta \geq 10\%$, loop cost ~ 5 –14 W/W with recovery), Tier II is the break-even loop ($\text{COP}_{\text{net}} \rightarrow 1$ at $\eta_P \eta_{\text{LPC}} \eta_{\text{cpl}} \simeq 0.75$), and Tier III is the self-powered chip.

8.3.5 Anti-Stokes efficiency vs. pump red-shift

The per-photon quantum-defect efficiency — the ceiling on η_{ASF} when every absorbed photon is re-emitted at the mean fluorescence frequency — is

$$\eta_q = \frac{\bar{\omega}_f - \omega_p}{\omega_p} = \frac{\lambda_p - \bar{\lambda}_f}{\bar{\lambda}_f}, \quad (8.12)$$

with cooling requiring $\lambda_p > \bar{\lambda}_f$ (we write η_q rather than the η_c of the older literature to avoid the clash with the wall-plug efficiency η_c of Table 1.2). Pushing the pump into the red tail widens the shift per absorbed photon, but two penalties grow with it: $\sigma_a(\lambda_p)$ falls exponentially, so the dye's share of the absorbed pump $\eta_{\text{abs}} = \alpha_r/(\alpha_r + \alpha_b)$ falls against the fixed solvent / host background, and the transparency ceiling (8.8) collapses. The realised efficiency is $\eta_{\text{cool}} = \eta_{\text{abs}} \eta_{\text{EQE}}(1 + \eta_q) - 1$ of (8.6), and it has an interior optimum in λ_p .

Table 8.3 tabulates η_q for R101/R640 ($\bar{\lambda}_f = 618$ nm, the intrinsic short-path mean emission of Section 8.3.1) and R6G ($\bar{\lambda}_f = 580$ nm) as the pump sweeps into the red tail, together with the energy removed per photon. Note the energy scale: at $\lambda_p = 680$ nm the shift is 183 meV, i.e. $\sim 7 k_B T_h$ at room temperature — an order of magnitude larger than the one- $k_B T$ shift of Yb:YLF (24 meV). That is precisely why the Boltzmann-weighted tail absorption at 680 nm is $\sim 10^{-4}$ of the peak and why the useful operating point lies much closer to the emission line than the bare η_q column suggests.

λ_p (nm)	η_q per absorbed photon (%)		$\hbar\omega_f - \hbar\omega_p$ (meV)	
	R101/R640 ($\bar{\lambda}_f = 618$ nm)	R6G ($\bar{\lambda}_f = 580$ nm)	R101/R640	R6G
600	—	3.4	—	71
620	0.3	6.9	6	138
640	3.6	10.3	69	200
651	5.3	12.2	102	233
660	6.8	13.8	128	259
680	10.0	17.2	183	314
700	13.3	20.7	235	367
720	16.5	—	284	—

Table 8.3: Per-absorbed-photon quantum-defect efficiency (8.12) and energy removed per photon as a function of pump wavelength for R101/R640 ($\bar{\lambda}_f = 618$ nm, intrinsic short-path mean emission) and R6G ($\bar{\lambda}_f = 580$ nm). Pumping requires $\lambda_p > \bar{\lambda}_f$; “—” marks Stokes / no-cooling operation. These are *ceilings*: the realised η_{cool} is reduced by η_{abs} and η_{EQE} and, with the measured absorption tail, peaks near 651 nm (+5.3%, 10^{-2} M, 300 K) rather than growing without bound into the red. In a 1 cm cuvette the mean fluorescence red-shifts to $\bar{\lambda}_f \sim 660$ nm at 10^{-2} M, collapsing η_q at fixed λ_p ; a thin film with a sub-extinction-length escape path retains most of the intrinsic column, which is the appropriate one for design once film emission spectra confirm it.

Where the optimum actually sits. Applying (8.6) with the measured R101 extinction spectrum, its anchored exponential red tail (Section 9.3.2), and the Kedenburg ethanol background gives, for $\eta_{EQE} = 1$ and $\bar{\lambda}_f = 610$ nm:

- at 10^{-2} M, 300 K: peak $\eta_{cool} = +5.3\%$ at $\lambda_p = 651$ nm, with $\eta_{abs} \simeq 0.98$; net cooling requires $\eta_{EQE} \geq 0.950$ at that wavelength and is possible only for $\lambda_p \simeq 610$ –669 nm;
- at 10^{-1} M, 300 K: peak +8.7% at 672 nm, $\eta_{EQE} \geq 0.92$;
- raising the tail temperature to 400 K (tail steepness $\sigma = 1$) moves the optima to 674 nm (+8.5%) and 702 nm (+13.0%) respectively, with the required EQE falling to 0.885 in the latter case.

The tail model is anchored to a measured value and enters only as a multiplicative factor, so an order-of-magnitude error in the assumed tail absorption moves the peak η_{cool} by $\pm 30\%$ relative and shifts the optimum by ~ 20 nm — real, but far smaller than the input uncertainty. Direct measurement of $\varepsilon(\lambda)$ beyond 620 nm on high-purity R640 is the single most valuable experiment for pinning these numbers.

Two operating points must be kept distinct. The *near-term experimental point* that follows from Table 8.3, Table 8.1 and the tail model (rungs 0–2 of Table 8.2) is: pump at $\lambda_p \simeq 650$ –680 nm (the longer end for the hotter tile) into a 5–20 μm SMILES-R640-in-polymer film at $N_t \simeq 10^{-1}$ M, complete pump absorption through a critically coupled resonator of $Q \sim 10^3$ – 10^4 designed at the operating excitation (Section 9.7.2), single-mode fluorescence collection at $\bar{\lambda}_f \simeq 620$ –630 nm, and a GaP-class LPC matched to that line. The expected performance is $\eta_{cool} \simeq 3$ –8% ($\eta_{EQE} \geq 0.92$ –0.95 required), $P_{cool}/A \simeq 0.3$ –2 W/mm² per 5 μm depending on die temperature, and an incident pump flux of ~ 10 –30 W/mm² per W/mm² of heat lifted. The *design point* of the book is rungs 3–7 and the Tier II/III cascade: $\gtrsim 10^3$ W/mm² at $\eta_{cool} \geq 10\%$, rising to 30–60% with cascaded stages on 600 K tiles. The near-term point is a hot-spot razor whose loop is a pump recycler ($\eta_{ASF} \lesssim 0.08$ puts self-powering out of reach at any temperature a polymer survives); the design point is where Chapter 10’s net-generation regime lives.

8.3.6 Aggregate and solid-state organic emitters: superradiant J-aggregates and aggregation-induced emission

The ladder of Section 8.3.4 treats n_t and Γ_{tot} as levers to be pulled by isolation chemistry and by photonic LDOS respectively, because for monomeric dyes aggregation is a loss: H- and J-aggregates quench, and the concentration wall of Figure 8.4 is where they appear. Two classes of organic emitter invert that rule and pull the same two levers from the materials side.

Superradiant J-aggregates. In a J-aggregate the head-to-tail packing couples the transition dipoles constructively: the lowest exciton carries the oscillator strength of N_c coherently coupled molecules, so the radiative rate is $\Gamma_r \simeq N_c \Gamma_r^{\text{mono}}$ — a Purcell-like enhancement with no cavity — and the J-band is red-shifted and motionally narrowed. Lee and Kim [69] have shown that this superradiant emission survives hot-band (anti-Stokes) excitation, giving bright anti-Stokes fluorescence with a ~ 6 nm linewidth and naming optical cooling of organic solids as the application; the historical objection — low, irreproducible quantum yield from disorder — is answered by silica-encapsulated TDBC aggregates with $\eta_{\text{IQE}} \simeq 0.98$ and $\tau = 234$ ps [70]. In the language of (8.11) that lifetime is $\Gamma_{\text{tot}} \simeq 4 \times 10^9 \text{ s}^{-1}$, a factor ~ 20 above R640 and equal to rung 4's $\bar{F}_P = 30$ before any photonic structure is added; the narrow line reduces the self-absorption that sets the escape wall; and the solid, disorder-suppressed matrix is directly compatible with the thin-film tile. Three cautions. Narrow emission means a narrow hot band: x_{max} at a given η_q still obeys the McCumber relation (5.7), so the pump must sit in a thermally populated tail that is now only a few $k_B T$ wide, and the achievable η_q per photon is correspondingly small — the J-aggregate is a natural *cascade stage* (Tier II) rather than a single-stage extractor. Exciton–exciton annihilation replaces concentration quenching as the density-dependent loss, and sets its own ceiling on x . And N_c is temperature-dependent (dephasing shrinks the coherence length), so the 400–600 K rungs must be re-evaluated per material. NIR/SWIR J-aggregates built from heptamethine cyanines [71] extend the class toward the 980 nm pump line where laser efficiency η_P is highest. On the assumptions of Table 1.1 (10^{20} cm^{-3} , $F_P = 10$ on top of the superradiant rate, 400 K, pump 60 meV into the hot band of a 590 nm J-band) the McCumber cap alone allows $x_{\text{max}} = 0.15$ and $6 \times 10^6 \text{ W/mm}^3$; an annihilation-limited $x \simeq 0.02$ ($\gamma_{\text{EEA}} \sim 10^{-8} \text{ cm}^3/\text{s}$) still gives $\sim 10^6 \text{ W/mm}^3$, i.e. 10^3 W/mm^2 from a 1 μm film — the design-point density from a film thinner than any other family needs, at $\eta_q \simeq 3\%$. That is the semiconductor's profile, and the J-aggregate's role is the semiconductor's: the dense, bluest, terminal stage of a cascade whose efficiency is carried by the broader-band stages below it.

Aggregation-induced emission (AIE). The complementary route keeps the monomer's spectrum but removes its non-radiative channel: AIE luminogens are dark in dilute solution because peripheral rotors provide a fast non-radiative path, and become near-unity emitters when aggregation restricts that motion [72]. The enhancement is intramolecular, so it does not require the specific packing of a J-aggregate, and it is strongest in exactly the rigid polymer or sol-gel host the tile wants. Hot-band anti-Stokes fluorescence from AIE dots under 980 nm excitation has been demonstrated, with the anti-Stokes signal *increasing* with concentration [73] — the inverse of the rhodamine wall. For the ladder, AIE is an alternative to SMILES for rung 1 (dense n_t at preserved IQE) whose design rules for Stokes shift, linewidth and yield are already extensive; whether an AIE emitter can be designed for the mirror-image, small-Stokes-shift band the cooling application prefers is the open question, and the one the materials-exploration level of Section 9.3.2 should ask first, since (8.10) ties the radiative rate of any such design to its absorption band.

Platform	$\sigma_a(\omega_p)$ (cm^2)	τ	η_{EQE} (measured)	n_t	$p_{\text{cool}}^{\text{max}}$ (W/mm^3)	P_{cool}/A at reference d (W/mm^2)
Yb:YLF crystal (modular)	8×10^{-21}	1.8 ms	> 0.97	$5 \times 10^{19} \text{ cm}^{-3}$ (5%)	~ 0.03	3×10^{-4} (10 μm); 0.03 (1 mm)
Cr:LiSAF crystal (modular)	5×10^{-20}	67 μs	~ 0.9 (> 0.97 needed)	$1 \times 10^{21} \text{ cm}^{-3}$	~ 20	0.2 (10 μm)
GaAs/GaInP DH (epitaxial, monolithic)	$\alpha \simeq 100 \text{ cm}^{-1}$ (Urbach)	ns	0.96–0.99 (LT)	$N^* = 6 \times 10^{17} - 2 \times 10^{18} \text{ cm}^{-3}$	80 ($\eta_e = 0.25$) to 4×10^3 ($\eta_e = 0.9$)	0.8–40 (10 μm)
Ge nanocrystal (CMOS-compatible)	structured	$\gtrsim \text{ns}$	moderate	10^{-3} M-class	n/a	n/a
R101/R640 in EtOH (mm cuvette, legacy)	3×10^{-20} (651 nm)	4.3 ns	0.75 at 10^{-3} M	$\lesssim 10^{-3}$ M	$\lesssim 2$	EQE-limited: none above $\sim 10^{-4}$ M
SMILES-R640 polymer film (5–20 μm)	$3 \times 10^{-20} - 1.5 \times 10^{-19}$	ns	0.81 (soln., 10^{-2} M); ≥ 0.9 projected	$10^{-2} - 10^{-1}$ M	6–18 (300 K, 10^{-2} M); 60–400 (400 K, 10^{-1} M)	0.03–2 (5 μm)

Table 8.4: Platform comparison for the extractor layer, with cooling density stated first as the volumetric transparency / Auger ceiling $p_{\text{cool}}^{\text{max}}$ of Section 8.3.3 ($\eta_{\text{EQE}} = 1$) and then as an areal density at a reference thickness. All areal entries scale with the realised η_{cool} , which is positive only when $\eta_{\text{EQE}} > (1 + \eta_q)^{-1}$; the R101-in-EtOH row fails this test above $\sim 10^{-4}$ M, and Cr:LiSAF fails it at its demonstrated IQE. The direct-bandgap semiconductor remains the highest- p_{cool} platform but at $\eta_{\text{ASF}} \simeq 0.3\text{--}1\%$; the SMILES-R640 film reaches the W/mm^2 -per-5 μm class only at $\gtrsim 400$ K and $\gtrsim 10^{-1}$ M, where it does so at $\eta_{\text{ASF}} \simeq 3\text{--}8\%$ without epitaxy. Yb:YLF is a thick-crystal platform: its ceiling per 10 μm is three orders below the others because of its millisecond lifetime.

8.3.7 Nanoparticle and quantum-dot suspensions

Colloidal nanocrystal suspensions (CdSe/CdS, InP, perovskites) and hyperbolic-metamaterial-coupled dye/NC composites [74] provide tunable bandgaps and high quantum yields. Integration issues parallel those of the dyes, with the added complication of agglomeration and ligand chemistry for long-term stability.

8.3.8 Polyatomic molecular cooling

A frontier platform, direct laser cooling of polyatomic molecules [75], is mentioned for completeness; it is currently best suited to trapped or cryogenic gas-phase experiments rather than on-chip cold plates.

8.4 Platform comparison and design-selection rules

The choice between platforms follows the closed-loop master equation (1.11):

- If the mission is **maximum COP at moderate cooling power** (laboratory refrigeration, cryogenic modules), choose Yb:YLF or Yb:silica. The photonic network pushes η_{ASF} via Purcell enhancement; LPC is single-junction at $\sim 1 \mu\text{m}$.
- If the mission is **maximum cooling power density in a mature, epitaxy-based platform** (HPC hot-spot cold plate via direct integration with III-V processes), direct-bandgap semiconductors (GaAs, InGaAs) offer the highest volumetric ceiling ($10^2\text{--}10^3 \text{ W}/\text{mm}^3$) and share the epitaxy of a co-designed multi-junction LPC. The costs are manufacturability (MBE / MOCVD lattice-matched growth, a substrate constraint on form factor), a per-temperature pump wavelength that must track the Varshni gap shift (Section 9.3.2), and above all a quantum-defect efficiency of only $\sim 3\%$: the pump power is $30\text{--}300\times$ the heat lifted, so this platform lives or dies on $\eta_P \eta_{\text{LPC}}$ (Chapter 7).

- If the mission is **hot-tile cooling in a solution-processable, substrate-agnostic platform** (chip-scale ASF cooling with polymer thin films on $\gtrsim 400$ K compute zones), choose **SMILES-R640-in-polymer** (Section 8.3.2). At 10^{-1} M and 400 K a 5 μm film reaches ~ 2 W/mm² at $\eta_{\text{cool}} \simeq 3\text{--}8\%$ — a 5–10 $\times$ better pump economy than GaAs at a 10–100 $\times$ lower volumetric ceiling — while deploying on any planar substrate and allowing zone-by-zone extractor selection under the heterogeneous-thermal template of Section 10.9. The gating experiment is $\eta_{\text{EQE}} \geq 0.92\text{--}0.95$ in the film at $\geq 10^{-2}$ M; the acid-free R640 chemistry (81% IQE in solution) makes this plausible but not yet demonstrated.
- If the mission is **rapid field prototyping / swappable extractor**, solvent-phase dye dispersions remain useful: the fluidic extractor can be iterated without rebuilding the photonic stack. This is where the legacy R101/EtOH form factor (Figure 8.3) still fits, subject to the $N_{\text{t}} \lesssim 10^{-3}$ M wall of Figure 8.4.

Operational-temperature range as a first-class design input

The comparison above ranks platforms by η_{EQE} , cross-section, and lifetime at nominal room-temperature operation. The heterogeneous-thermal architecture of Section 10.9 adds a second, equally important axis: the extractor material’s *operational-temperature range* $[T_{\text{min}}, T_{\text{max}}]$, inside which η_{EQE} stays above some architectural threshold (typically ≥ 0.9 of the peak). Zone-by-zone the target windows are:

- **Cold storage / control tiles** ($T \sim 150\text{--}300$ K). Yb:YLF and Yb:silica are ideal: η_{EQE} is unchanged or slightly improved on cooling, and the sub-ambient regime collapses subthreshold leakage in the underlying silicon. This is the family for which the theory chapters quote their reference numbers; it is a low-power-density, high-COP platform (Table 8.4), which is exactly what a leakage-suppression tile needs.
- **Warm interconnect / uncore tiles** ($T \sim 300\text{--}400$ K). Yb:YLF still works but concentration quenching begins to erode the useful window near the upper end. Yb-doped ZBLAN or fluoride glasses are the swap-in for the upper-400 K corner.
- **Hot compute tiles** ($T \sim 400\text{--}600$ K). Yb-based hosts are past their quenching temperature. Ho³⁺- and Tm³⁺-doped fluoride hosts hold η_{EQE} further into this range because their upper-manifold phonon sidebands are less thermally populated at these temperatures [6]. Cr³⁺-doped colquiriites are the allowed-transition alternative when higher cooling-power density is more important than absolute η_{EQE} . At the lower end of this window (400–450 K, the limit of common optical polymers) the SMILES-R640 film is the natural candidate: its transparency ceiling and tail absorption both grow exponentially with T (Section 8.3.3), so a hot compute tile is where the molecular-dispersion platform is *strongest*, not weakest.
- **Exotic hot-corner tiles** ($T > 600$ K). Wide-bandgap-hosted rare-earth complexes (SiC:Er, GaN:Yb) are the active research frontier here. These materials are the natural pairing for the SiC / GaN compute devices that dominate the reliability envelope in this regime.

The consequence for the extractor design is that a chip with heterogeneous thermal zones will *not* use a single material across its full area. Each zone’s extractor is selected from a different row of the platform table, matched to that zone’s target T_{j} and to the underlying device technology’s operating envelope. This is the material-side counterpart of the architectural template of Section 10.9: photonic microrefrigeration enables per-tile temperature targeting; the material platform must then enable per-tile extractor selection.

8.5 Chapter summary

Each material platform partitions $\{\sigma_a, \sigma_e, \tau_r, \gamma_{nr}, \alpha_b\}$ differently, driving different photonic-network, coupler, back-reflector, and sensor designs. The inverse design of the extractor is therefore inseparable from the surrounding stack — as anticipated in Section 5.16 and visible throughout the survey above. The volumetric route of Section 8.3.3 is the common currency: every platform is characterized by a ceiling p_{cool}^{max} in W/mm^3 — set by pump transparency for dyes and ions, by Auger roll-over for semiconductors — and by the quantum-defect efficiency that fixes how much pump the loop must recycle per watt lifted. Concrete numerical targets for each platform, the pump-resonator designs that make red-tail absorption complete, and the companion photonic-network figures of merit are the subject of Chapter 9.

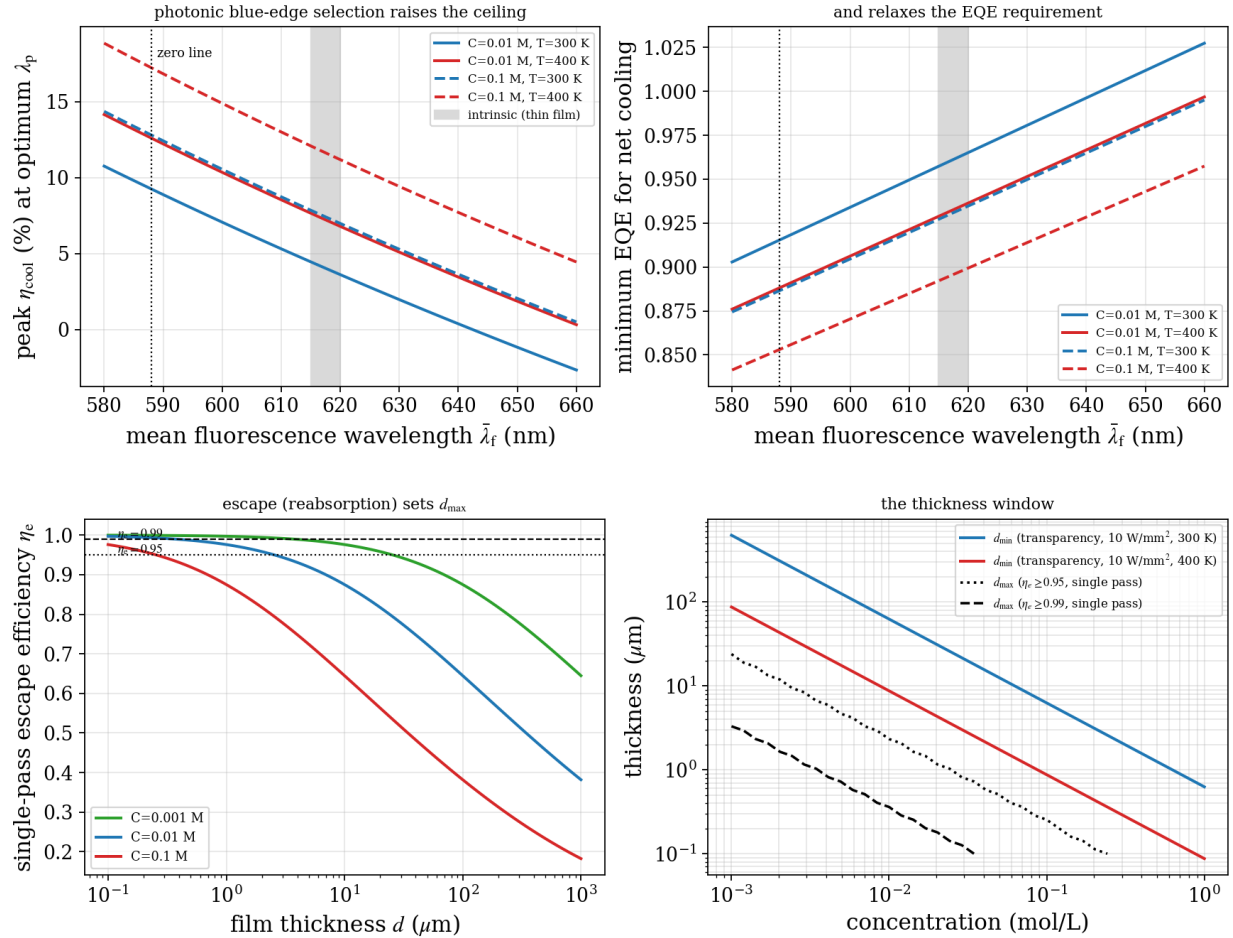


Figure 8.5: Top: response of the R101/R640 extractor to a photonic shift of the mean fluorescence wavelength (Mo Chen’s tail model, $\eta_{\text{EQE}} = 1$, optimum λ_p at each $\bar{\lambda}_f$): peak η_{cool} (left) and the minimum EQE for net cooling (right). The grey band is the intrinsic thin-film mean; the dotted line the zero line. Bottom: the film-thickness window. Left, single-pass escape efficiency of an isotropically emitting slab vs. thickness for three concentrations (no index trapping); right, the resulting d_{max} for $\eta_e \geq 0.95$ and 0.99 against the transparency floor d_{min} (8.9) at 10 W/mm² incident. The single-pass window is closed at every concentration: the design-point film operates in the photon-recycling regime and its EQE is set by IQE and by Purcell-directed extraction, not by thickness.

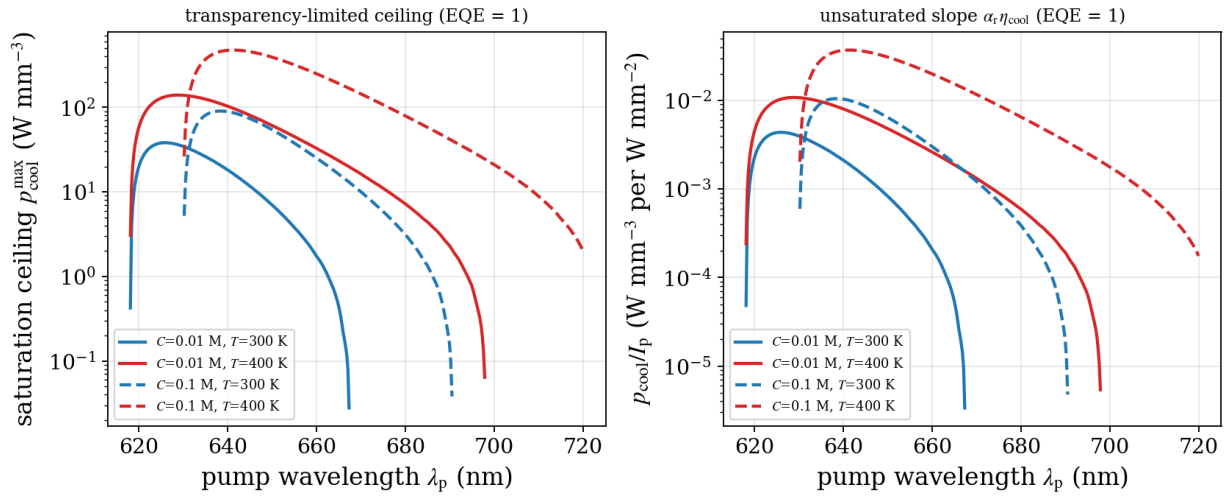


Figure 8.6: Volumetric cooling-power density of an R101/R640 extractor from (8.7)–(8.8) ($\eta_{\text{EQE}} = 1$; $\bar{\lambda}_f = 618 \text{ nm}$ at 10^{-2} M , 630 nm at 10^{-1} M ; Kedenburg ethanol background). Left: the pump-transparency ceiling $p_{\text{cool}}^{\text{max}}$; right: the unsaturated slope $\alpha_r \eta_{\text{cool}}$, i.e. the cooling density per unit local pump intensity. Curves end where η_{cool} changes sign. Both panels improve by 5–10 $\times$ from 300 K to 400 K and by 10 $\times$ per decade of concentration.

Part V

Integration: Inverse Design and the Full-Stack Loop

Chapter 9

Inverse Design and the Full-Stack Optimization Loop

The preceding chapters built a self-consistent theory of the thin-film anti-Stokes fluorescence (ASF) photonic cold plate. Chapter 1 fixed the thermodynamic ledger that the device must satisfy. Chapters 4 and 5 translated that ledger into a coupled cavity / electro-optic model whose figures of merit F_P , Q , V and the resonance overlap of Section 6.8 carry every electromagnetic constraint. Chapters 6 and 7 closed the loop back to the ambient by defining η_{ASF} and η_{LPC} under their exergy and bandwidth limits. Chapter 8 then surveyed the three material-platform classes — doped-host, semiconductor, molecular-dispersion — whose constitutive parameters $\{\sigma_a, \sigma_e, \gamma_r, \gamma_{nr}, \alpha_b, N\}$ plug into the model.

This chapter takes the next, integrative step: it asks how a complete, manufacturable cold-plate design is actually *found*, given the high-dimensional search space the previous chapters expose. The target is the design point of Section 8.3.4 — a tile in which every factor of the bulk cooling-density ceiling (8.11) has been lifted by photonic engineering to reach $\gtrsim 10^3 \text{ W/mm}^2$ at $\eta_{cool} \geq 10\%$ (Tier I), 30% (Tier II) and 60% (Tier III) — and the chapter should be read as the procedure by which the ladder’s rungs are found, not as a description of what has been built. The answer, which we call the *full-stack optimization loop*, is a hierarchical adjoint-optimization procedure that composes a per-material extractor optimization, a joint photonic-network / coupler / back-reflector / sensor inverse design, a coupled laser-power-converter design step, and a population-level evolutionary search across material platforms and manufacturability constraints. The loop is what binds the theory of Chapters 1 to 8 into a single design plan that can be released to fabrication with high-performance-computing (HPC) hot-spot cooling as the target application.

The chapter has three parts. Section 9.1 defines the HPC-cooling design target and the figures of merit a successful design must hit. Section 9.2 sets out the four-level optimization hierarchy and the role each level plays. Sections 9.3 through 9.5 are the design-step companions: the materials-exploration framework that provides Level-0 candidates; the extractor / coupler / back-reflector / sensor inverse designs of Level 1 and Level 2.A; the LPC inverse design of Level 2.B; and the outer evolutionary loop of Level 3. Section 9.7 illustrates the procedure with representative back-reflector and molecular-dispersion-platform results, and Section 9.8 closes the chapter with the integration view; Chapter 10 then carries the resulting tile capabilities into the chip-architecture question.

9.1 The HPC-cooling design target

The design target for this volume is hot-spot cooling in high-performance computing. Three numerical anchors set the target. First, the heat flux at a modern AI-accelerator hot spot exceeds 1 kW/cm^2 locally, projected to grow [76, 77]. Second, the thermal budget is dominated not by the cold-plate fluid loop but by the near-junction conduction path; a die-side cold plate that intercepts heat within the top $100 \mu\text{m}$ of silicon eliminates much of the parasitic resistance of conventional liquid-cooled solutions. Third, the metric of value is not COP alone but $P_{\text{cool}} / P_{\text{net}}$ at the constrained *form factor* of the chip — areal cooling power density P_{cool}/A in W/mm^2 at fixed total stack height, with manufacturability and reliability priced in.

The thermodynamic ceiling on this target was set by Equation (1.11) and the Carnot bound of Equation (1.13); the electromagnetic ceilings by the two absorption-limited Q 's of Section 4.4 and Chapter 8 (low- Q fluorescence mode, high- Q pump mode); the material ceiling by the transparency / Auger limit of Section 8.3.3; and the LPC-recycling ceiling by the bandwidth / entropy bound of Section 7.3. The concrete requirements are the rungs of Table 8.2: complete red-tail pump absorption in a $20\text{--}50 \mu\text{m}$ extractor through a resonator designed at the operating excitation; blue-edge emission selection that moves the mean emission to the zero line; a band-averaged Purcell factor of $30\text{--}100$ on the total decay rate; $\eta_{\text{EQE}} \geq 0.99$ per stage; and, for Tiers II–III, a cascade of N stages whose emission lines are chained through the photonic network into successive extractors and finally into a matched multi-junction LPC. None of these is reached by a hand design. The full-stack optimization loop is the framework that finds the manufacturable design that approaches them, and its search space is deliberately the largest one the physics allows: all candidate materials, all pump, emission and LPC-recovered wavelengths, and the number and ordering of stages — a space that is tractable only with extreme-scale computing, which the loop assumes.

9.2 The hierarchical optimization loop

The design is constructed by a four-level loop. Each level operates on a quantity defined in the chapters that precede this one and produces the decision variables consumed by the next level.

Level 0 — material-platform class and stage count. One of {doped-host, semiconductor, molecular-dispersion} is selected for each of $N \geq 1$ cascaded extractor stages (Section 8.3.4); the choice fixes the constitutive-model family of Section 5.4 used in Level 1, the manufacturability model used in Level 3, and the integration mode for the photonic network and (where applicable) the fluid loop. For $N > 1$ the stages are ordered by zero-line energy so that each stage's emission falls in the red tail of the next.

Level 1 — per-material extractor optimization. For each candidate material drawn from the platform class, the framework optimizes (i) the extractor's structuring and thickness; (ii) the pump wavelength ω_p^* at the target operating temperature; and (iii) the conditional coupler material assumed for the in-loop fluorescence-response model. The output is a baseline cooling-power density $P_{\text{cool}}^{(\text{L1})}$ per candidate.

Level 2.A — joint coupler / back-reflector / sensor inverse design. With the Level-1 extractor and ω_p^* held fixed, the surrounding stack — structured coupler, back-reflector, and sensor layer — is jointly inverse-designed at ω_p^* and at the mean fluorescence wavelength ω_f^* implied by the L1 extractor. The output is a full-stack cooling-power density $P_{\text{cool}}^{(\text{L2.A})}$; enhancement over $P_{\text{cool}}^{(\text{L1})}$ is set by the inner-loop iteration count and by the random restarts permitted in Level 3.

Level 2.B — coupled LPC inverse design. With the L2.A stack held fixed, the LPC stack is inverse-designed at ω_f^* subject to the bandwidth / entropy limit of Section 7.3. Junction

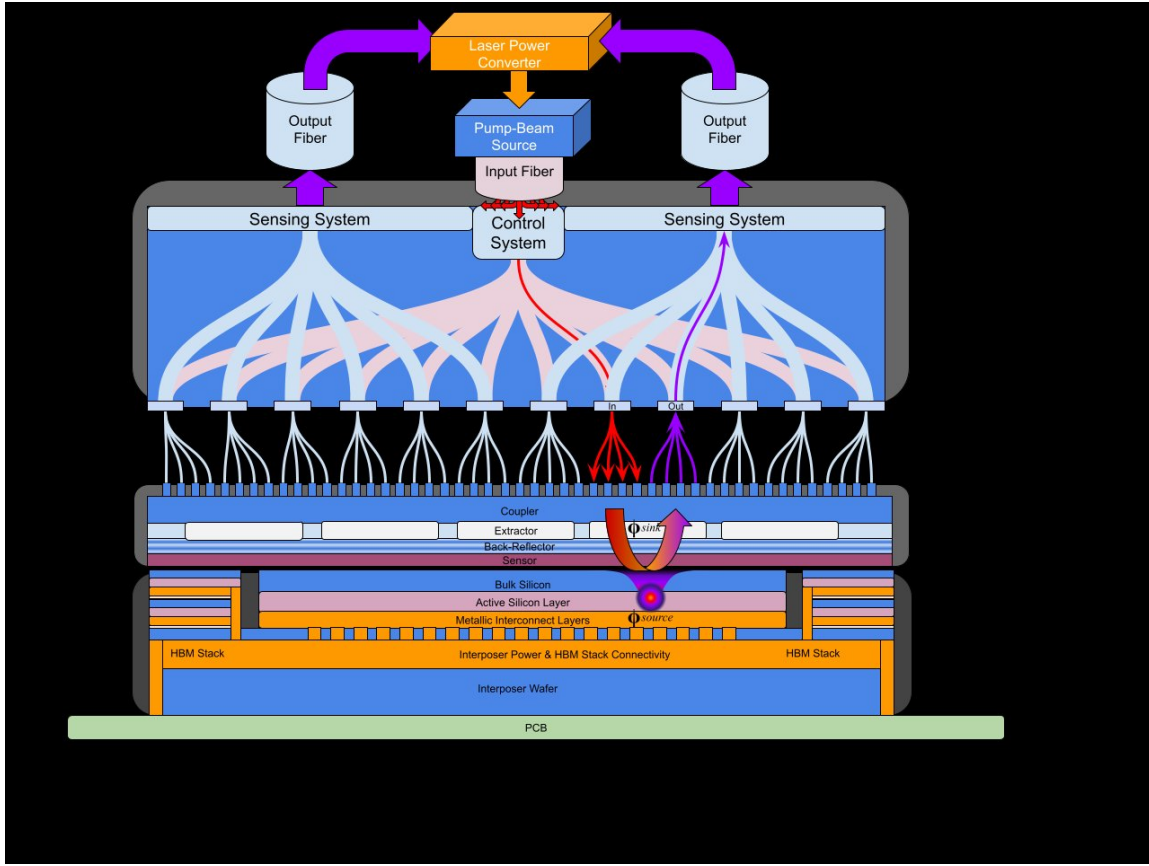


Figure 9.1: Full-stack reference architecture. A pump-laser source and an LPC subsystem are coupled through a splitter into a photonic network of waveguides or hollow-core fibers, which distribute pump light into and collect fluorescence out of an array of photonic cooling tiles tiled across the cooled body. Each tile is the cold-plate unit cell — coupler / extractor / back-reflector / sensor — on which the inverse-design loop of this chapter operates.

partitioning, AR / notch coatings, and integration mode (external recovery; monolithic backside; LPC-on-back-reflector) are decision variables.

Level 3 — outer evolutionary loop. Each fully optimized Level-1 + Level-2 outcome is one *individual*; a population of individuals is evolved over genes that encode (a) inverse-design parameters used in Levels 1 and 2.A; (b) fabrication constraints; (c) per-layer materials; (d) per-layer thicknesses; and (e) the threshold Q -enhancement at which the inner loop terminates. Selection, crossover, and mutation produce successive generations until convergence or a generation budget; the final population is rank-ordered by manufacturability as in Section 9.6.1.

Figure 9.2 renders the loop as a flow. The reader will note that *every* ingredient of the per-individual evaluation is one of the three coupled physics modules established earlier — optical (Chapter 4), quantum-dynamics (Chapter 5), and thermodynamic (Chapter 1 and 6). Inverse design provides the search procedure; the chapters preceding this one provide the objective and constraint functions.

Remark 9.1. The decomposition is not arbitrary: it tracks the energy-scale ordering of the underlying physics. Level 1 sets the slowest, materials-driven quantities; Level 2.A sets the photonic-mode geometry that converts those quantities into device performance; Level 2.B routes the resulting fluorescence into electrical work; Level 3 explores the discrete, combinatorial choices —

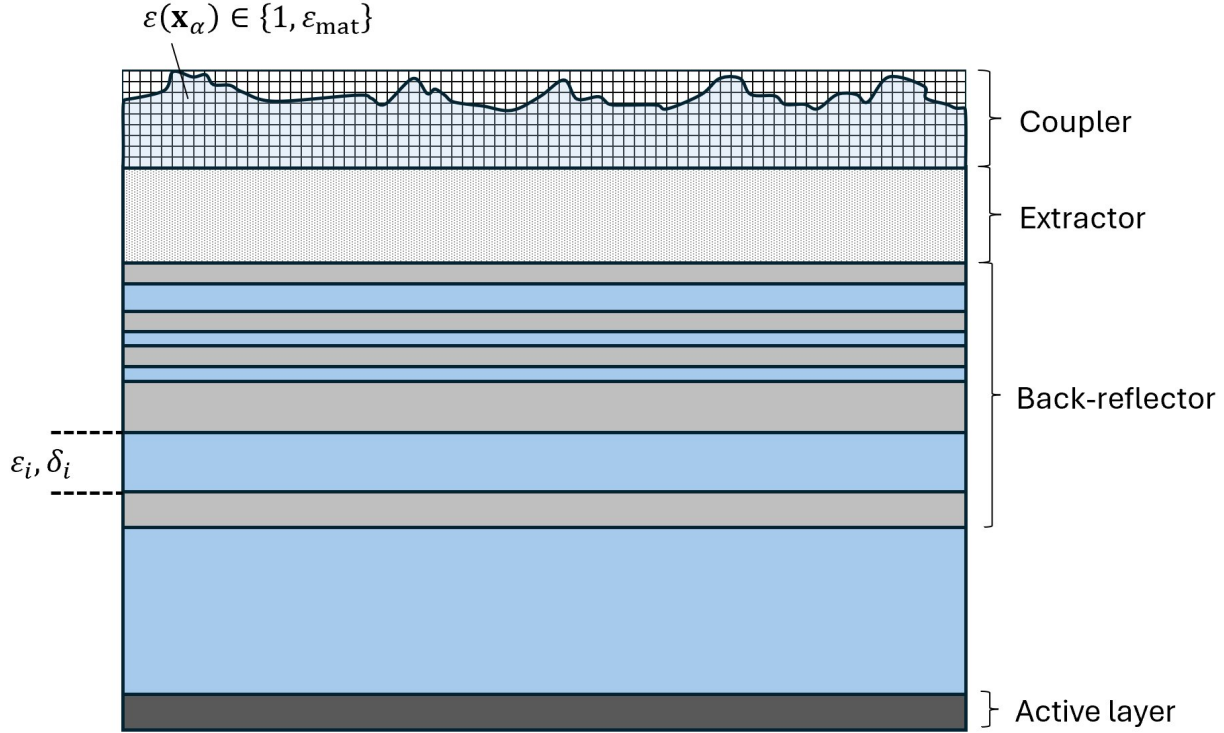


Figure 9.2: The full-stack optimization loop. Level 0 selects a material platform; Level 1 runs a per-material adjoint optimization of the extractor and pump wavelength; Level 2.A jointly designs the coupler, back-reflector, and sensor anchored to the L1 extractor; Level 2.B inverse-designs the LPC at the resulting mean fluorescence wavelength; Level 3 evolves a population of full-stack individuals against a multivariate manufacturability-aware fitness. Closed-loop sensor telemetry from manufactured units updates priors, closing the loop end-to-end.

which material, which family, which fab process — that the continuous adjoint cannot. Trying to short-circuit a level (e.g., co-optimizing the LPC with the extractor before the extractor has been pinned by Level 1) couples the quantum-defect-limited quantum-dynamics scale to the joule-scale electrical-recovery scale and loses the conditioning that makes adjoint methods tractable.

Per-tile operating temperature as a design variable

The description above pins each tile’s operating temperature T_h as an inherited input, effectively assuming a uniform- T_h die. The heterogeneous-thermal architecture of Section 10.9 calls for a further generalization: $T_h^{(i)}$ is itself a per-tile design variable, and the inverse-design objective should accept it as such. Concretely, Level 0’s material class becomes a *per-tile* choice matched to a target $T_h^{(i)}$ window (Section 8.4), Level 1’s optimal pump wavelength $\omega_p^{*,(i)}$ shifts because the anti-Stokes shift and $\eta_{\text{ASF}}(T_h^{(i)})$ both track temperature (Section 3.4), and Level 2.B’s LPC exergy bound becomes zone-specific because the Carnot factor $(1 - T_0/T_h^{(i)})$ is per-tile. The Level 3 evolutionary loop then optimizes not a single scalar P_{cool}/A but a spatial allocation $\{(T_h^{(i)}, m^{(i)}, \omega_p^{*,(i)}, \dots)\}_i$ across the tile grid, subject to (a) the packaging-imposed gradient limit between neighboring tiles, (b) the underlying device technology’s reliability envelope in each zone (Cu-Si vs. SiC / GaN), and (c) the extractor material’s operational- T range.

The fitness function to maximize is the zone-additive

$$F = \sum_i w_i P_{\text{cool}}^{(i)}(T_h^{(i)}, m^{(i)}, \dots) - \lambda \sum_i [Q_{c,i}/\text{COP}_i - \Delta P_{\text{leak},i}(T_h^{(i)})], \quad (9.1)$$

with w_i the workload-weighted importance of tile i (from architecture / activity maps of Chapter 10) and λ the marginal cost of loop power. In this form the optimizer will naturally push high-activity compute tiles hot (large Carnot factor, wider ϕ_i , large recovered exergy) and static-power-dominated cache / control tiles cold (exponential collapse of ΔP_{leak}), producing the three-zone template of Table 10.8 without any need to prescribe it by hand. The generalization is orthogonal to the four-level decomposition above — each level’s inner objective inherits a per-tile $T_h^{(i)}$ as an additional degree of freedom — and does not disturb the conditioning arguments of the preceding remark.

One integrated loop: performance, materials, thermal feedback, and manufacturability under a single objective

A recurring temptation in optical-cooling device work is to run each of the loop’s four decision axes in isolation — pick a material from a data-sheet, optimize the photonic stack against that fixed material, hand the design to a fab that reports back what is manufacturable, and only then discover that the resulting tile does not clear the chip-level thermal target. The Maxwell IP framing [78] is the opposite: all four axes are coupled inside a *single* optimization loop, and none of them is allowed to converge in isolation. Concretely, the loop of Figure 9.2 closes over:

1. *Performance optimization* — adjoint / evolutionary search over photonic-network, coupler, back-reflector, and sensor geometry, driven by the multiphysics coupling of Chapters 5 and 6 and evaluated against the P_{cool}/A target of Section 9.1. This is the Level 2.A / Level 2.B inner loop.
2. *Materials exploration and down-selection* — a per-candidate sweep over the extractor material’s constitutive parameters ($\sigma_a, \sigma_e, \gamma_r, \gamma_{nr}, \alpha_b, N_t$, temperature dependence) drawn from the platform class of Chapter 8. Level 0 selects the class; Level 1 picks the candidate and its optimal pump wavelength. Crucially, the materials layer is not a static pre-optimization: the outer loop can retire a candidate (or an entire platform class) if downstream constraints make it infeasible.
3. *Chip thermal modelling feedback* — the tile’s realised cooling-power density and hotspot capacity flow back into the chip-level thermal model of Chapters 2 and 10, and the per-tile $T_h^{(i)}$ target of Section 9.2 is itself a decision variable set by that feedback. If a hot compute zone’s $T_h^{(i)}$ target is not achievable at the tile’s current $\eta_{\text{EQE}} / \eta_{\text{ASF}}$, the loop shifts the target down, picks a different extractor material, or allocates additional tile area — each of these is a legitimate outer-loop move.
4. *Manufacturability and supply-chain constraints* (Section 9.6.1) — fabrication-process compatibility, minimum feature size, layer-count budget, material-availability filters, and yield priors are first-class terms in the Level 3 fitness function. A candidate that hits the physics target but requires a fabrication process the fab cannot deliver, or a rare material the supply chain cannot supply, is scored down the same way a physics-infeasible candidate is.

Concretely, the Level 3 fitness function of (9.1) generalizes to

$$F_{\text{tot}} = \underbrace{\sum_i w_i P_{\text{cool}}^{(i)}(T_h^{(i)}, m^{(i)}, g^{(i)})}_{\text{performance}} - \underbrace{\lambda_P \sum_i [Q_{c,i}/\text{COP}_i - \Delta P_{\text{leak},i}(T_h^{(i)})]}_{\text{thermal cost}} - \lambda_M C_{\text{mfg}}(\{g^{(i)}, m^{(i)}\}) - \lambda_S C_{\text{sup}}(\{m^{(i)}\}), \quad (9.2)$$

with $m^{(i)}$ the per-tile material choice, $g^{(i)}$ the per-tile photonic geometry, C_{mfg} the manufacturability penalty (process incompatibility, feature-size overrun, layer-count blowup), and C_{sup} the supply-chain penalty (rare-earth availability, epitaxy tool scarcity, precursor cost). The Lagrange weights $\lambda_P, \lambda_M, \lambda_S$ express the program's trade between performance and deliverability. Setting them consistently across all four axes is what turns the loop from four disjoint optimizations into one integrated one.

The chapter that follows describes each level in isolation for clarity, but every design run reported in Section 9.7 was executed with all four axes coupled through (9.2).

9.3 Level 0 / Level 1: materials exploration and the extractor

Level 0 picks a platform class; Level 1 picks a candidate material from that class and optimizes the extractor against it. The platform classes were the subject of Chapter 8; the materials-exploration framework that supplies Level-1 candidates and their constitutive parameters is the subject of this section.

9.3.1 Platform-specific specialization of the inverse-design loop

The four-level loop is the same architecture across all three material families of Chapter 8, but the *search space*, the *binding constraints*, and the *manufacturability priors* each Level operates on are platform-specific. Recognizing this up front sharpens both the Level-0 platform selection and the way each Level's cost function is parameterized. Table 9.1 summarizes the three specializations; the paragraphs that follow expand on each.

Doped-crystal branch. The pump wavelength is essentially pinned by the ion transition, so Level 1 has one fewer continuous knob than the other platforms. The absorption cross-section is small ($\sigma_a \sim 10^{-20} \text{ cm}^2$ for Yb:YLF), so Level 2.A must deliver a strong resonant enhancement to hit $\eta_{\text{abs}} \rightarrow 1$; this is where the (Q/V) -controlled Purcell factor of Chapter 5 does the heavy lifting. Manufacturability is dominated by Czochralski / hydrothermal single-crystal growth and dopant-uniformity tolerance; supply-chain risk is set by rare-earth availability (Yb, Ho, Tm are moderately concentrated in the global market). The Level 3 evolutionary loop moves through discrete host / dopant / concentration triples with modest crossover diversity.

Direct-bandgap-semiconductor branch. Absorption is trivially strong ($\sigma_a \sim 10^{-16} \text{ cm}^2$), so Level 2.A does not need to work as hard on the pump-side cavity; the effort shifts to the fluorescence-collection cavity and to Purcell-enhanced mode selection for exergy extraction. Level 1's pump wavelength is a real continuous knob, tunable by bandgap engineering (III-V compositional gradients, quantum-well thickness, quantum-dot size). Manufacturability is the binding Level-3 constraint: MBE / MOCVD lattice-matched epitaxy is a narrow, capital-intensive process; the substrate constraint limits the deployable form factor; and low-temperature operation is often required to reach the $\eta_{\text{EQE}} \rightarrow 0.99$ window. The Level 2.B LPC can be lattice-matched and monolithically co-integrated, which is a real fitness bonus in (9.2).

SMILES-in-polymer branch. This is the platform with the richest continuous search space at Level 0: the dye is a molecular choice from a design-space of many rhodamine and non-rhodamine candidates, the loading fraction is a continuously-variable parameter, and the polymer host can be selected almost independently. The pump wavelength is a continuous knob determined by

Loop level	Doped crystals (Yb:YLF, Cr:LiSAF)	Direct-bandgap semi-conductors (GaAs, InGaAs)	SMILES-R640-in-polymer thin film
<i>L0: search dimension</i>	discrete: host $\times$ dopant $\times$ concentration	discrete: III-V system $\times$ QW / QD geometry	continuous: dye choice $\times$ loading fraction $\times$ polymer host
<i>L1: pump wavelength ω_p^*</i>	fixed by ion transition (~ 1020 nm Yb)	tunable via bandgap engineering (700–1700 nm); must track $E_g(T)$ (Section 9.3.2)	set by tail optimum, 650–700 nm, T -dependent (Section 9.3.2)
<i>L1: absorption</i>	low σ_a ; requires thick film + resonant enhancement	Urbach-tail $\alpha \sim 100 \text{ cm}^{-1}$; $\sim 10\%$ single pass per $10 \mu\text{m}$; resonant enhancement still needed	small σ_a in tail; needs high $N_t + Q \sim 10^4$ pump resonator; thickness bounded below by (8.9)
<i>L1: extractor geometry</i>	bulk crystal or thick glass film (10s of μm)	epitaxial QW / QD stack (nm-scale layer control)	1–10 μm structured polymer film; substrate-agnostic
<i>L2.A: back-reflector spec</i>	DBR at $\omega_p, \bar{\lambda}_f \sim 1000$ nm	DBR at $\omega_p, \bar{\lambda}_f \sim 850$ nm	Ag mirror + inverse-designed GaP/SiN slab, $Q \sim 10^4$ at ω_p ; $\bar{\lambda}_f \sim 620$ – 630 nm (Section 9.7.2)
<i>L2.B: LPC bandgap</i>	GaAs or GaAs/GaInP tandem near 1000 nm	lattice-matched, co-integrated	GaP or GaP/AlGaInP near $\bar{\lambda}_f$
<i>L3: manufacturability</i>	Czochralski growth, dopant-uniformity constraints	MBE / MOCVD; lattice-match; substrate cost	spin coat / dip coat / imprint; substrate-agnostic
<i>L3: supply chain</i>	rare-earth availability (Yb, Ho, Tm)	III-V precursor supply	rhodamine + cyanostar; commodity polymers

Table 9.1: How the same four-level inverse-design loop specializes across the three material platforms of Table 8.4. The search space, the wavelength anchors that Level 1 optimizes around, and the manufacturability / supply-chain priors that Level 3 penalizes all shift with platform choice — Level 0’s decision is therefore not a formality but sets the topology of every downstream cost function.

red-tail-pump depth (Table 8.3); pumping at $\lambda_p \geq 680$ nm lifts the quantum-defect ceiling η_q to 5–10%, but the realised η_{cool} peaks near 650 nm at 300 K (+5%) and moves to 670–700 nm on a 400 K tile (Section 9.3.2); the pump wavelength is therefore a per-temperature variable exactly as for the semiconductor branch. Level 2.A must deliver resonant enhancement on the red-tail pump wavelength to compensate the small $\sigma_a(\lambda_p)$ ($Q \sim 10^3$ – 10^4 , Section 9.7.2); this is architecturally similar to the doped-crystal case but with the concentration lift ($N_t \gtrsim 10^{-2}$ M under SMILES) reducing the burden, and with the transparency ceiling (8.9) setting a *minimum* dye-layer thickness. Level 2.B’s LPC is a GaP or GaP/AlGaInP device tuned to $\bar{\lambda}_f \sim 620$ – 630 nm — distinct from both the Yb-band and the GaAs-band LPCs. Manufacturability is the platform’s decisive advantage: solution-processable deposition (spin coat, dip coat, nanoimprint) on essentially any planar substrate, no lattice-matching requirement, and commodity supply chain for the polymer host and the rhodamine chromophore. Level 3’s manufacturability penalty C_{mfg} is correspondingly small, and the fitness weight λ_M — which for the semiconductor branch often dominates the outer-loop selection — has room to be relaxed.

Consequence for Level 0 selection. Table 9.1 makes clear that Level 0 is not a formality: picking a platform sets which of Level 1’s knobs are continuous vs. discrete, which of Level 2.A’s

optimizations is easier, whether Level 2.B can be monolithically integrated with the extractor, and how heavy the Level 3 manufacturability penalty is. In an integrated loop ((9.2)) the outer evolutionary search should be allowed to retire platform choices between generations; “Level 0 first, then run the rest” is not the right ordering because the outer loop’s manufacturability and thermal-feedback signals are what tell the search when to give up on a platform. This is what distinguishes the unified-loop framing from the traditional *material* $\rightarrow$ *device* $\rightarrow$ *manufacturability* pipeline that dominates optical-cooling practice today.

9.3.2 Modified Van Roosbroeck–Shockley emission and the Urbach tail

The framework requires, per candidate material, an emission spectrum $R_{\text{sp}}(\omega_f)$ consistent with the absorption coefficient $\alpha(\omega_p)$. Detailed balance ties the two via the Van Roosbroeck–Shockley relation [79],

$$R_{\text{sp}}(\omega) = \frac{n^2(\omega) \alpha(\omega) \omega^2}{\pi^2 c^2} \frac{1}{\exp[(\hbar\omega - \Delta\mu)/k_B T] - 1}, \quad (9.3)$$

in which $\Delta\mu$ is the quasi-Fermi-level splitting (zero in thermal equilibrium, positive under pump). For anti-Stokes operation the framework uses the *modified* VRS relation that accounts for self-absorption, the adapted Urbach rule for sub-bandgap absorption, and photo-excited carrier-temperature corrections [80]; sub-bandgap absorption is modelled with the Urbach tail

$$\alpha(E) = \alpha_0 \exp((E - E_g)/E_u), \quad E < E_g, \quad (9.4)$$

with Urbach energy E_u typically in the 5 meV to 50 meV range for semiconductors. The extension is essential: the pump for anti-Stokes operation lies in this tail by construction.

The molecular-dispersion constitutive model: an anchored exponential red tail

For a dye the same construction is used, but the anchor is a measured molar extinction rather than a band edge. Integrating-sphere and spectrophotometer data for R101 are trustworthy only to a crossover wavelength $\lambda_0 \simeq 600$ nm; beyond it the signal is buried in instrument noise and, for less-pure samples, in impurity absorption. The framework therefore retains the measured $\varepsilon(\lambda)$ for $\lambda < \lambda_0$ and continues it with

$$\varepsilon(E) = \varepsilon(E_0) \exp\left[\frac{\sigma(E - E_0)}{k_B T}\right], \quad E < E_0 = hc/\lambda_0, \quad \varepsilon(E_0) = 4290 \text{ L mol}^{-1} \text{ cm}^{-1}, \quad (9.5)$$

i.e. an Urbach tail with $E_u = k_B T/\sigma$; $\sigma = 1$ is the pure Boltzmann limit expected for a thermalized vibronic manifold (Section 5.8), $\sigma > 1$ a steeper tail. The dye absorption coefficient is $\alpha_r = \ln 10 C \varepsilon$, the host (ethanol or polymer) contributes a measured background $\alpha_b(\lambda)$, and the Level-1 objective is the realised cooling efficiency of (8.6), $\eta_{\text{cool}} = \eta_{\text{abs}} \eta_{\text{EQE}} \lambda_p / \bar{\lambda}_f - 1$ with $\eta_{\text{abs}} = \alpha_r / (\alpha_r + \alpha_b)$. Figure 9.3 shows the anchored spectrum at three tail temperatures. Two properties of the model make it robust as a design input. First, the anchor error enters as one multiplicative factor and cannot compound with wavelength; a $10\times$ error in the assumed tail moves the peak η_{cool} by only $\sim 30\%$ relative and the optimum λ_p by ~ 20 nm, because η_{abs} is already deep in saturation ($\simeq 0.98$) at the operating points. Second, the T in (9.5) is the temperature of the *unmeasured tail* only — the measured part is held fixed — so sweeping it is a bounded sensitivity test, not a prediction of the full spectrum’s thermal response; direct measurement of ε beyond 620 nm on high-purity R640 remains the priority experiment. The results (peak $\eta_{\text{cool}} = +5.3\%$ at 651 nm for 10^{-2} M, $+8.7\%$ at 672 nm for 10^{-1} M, $\eta_{\text{EQE}} \geq 0.95 / 0.92$ required, all at 300 K and $\eta_{\text{EQE}} = 1$; $+13\%$ at 702 nm for

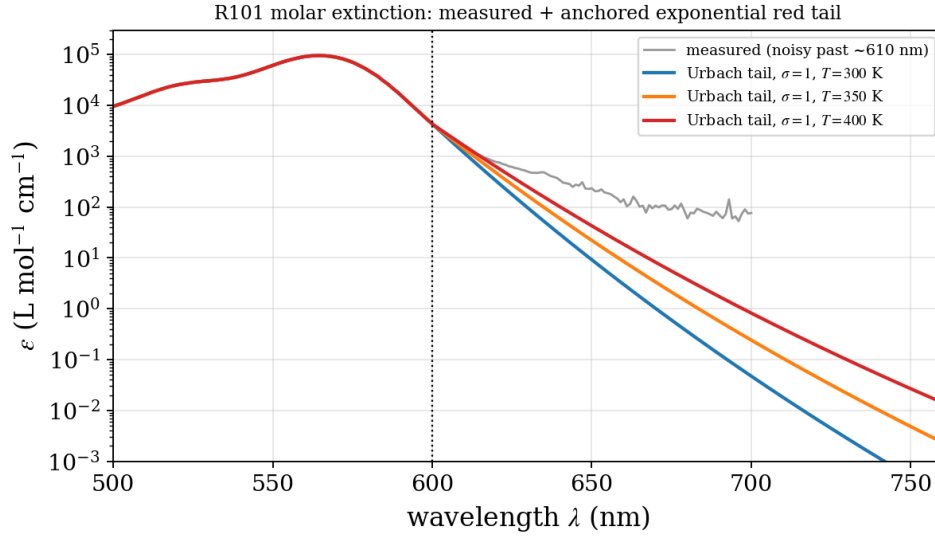


Figure 9.3: R101 molar extinction: measured spectrum (grey; noisy and impurity-dominated beyond ~ 610 nm) and the anchored exponential red tail (9.5) for tail temperatures of 300, 350 and 400 K ($\sigma = 1$). The dotted line marks the 600 nm anchor. At 651 nm the tail gives $\varepsilon \simeq 8$ (300 K) to $39 \text{ L mol}^{-1} \text{ cm}^{-1}$ (400 K); at 680 nm, 0.35 to 3.7. The inverse-design assumption $\kappa \simeq 10^{-4}$ at 10^{-2} M (Section 9.7.2) corresponds to $\varepsilon \simeq 850$, i.e. a pump near 615 nm.

10^{-1} M with a 400 K tail) are collected in Section 8.3.5, and the volumetric densities they imply in Table 8.1. The same $\alpha_r(\lambda_p, T)$ is the loss the Level-2.A pump resonator of Section 9.7.2 must be critically coupled to.

The pump wavelength is a per-temperature variable

Both constitutive models make ω_p^* temperature-dependent, and in opposite ways that the Level-1 optimizer must track. For a semiconductor the gap follows Varshni, $E_g(T) = E_g(0) - aT^2/(T + b)$ (GaAs: $E_g(0) = 1.519 \text{ eV}$, $a = 5.4 \times 10^{-4} \text{ eV/K}$, $b = 204 \text{ K}$), i.e. -0.45 meV/K near room temperature. A pump fixed at the 300 K design point of Table 9.2 (890 nm, 30 meV below $E_g(300 \text{ K})$) sits only 11 meV below the gap at 340 K and *above* it at $T \gtrsim 365 \text{ K}$; because the mean fluorescence redshifts with the gap, η_q collapses well before the crossing (from 3.5% to $\sim 1\%$ at 340 K) and net cooling is lost once $\eta_{\text{EQE}} < 1/(1 + \eta_q)$. In a chip-level simulation with a hot-branch die profile, a fixed 890 nm pump drove three of the GaAs tiles into heating while a per-tile retuned pump drove none. The requirement is therefore $\omega_p^*(T)$ such that $\hbar\omega_p^* = E_g(T) - \Delta_p$ with a temperature-independent (or optimized) tail depth Δ_p ; the Urbach absorption at that depth grows with T ($E_u \propto T$ in the phonon-dominated regime), which partly compensates the loss of η_e at higher T . For a dye the sign is reversed: the tail steepness $k_B T/\sigma$ and the transparency ceiling (5.7) both *improve* with T , pushing ω_p^* to the red (651 $\rightarrow$ 674 nm from 300 to 400 K at 10^{-2} M) and raising the achievable density. In both cases the photonic network's pump source must be tunable over $\sim 30\text{--}50 \text{ nm}$, or partitioned into wavelength channels matched to the thermal zones of Section 10.9; a single fixed-wavelength pump is a design error over a non-isothermal die.

Table 9.2: GaAs reference parameters used as the materials-exploration framework’s 300 K benchmark.

Parameter	Value	Source
A (SRH)	$5 \times 10^4/\text{s}$	[81]
B (radiative)	$4 \times 10^{-10} \text{ cm}^3/\text{s}$	[82]
C (Auger)	$4 \times 10^{-30} \text{ cm}^6/\text{s}$	[82]
α (band-edge)	$10^4/\text{cm}$	Adachi 1999 [83]
α (Urbach tail, 890 nm)	$100/\text{cm}$	Adachi 1999
λ_p	890 nm	Urbach-tail region
λ_f	860 nm	VRS-derived
η_e (base)	0.25	[82]
N_{opt}	$5.7 \times 10^{17}/\text{cm}^3$	at base parameters
$p_{\text{cool}}^{\text{opt}}$ (bulk)	$80 \text{ W/mm}^3 (= 8 \times 10^4 \text{ W/cm}^3)$	(8.2) at N_{opt} ; $\eta_{\text{ASF}} \simeq 0.3\%$
Max passivation gain	$100 \times (\text{InGaP DH})$	[84]

9.3.3 The breakeven A_0 constraint

For semiconductor extractors, the framework recasts the cooling-power calculation in terms of the standard A, B, C recombination coefficients. Define the per-photon quantum defect $\eta_q = (\nu_f - \nu_p)/\nu_p$, its background-corrected value $\tilde{\eta}_q = \eta_q - \alpha_b/\alpha$ (the fraction of each absorbed pump photon’s energy that is available for lifting after the host’s parasitic share is paid; this is *not* the pump-absorption fraction η_{abs} of Table 1.2), and the free-carrier-augmented effective Auger coefficient $C' = C + \sigma_{\text{fca}}\eta_e B/\alpha$. The volumetric cooling power is (8.2), $p_{\text{cool}}(N) = \hbar\omega_p[\tilde{\eta}_q\eta_e B N^2 - AN - C'N^3]$, and its maximum over N is

$$p_{\text{cool}}^{\text{opt}} = \frac{4}{27} \tilde{\eta}_q^3 \frac{(\eta_e B)^3}{C'^2} \hbar\omega_p \quad (A \rightarrow 0), \quad p_{\text{cool}}^{\text{opt}} \propto \frac{(\eta_e B)^3}{C'^2}. \quad (9.6)$$

(The prefactor is $4/27$, not the $4/9$ that appeared in earlier drafts and in some framework outputs, which overstates the optimum by exactly $3\times$; $\tilde{\eta}_q\hbar\omega_p$ is the energy lifted per escaped photon, $\hbar\bar{\omega}_f - \hbar\omega_p$.) The breakeven SRH rate at which cooling is just possible is

$$A_0 \equiv \frac{\tilde{\eta}_q^2 (\eta_e B)^2}{4C'}, \quad \text{cooling requires } A < A_0, \quad (9.7)$$

since $\max_N[\tilde{\eta}_q\eta_e B N - C'N^2] = A_0$ is the largest SRH rate that $p_{\text{cool}} > 0$ can absorb. (The weaker condition $A < \frac{4}{3}A_0$ only guarantees that $p_{\text{cool}}(N)$ has a stationary point; for $A_0 < A < \frac{4}{3}A_0$ that stationary point is a local maximum of a *negative* — heating — function.) A candidate material that fails (9.7) at the achievable passivation level is rejected before any structural inverse design is attempted. Equation (9.6) also makes explicit the two dominant levers Level 2.A is asked to pull: the cubic gain in $(\eta_e B)$ from photonic enhancement of the radiative rate and the extraction efficiency, and the inverse-square gain in C'^{-2} from cryogenic operation, where $C'(T) \propto \exp(-2.4(300/T - 1))$ suppresses Auger recombination sharply.

Result 9.2 (Optimal carrier density). The carrier density that maximizes $p_{\text{cool}}(N)$ of (8.2), for $A < A_0$, is

$$N_{\text{opt}} = \frac{\tilde{\eta}_q \eta_e B}{3C'} \left[1 + \sqrt{1 - 3A/(4A_0)} \right]. \quad (9.8)$$

For GaAs at room temperature with the reference parameters of Table 9.2 ($\tilde{\eta}_q = \eta_q = 0.0349$, $A/A_0 = 0.066$), $N_{\text{opt}} \simeq 5.7 \times 10^{17} \text{ cm}^{-3}$ and the bulk $p_{\text{cool}}^{\text{opt}} \simeq 81 \text{ W/mm}^3$ ($8.1 \times 10^4 \text{ W/cm}^3$), with

Table 9.3: Family-specific passivation gain limits used by the materials-exploration framework. Each entry is the maximum reduction of the bulk-SRH coefficient A achievable by a realistic passivation technology for that family.

Material family	Max A reduction	Best passivation technology
GaAs-based III-V	100×	InGaP lattice-matched DH [84]
InP-based III-V	50×	InAlAs passivation
Other III-V	20×	AlGaIn-style and similar
II-VI	10×	MgZnO; limited options
IV-VI (PbX)	5×	poor surface chemistry
MP-screened / other	2×	no demonstrated interface technology

the Sheik-Bahae / Epstein A, B, C coefficients [82]; the $A \rightarrow 0$ form (9.6) gives 88 W/mm^3 . The corresponding absorbed pump density is $\hbar\omega_p(AN + BN^2 + C'N^3) \simeq 3 \times 10^4 \text{ W/mm}^3$, so $\eta_{\text{ASF}}^{\text{opt}} \simeq 0.27\%$ at $\eta_e = 0.25$; the figure of merit the loop actually maximizes is therefore p_{cool} *per unit pump*, which improves as $\eta_e \rightarrow 1$ and $\tilde{\eta}_q$ toward its temperature-set ceiling.

From the volumetric optimum to a tile. The areal cooling density of a layer of thickness d at uniform N_{opt} is $p_{\text{cool}}^{\text{opt}} d$: 0.16 W/mm^2 for $2 \mu\text{m}$ and 0.8 W/mm^2 for $10 \mu\text{m}$ of bulk GaAs, against a 300 W/mm^2 absorbed pump per $10 \mu\text{m}$. The $\sim 10 \text{ W/mm}^2$ (1 kW/cm^2) hot-spot target of Section 9.1 is reached only through the cubic lever in (9.6): raising the extraction efficiency from 0.25 to 0.9 multiplies $p_{\text{cool}}^{\text{opt}}$ by 47 (to $\sim 4 \times 10^3 \text{ W/mm}^3$, 40 W/mm^2 per $10 \mu\text{m}$) at $N_{\text{opt}} \simeq 2 \times 10^{18} \text{ cm}^{-3}$ and $\eta_{\text{ASF}} \simeq 1\%$; a further Purcell enhancement of the radiative rate by $5\times$ adds $125\times$. The chain is explicit so that its costs are: every factor of $\eta_e B$ raises N_{opt} linearly and the absorbed pump density quadratically, so the semiconductor tile’s binding constraint at high density is not the extractor but the pump flux the network can deliver and the fluorescence flux the LPC must convert. Cryogenic operation ($C(T)$ suppression) is a third lever but conflicts with the hot-tile role of Section 10.9; it is the right lever for cold storage tiles, not for compute tiles.

9.3.4 Family-specific passivation limits

The maximum achievable reduction of the SRH coefficient A by surface passivation is materials-family-specific, set by the available interface-engineering technology. The framework caps $A_{\text{eff}} = A_{\text{base}} \cdot \max(A_{\text{mult}}, 1/\max_passivation)$ per Table 9.3. These caps enter the Level-1 design constraints and the Level-3 manufacturability fitness term.

9.3.5 The materials-exploration data pipeline

The framework draws candidate constitutive parameters from a four-tier data hierarchy. In decreasing accuracy and increasing coverage:

1. the Adachi (1999) experimental optical-constants compilation for ~ 30 reference III-V and II-VI compounds [83];
2. the `refractiveindex.info` (RII) experimental database for ~ 100 additional materials;
3. the Materials Project (MP) DFT database for ~ 3000 direct-bandgap compounds, with the standard DFT-bandgap underestimation corrected against `matminer` experimental bandgaps where available;

4. the `matminer` experimental-bandgap dataset (~ 4600 materials) used as a correction table for (3).

For each candidate, the absorption spectrum is constructed from the highest-priority available data, augmented near the band edge with the Elliott formula and below it with the Urbach tail of (9.4). The mVRS relation (9.3) then yields $R_{\text{sp}}(\omega_{\text{f}})$ and the mean fluorescence wavelength $\bar{\lambda}_{\text{f}}$. The framework optimizes the pump wavelength λ_{p} over the candidate’s Urbach tail to maximize $p_{\text{cool}}(N_{\text{opt}})$ subject to (9.7).

Remark 9.3. The same pipeline applies, with platform-specific constitutive models, to doped-host and molecular-dispersion candidates. For doped-host extractors the constitutive variables are host-lattice parasitic absorption, dopant concentration, and radiative lifetime (Section 8.1). For molecular-dispersion extractors they are solvent absorption, molecular IQE, and concentration-dependent quenching/aggregation thresholds (Section 8.3.1). In every case the L1 output is the same triple — optimized extractor structuring, ω_{p}^* , and a baseline P_{cool} — handed forward to L2.

9.4 Level 2.A: joint coupler, back-reflector, and sensor inverse design

With the L1 extractor and ω_{p}^* held fixed, the surrounding stack is jointly inverse-designed. Each of the three layers below operates on quantities introduced in Chapters 4 and 5; the inverse-design step searches the corresponding geometric parameter space against an objective expressed in those chapters’ units.

9.4.1 The coupler

The coupler delivers pump light into the extractor and extracts anti-Stokes fluorescence out of it while suppressing Fresnel reflection of pump, parasitic absorption, and total-internal-reflection trapping. Three embodiments span the design space:

- a planar index-matched thin-film stack (graded-index or multilayer AR coating);
- a dome-shaped index-matched structure bonded to the extractor, which lowers the effective critical angle for outgoing fluorescence and provides a thermal-spreading path [85];
- a sub-wavelength grating / metasurface achieving polarization- and angle-selective injection.

The dome is a specific embodiment of the coupler layer, not a replacement for it. Decision variables are layer count, per-layer n, d pairs, dome radius, bond-line thickness, and grating period and duty cycle.

9.4.2 The back-reflector

The back-reflector recirculates pump and fluorescence and suppresses downward leakage into the cooled body. It is the most extensively inverse-designed layer of the stack and the layer for which the most informative 2D inverse-design dataset is available; we therefore devote Section 9.7.1 to its illustrative results. The embodiment classes the framework supports are: (i) a multi-material distributed Bragg reflector (DBR); (ii) a reflective metal thin-film (Ag- or Au-class with protective dielectric); (iii) a hybrid DBR-on-metal stack; and (iv) a structured (patterned) reflector co-inverse-designed with the extractor lateral structuring in L2.A.

9.4.3 The sensor layer

The sensor layer feeds the closed-loop control of the deployed cold plate and provides ground-truth data that updates the materials- exploration priors used by L0. Embodiments include:

- a thermo-optic thin film interrogated by an idler / probe beam: VO₂ operated near its insulator–metal transition at ~ 340 K; GST phase-change films; SiO₂/Si thermo-optic resonators; doped-semiconductor band-edge films;
- thin-film thermocouples or RTD arrays patterned at the cold face of the extractor;
- on-chip photodiode flux taps and AWG-based micro-spectrometers for fluorescence-flux and spectral-drift monitoring.

Decision variables are sensor-layer thickness (minimized to suppress parasitic absorption and thermal mass), probe wavelength (off-resonance with the extractor; e.g., 1310 nm to 1550 nm telecom bands for VO₂), required probe power, and integration mode (extractor cold-face / back of back-reflector / coupler-output tap / waveguide-coupled in photonic-network embodiments).

9.5 Level 2.B: coupled LPC inverse design

With the L2.A stack pinned, the LPC stack is inverse-designed at the mean fluorescence wavelength ω_f^* of the L2.A output. The LPC inverse design is conducted under the bandwidth and entropy bound of Section 7.3 — i.e., a single-junction LPC sees a fundamental $\eta_{\text{LPC}} \leq 1 - T_0 S_{\text{in}}/E_{\text{in}}$ and recovers efficiency only by sub-band partitioning (multi-junction stacked, laterally segmented, or dichroic-split parallel architectures). The decision variables include the per-junction material assignment, the junction count, the AR / notch coatings, the series/parallel interconnect (with per-line MPPT preserving per-junction voltage when parallel), the integration mode (external recovery; monolithic backside on a semiconductor-platform stack; LPC tiles on the back-reflector substrate), and the closed-loop control law that routes recovered power back to the pump driver. Satisfying the self-powered condition Equation (1.17) — which, with η_{ASF} bounded by the quantum defect, amounts to $\eta_{\text{P}} \eta_{\text{LPC}} \eta_{\text{cpl}} \gtrsim 0.9$ — would let the loop cool the heat-generating body and export net electrical power simultaneously; short of it, the LPC’s role is to reduce the loop’s electrical cost per watt lifted from $1/(\eta_{\text{P}} \eta_{\text{ASF}})$ to $[1/\eta_{\text{P}} - \eta_{\text{LPC}} \eta_{\text{cpl}}(1 + \eta_{\text{ASF}})]/\eta_{\text{ASF}}$ (Chapter 7). Either way the recovered fraction is a Level-2.B output, not a Level-1 input.

9.6 Level 3: outer evolutionary loop

Each completed Level-1 + Level-2 outcome is one individual. The genes of each individual are $g = \{\theta_{\text{ID}}, \theta_{\text{fab}}, \{m_\ell\}, \{d_\ell\}, Q_{\text{thr}}\}$, where θ_{ID} are the inverse-design parameters used in Levels 1 and 2.A, θ_{fab} are the fabrication constraints, m_ℓ and d_ℓ are the material identity and thickness of the ℓ th layer, and Q_{thr} is the threshold cavity-Q at which the inner loop terminates. The fitness is multivariate,

$$F(g) = w_1 P_{\text{cool}}/A + w_2 \eta_{\text{ASF}} + w_3 \eta_{\text{LPC}}(\bar{\lambda}_f) + w_4 M_{\text{laser}}(\omega_p^*) - w_5 \text{Risk}(\theta_{\text{fab}}), \quad (9.9)$$

where $M_{\text{laser}}(\omega_p^*)$ is the manufacturability score for commercial pump-laser availability and wall-plug efficiency near ω_p^* and $\text{Risk}(\theta_{\text{fab}})$ is the Level-3 fab-risk penalty discussed in Section 9.6.1. The weights w_i parametrize the design tier (laboratory refrigeration vs. HPC-cooling deployment vs. self-powered-regime demonstration). Standard selection / crossover / mutation operators evolve the population for a generation budget; the final population is rank-ordered and analyzed for manufacturability per Section 9.6.1.

9.6.1 Manufacturability and supply-chain analysis

The Level-3 fitness term $\text{Risk}(\theta_{\text{fab}})$ prices in five physical and three economic factors:

- physical: minimum lithographic feature size; per-layer thickness budget; etch-tolerance over feature height; defect tolerance during epitaxial growth or dye purification; and layer-bonding compatibility (lattice mismatch and interface absorption);
- economic: number of fab / wafer-processing steps; supply-chain maturity for each constituent material; and foreign-sourcing dependence for each constituent material and the pump-laser candidate.

The output of Level 3 is therefore not a single optimum but a *ranked list* of design plans, each comprising geometry, materials BOM, process flow, and sensor layout, that manufacturing operations can release directly to a fabrication run. Sensor-layer telemetry from manufactured units feeds back into the materials-exploration priors used by Level 0, closing the loop end-to-end.

9.7 Illustrative results

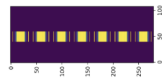
We illustrate the loop with three representative design studies: an inverse-designed back-reflector for the semiconductor platform; a critically coupled pump resonator for a 10^{-2} M dye layer, which is the Level-2.A problem the molecular-dispersion platform makes hardest; and a molecular-dispersion-platform photonic cooling tile that exercises the fluid-loop hardware as a Level-3 decision variable.

9.7.1 Back-reflector inverse-design results

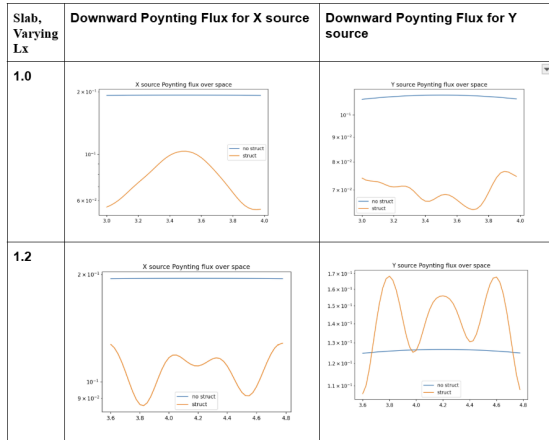
A 2D inverse-design study by Chao [86] sweeps back-reflector embodiments at unit (vacuum) wavelength $\lambda \approx 970$ nm for Si embedded in SiO_2 , with the objective minimising the spatially averaged downward Poynting flux for an isotropically polarized point- source distribution. The four classes of structure — slab grating, multilayer DBR, hybrid grating-on-multilayer, and pixel-wise topology-optimized 2D photonic crystal — span the engineering trade space the framework explores.

The takeaways have direct consequences for the hierarchy. First, slab geometries are unit-order ineffective and are eliminated by Level 2.A’s own gradient before the genetic operator of Level 3 ever has to score them. Second, a 1D multilayer at $\sim 1.2\lambda$ of total thickness reaches >20 reduction; this is the manufacturable “baseline” choice and the one reported as Embodiment A in the companion patent disclosure. Third, the hybrid (multilayer + 0.2λ slab grating top) reaches the same >20 reduction at 1.0λ total thickness with an upward/downward flux ratio of ~ 50 , a result robust to additional top-coupler structuring — Level 2.A’s joint co-design of coupler and back-reflector inherits this robustness by construction. Fourth, full pixel-wise topology optimization yields 2D rectangular photonic crystals achieving >20 reduction at 0.5λ of total thickness, providing the thinnest known back-reflector for the same target. Finally, an Ag (or Cu) thin film immediately below the dielectric multilayer trades a small absorption penalty for additional broadband reflectivity at the fluorescence band; the exact trade is design-dependent and is one of the most consequential Level-2.A decisions.

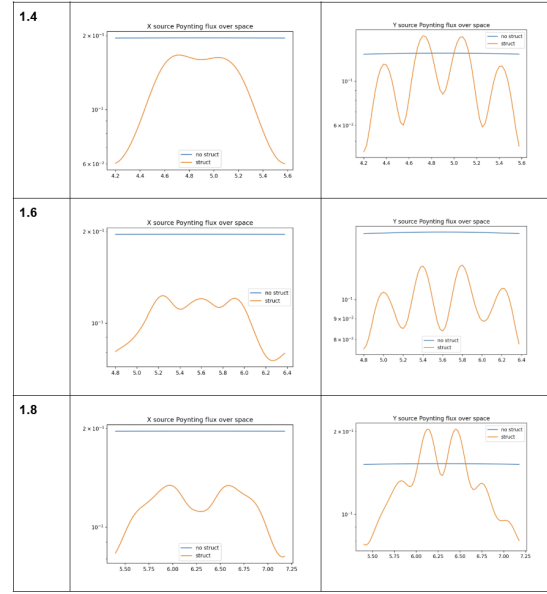
A grown multilayer: GaAs/AlGaAs DBRs for a Yb:YLF extractor. Figure 9.8 shows the first measured back-reflectors in this book: three GaAs/AlGaAs distributed Bragg reflectors grown at Sandia National Laboratories (R. Sarma) as the back-mirror of a Yb:YLF-based extractor, i.e. the multilayer class of Figure 9.5 realized epitaxially at $\lambda \simeq 1$ μm . Three lessons carry directly into the Level-2.A search. First, the *stop-band width is fixed by index contrast*: for a quarter-wave stack, $\Delta\lambda/\lambda_0 = (4/\pi) \arcsin[(n_H - n_L)/(n_H + n_L)]$, which for GaAs/Al_{0.9}Ga_{0.1}As (3.5/3.0) gives



A slab-like grating with vertical side walls is generally ineffective for both polarizations, and changing the period or thickness does not help. The below table illustrates performance of slab gratings of a fixed thickness of 0.5λ wavelengths for varying period size L_x . Compared with no reflector at all, only a reduction on the order of unity is achievable.



(a) Slab-grating reflector with vertical sidewalls. Period sweep at fixed thickness 0.5λ . Reduction is unity-order; period sweep does not help.



Increasing the slab thickness does not help: see flux plots right below for inverse-designed slab with period 1.0 and thickness 0.8

(b) Slab thickness variation. Increasing thickness alone does not produce strong polarization-balanced suppression of downward flux.

Figure 9.4: Slab-class back-reflector geometries. Geometry, not bulk thickness, sets the achievable reduction factor for this class.

~ 95 nm — exactly the ~ 90 – 100 nm plateau measured. A doped-crystal extractor fits inside that window: the Yb:YLF pump line (1020 – 1030 nm) and the fluorescence band that must also be returned ($\bar{\lambda}_f \simeq 995$ nm, extending to ~ 950 nm) span ~ 80 nm, and sample W1800 places its plateau (~ 945 – 1040 nm, $R \gtrsim 0.97$) over both. A dye or Cr:LiSAF extractor, whose Stokes shift plus emission band exceed 100 nm (Section 8.3.1), does *not* fit, which is the quantitative reason the hybrid metal-underlayer or high-contrast (Si/SiO₂) embodiments of Figures 9.7a and 9.7b are needed there. Second, *growth non-uniformity is a design variable*: the centre-to-edge stop-band shift is ~ 30 nm for W1801 and ~ 60 nm for W1797 (whose edge plateau, at 1000 – 1130 nm, has slid off the fluorescence band entirely), so the tolerance analysis of Level 3 (Section 9.6) must include a radial thickness gradient, and a target that sits at the plateau centre rather than its edge — W1800 — is the robust choice. Third, the plateau's short-wavelength edge is where the loss budget is decided: $1 - R$ at 950 nm is the fraction of the blue tail of the fluorescence (the photons carrying the most heat per photon, Section 8.3.4) lost through the substrate on every pass. Epitaxial DBRs of this kind are also the natural back-reflector for the monolithic semiconductor platform of Section 9.7.5, where the same growth run can carry the GaAs extractor and its mirror.

9.7.2 A critically coupled pump resonator for the molecular-dispersion platform

The second design study is the Level-2.A problem that the molecular-dispersion platform makes hardest: absorbing essentially all of a red-tail pump in a dye layer whose extinction coefficient at the pump wavelength is $\kappa \sim 10^{-4}$ (10^{-2} M R101 at 615 – 640 nm; Figure 9.3) — two orders of magnitude too small for single-pass absorption in any film thin enough to keep its escape efficiency, and two

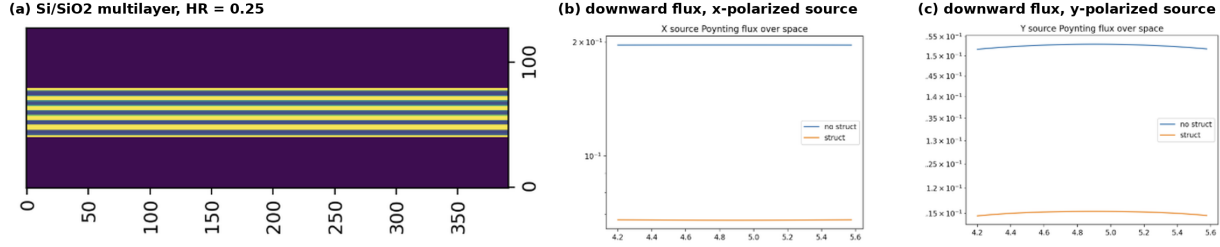


Figure 9.5: Multilayer (alternating Si / SiO₂) back-reflector. (a) Unit cell at total thickness $HR = 0.25\lambda$; (b, c) downward Poynting flux across the cell for an x - and a y -polarized point source, with (blue) and without (orange) the structure. At this thickness the reduction is only ~ 4 – $10\times$; across the sweep 0.25 – 1.5λ of the source study the effectiveness scales with thickness and, for the Si/SiO₂ contrast (~ 11), $\sim 1.2\lambda$ is needed for both polarizations to exceed $20\times$ average reduction.

orders too small for the 10^{-4} M solution-phase baseline to be rescued by any fabricable resonator ($\kappa \sim 10^{-6}$ would need $Q \sim 10^6$). The study (J. Mohajan, 2026) therefore fixes the concentration at 10^{-2} M, accepting the IQE penalty of Section 8.3.1, and asks for a mode of moderate $Q \sim 10^4$ — the pump-side absorption ceiling $n/2\kappa$ of (4.5) — that is critically coupled to a normally incident plane wave.

Geometry. The unit cell (Figure 9.9a) is a square lattice of period $a_x = a_y \simeq 420$ nm with, from top to bottom: air; the dye layer (ethanol + R101, $n \simeq 1.36$, $h \simeq 600$ nm); a *device layer* of thickness $t \simeq 320$ – 420 nm whose in-plane permittivity distribution is the inverse-design variable, filled with the liquid where the high-index material is absent; a quartz spacer of $H \simeq 600$ – 640 nm; and a silver back-mirror. Two device-layer materials were run: GaP ($n \simeq 3.3$) for the first designs at $\lambda = 617$ nm, and SiN ($n \simeq 2.0$) for the smaller-period re-runs at 640 nm. The optimizer’s density variable $\rho \in [0, 1]$ on a 70×70 grid (6 nm pixels) is binarized at convergence; the converged design of Figure 9.9a is essentially a *stack of three high-index bars* (fill fractions 0.4–0.5 per row) separated by liquid-filled gaps, i.e. a coupled multi-slab guided-mode resonance rather than a free-form scatterer — the structure the coupled electro-optic model of Chapter 5 would have suggested by hand, but with the bar thicknesses and gap widths tuned to put the field maximum in the liquid.

Results. At the design wavelength the absorbed power equals the incident plane-wave power to within the discretization error (101%, converging to 100% with resolution): complete absorption of a normally incident pump in a 600 nm dye layer with $\kappa = 10^{-4}$. The field and mode profiles coincide — the incident wave couples to a single mode of $Q \simeq 3500$ in the first (GaP, 617 nm) design. Spectral scans of the periodic structure over 550–640 nm (Table 9.4) show the design mode at 640 nm with $Q = 1.2$ – 1.3×10^4 and $\sim 100\%$ absorptivity, surrounded by a comb of parasitic modes with absorptivities of 0.2–41% and Q from 10^2 to 9×10^4 . Three lessons carry into the loop:

1. *Critical coupling is fragile under perturbation.* Perturbing the geometry does not merely shift the resonance — it breaks the rate match between radiative and absorptive loss, and the absorption collapses. Level-3 must therefore score these designs with a fabrication-tolerance term, and the sensor layer’s fluorescence-flux tap becomes the in-service lock that keeps the pump on the (thermally drifting) resonance.
2. *The parasitic-mode comb is a heating hazard.* Many of the secondary modes originate at the silver interface; they absorb pump power in the metal or, if they overlap the fluorescence band, provide red-shifted emission channels that return photons to the dye. The authors’ remedies —

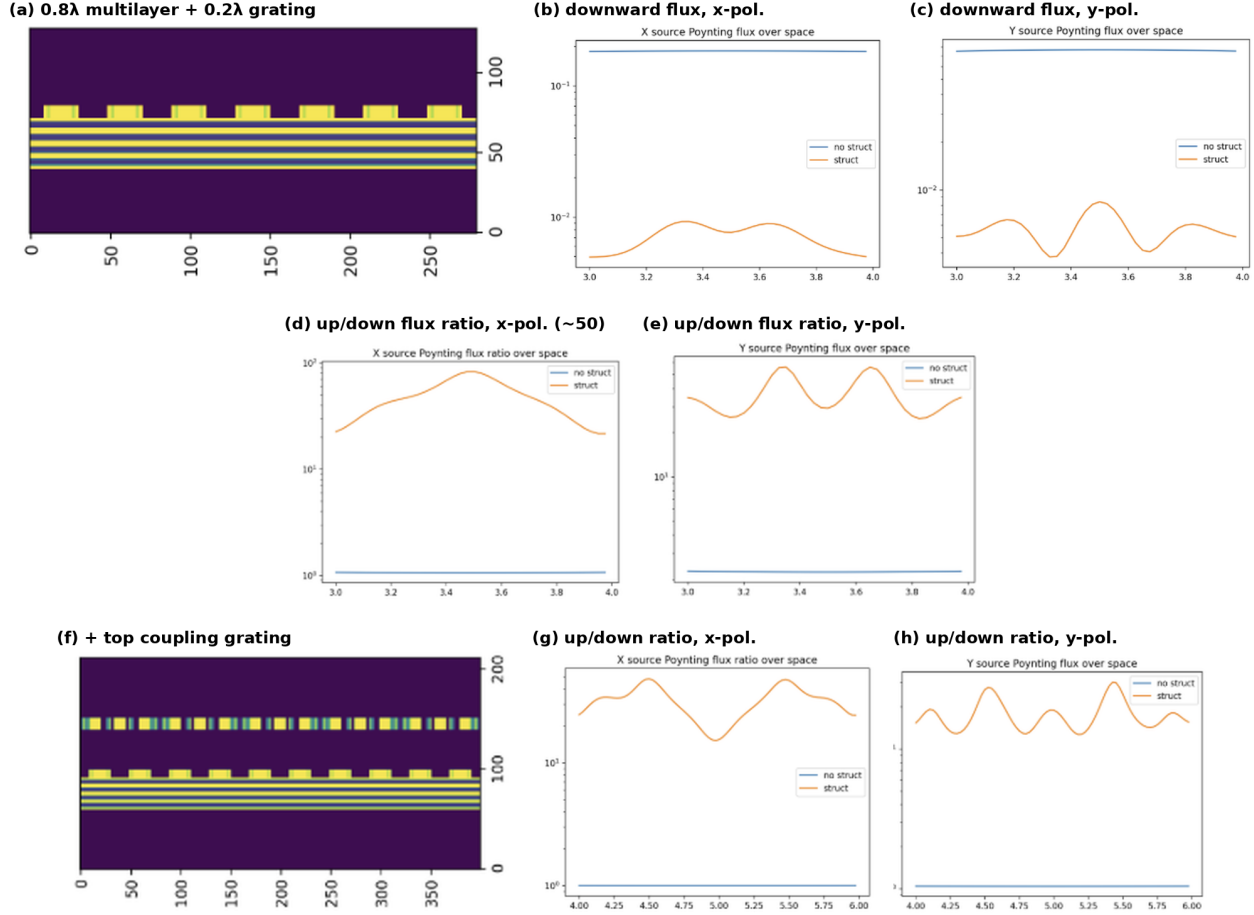
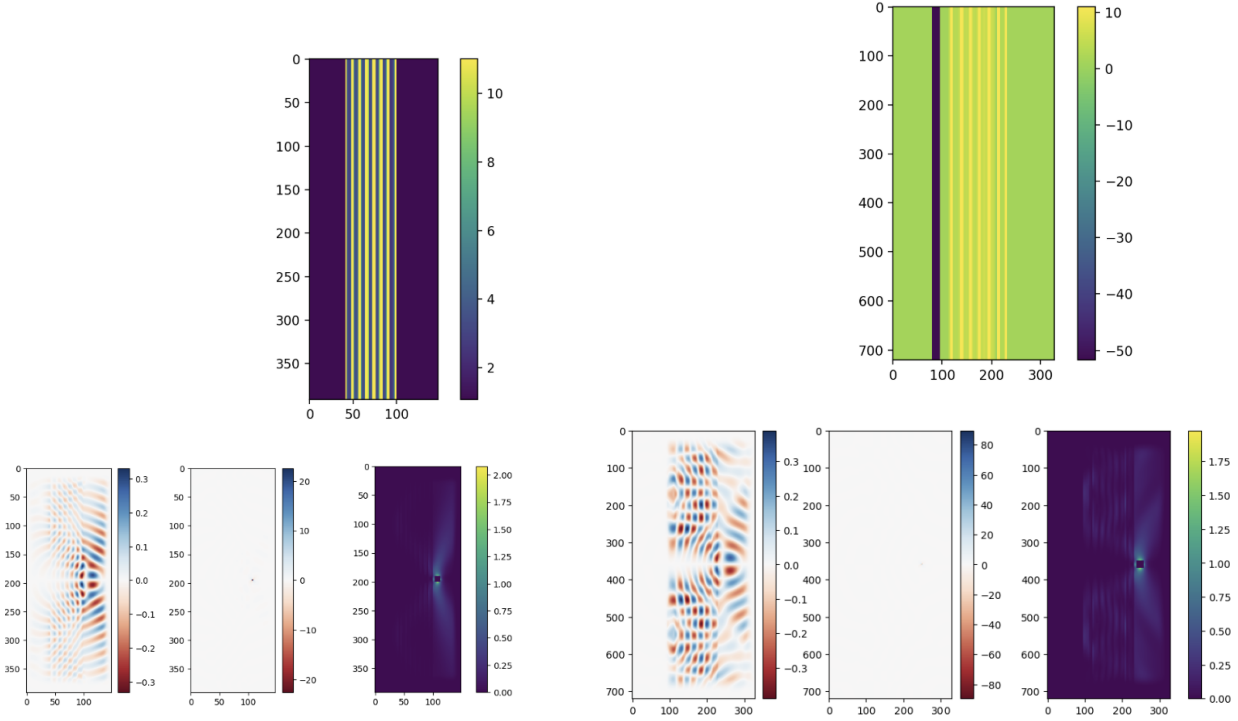


Figure 9.6: Hybrid back-reflector embodiments. (a–c) A 0.8λ multilayer capped by a 0.2λ slab grating (total 1.0λ) suppresses the downward flux by $>20\times$ for both polarizations; (d, e) the resulting up/down flux ratio is ~ 50 on average. (f–h) Adding a top coupling grating (for pump in-coupling / LDOS enhancement) leaves the ratio intact, which is why Level 2.A treats coupler and back-reflector as one jointly inverse-designed stack.

push the mirror further away, shrink the period, and confine absorption to the liquid above the device layer — are all moves that reduce the parasitic α_b of (8.6).

3. *Thicken the liquid and lower Q .* The recommendation to trade Q for path length is the transparency ceiling of Section 8.3.3 seen from the electromagnetic side: at 600 nm of 10^{-2} M dye at 300 K the film can absorb at most $\sim 0.3 \text{ W/mm}^2$ before stimulated emission at ω_p bleaches it and detunes the critical coupling, so a hot-spot tile needs $\gtrsim 20 \mu\text{m}$ of liquid at that concentration (or $\sim 3 \mu\text{m}$ at 400 K, $\sim 0.3 \mu\text{m}$ at 10^{-1} M and 400 K; (8.9)). The resonator should be designed at the operating excitation, with $\kappa_{\text{eff}} = \kappa[1 - x(1 + \sigma_e/\sigma_a)]$.

Where this sits in the loop. The absorption-only study above is the first rung of the Level-2.A programme; it establishes that critical coupling of a red-tail pump into a $\kappa \sim 10^{-4}$ layer is fabricable at $Q \sim 10^4$. The same code base is now being extended to a *joint absorption–emission* multi-objective optimization: maximizing the power emitted by an ensemble of uncorrelated dipoles into a single blue-shifted channel (a Rayleigh-quotient problem on the imaginary part of the Green’s function, i.e. the LDOS) while *minimizing* it at red-shifted frequencies where emission would heat.

(a) Full-dielectric Si/SiO₂ multilayer, HR = 1.5λ.

(b) Same multilayer on a 0.2λ Ag underlayer: extra broadband fluorescence-band reflectivity at the cost of stronger coupling into in-plane guided modes.

Figure 9.7: Full-dielectric versus hybrid metal-underlayer embodiments (top: permittivity map of the unit cell; bottom: field, flux and intensity maps for a point source). Level 2.A trades the metal-induced absorption against the broadband-reflectivity gain on a per-design basis.

In the language of Table 8.2 this is the search for rungs 3 and 4 simultaneously — blue-edge selection of the mean emission and a band-averaged Purcell factor on the total decay rate — under the constraint that the pump-side critical coupling of rung 6 survives. First unfiltered runs prioritize a very high pump Q ($\sim 4 \times 10^4$) with a modest fluorescence Q (~ 160) — the two-ceiling rule of Chapter 8 rediscovered by the optimizer — but also produce undesired red-shifted peaks, which is what the LDOS-suppression objective is being added to remove. Because the Green’s function is analytic in the upper half-plane, the red-shifted suppression can be evaluated as a bandwidth integral by the complex-frequency trick, so one or two additional solves suffice; the remaining challenge is the Brillouin-zone integration needed to represent truly uncorrelated emitters in a periodic structure. From there the programme expands, under the assumption of continued access to extreme-scale compute, to the full loop of Section 9.2: the joint objective is evaluated across the set of all candidate materials, all pump and emission wavelengths, the cascade stage count, and the LPC-recovered wavelengths, with Level 3 pricing manufacturability against the tier targets. The ladder table is the loop’s scoreboard: each generation is scored by which rung it has climbed.

9.7.3 The cascaded extractor and its inter-stage couplers

Tiers II and III of Section 8.3.4 require N extractor stages in series, each performing a small anti-Stokes lift near its own transparency cap and handing its fluorescence to the next. Figure 9.10 shows the arrangement the loop must design and defines the Level-2.A objective for the couplers

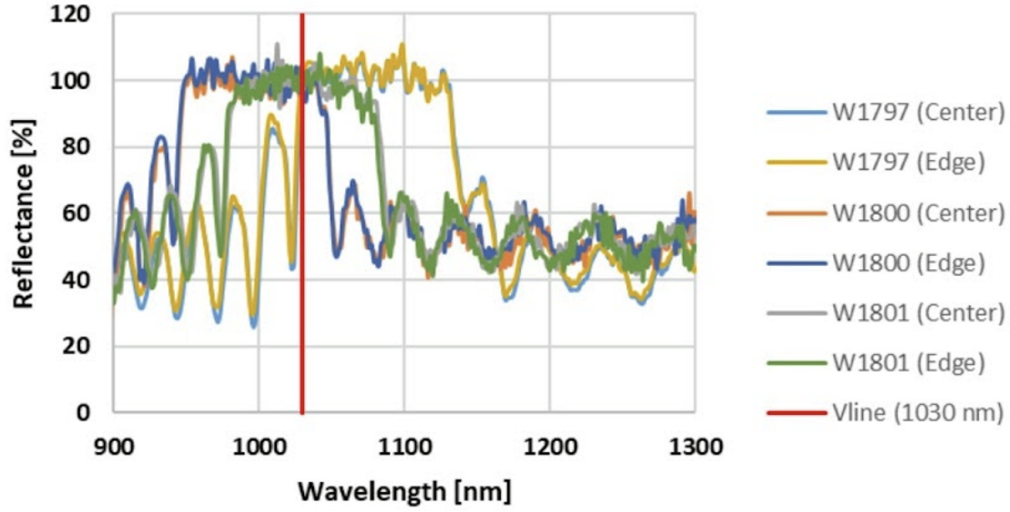


Figure 9.8: Measured reflectance of three GaAs/AlGaAs DBRs (wafers W1797, W1800, W1801; centre and edge of each wafer) grown as back-reflectors for a Yb:YLF extractor; the red line marks the 1030 nm pump. Values above 100 % are a reference-mirror normalization artefact. W1800 is the design point: its ~ 95 nm stop-band covers both the pump and the fluorescence band down to ~ 950 nm at centre and edge alike. Data courtesy of R. Sarma, Sandia National Laboratories [87].

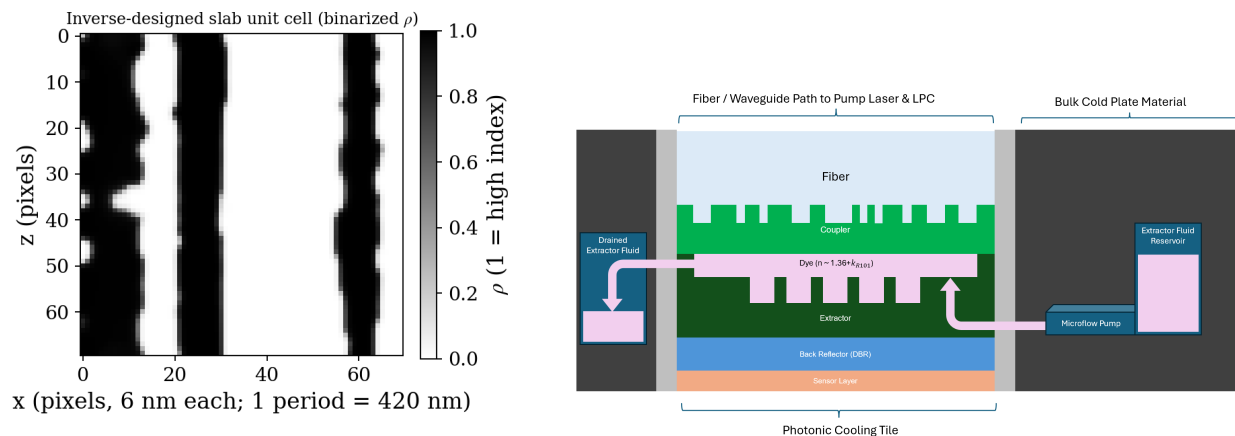
between stages.

Stage ordering. Each stage is characterized by its zero-line energy $E_{00}^{(i)}$, Stokes offset $\delta_i = E_{00}^{(i)} - \hbar\bar{\omega}_{f,i}$ (reduced by blue-edge selection, rung 3), and tail steepness. Stage $i + 1$ must be pumped at $\hbar\bar{\omega}_{f,i}$, a depth $\Delta_{i+1} = E_{00}^{(i+1)} - \hbar\bar{\omega}_{f,i}$ into its tail; its transparency cap is $x_{\max}^{(i+1)} = [1 + e^{\Delta_{i+1}/k_B T}]^{-1}$ and its lift is $\eta_{q,i+1} = (\Delta_{i+1} - \delta_{i+1})/\hbar\bar{\omega}_{f,i}$. The design rule is therefore a constant tail depth per stage, $\Delta_i \simeq 3\text{--}5 k_B T$ ($x_{\max} \sim 10^{-2}$), which with $\delta_i \rightarrow k_B T$ gives $\eta_{q,i} \simeq 3\text{--}6\%$ per stage and fixes the zero-line spacing at $E_{00}^{(i+1)} - E_{00}^{(i)} \simeq \Delta - \delta \simeq 50\text{--}100$ meV: a Tier-II cascade spans ~ 0.5 eV over eight to ten emitters (e.g. a rhodamine / coumarin / pyrromethene sequence, or a graded-composition semiconductor stack), which is why the Level-0 search must range over *all* candidate materials and not one platform family.

Inter-stage coupler objective. For the coupler $C_{i,i+1}$ the Level-2.A objective is the routed fraction

$$\eta_{\text{route},i} = \int d\omega \frac{S_i(\omega) T_{i \rightarrow i+1}(\omega) \mathcal{A}_{i+1}(\omega)}{\int d\omega' S_i(\omega')}, \quad \eta_{\text{cool}}^{(N)} = \prod_{i=1}^N [(1 + \eta_{q,i}) \eta_{\text{EQE},i} \eta_{\text{route},i}] - 1, \quad (9.10)$$

where S_i is the Purcell-shaped emission spectrum of stage i , $T_{i \rightarrow i+1}$ the coupler transmission and $\mathcal{A}_{i+1}$ the absorptivity of stage $i + 1$'s pump resonator. Two constraints make this a genuine inverse-design problem rather than a filter design: the coupler must be *spectrally asymmetric* — transparent to stage i 's emission band but reflective at stage i 's own absorption band, so that photons emitted toward the coupler are not returned into stage i 's reabsorption — and it must present to stage $i + 1$ a critically coupled resonance over the *whole* emission band of stage i (~ 50 nm), not the single line the pump resonator of Section 9.7.2 was designed for. The second requirement is what pushes the



(a) Converged, binarized design variable ρ over one 70×70 -pixel unit cell (6 nm pixels, $a \simeq 420$ nm); dark = high-index device material, white = liquid. The optimizer converges to three coupled slabs of ~ 40 – 50% fill.

(b) Position of the pump resonator in the tile stack (Figure 9.12): the structured device layer is the coupler/extractor interface, the quartz spacer and Ag mirror the back-reflector.

Figure 9.9: Inverse-designed pump resonator for a 10^{-2} M R101 dye layer ($\kappa \simeq 10^{-4}$).

inter-stage resonator toward moderate Q with multiple modes, or toward the broadband-LDOS structures of rung 4, and it is the reason the cascade’s per-stage $\eta_{\text{route},i} \eta_{\text{EQE},i} \geq 0.99$ is a photonic rather than a chemical requirement. With $N = 10$ the tolerance is unforgiving — a 0.98 per stage costs 18% of the lift — so the Level-3 fitness for a cascade individual must score the product, not the mean, of its stage efficiencies. This coupler problem, together with the joint absorption–emission objective above, is the next expansion of the multi-objective inverse design.

9.7.4 A molecular-dispersion-platform photonic cooling tile

The molecular-dispersion platform exercises the fluid loop — extractor fluid reservoir, microflow pump, drained-fluid reservoir — as a Level-3 decision variable, in addition to the standard layer-stack genes. Figure 9.11 shows the full-system embodiment and Figure 9.12 shows the photonic cooling tile in cross-section, with explicit labelling of the layer stack established by the L0–L2 design steps and of the fluid path.

The fluid-loop genes encode flow-rate, channel geometry, and reservoir volumes; their fitness contribution is set by photobleaching kinetics and by the thermal load the fluid stream removes. The advantages of the molecular-dispersion platform are operational rather than performance-limited: the extractor is refreshable without rebuilding the photonic stack, and the fluidic substrate makes Level-1 candidate swapping nearly free relative to a solid extractor’s epitaxial re-growth.

Cooling-feasibility maps for a Rhodamine 101 / ethanol extractor

To make the Level-1 design space concrete for the molecular-dispersion platform, Figures 9.13 and 9.14 render the required escape efficiency η_e^{req} for net cooling of a Rhodamine 101 (R101) / ethanol extractor as a function of pump wavelength λ_p and mean fluorescence wavelength λ_f , at three molar concentrations $M \in \{10^{-4}, 10^{-3}, 10^{-2}\}$ M (columns) and three internal quantum efficiencies $\eta_{\text{IQE}} \in \{0.950, 0.975, 0.990\}$ (rows). Black pixels mark (λ_p, λ_f) pairs at which no escape

Optimized from half-slab				Optimized from full slab			
λ_{peak} (nm)	λ_{mode} (nm)	Absorptivity (%)	Q	λ_{peak} (nm)	λ_{mode} (nm)	Absorptivity (%)	Q
550.75	550.73	32.9	13 100	556.50	556.39	11.6	3 780
575.25	575.24	22.9	20 600	586.50	586.51	31.2	4 680
576.25	576.18	14.6	1 590	595.75	595.67	6.8	4 360
598.25	598.31	20.8	4 310	608.00	608.09	15.6	4 190
608.25	608.23	24.5	2 930	617.00	617.19	0.5	43
628.00	627.98	41.5	3 330	633.00	633.03	27.4	15 400
637.25	637.29	17.2	15 200				
640.00	640.00	100.2	13 300	640.00	640.00	99.9	12 100

Table 9.4: Resonant modes of the two 640 nm inverse-designed pump resonators (SiN device layer, $a \simeq 420$ nm), from spectral scans over 550–640 nm; modes with absorptivity below 2% are omitted. The design mode (bold) reaches complete absorption at $Q \simeq 1.2\text{--}1.3 \times 10^4$, consistent with the pump-side ceiling $Q_{\text{abs}} = n/2\kappa \simeq 7 \times 10^3$ at $\kappa = 10^{-4}$ once the field-fill fraction in the liquid is accounted for. The secondary modes are parasitic absorption channels (largely at the Ag interface) that the follow-on designs suppress by moving the mirror and shrinking the period.

efficiency $\eta_e \leq 1$ yields net cooling — i.e., the parasitic heat load from solvent absorption and residual non-radiative decay cannot be overcome even if every photon that reaches a surface leaves (infeasible region). Coloured pixels report the smallest η_e consistent with net cooling at the panel’s IQE, on a colour scale from 0.25 (deep blue, comfortable) to 1.0 (red, marginal). The quantity plotted is the *escape* efficiency, not the EQE: net cooling always requires $\eta_{\text{EQE}} \geq \lambda_f/\lambda_p \gtrsim 0.9$, and a low required η_e simply means that at high IQE the reabsorption–re-emission cascade recovers most trapped photons. (Note also that the measured IQE of 0.81 for acid-free R640 at 10^{-2} M, and 0.62 for acidified R101, both fall *below* the lowest row of the grid: the 10^{-2} M column is realizable only with SMILES-protected IQE.) Each map is presented in two solvent-absorption variants: “Original” uses the historical R101-in-ethanol parasitic absorption baseline, while “Kedenburg” uses the lower solvent-absorption values reported in the Kedenburg *et al.* low-loss measurements [88]; the comparison brackets the uncertainty in α_b of the host solvent and quantifies how much the feasible region grows as solvent purity improves. Figure 9.14 replaces the earlier free extrapolation of the dye spectra with the anchored-tail model of Section 9.3.2, which fixes the red-pump window quantitatively and exposes its two remaining uncertainties (tail steepness and tail temperature); the response of the same model to a photonically shifted $\bar{\lambda}_f$, and the film-thickness window it implies, are in Figure 8.5.

Three observations follow directly from these maps and they fix the shape of the molecular-dispersion-platform Level-1 search:

1. *Concentration is the dominant lever.* At $M = 10^{-4}$ M the entire (λ_p, λ_f) plane is infeasible regardless of IQE: the absorbed pump fraction is too low to outrun parasitic heating. At $M = 10^{-3}$ M a thin diagonal feasible band emerges along the small anti-Stokes shift; at $M = 10^{-2}$ M the feasible region opens broadly and η_e^{req} drops below 0.5 over a substantial portion of the plane. Concentration is therefore the first decision variable Level 1 tightens, capped from above by concentration-quenching and aggregation thresholds (Section 8.3.1).
2. *IQE expands the feasible region multiplicatively.* Stepping η_{IQE} from 0.950 to 0.990 at fixed $M = 10^{-2}$ M is roughly equivalent — in terms of feasible-region area — to a tenfold increase in concentration. This translates the dye-chemistry literature’s focus on red-shifted, sterically protected R101 derivatives into a quantitative Level-1 fitness gain.
3. *Solvent absorption sets the ceiling.* Comparing the “Original” and Kedenburg panels at fixed

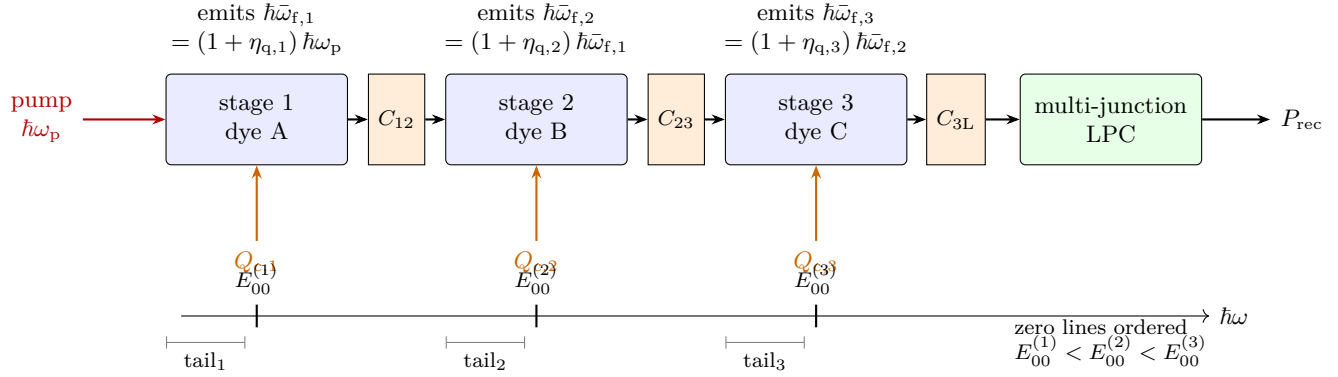


Figure 9.10: Cascaded anti-Stokes extractor. Stage i has zero-line energy $E_{00}^{(i)}$, ordered so that the mean emission of stage i (a few $k_B T$ below its own zero line, but above the pump that fed it) falls in the red tail of stage $i + 1$. Each stage lifts $Q_{c,i}$ at a small $\eta_{q,i}$ near its transparency cap; the couplers $C_{i,i+1}$ are Level-2.A objects that must (i) extract stage i 's emission with $\eta_e \rightarrow 1$, (ii) deliver it into a critically coupled pump resonator of stage $i + 1$, and (iii) keep stage i 's emission out of its own reabsorption band. The final coupler feeds a multi-junction LPC matched to the last line (Section 7.6.3).

M and η_{IQE} shows that improving the solvent's α_b by the factor reported in [88] expands the feasible λ_p window by tens of nm and lowers η_e^{req} by 0.1–0.2 over much of the feasible region. Solvent purity is therefore not a fixed cost the platform pays; it is an ungated decision variable whose impact rivals concentration and IQE.

The role of these maps in the optimization hierarchy is direct: they visualize the Level-1 objective surface against which the inner adjoint loop optimizes. Pixels in the comfortable-blue region are Level-1 *candidate seeds* for the L2.A coupler / back-reflector / sensor inverse design; pixels in the marginal-red region are admissible only when the Level-2.A photonic engineering can deliver the required η_e within the absorption-limited Q ceiling of Section 4.4; black pixels are excluded by Level-0 before the inner loop runs. The one-dimensional tail model of Section 9.3.2 is the λ_f -marginalized cut through these maps and supplies the checkable numbers (+5.3% at 651 nm, $\eta_{EQE} \geq 0.95$) that the colour scale only indicates. Used this way, the maps make precise what Section 8.3.1 could only say qualitatively: that the molecular-dispersion platform's competitive regime is high- M , high-IQE, low- α_b -solvent, with a pump-wavelength window set by the dye absorption tail rather than by the dye peak.

9.7.5 Doped-host- and semiconductor-platform tile reference geometries

For completeness, Figures 9.15 and 9.16 show, respectively, the doped-host- and semiconductor-platform reference tile geometries that the loop produces. In both, the layer order coupler / extractor / back-reflector / sensor of Section 9.4 is preserved; the platform-specific differences are confined to the extractor (Yb:YLF crystal vs. GaAs/GaInP DH) and to the integration mode of the LPC (external recovery vs. monolithic backside).

9.8 Chapter summary

The three-platform ASF photonic cold plate is, in the integration view of this chapter, a single multi-physics object whose design is read off a hierarchical adjoint procedure rather than constructed by hand. The thermodynamic ledger of Chapter 1 is the ledger Level 1 and Level 2.B settle. The

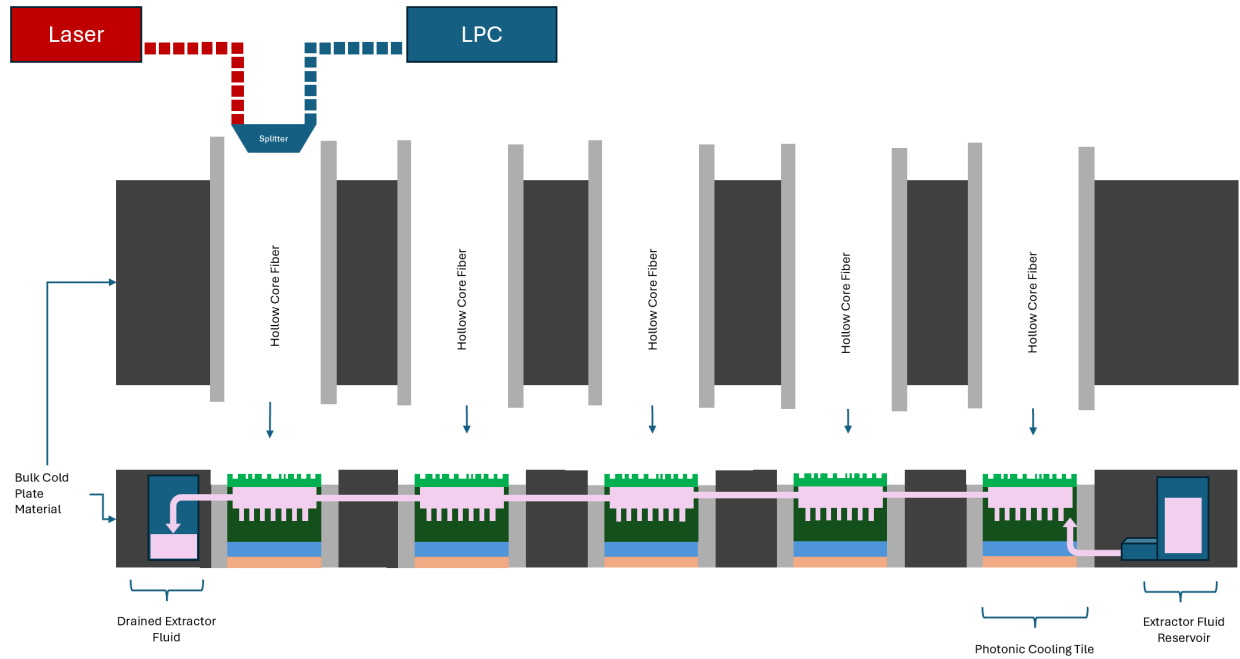


Figure 9.11: Full-system molecular-dispersion-platform embodiment. A remote pump-laser source and LPC are coupled through a splitter into a hollow-core-fiber photonic network that distributes pump light into, and collects fluorescence out of, an array of photonic cooling tiles arrayed across a bulk cold plate. An extractor fluid reservoir feeds dye fluid into the tile via a microflow pump; spent fluid is collected in a drained-extractor-fluid reservoir on the opposing side.

coupled cavity / electro-optic model of Chapter 5 provides Level 1 and Level 2.A’s objective and constraint functions. The exergy and bandwidth limits of Chapter 6 bound Level 2.B. The materials of Chapter 8 are Level 1’s input alphabet and Level 0’s choice. And inverse design — the topic of this chapter — is the search procedure that converts the alphabet into a manufacturable plan.

For HPC hot-spot cooling specifically, the loop’s selection depends on the tile’s thermal zone (Section 10.9). For the $\sim 1 \text{ kW/cm}^2$ (10 W/mm^2) areal target on a compute tile, the semiconductor platform (GaAs / GaInP DH extractor with photonic extraction $\eta_e \rightarrow 0.9$; hybrid DBR-with-Ag-underlayer back-reflector; VO_2 thermo-optic sensor; monolithic backside multi-junction LPC; per-temperature pump wavelength) has the highest volumetric ceiling, at the price of $\eta_{\text{ASF}} \sim 1\%$ and hence a loop that stands or falls on $\eta_P \eta_{\text{LPC}}$. On $\gtrsim 400 \text{ K}$ tiles the SMILES-R640-in-polymer film at $\gtrsim 10^{-1} \text{ M}$ reaches the W/mm^2 -per- $5 \mu\text{m}^2$ class with $5\text{--}10\times$ better pump economy, no epitaxy, and a fully inverse-designable pump resonator (Section 9.7.2) — provided the film EQE clears $0.92\text{--}0.95$, which the acid-free chemistry of Section 8.3.1 makes plausible. The Yb:YLF doped-host platform is selected for sub-ambient storage tiles, for laboratory refrigeration and for the $>97\%$ external-quantum-efficiency demonstrations of Section 8.1.1; the solvent-phase dye tile remains the rapid-prototyping vehicle. Each is the output of the same loop run with different Level-3 weights and a different per-tile T_h .

That the loop produces qualitatively different but physically consistent designs from the same theoretical framework is, in the authors’ view, the integration result this book has been building toward. The hierarchical adjoint procedure is not an afterthought to the physics: it is the operational expression of the physics. What the loop hands to the chip architect is a per-tile capability — a cooling density in W/mm^2 , a pump economy $1/\eta_{\text{ASF}}$, an LPC-recovered fraction, and a spatial /

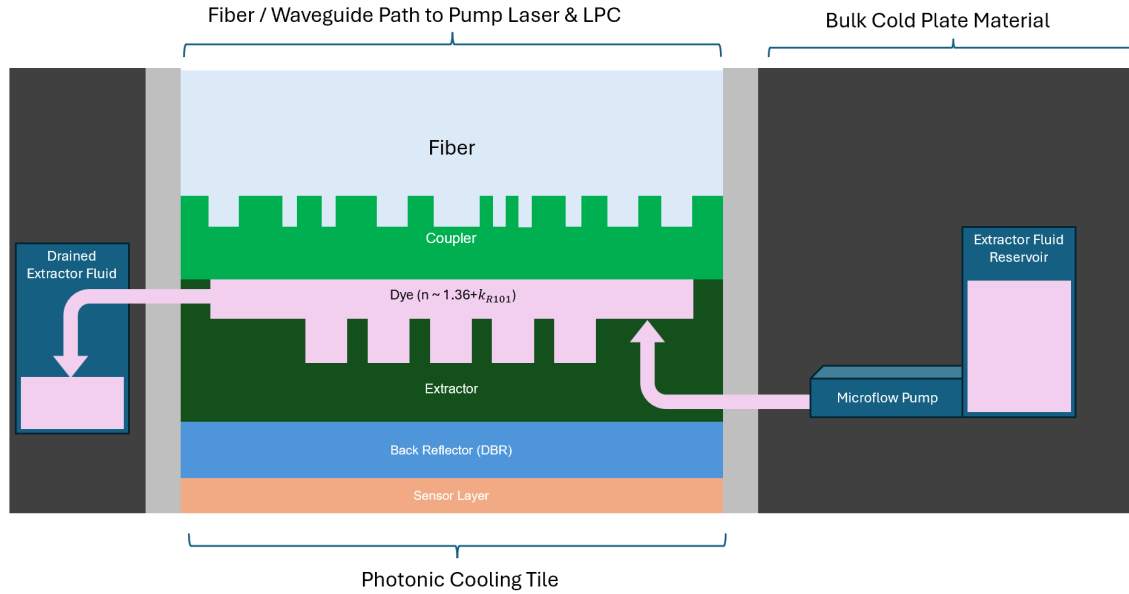
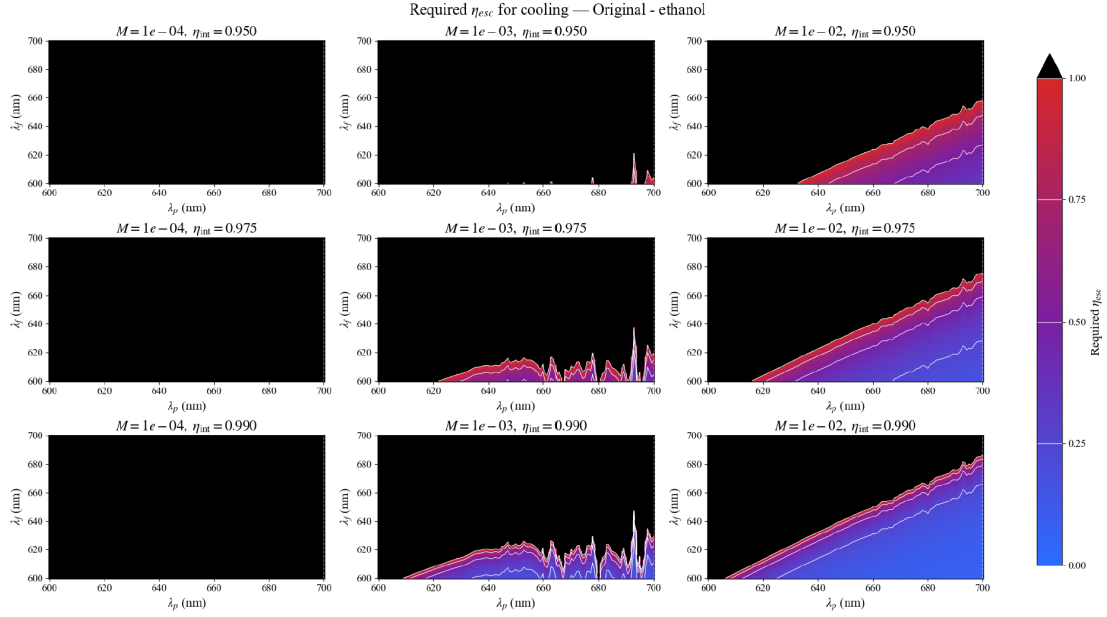


Figure 9.12: Photonic cooling tile detail (molecular-dispersion platform). Pump and fluorescence are routed by a hollow-core fiber / waveguide to a structured coupler at the top of the tile; the coupler is followed by a dye-bearing layer (e.g. Rhodamine 101 in solvent, $n \simeq 1.36 + i k_{R101}$) flowing through structured channels of an extractor host; the extractor sits on a multi-material distributed Bragg reflector back-reflector; the back-reflector is followed by a sensor layer (VO_2 or other thermo-optic thin film). The structured photonic features at the coupler/extractor and back-reflector/extractor interfaces are co-inverse-designed in Level 2.A.

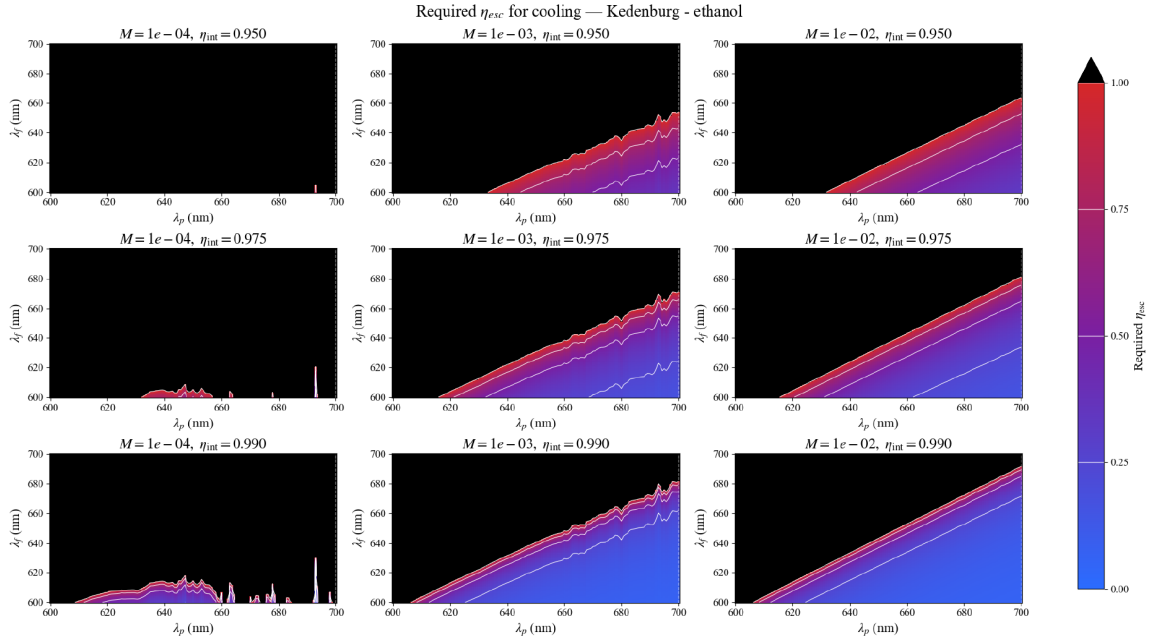
temporal addressing granularity — and Chapter 10 asks what a processor does with it. Reduction-to-practice of the resulting designs, and the experimental demonstration of W/mm^2 -class hot-spot cooling with high-fraction pump recovery, is the work of the manufacturing program for which this book is the theoretical scaffold.

“Original”-Ethanol



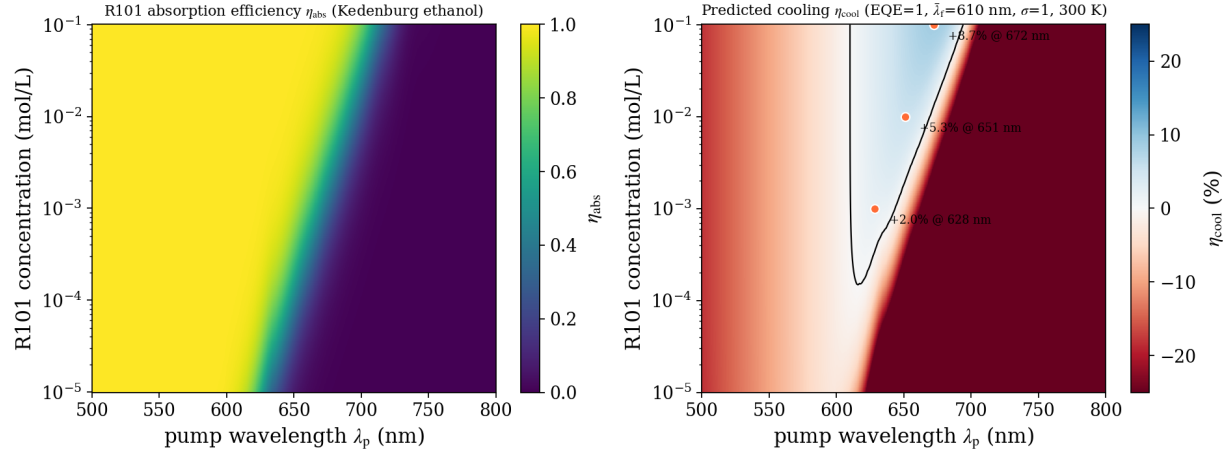
(a) “Original” R101-in-ethanol baseline solvent absorption.

Kedenburg -Ethanol

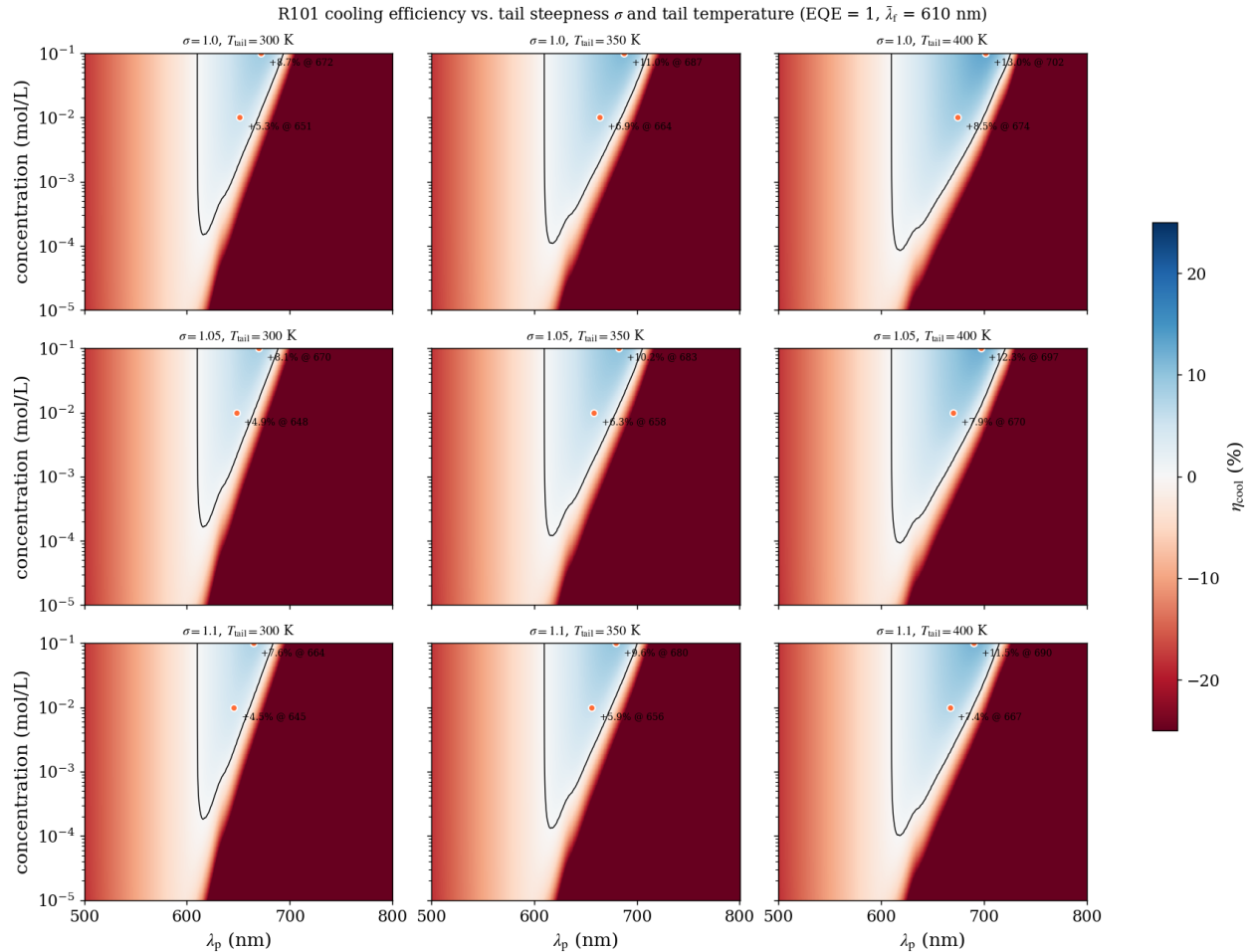


(b) Kedenburg *et al.* low-loss ethanol absorption [88]; reduced α_b expands the feasible (coloured) region noticeably, especially at the lower molar concentrations.

Figure 9.13: Required photon-escape efficiency η_e^{req} for net cooling of an R101/ethanol extractor. Each panel is a 3×3 grid in (M, η_{IQE}) ; axes within each subplot are (λ_p, λ_f) in nm. Black: infeasible at any escape efficiency ≤ 1 . Coloured: minimum η_e at the panel’s IQE (deep blue ~ 0.25 to red ~ 1.0); the corresponding required EQE follows from the recycling identity $\eta_{\text{EQE}} = \eta_{\text{IQE}}\eta_e / [1 - (1 - \eta_e)\eta_{\text{IQE}}]$ and is always $\geq \lambda_f/\lambda_p \gtrsim 0.9$ — an escape efficiency of 0.25 at $\eta_{\text{IQE}} = 0.99$ corresponds to $\eta_{\text{EQE}} = 0.96$. These maps are the molecular-dispersion-platform companion to Table 9.2 for semiconductors and Figure 8.1 for rare-earth-doped hosts.



(a) Absorption efficiency $\eta_{\text{abs}} = \alpha_r / (\alpha_r + \alpha_b)$ (left) and predicted η_{cool} (right) over pump wavelength and concentration, with the $\eta_{\text{cool}} = 0$ contour and the optimum of each decade marked ($\eta_{\text{EQE}} = 1$, $\bar{\lambda}_f = 610$ nm, $\sigma = 1$, 300 K, Kedenburg ethanol).



(b) Sensitivity of the cooling map to the two unknowns of the tail model: steepness σ (rows) and tail temperature (columns). The measured spectrum below 600 nm is held fixed; only the extrapolated tail changes. The feasible region moves red and widens with temperature and narrows with σ ; the optimum η_{cool} at 10^{-1} M ranges from +7.6% ($\sigma = 1.1$, 300 K) to +13% ($\sigma = 1$, 400 K).

Figure 9.14: Cooling-feasibility maps from the anchored-tail constitutive model of Section 9.3.2 (M. Chen, `dye_cooling_model.py`), replacing the earlier free extrapolation of the dye spectra. Compared with Figure 9.13, the model makes the long-wavelength window quantitative: at 10^{-2} M it closes at 669 nm, and the far-red points of the old maps are outside it.

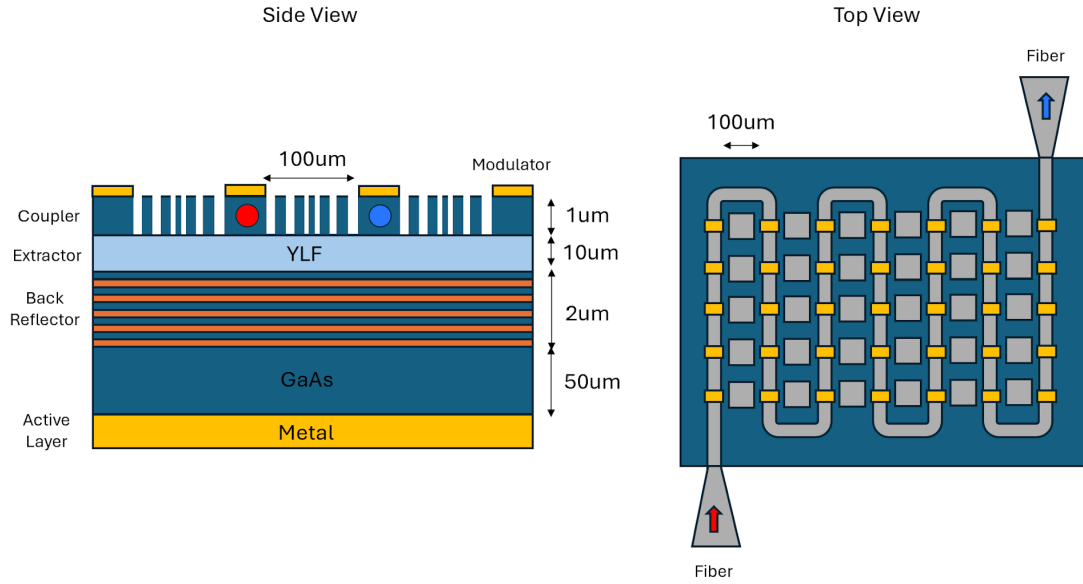


Figure 9.15: Doped-host platform tile reference geometry. Yb:YLF extractor between a structured fused-silica coupler (with optional domed embodiment) and a multi-material DBR back-reflector. The thin-film RTD sensor is patterned on the back-face of the back-reflector. The pump arrives through a single-mode fiber; the LPC is in external recovery.

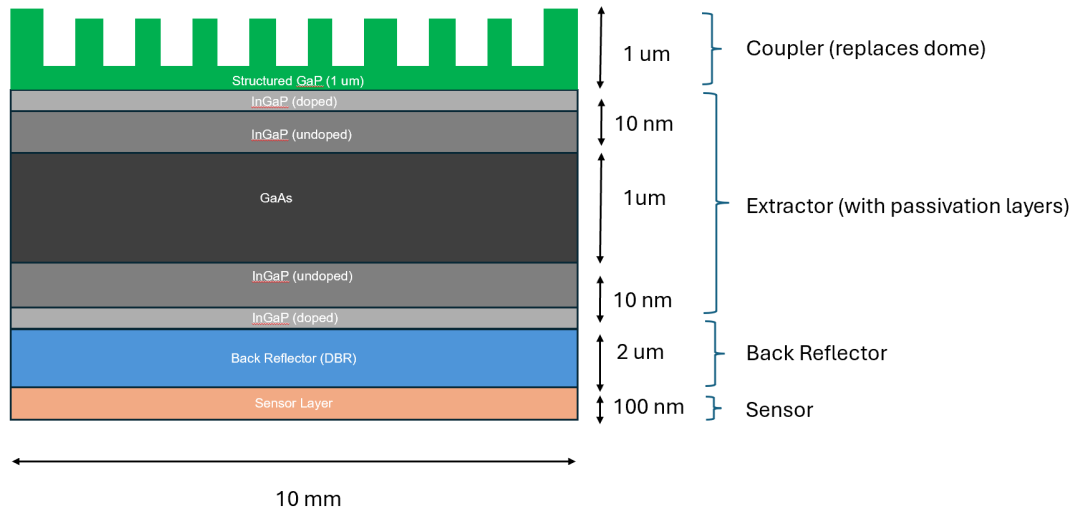


Figure 9.16: Semiconductor-platform tile reference geometry. GaAs extractor with lattice-matched GaInP / AlGaAs passivation windows in a double-heterostructure (DH) configuration; the epitaxial DBR back-reflector and the monolithic backside InGaAs / GaAs multi-junction LPC share the same epitaxial growth as the extractor. A VO_2 thermo-optic sensor at the cold face is interrogated by a telecom-band probe.

Part VI

Architectural Consequences

Chapter 10

Architectural Impacts of Localized Photonic Cooling

10.1 Scope

Chapter 1 introduced the architectural budget inequality (1.32); Chapter 2 showed how photonic microrefrigeration moves the wall; the device chapters (Chapters 4 to 8) built the physics that decides whether and where the move is feasible; and Chapter 9 folded those constraints into the integrated design problem. This final chapter closes the loop on the *architectural* side: given a chip whose package can remove $P_{\text{chip}}^{\text{max,hyb}}(s)$ of (1.33), what does the chip designer do differently? Three threads are unpicked:

- how the dynamic / static / leakage decomposition (1.23) reorganises under tighter T_j control;
- how wire-density and transistor-density budgets evolve when the thermal cap is no longer binding;
- what the *practical* performance-scaling limits are, comparing pure laser cooling against hybrid photonic / convective architectures.

These are engineering judgments built on the physics of the preceding chapters; we treat the formulas as design rules rather than as new derivations.

10.2 The dynamic / static rebalance

Recall the chip-power split (1.23)–(1.26) and the thermal-budget inequality (1.32). With the operating-point junction temperature T_j suppressed by ΔT_{lift} thanks to localised photonic cooling, three knobs reorganise.

Leakage suppression

Subthreshold leakage scales as $I_{\text{leak}} \propto e^{-V_T/(nk_B T_j/q)}$, roughly doubling per ~ 10 K of T_j . A 40 K reduction in junction temperature thus suppresses subthreshold leakage by $2^{40/10} = 16\times$, and gate-oxide leakage by a smaller but still significant factor. Plugging into (1.26):

$$P_{\text{static}}(T_j - \Delta T_{\text{lift}}) \approx P_{\text{static}}(T_j) 2^{-\Delta T_{\text{lift}}/10 \text{ K}}. \quad (10.1)$$

At $\Delta T_{\text{lift}} = 40$ K, the chip's static power drops by an order of magnitude. In modern 5 nm-class chips, P_{static} is ~ 20 – 40 % of total chip power at full clock; suppressing it by $16\times$ recovers ~ 18 – 38 % of the thermal budget back into headroom for P_{dynamic} , *without any process-node change*.

Voltage-frequency reallocation

The recovered headroom flows into either of two architectural moves allowed by (1.30):

1. **Lower- V_{dd} operation at fixed f .** Dropping V_{dd} by 10 % reduces P_{dynamic} by ~ 19 %; if the cooling-recovered budget is partly invested in threshold-voltage scaling, the same workload runs at lower power. This is the energy-efficient operating point.
2. **Higher f at fixed V_{dd} .** Staying at the rated V_{dd} and cashing the headroom in clock frequency: at fixed V_{dd} , each 1 % of recovered budget buys 1 % of clock — the linear regime of (1.30). This is the max-throughput operating point.

The right choice depends on workload mix and is what the architecture’s DVFS controller is optimising over. The point of microrefrigeration is to widen the choice space.

Boost-clock recovery

Modern chips expose a max boost frequency f_{boost} that is reachable only when the worst-case node is below the thermal limit. Without microrefrigeration, this constraint forces $f_{\text{boost}} < f_{\text{rated}}$ at hot operating points (“thermal throttling”). With localised photonic cooling targeting the boost regions (Section 2.7), the boost regime is sustained as long as the photonic loop has the COP_{MR} and the $P_{\text{gen}}^{\text{max}}$ of (1.19) can fund the extra dissipation. Architecturally, this is the difference between a chip that *has* a boost regime and a chip that can *stay* in its boost regime under realistic workloads.

10.3 Wire density, transistor density, and the C_{eff} budget

The effective per-transistor capacitance C_{eff} in (1.24) is a sum of gate, source / drain, and interconnect contributions:

$$C_{\text{eff}} = C_{\text{gate}} + C_{\text{junction}} + C_{\text{wire}}, \quad (10.2)$$

with C_{wire} often dominating at modern nodes ($\gtrsim 60\%$ of the total in ≤ 7 nm processes). The wire capacitance is *set by the architecture*: by the fraction of die area dedicated to interconnect, the average wire length, the metal-pitch hierarchy, and the choice of low- κ dielectrics. When the thermal cap is binding, the chip designer is forced into wire-density configurations that minimise P_{dynamic} at the cost of layout efficiency (longer wire runs, more buffering). Once the cap is relaxed by s , the architect can:

- move closer to a higher-density wire layout, paying more C_{wire} per transistor for shorter, faster signal paths — improving *latency* at fixed throughput;
- integrate larger on-die memories (SRAM, last-level cache), which carry high C_{eff} per bit but improve cache hit rates, indirectly raising the effective compute-per-watt;
- adopt 3D stacking with tight thermal coupling — previously forbidden because the buried layers had no escape path for heat (Section 2.6).

Each of these is a re-allocation of the thermal-budget margin into a layout choice; each is invisible until the margin exists.

Transistor-density correlate. Symmetrically, the transistor density $N_{\text{tr}}/A_{\text{die}}$ sits inside (1.32) as a multiplier on $\alpha C_{\text{eff}} V_{\text{dd}}^2 f$. Pushing the density up by a process-node generation (e.g. from 5 nm to 3 nm, $\sim 1.7\times$ density at constant die area) at constant per-transistor energy raises P_{chip} by $\sim 1.7\times$ before any architectural offset. Without microrefrigeration, the architect must compensate

by lowering α (more dark silicon), lowering V_{dd} (lower clocks), or accepting a higher R_{die} -induced thermal-throttle floor. With s , (1.33) absorbs the increase directly.

10.4 Practical scaling limits: pure laser vs. hybrid

The architectural budget moves with s in (1.33); the question is how far s can practically be pushed. Three regimes:

Hybrid regime ($0 < s < 1$)

Photonic cooling intercepts hot-spot bursts; conventional convection handles the bulk. This is the operating regime of Figure 2.5 and the regime that (1.17) most easily sustains. Scaling limits:

- Convective wall on bulk: $\bar{q}_{bulk} \leq q_{wall}$ of (2.2);
- ASF wall on hot spots: bounded by the absorption ceiling (4.5) and the fluorescence-exergy bound (1.14);
- Aggregate: P_{chip}^{max} bounded by the *larger* of the two walls' contributions, set by which nodes the photonic network can reach.

For a 2D heat-flux landscape with hot-spot-to-mean flux ratios $\alpha/\beta \sim 5\text{--}10$ (Chapter 2) and microrefrigeration deployed at the worst-case nodes, the achievable $P_{chip}^{max,hyb}$ is $\sim 3\text{--}5 \times P_{chip}^{max,conv}$ without changing the package's bulk convective design.

Pure laser-cooled regime ($s \rightarrow 1$)

In the limit $s \rightarrow 1$ every joule of chip dissipation leaves through the photonic network, and the convective system handles only laser / LPC waste heat (a lower-grade thermal load). Equation (1.33) predicts an unbounded budget, but two practical ceilings appear:

1. **Photonic-network throughput.** Each cold-plate tile has a finite cooling density set by its extractor's transparency or Auger ceiling (Table 8.4: $0.1\text{--}10\text{ W/mm}^2$ depending on platform, thickness and tile temperature) and must be fed $1/\eta_{ASF} \simeq 10\text{--}100$ times that in pump flux, i.e. $10\text{--}10^3\text{ W/mm}^2$ of optical power through the network, the coupler and the resonator — well inside the damage envelope of dielectric and polymer films only with a duty cycle (Section 10.8). At $s \rightarrow 1$, every chip-electrical watt must pass through the photonic loop, so the per-tile density and the pump-flux limits translate to a chip-electrical ceiling.
2. **Loop-feasibility binding.** Every additional watt of COP_{MR}^{-1} overhead is electrical input the loop must source. When (1.17) fails (Carnot factor too small at T_h , or η_{ASF} too small), the loop draws on external electrical power that the convective system would have absorbed for free. The trade is not always favourable, and at low $T_h - T_0$ the hybrid regime is decisively better.

Pure laser cooling is therefore the long-horizon endpoint, not the near-term operating regime. Hybrid is the design-current sweet spot.

Net-generation regime ($s > 1$ effective)

When the loop crosses the self-powered threshold of (1.18) and exports $P_{gen} = -P_{net} > 0$, the chip + cold-plate combination acts as a heat-engine / refrigerator hybrid (Section 1.16). The architectural implication is striking: in this regime, raising P_{chip} does *not* simply add cost — it adds exergy that the LPC partially converts back into the supply. Equivalently, the chip's effective electrical-supply cost is $P_{chip}(1 - P_{gen}/P_{chip})$, with the bracket shrinking as T_h rises. Chapter 8

fixes where the door is: with $\eta_P \eta_{\text{cpl}} \simeq 0.75$, (1.17) requires $\eta_{\text{ASF}} \geq 0.33/\phi$, i.e. $\eta_{\text{ASF}} \geq 0.67$ on a 600 K tile and no attainable value on a 350 K one. The single-stage extractors of the present experiments ($\eta_{\text{ASF}} \lesssim 0.1$) are therefore pump recyclers whose electrical cost per watt lifted the LPC brings down to the few- $\times$ range (Section 10.8); the cascaded Tier-III design point of Section 8.3.4 ($\eta_{\text{ASF}} \geq 0.6$) on a wide-bandgap 600 K compute zone is the configuration in which the loop pays for itself. Net generation is thus a joint property of the extractor tier *and* the thermal-zone architecture of Section 10.9: it is reachable, and it is reachable only by designing the chip and the cold plate together.

10.5 The CMOS power equation, revisited

Before separating near-term from long-term implications, recall the chip-power equation that Section 1.18 introduced:

$$P_{\text{chip}} = P_{\text{dynamic}} + P_{\text{static}} \quad (10.3)$$

$$P_{\text{dynamic}} = \alpha N_{\text{tr}} C_{\text{eff}} V_{\text{dd}}^2 f, \quad (10.4)$$

$$P_{\text{static}} \propto N_{\text{tr}} V_{\text{dd}} \exp\left(-\frac{V_{\text{th}}}{nk_{\text{B}}T_{\text{j}}/q}\right). \quad (10.5)$$

Three architectural consequences follow that the rest of the chapter analyses:

- *Cubic frequency scaling* (Equation (1.30)): $P_{\text{dynamic}} \propto f^3$ when V_{dd} tracks f ; the architect cannot push frequency past the thermal cap without paying $\propto f^3$. Photonic cooling moves the cap, but the cubic exponent on *recovered* headroom means a $2\times$ cap-improvement buys $2^{1/3} \sim 1.26\times$ of f (cubic regime) or $2\times$ of f (linear regime).
- *Linear regime determined by V_{th}* : the linear regime is the corner of the (1.28) curve where $V_{\text{dd}} \sim V_{\text{th}}$ — gates barely switch, drive current is small, and $f \propto I_{\text{drive}} \propto (V_{\text{dd}} - V_{\text{th}})^{\alpha_{\text{MOS}}}$. The linear regime ends when V_{dd} nears the threshold; below that, the chip stops working. Photonic cooling does *not* move V_{th} directly, but it relaxes the lower bound on V_{th} (Section 10.2) by suppressing leakage at lower T_{j} , which lets the architect select lower- V_{th} cells where they help.
- *Capacitance and leakage trade-offs*: C_{eff} is set by gate, drain-junction, and wire contributions, with C_{wire} dominating at modern nodes; lower T_{j} reduces leakage exponentially but does not change C_{eff} directly — it just reallocates the thermal budget so that the architect can afford *higher* C_{eff} (denser logic, more on-die memory). The lever is indirect, but compounding.

The first two threads — cubic frequency scaling and the V_{th} trap — determine *near-term* effects (Section 10.7); the wire / capacitance / 3D-stacking arguments determine *long-term* effects (Section 10.10). Before separating those two horizons, Section 10.6 folds them into a single predictive ROI model.

10.6 From thermal limit to ROI: marginal performance-per-total-watt

The most useful question for a chip architect is not “does photonic cooling save power?” It is “does photonic cooling move the chip to a better marginal performance-per-watt frontier than conventional cooling plus voltage-frequency scaling?” This section builds the model that answers that question. Everything collapses to one comparison: the marginal performance-per-total-watt with active-layer

Chip class	Thermal-derived fraction F_T	Hotspot fraction h	Cooling ROI
Thin mobile CPU / SoC	0.60–0.85	0.25–0.55	high
Desktop CPU	0.35–0.65	0.20–0.45	medium-high
Server CPU	0.25–0.50	0.15–0.35	medium
Datacenter GPU / AI accel.	0.25–0.60	0.35–0.70	high if hotspot-limited
HBM / 2.5D / 3D stack	0.50–0.80	0.40–0.80 (local)	high if localized
Near-threshold / IoT	0.10–0.35	0.05–0.20	low

Table 10.1: Starting-point priors for the thermal-derived fraction F_T of the power-injection cap (10.9) and the hotspot-power concentration h for representative chip classes. Values are technical starting estimates, not calibrated constants; they synthesize HotGauge / Boreas hotspot data, dark-silicon scaling [89], Intel / AMD turbo-boost limit structures [90, 91], and GPU power-cap studies of recent H100-class accelerators [92].

microrefrigeration, η_{cool} , versus the marginal performance-per-total-watt available from conventional convective cooling combined with voltage / frequency boost, η_{conv} :

$$\eta_{\text{cool}} = \frac{\Delta \text{Perf}}{\Delta P_{\text{inj}} + Q_c / \text{COP} - \Delta P_{\text{leak}} - \Delta P_{\text{PDN}}}, \quad \eta_{\text{conv}} = \frac{\Delta \text{Perf}_{\text{conv}}}{\Delta P_{\text{inj,conv}} + \Delta P_{\text{fan/pump}}}, \quad (10.6)$$

with the figure of merit

$$\text{FOM} \equiv \frac{\eta_{\text{cool}}}{\eta_{\text{conv}}}, \quad \text{microrefrigeration wins when FOM} > 1. \quad (10.7)$$

The point of the rest of this section is to expand every term in (10.6) into chip-specific, measurable quantities so that the ratio can be evaluated on real silicon.

10.6.1 Full-stack chip-power model

For a hotspot or block i ,

$$P_i = P_{\text{dyn},i} + P_{\text{sc},i} + P_{\text{leak},i} + P_{\text{PDN},i}, \quad (10.8)$$

with the usual CMOS expansions $P_{\text{dyn},i} = \alpha_i C_i V_i^2 f_i$, $P_{\text{leak},i} = V_i I_{\text{leak},i}(T_i, V_i, V_{\text{th}})$, and the IR-loss term $P_{\text{PDN},i} = I_i^2 R_{\text{PDN},i}(T_i)$. The maximum *useful* injected power into block i is then the minimum of an entire stack of caps:

$$P_{0,i} = \min \left[P_{\text{thermal}}, P_{\text{timing}}(T), P_{\text{leak/run}}(T), P_{\text{EM}}(T), P_{\text{droop}}, P_{\text{pkg/I}}, P_{\text{firm/plat}} \right]. \quad (10.9)$$

This is the central observation behind the model: microrefrigeration only relaxes the *temperature-coupled* terms $P_{\text{thermal}}, P_{\text{timing}}(T), P_{\text{leak/run}}(T), P_{\text{EM}}(T)$. Hard electrical caps — package current, VRM current, motherboard IR drop, firmware power policy, rack or battery limits — are unmoved by cooling. Whether photonic cooling has ROI on a given chip depends on which line of (10.9) is actually binding for that chip.

Thermal-fraction estimates per chip class. A coarse but useful summary of how much of each chip class’s injection limit is temperature-derived (denoted F_T , the fraction of P_0 recoverable by lowering hotspot temperature) is given in Table 10.1. Thin / mobile parts are overwhelmingly thermally limited; server CPUs are bound by socket and rack constraints long before thermal; GPUs / AI accelerators are workload-dependent; HBM / 3D stacks are spreading-resistance limited.

Mechanism	What it changes	Coupling to ΔT
Leakage suppression	static power P_{leak}	$\Delta P_{\text{leak}} \propto s_{\text{leak}} \Delta T$
Timing-guardband relief	f_{max}	$\Delta f / f \propto s_f \Delta T$
Throttle avoidance	DVFS clipping residency	step in L_{throttle}
Turbo / boost recovery	sustained V, f headroom	via $\Delta P_{\text{thermal}}$
EM / reliability uplift	allowable j at fixed MTTF	j up via Black's law
PDN-loss reduction	$I^2 R$ interconnect loss	$\Delta P_{\text{PDN}} \propto s_R \Delta T$
Local spreading bypass	directly attacks $R_{\theta,s}$	$T_{\text{peak}} \rightarrow T_{\text{peak}} - Z_{\theta,c} Q_c$

Table 10.2: Mechanisms through which ΔT drives performance gain or power saving. Each appears in the per-mechanism expressions (10.14)–(10.20).

10.6.2 Thermal model and the convection wall

The peak local temperature at hotspot i under conventional cooling is

$$T_{\text{peak},i} = T_{\text{amb}} + R_{\theta,g} P_{\text{chip}} + R_{\theta,s,i} P_{\text{hot},i}, \quad (10.10)$$

with the hotspot power concentration written as

$$P_{\text{hot},i} = h_i P_{\text{chip}}, \quad (10.11)$$

where h_i is the fraction of total chip power that lands in hotspot-prone structures — vector / tensor units, register-file ports, clock generators, scheduler / rename logic, SRAM ports, HBM PHYs, NoC arbiters. $R_{\theta,g}$ is the package-to-ambient global thermal resistance set by the cold plate and lid; $R_{\theta,s,i}$ is the *local spreading resistance* from the hotspot down to the bulk silicon and the spreader.

The convection wall appears when

$$R_{\theta,s,i} P_{\text{hot},i} \gg R_{\theta,g} P_{\text{chip}}. \quad (10.12)$$

At that point, adding fan / pump power lowers $R_{\theta,g}$ but barely shifts the binding term. This is exactly the regime called out by the HIR thermal roadmap, which projects that future thermal solutions must cover hotspot density envelopes roughly $2\text{--}4\times$ *the average supported power density* — conventional convection sees the average; the architect hits the wall at the local peak [93]. The 2024 micro-TED work reports ALU / clock-generator average fluxes around $\sim 1 \text{ kW/cm}^2$ with local peaks several times the surrounding regions, and demonstrates active first-harmonic hotspot oscillation cancellation up to 2.35 K — a direct experimental signature of this regime [4].

10.6.3 What microrefrigeration changes

Active-layer cooling lowers the local hotspot temperature by

$$\Delta T_i = Z_{\theta,\text{cool},i} Q_{c,i}, \quad (10.13)$$

where $Z_{\theta,\text{cool},i}$ is the local cooling impedance (steady-state limit: $Z_{\theta,\text{cool},i} \approx R_{\theta,\text{local},i}$; transients: $\Delta T_i(\omega) = |Z_{\theta,i}(\omega)| Q_{c,i}(\omega)$). That single temperature reduction propagates through five quantitatively distinct mechanisms, summarized in Table 10.2.

Leakage saved.

$$\Delta P_{\text{leak},i} \approx P_{\text{chip}} f_{\text{leak}} f_{\text{cooled},i} s_{\text{leak}} \Delta T_i, \quad (10.14)$$

Block class	s_f (%/°C)
CPU scalar core	0.1–0.4
AVX / vector unit	0.2–0.8
GPU SM / tensor block	0.1–0.5
SRAM / cache path	0.05–0.3
HBM PHY / controller	0.1–0.5

Table 10.3: Per-block timing sensitivity s_f used in (10.17). These capture timing guardband relief as well as transistor-physics $f_{\max}(T)$, and are starting priors rather than calibrated constants.

with $f_{\text{leak}} = P_{\text{leak}}/P_{\text{chip}} \sim 0.10\text{--}0.35$ depending on node, voltage and workload; $f_{\text{cooled},i}$ the fraction of leakage-active area actually cooled; and the per-degree sensitivity $s_{\text{leak}} \sim 0.03\text{--}0.08/^\circ\text{C}$ at modern nodes. Worked instance: a 300 W GPU with $f_{\text{leak}} = 0.15$, $f_{\text{cooled}} = 0.10$, $s_{\text{leak}} = 0.05/^\circ\text{C}$, and $\Delta T = 10\text{ K}$ gives $\Delta P_{\text{leak}} \approx 2.3\text{ W}$ — a real contribution to the denominator of (10.6).

PDN / interconnect loss saved.

$$\Delta P_{\text{PDN},i} \approx P_{\text{PDN},i} f_{\text{PDN cooled},i} s_R \Delta T_i, \quad (10.15)$$

with copper-resistance temperature coefficient $s_R \approx 0.0039/^\circ\text{C}$. Smaller than ΔP_{leak} at typical ΔT , but non-negligible in dense backside-PDN, bump, TSV, and high-current GPU / AI grids.

Thermal headroom recovered. The extra injectable power that the same thermal cap allows after cooling is

$$\Delta P_{\text{thermal},i} = \frac{\Delta T_i}{R_{\theta,g} + R_{\theta,s,i} \psi_i}, \quad (10.16)$$

where $\psi_i \in [0, 1]$ is the fraction of newly injected power that lands back in the same hotspot (we reserve ϕ for the Carnot factor $1 - T_0/T_h$). Typical priors: $\psi \sim 0.3\text{--}0.7$ for core-wide boost, $0.7\text{--}1.0$ for tight FU / vector boost, and $0.1\text{--}0.3$ for memory-bound workloads.

Timing / frequency gain.

$$\Delta \text{Perf}_{\text{timing},i} \approx U_i s_{f,i} \Delta T_i, \quad (10.17)$$

with utilization U_i of the cooled block and the empirical per-degree sensitivity $s_f = (1/f)(\partial f_{\max}/\partial T)$ falling in the ranges of Table 10.3.

Throttle avoidance. Often the dominant term. If the chip has measured throttle residency or frequency-cliff loss $L_{\text{throttle},i}$,

$$\Delta \text{Perf}_{\text{throttle},i} \approx W_i \min \left[L_{\text{throttle},i}, \frac{\Delta T_i}{T_{\text{peak},i} - T_{\text{throttle},i}} L_{\text{throttle},i} \right], \quad (10.18)$$

where W_i is the workload’s sensitivity to that block’s clipping. If ΔT is large enough to drag the peak below the throttle threshold, the entire L_{throttle} loss is recovered — a discrete cliff, not a smooth gain.

Regime	ϵ_P
CPU turbo region	0.3–0.8
CPU near $f_{\max}$	0.1–0.4
GPU compute-bound	0.3–0.7
GPU memory-bound	0.05–0.25
HBM-throttled	0.4–1.0
Server throughput at efficiency knee	0.05–0.3

Table 10.4: Marginal performance elasticity to power ϵ_P used in (10.20) (written ϵ_P , not η_P , to avoid the pump wall-plug efficiency η_P of Table 1.2). In the cubic regime of (1.30), ϵ_P saturates as V_{dd} tracks f .

Metric	Source	Used in
P_{chip} , block power P_i	telemetry / TDP	(10.14),(10.20)
Hotspot fraction h_i	block power maps / PMU	(10.11),(10.10)
$T_{\text{peak}}, T_{\text{throttle}}, T_{\text{max}}$	on-die sensors / datasheet	(10.10),(10.18)
Throttle residency, requested-vs-actual f	PMU / DVFS log	(10.18)
Leakage fraction f_{leak}	silicon char. at V, T	(10.14)
Current droop / VRM headroom	PDN model + telemetry	(10.19)
Package / firmware / rack caps	platform documentation	(10.9)
Block utilization U_i	PMU	(10.17)
$R_{\theta,g}, R_{\theta,s,i}, Z_{\theta,\text{cool},i}$	thermal FEM / IR map	(10.10),(10.13)
$Q_{c,i}$, COP	microcooler spec	(10.6)

Table 10.5: Plug-in quantities required to evaluate (10.23) on a real chip. All are either directly measurable in production telemetry or estimable from layout / thermal models.

Boost / extra injected power. The extra usable injected power is the minimum of the temperature-coupled and the hard caps:

$$\Delta P_{\text{inj},i} = \min[\Delta P_{\text{thermal},i}, \Delta P_{\text{timing},i}, \Delta P_{\text{EM},i}, \Delta P_{\text{droop},i}, \Delta P_{\text{plat},i}], \quad (10.19)$$

and the performance from spending it is

$$\Delta \text{Perf}_{\text{boost},i} = \epsilon_{P,i} \frac{\Delta P_{\text{inj},i}}{P_i}, \quad (10.20)$$

with $\epsilon_{P,i}$ the marginal performance elasticity to power at the current operating point (priors in Table 10.4).

10.6.4 Unified figure of merit

Summing the gain mechanisms in the numerator and collecting power savings into the denominator of (10.6):

$$\Delta \text{Perf} = \Delta \text{Perf}_{\text{boost}} + \Delta \text{Perf}_{\text{timing}} + \Delta \text{Perf}_{\text{throttle}} + \Delta \text{Perf}_{\text{TDP}} + \Delta \text{Perf}_{\text{latency}}, \quad (10.21)$$

$$\Delta P_{\text{total}} = \Delta P_{\text{inj}} + \frac{Q_c}{\text{COP}} - \Delta P_{\text{leak}} - \Delta P_{\text{PDN}}, \quad (10.22)$$

so the predictive form of (10.6) becomes

$$\eta_{\text{cool},i} = \frac{\Delta \text{Perf}_{\text{boost}} + \Delta \text{Perf}_{\text{timing}} + \Delta \text{Perf}_{\text{throttle}} + \Delta \text{Perf}_{\text{latency}}}{\Delta P_{\text{inj}} + Q_c/\text{COP} - \Delta P_{\text{leak}} - \Delta P_{\text{PDN}}}. \quad (10.23)$$

The model becomes predictive once every ΔP_* and ΔPerf_* on the right is computed from the chip-specific quantities of Table 10.5.

Chip class	Dominant binding limit	Predicted $\eta_{\text{cool}}/\eta_{\text{conv}}$
Thin mobile CPU / SoC	form-factor cooling, local turbo	1.5–4.0
Desktop CPU	boosted core / AVX hotspot	1.1–2.5
Server CPU	socket / rack power	0.8–1.8
Datacenter GPU (compute-bound)	tensor / SM hotspots	1.2–3.0
Datacenter GPU (memory-bound)	board power, HBM bandwidth	0.7–1.2
HBM / 2.5D / 3D stack	buried spreading resistance	1.5–5.0
Near-threshold / IoT	voltage floor / leakage minimum	0.5–1.1

Table 10.6: Predicted ratio of cooling-enabled to convection-enabled marginal performance-per-watt for representative chip classes. Model outputs from (10.23) given the priors of Table 10.1.

10.6.5 Architecture-class predictions

Combining the priors of Table 10.1 with the gain decomposition (10.21) and the cost (10.22) yields predicted ratios $\eta_{\text{cool}}/\eta_{\text{conv}}$ per chip class (Table 10.6). The ranges reflect workload sensitivity: the high end applies when the workload is hotspot-bound, the low end when it is package-current- or memory-bandwidth-bound.

10.6.6 Worked example: GPU tensor hotspot, COP = 0.3

To show that even very low-COP active-layer cooling can be ROI-positive when applied as a hotspot razor rather than a refrigerator, take a datacenter GPU with the following observable state:

- $P_{\text{chip}} = 700 \text{ W}$, tensor-block hotspot $P_{\text{hot}} = h P_{\text{chip}}$ with $h \sim 0.5$;
- active heat removed $Q_c = 2 \text{ W}$ at COP = 0.3, so $Q_c/\text{COP} = 6.67 \text{ W}$;
- local cooling impedance $Z_{\theta, \text{cool}} = 4 \text{ K/W}$ giving $\Delta T = 8 \text{ K}$;
- leakage saved $\Delta P_{\text{leak}} = 700 \times 0.15 \times 0.10 \times 0.05 \times 8 \approx 4.2 \text{ W}$ per (10.14) with $f_{\text{leak}} = 0.15$, $f_{\text{cooled}} = 0.10$, $s_{\text{leak}} = 0.05/^\circ\text{C}$;
- PDN saved $\Delta P_{\text{PDN}} \approx 0.5 \text{ W}$;
- extra usable injected power $\Delta P_{\text{inj}} = 20 \text{ W}$;
- realised performance gain $\Delta \text{Perf} \approx 5\%$ (boost + recovered throttle residency).

Plugging into (10.22) and (10.23),

$$\Delta P_{\text{total}} = 20 + 6.67 - 4.2 - 0.5 = 21.97 \text{ W}, \quad \eta_{\text{cool}} = \frac{5\%}{21.97 \text{ W}} \approx 0.228 \text{ \%/W}. \quad (10.24)$$

A conventional boost path on the same GPU — spend 40 W of board power for 3% gain — yields $\eta_{\text{conv}} \approx 0.075 \text{ \%/W}$, so the figure of merit is

$$\text{FOM} = \frac{0.228}{0.075} \approx 3.0, \quad (10.25)$$

i.e. at COP = 0.3, a 2 W active-layer hotspot razor outperforms conventional boost by $\sim 3\times$ in marginal performance-per-watt — *provided* the 2 W is deposited on the right active-layer hotspot. The same 2 W spread over a broad region without timing or boost leverage *fails*: the model is exquisitely sensitive to hotspot localisation. This is the central design rule for the low-COP regime: Q_c need only be ~ 0.3 – 1.5% of chip power, but it must unlock 2–5% of performance. Note also what COP = 0.3 implies at the tile: with $\eta_{\text{ASF}} \simeq 0.05$ (Chapter 8) the gross wall-plug COP of the loop is $\eta_P \eta_{\text{ASF}} \simeq 0.035$, and 0.3 is reached only with $\eta_P \eta_{\text{LPC}} \eta_{\text{cpl}} \simeq 0.85$ of pump recovery (Section 10.8); the example is therefore a recovery-enabled operating point, and at the gross value the same 2 W razor would cost 57 W and still return FOM $\simeq 0.9$.

10.6.7 Time-forward scaling and the convection wall

The ROI of microrefrigeration scales over time exactly the way the convection wall scales *against* conventional cooling. With $h(t)$ the hotspot power fraction (10.11) and $R_{\theta,s}(t)$ the local spreading resistance, the trend lines that matter are

$$\frac{dh}{dt} > 0, \quad \frac{dR_{\theta,s}}{dt} > 0, \quad \frac{d\eta_{\text{conv}}}{dP_{\text{chip}}} < 0. \quad (10.26)$$

The first two are forced by wider tensor / matrix engines, denser SRAM and register-file ports, chiplet adjacency, HBM stacking, 3D integration, and backside power delivery, all of which put more of the chip's power into a localised peak. The third is forced by the cubic-frequency-scaling argument (1.30): each additional watt of package power buys less performance because of the $P_{\text{dyn}} \propto V_{\text{dd}}^2 f$ cost of boost. Combined, (10.26) guarantees that

$$\boxed{\eta_{\text{cool}} > \eta_{\text{conv}} \quad \text{when} \quad R_{\theta,s,i} P_{\text{hot},i} \gg R_{\theta,g} P_{\text{chip}},} \quad (10.27)$$

i.e. when convection's leverage is on the wrong term. The practical consequence is that microrefrigeration's ROI *rises with time* on exactly the architectures the industry is converging toward — AI accelerators, HBM-stacked logic, chiplet-dense GPUs, form-factor-constrained client devices — regardless of whether the absolute COP improves. This is the central scaling thesis of the chapter.

10.7 Near-term implications: current chips, current packages

What changes when photonic cooling is dropped on a chip that already exists

In the near-term scenario, the chip's process node, transistor geometry, and V_{th} cells are fixed. Only the package changes. The microrefrigeration cap of (1.33) permits a larger sustained P_{chip} without exceeding $T_{\text{j,max}}$. The near-term wins are immediate and redistributive:

1. **Sustained boost-clock recovery.** Modern CPUs and GPUs expose a max boost frequency f_{boost} that is reachable only when the worst-case node is below the thermal limit. Without microrefrigeration, this constraint forces the chip into thermal throttling: boost is sustained only for short bursts before T_{j} hits the trip point and f is dropped. Photonic cooling targeting boost-active functional units (Section 2.7) sustains f_{boost} under realistic workloads. Architecturally, this is the difference between a chip that *has* a boost regime and a chip that can *stay* in its boost regime; no process-node change required.
2. **Leakage-power recovery.** Per (10.1), a 40 K reduction in T_{j} suppresses subthreshold leakage by $16\times$ at fixed V_{th} . In modern 5 nm-class chips this recovers $\sim 18\text{--}38\%$ of the thermal budget back to P_{dynamic} . The recovered budget can be cashed either as $\sim 10\%$ lower V_{dd} at fixed f (energy-efficient operating point) or as $\sim 10\%$ higher f at fixed V_{dd} (throughput operating point). The DVFS controller picks based on workload mix; the architect's job is to expose the choice.
3. **Dark-silicon recovery.** When the average per-area power density exceeds the package's removal capacity, the architect gates idle functional units to keep $\langle q \rangle$ below q_{passive} . This is dark silicon, an architectural response to (1.32). Photonic cooling at the active functional units eliminates the need for the gating: instead of dimming the chip globally, cooling can be routed to the locally-active regions in real time (Section 2.7). Unlocking this can recover $\sim 20\text{--}40\%$ of nominal silicon area for compute — a near-term performance lever that no process node can match.

4. **Compute-efficiency improvement.** Sustained boost is throughput / latency; leakage recovery is energy efficiency. Combined,

$$\eta_{\text{compute}}^{\text{photonic}} \sim \eta_{\text{compute}}^{\text{conv}} \frac{P_{\text{dyn, recovered}}}{P_{\text{dyn, baseline}}} \frac{f_{\text{boost, sustained}}}{f_{\text{rated}}}, \quad (10.28)$$

which for current AI accelerators yields $\sim 1.5\text{--}2\times$ compute-per-watt at the same process node — a process-node-equivalent improvement without the multi-billion-dollar fab investment.

What does *not* change in the near term

The architecture itself. The chip’s transistor count, microarchitecture (pipeline depth, ROB width, cache hierarchy), and ISA are all fixed. The cooling cap moves; the chip uses the moved cap. No-rebuild deployment is the near-term value proposition: existing chips, existing fabs, existing software stacks — only the package changes.

10.8 Pump economy, power recovery, and what targeting buys

The ROI model above treats the tile as a black box with a COP. The device chapters fix what is inside the box, and three of their conclusions reshape the architectural picture enough to be restated here in the architect’s units.

The loop is a pump recycler, not a heat engine

Every platform in Table 8.4 has an anti-Stokes efficiency bounded by its quantum defect, $\eta_{\text{ASF}} \leq \bar{\omega}_f/\omega_p - 1$: $\simeq 0.02\text{--}0.03$ for Yb:YLF, $\simeq 0.01\text{--}0.035$ for GaAs, $\simeq 0.03\text{--}0.08$ for a red-tail-pumped rhodamine. A tile that lifts Q_c therefore absorbs $Q_c/\eta_{\text{ASF}} \simeq 12\text{--}50 Q_c$ of pump, and re-emits $(1 + \eta_{\text{ASF}})Q_c/\eta_{\text{ASF}}$ as fluorescence. Where that fluorescence goes decides the loop’s economics:

$$\frac{P_{\text{elec}}}{Q_c} = \frac{1}{\eta_{\text{ASF}}} \left[\frac{1}{\eta_P} - \eta_{\text{LPC}} \eta_{\text{cpl}} (1 + \eta_{\text{ASF}}) \right], \quad \text{COP}_{\text{net}} = Q_c/P_{\text{elec}}. \quad (10.29)$$

Without recovery ($\eta_{\text{LPC}} = 0$) the cost is $1/(\eta_P \eta_{\text{ASF}}) \simeq 30\text{--}70$ electrical watts per watt lifted — a COP of $0.015\text{--}0.035$, an order of magnitude below a μTEC . With $\eta_P = \eta_{\text{LPC}} = 0.7$, $\eta_{\text{cpl}} = 1$ and $\eta_{\text{ASF}} = 0.05$ it is 14 (COP = 0.07); with $\eta_P = \eta_{\text{LPC}} = 0.85$ it is 5.7 (COP = 0.18); the COP = 0.3 of the worked example needs $\eta_P \eta_{\text{LPC}} \eta_{\text{cpl}} \simeq 0.85$. Three architectural facts follow. First, the LPC is not an efficiency refinement; it is the difference between a device that is uncompetitive with a μTEC and one that beats it. Second, the recovered electrical power is of order Q_c/η_{ASF} , i.e. $10\text{--}50\times$ larger than the cooling it enables: a 2 W razor circulates ~ 40 W of optical power and returns ~ 30 W electrically. The LPC, its thermal management (it dissipates $(1 - \eta_{\text{LPC}})$ of that flow, away from the die) and its power electronics are therefore first-class system components, sized by the pump flow and not by Q_c . Third, the pump laser’s wall-plug efficiency η_P enters the cost bracket as $1/\eta_P$ and is the single most leveraged number in the loop; a per-temperature tunable pump (Section 9.3.2) must not be bought at the price of η_P .

Table 10.7 evaluates (10.29) at the three efficiency tiers of Section 8.3.4, which is the form in which the device chapters hand their result to the architect. The transition between tiers is not gradual: at Tier I the cold plate is a performance feature bought with power; at Tier II it is power-neutral and the ROI model of Section 10.6 loses its denominator’s largest term; at Tier III on a 600 K zone the compute tile is a net electrical source and the thermal-zone template of Section 10.9 becomes an energy-harvesting floorplan.

Tier	η_{ASF}	P_{elec}/Q_c at $\eta_P = 0.75$, $\eta_{\text{cpl}} = 0.95$	$\eta_P \eta_{\text{cpl}}$ required for self-powering at 600 K (ideal LPC)	Architectural regime
Bulk (today)	0.05	12	0.98 (beyond present lasers)	hot-spot razor; LPC is the difference from a μTEC
I	0.10	5.5	0.95 (beyond present lasers)	$\gtrsim 10^3 \text{ W/mm}^2$ targeted, pulsed tiles; cooling as a scheduled pump-joule budget
II	0.30	1.4	0.87	near-neutral loop; sustained boost sized to tile, not package; 3D interlayer tiles
III	0.60	0.32 ($\rightarrow 0$ at $\eta_P \eta_{\text{LPC}} \eta_{\text{cpl}} = 0.63$)	0.77	self-powered compute zones; chip as partial supplier; PUE accounting inverted

Table 10.7: Loop economics by extractor tier from (10.29) and (1.17) ($\phi = 0.5$ at 600 K, $T_0 = 300$ K). The tiers are those of Table 8.2; each is reached by a different rung of the photonic ladder, and each changes the architectural role of the cold plate qualitatively.

Optical flux, not cooling density, sets the tile’s envelope

A hot spot of $1\text{--}10 \text{ W/mm}^2$ cooled by a dye or GaAs tile requires $10\text{--}300 \text{ W/mm}^2$ of pump flux to be delivered into a resonator whose absorbing layer is a few micrometres thick, and the same flux plus η_{ASF} to be extracted as fluorescence and routed to the LPC. These fluxes are two orders of magnitude above what a continuous-wave dielectric or polymer film tolerates over a product lifetime, and they exceed the transparency ceiling of a thin dye layer (Section 8.3.3) unless the layer is thickened. The architectural consequence is that the tile envelope is set by *optical* flux and thermal-bleaching limits, and that these are best respected by *not* running the tile continuously:

- *Spatial targeting* concentrates the available pump on the $\sim 1\text{--}5\%$ of die area that actually binds the thermal budget (Section 10.6.6); the network delivers tens of watts to a few square millimetres rather than kilowatts to a die.
- *Temporal targeting* exploits the mismatch between the die’s thermal time constants (tens of microseconds for a $10 \mu\text{m}$ hot spot, milliseconds for the package) and the extractor’s response (nanoseconds). A tile driven at duty cycle D delivers the same time-averaged Q_c at a peak flux $1/D$ higher during the burst and zero between bursts, which is the wrong direction for damage; the right use of temporal addressing is the converse — run the tile only while the hot spot exists (a tensor burst, a boost interval, a throttle-cliff approach) and let the passive path carry the rest. Because the leakage and throttle terms of (10.23) are nonlinear in ΔT , the performance return per pump joule is highest exactly in those intervals.
- *Per-tile temperature and wavelength* are coupled: the optimum pump wavelength shifts by $30\text{--}50 \text{ nm}$ across the $300\text{--}450 \text{ K}$ range for both dye and semiconductor extractors (Section 9.3.2), so the photonic network becomes a small WDM system — one channel per thermal zone — and the sensor layer’s temperature and fluorescence-flux readings close the loop on channel assignment as well as on pump power.

How chip architectures evolve to use this

Taken together, the device chapters predict a specific evolution of the processor around the cooling capability rather than a uniform lowering of T_j :

1. *Thermal zoning at floorplan time* (Section 10.9): compute islands designed to run hot (where a dye tile’s ceiling is highest and the Carnot factor widest), storage and control islands designed

- to run cold (where Yb:YLF’s sub-ambient regime collapses leakage), with thermal-isolation features and refractory metallization between them.
2. *Cooling as a scheduled resource*: the run-time system already schedules power (DVFS, boost budgets, power gating); a photonic cold plate adds a second budget — pump joules per interval — that is spent on the block whose $\Delta\text{Perf}/\Delta P$ is currently highest. The sensor layer supplies the state; the LPC’s recovered power is the budget’s replenishment. Thermal-aware scheduling, today a mitigation, becomes an allocation problem with a physical currency.
 3. *Boost regimes sized to the tile, not the package*: a block may be allowed to exceed the package-level thermal cap for as long as its tile can hold it, which turns the boost interval into a design parameter set by the tile’s pump budget and the thermal time constant of the hot spot rather than by a die-wide guardband.
 4. *3D stacks with interlayer tiles*, where a 5–20 μm extractor film and its resonator fit inside the bonding layer, the pump and fluorescence are carried by in-plane waveguides, and each layer’s temperature is an independent variable — the case in which conventional cooling has no answer and the tile’s modest absolute density is sufficient because the buried hot spots are small.
 5. *Pump-source and LPC integration on the board*: because the circulating optical power is $10\text{--}50\times$ the heat lifted, the laser diodes and LPCs are board-level components of tens to hundreds of watts, thermally coupled to the convective loop; the architecture’s “cooling subsystem” becomes an optical power bus with the chip as one load.

None of this is available to a convective or thermoelectric cold plate. Items 1–5 are Tier-I consequences: they depend on high-fraction pump recovery and on sub-millimetre, sub-millisecond addressing, not on net generation. Tier II adds sustained, package-independent boost as a steady state; Tier III on 600 K zones adds the inverted power budget, in which the hottest compute islands are also the chip’s power sources and the floorplan is drawn to maximize the exergy they hand to the LPC.

10.9 Thermal heterogeneity as a design knob

The near-term story of Section 10.7 keeps every part of the chip at the same operating temperature. That convention is inherited from convective cooling, not from device physics. Convective and micro-channel cooling can only remove average heat; they have no ability to hold one region of the die hot while another region sits cold, so the whole die must live within whatever reliability envelope the weakest structure tolerates. Photonic microrefrigeration breaks that assumption because it targets heat extraction with sub-100- μm spatial resolution (Section 2.7); the temperature of each tile is now an independent architectural variable. The resulting design space is fundamentally different from what any prior cooling technology permits, and the long-term architectural template that exploits it is what this section develops.

Two pieces of physics that pull in opposite temperature directions

The chip’s power decomposes into a dynamic piece (1.24) that scales as $\alpha N_{\text{tr}} C_{\text{eff}} V_{\text{dd}}^2 f$ and a static piece (1.26) that scales $\propto e^{-V_{\text{th}}/(nk_{\text{B}}T_{\text{j}}/q)}$. These two depend on temperature in opposite ways — and, more importantly, their *system-level value* depends on temperature in opposite ways.

Compute (dynamic-power dominated) wants to be hot. The exergy the LPC can recover from chip heat is bounded by $(1 - T_0/T_{\text{h}})$ per (1.15). With $T_0 = 300\text{ K}$: at $T_{\text{h}} = 350\text{ K}$ this Carnot factor is $\phi = 0.14$; at $T_{\text{h}} = 500\text{ K}$ it is 0.40; at $T_{\text{h}} = 600\text{ K}$ it is 0.50. A compute tile lifted from a nominal

350 K to 500 K nearly triples the recoverable exergy per dissipated joule and correspondingly relaxes the self-powered feasibility condition (1.17) for that tile. The relaxation must be read against the quantum-defect ceiling on η_{ASF} ($\lesssim 0.1$): even at $\phi = 0.5$ the condition requires $\eta_P \eta_{\text{LPC}} \eta_{\text{cpl}} \gtrsim 0.95$, so the hot tile does not make the loop a net generator — it makes the loop’s pump recycling *cheaper*, and it raises the extractor’s own ceiling (Section 8.3.3: both the red-tail absorption and the transparency limit of a dye grow exponentially with T). A hot compute tile is where the molecular-dispersion platform is strongest, and where the semiconductor platform needs its pump retuned (Section 9.3.2).

Storage / control (static-power dominated) wants to be cold. Subthreshold leakage (1.29) carries the exponential $\exp(-V_{\text{th}}/(nk_{\text{B}}T_j/q))$, which halves per roughly 10 K of T_j reduction. A cache / register-file region held at 200 K instead of 350 K suppresses subthreshold leakage by $\sim 2^{15} \approx 30,000\times$ at fixed V_{th} ; in practice the collected saving is smaller (gate-oxide and band-to-band tunnelling floors dominate below ~ 150 K), but a realistic 100–200 $\times$ reduction is on the table. For a chip whose static fraction is $\sim 30\%$ of total power at nominal temperature (Section 10.2), this is the difference between static being $\sim 30\%$ and $\sim 0.3\%$ — a full architectural redirection of budget from leakage into dynamic-power headroom.

The two effects are multiplicative, and both are only accessible with spatial selectivity

The perf/W win is the *product* of the two. Wider Carnot factor on the hot compute tiles *and* exponentially reduced static drag from the cold storage tiles both contribute independently to the loop’s net electrical balance (1.11), and neither is available to a uniform-temperature design. This is the sharpest sense in which photonic microrefrigeration is architecturally unique. Peltier μTECs cannot lift regions above the die average; convection cannot cool them below it; only the photonic cold plate’s per-tile addressing can put a 500 K compute island next to a 200 K SRAM bank on the same die. The prior chapter’s convection-versus-MR-versus-hybrid decision map (Table 2.2) is one axis of that architectural space; per-tile T_{h} is the second, and it is the axis that has not yet been exploited.

Design template: three thermal zones

The concrete instantiation is a three-zone die, with per-zone material and T_{h} targets tied to the loop-level economics:

Two second-order gains follow from this template. First, the $V_{\text{th}}/V_{\text{dd}}$ trap of (1.29) loses its grip in the cold zones: lower T tolerates a lower V_{th} without leakage catastrophe, which lets the timing-critical paths run at higher f at fixed V_{dd} . Second, silicon carrier mobility improves $\sim 2\text{--}3\times$ below 200 K, so the cold zones can be clocked meaningfully faster than their room-temperature siblings. Both effects widen the DVFS envelope beyond what a uniform-temperature chip can reach at any single operating point.

Three constraints that bound the practical reach

The pattern above is thermodynamically clean; the constraints that gate it are material, packaging, and reliability, and they should be flagged before this is oversold.

1. *Interconnect and dielectric wall at the hot end.* Standard Cu / low- κ interconnects have electromigration and TDDB lifetimes that fall exponentially with temperature; continuous

Zone	Target T_j	Dominant power term	Preferred devices	Extractor host
Compute (hot)	500–600 K	dynamic (CV^2f)	SiC / GaN CMOS, refractory metallization	Ho- or Tm-doped wide- T hosts; cascaded Tier-III extractor (Section 8.3.4)
Interconnect / un-core	300–400 K	mixed dynamic + wire loss	standard Cu-Si interposer	Yb:YLF thin film; SMILES-R640 film at the 400 K end; GaAs DH with retuned pump
Storage / control (cold)	150–250 K	static (I_{leak})	standard Si CMOS at low V_{dd} , high mobility	Yb:YLF (sub-ambient cooling regime)

Table 10.8: Three-zone thermal architecture enabled by photonic microrefrigeration. Each zone’s T_j is set by whichever physics dominates its perf/W contribution; the extractor material and device technology are chosen to support that zone’s operating window. Interconnects crossing between zones require thermally-decoupled routing (see caveats).

operation above ~ 400 K is not viable in today’s BEOL stack. The hot-compute zone as described requires *wide-bandgap devices* (SiC or GaN, viable to ~ 600 – 800 K), refractory metallization (W, Mo, Ta), and either high-temperature ceramic dielectrics or aggressive wear-levelling to hit commercial lifetime targets. This is an interconnect- and metallization-first story before it is a device story.

2. *Extractor material wall.* Yb:YLF’s ASF efficiency rolls off above ~ 400 K through concentration quenching and thermal population of the upper manifold’s phonon sidebands. Ho³⁺- and Tm³⁺-doped hosts hold up better into the 500–700 K range (Section 8.1.1), and SiC-hosted rare-earth complexes are an active research area for even hotter regimes. Each thermal zone therefore has to pick an extractor host from a different material family; the extractor design becomes zone-specific rather than uniform.
3. *Thermal-gradient / packaging wall.* A 500 K compute tile adjacent to a 200 K cache tile imposes a 300 K in-plane gradient over sub-mm distances. Interconnects crossing the boundary face thermal-stress fatigue and material mismatch; *thermal isolation trenches, refractory bridging layers, or deliberately-engineered spreading resistance* become first-class architectural elements. The photonic cold plate delivers spatial *heat extraction* selectivity; the packaging must deliver the matching spatial *heat conduction* selectivity, or the intended gradient collapses.

Consequences for the ROI model

The unified figure of merit (10.23) was derived assuming a single tile with a single T_h . Under the heterogeneous-temperature template it becomes zone-additive: each zone contributes its own ΔPerf_i (with per-zone ϵ_P and Carnot factor $\phi_i = 1 - T_0/T_h^{(i)}$) and its own cost $Q_{c,i}/\text{COP}_i - \Delta P_{\text{leak},i} - \Delta P_{\text{PDN},i}$. The system-level figure of merit is then

$$\eta_{\text{cool}}^{\text{het}} = \frac{\sum_i \Delta\text{Perf}_i(T_h^{(i)})}{\sum_i \Delta P_{\text{inj},i} + Q_{c,i}/\text{COP}_i - \Delta P_{\text{leak},i}(T_h^{(i)}) - \Delta P_{\text{PDN},i}(T_h^{(i)})}. \quad (10.30)$$

The multiplicative structure comes through the two-tier sum: the hot zones contribute large ΔPerf_i at wide ϕ_i , while the cold zones contribute large $-\Delta P_{\text{leak},i}$ that shrinks the denominator. Both terms are individually accessible with uniform-temperature photonic cooling, but only the heterogeneous-temperature architecture makes both simultaneously large.

What this means for the deployment story

The near-term deployment of Section 10.7 keeps a uniform T_h because it retrofits existing chips. The long-term deployment of Section 10.10 rearchitects around photonic cooling. The thermal-heterogeneity template of this section is the concrete instantiation of that rearchitecting; it is what a chip that is *designed for* photonic cooling looks like, as opposed to one that has photonic cooling bolted on. It is also the setting in which the loop-level self-powered regime of (1.13) moves from “possible at extreme boost” to “steady-state operating point of the compute zones” — which is the shift that makes photonic cooling a data-centre-level technology rather than a chip-level convenience.

10.10 Long-term implications: rearchitecting around photonic cooling

What changes when chips are designed for photonic cooling from the start

The long-term scenario is the opposite of the near-term one: the package’s cooling capacity is no longer the binding constraint, so the architect has freedom that conventional chips do not. Several choices that current architectures avoid become viable:

1. **3D stacking with tight thermal coupling.** Current 3D-stacked logic (e.g. AMD’s V-Cache) is constrained by the buried-layer heat-removal problem: heat generated in the lower layers has no efficient escape path through the upper layers. Photonic cooling can be embedded between layers in a 3D stack because it does not require fluid flow or large surfaces — $\sim 500\mu\text{m}$ of stack height is enough to host a coupler / extractor / back-reflector / sensor stack with optical fibres routed orthogonal to the layer plane. Long-term, ~ 8 -deep logic stacks become thermally viable.
2. **Higher-density wire layouts.** When the thermal cap is binding, designs sacrifice wire density (longer signal paths, more buffering) to lower C_{wire} and so P_{dynamic} . When the cap moves, the architect can run shorter / faster signal paths — improving *latency* at fixed throughput. Most valuable for memory-bandwidth-limited workloads (transformers, LLM inference) where wire delay through the interconnect fabric is the binding latency.
3. **Larger on-die memories.** SRAM caches are high- C_{eff} -per-bit but improve hit rates and indirectly raise effective FLOPS/W. Photonic cooling raises the C_{eff} ceiling, allowing larger LLCs and larger register files.
4. **Rebalanced $V_{\text{dd}}/V_{\text{th}}$ selection.** With lower T_j , the leakage trap of (1.29) is much less severe, so the architect can use lower- V_{th} cells in critical paths to push f further into the linear regime, leaving high- V_{th} cells in non-critical or rarely-used paths. Multi- V_{th} designs are common today; their effective design space widens.
5. **Self-powered / net-generation operating regime (Tier III).** When $T_h \gg T_0$ (600 K wide-bandgap compute zones) and the cascaded extractor reaches $\eta_{\text{ASF}} \geq 0.6$ (Section 8.3.4), the loop crosses the threshold of (1.17) at $\eta_P \eta_{\text{cpl}} \simeq 0.77$ and exports $P_{\text{gen}} > 0$. Long-term, this changes the data-centre power-budget calculation: the chip is not a pure power consumer but a partial supplier — a regime that PUE-driven data-centre economics has not yet learned to value. At Tier I–II the same architecture delivers the intermediate benefit: a cooling subsystem whose electrical draw is a small multiple of, or comparable to, the heat it lifts rather than $1/(\eta_P \eta_{\text{ASF}}) \sim 30\times$.

What does *not* change in the long term

A surprisingly large list:

- **Landauer floor.** The absolute lower bound on per-bit energy is $k_B T_{\text{chip}} \ln 2 \simeq 3.4 \text{ zJ}$ at 350 K; at $\sim 1 \text{ pJ/op}$ modern chips sit $\sim 10^8$ above this. Photonic cooling lowers T_{chip} modestly ($\sim 40 \text{ K}$) but does not change the order-of-magnitude gap. A future “below-Landauer” breakthrough requires reversible logic [9, 11], not better cooling.
- **Workload-entropy floor.** Per Gershenfeld [7], the irreducible information-erasure floor of a workload is set by the workload itself, not by the implementation. Cooling moves the implementation closer to the floor; it does not move the floor.
- **Speed-of-light limits on signal propagation.** Whether or not the chip can be cooled aggressively, signal wires propagate at finite speed ($\sim 1.5 \times 10^8 \text{ m/s}$ in modern interconnects). Long-distance on-die communication remains a latency bottleneck independent of thermal considerations.
- **Fabrication economics.** Photonic cooling does not change the cost per mm^2 of fabricated silicon; it changes how much that silicon can deliver. The chip’s BoM cost increases by the photonic-cold-plate / laser / LPC infrastructure; the value-per-cost trade depends on the workload’s thermal sensitivity.
- **ISA / microarchitecture-level decisions** (pipeline depth, register count, branch-predictor design). These are dominated by latency / throughput trade-offs that photonic cooling barely touches; what changes is the operating-point envelope, not the optimal point.

The honest performance-scaling limit

A useful summary metric is the chip-level performance-per-watt (FLOPS/W). Combining the architectural budget (1.32) with the dissipation-per-FLOP ϵ_{op} (technology-set), the chip’s compute throughput is bounded by

$$\text{FLOPS}_{\text{max}} \leq \frac{P_{\text{chip}}^{\text{max,hyb}}(s)}{\epsilon_{\text{op}}}. \quad (10.31)$$

With pure convection, $\epsilon_{\text{op}} \sim 1 \text{ pJ/op}$ for high-end AI accelerators and $P_{\text{chip}}^{\text{max,conv}} \sim 700 \text{ W}$, giving $\sim 700 \text{ TFLOPS}$ ceiling. With microrefrigeration delivering $s = 0.5$ the ceiling doubles to $\sim 1.4 \text{ PFLOPS}$; with $s = 0.9$, $\sim 7 \text{ PFLOPS}$. Current chips sit $\sim 10^8$ above the Landauer-style information-theoretic floor, so the cooling-cap improvement is the dominant performance lever for the next several generations.

Reality check. Equation (10.31) is optimistic in two ways. First, it assumes the photonic loop’s optical infrastructure (laser, fibre, LPC) does not consume so much electrical power that the gain is offset — this requires (1.17) to hold, which is non-trivial at modest $T_h - T_0$. Second, it assumes the workload can actually use the additional FLOPS, which depends on memory-bandwidth and inter-chip-communication scaling that photonic cooling does not address. The honest near-term claim is therefore $\sim 1.5\text{--}3\times$ compute-per-watt improvement; the long-term claim is the order-of-magnitude budget unlocked by the chip-redesign options enumerated above.

10.11 Hybrid vs. pure laser cooling: a recommendation

The three regimes of Section 10.4 suggest a clear deployment trajectory:

- **Year 0–3 (near-term): hybrid deployment.** Existing chips, existing fabs, photonic-cooling tile arrays bonded onto the package. The convective system carries the bulk; photonic cooling targets boost-region hot spots and recovers leakage. Sweet spot for retrofit and for new chips without architectural redesign.
- **Year 3–7 (mid-term): architecturally-aware hybrid.** New chips designed with photonic-cooling-aware floorplans: higher-density logic in regions covered by photonic tiles, legacy convective cooling for caches / I/O. The thermal envelope is co-designed with the photonic network.
- **Year 7+ (long-term): pure photonic / 3D-stacked.** Multi-layer logic stacks with embedded photonic cooling between layers. Convection handles only laser / LPC waste heat; the chip-electrical pathway is dominated by the photonic loop. Net-generation regime accessible at boost operation, with the chip partially funding its own thermal infrastructure.

The pure laser-cooled regime ($s \rightarrow 1$) is not an immediate goal — the per-tile cooling density ($\sim 1\text{--}10\text{ W/mm}^2$ with $10\text{--}100\times$ that in pump flux) and the loop-feasibility binding both demand a hybrid for at least the next chip generation. Hybrid is the correct operating-point baseline; pure laser is the long-horizon endpoint.

10.12 Chapter summary

Near-term wins (existing chips, existing fabs):

- Sustained boost-clock recovery: the photonic loop clips the worst-case node and lets boost frequencies sustain under realistic workloads.
- Leakage suppression: $16\times$ at $\Delta T = 40\text{ K}$ per (10.1), recovering $\sim 18\text{--}38\%$ of thermal budget.
- Dark-silicon recovery: spatial / temporal targeting unlocks $\sim 20\text{--}40\%$ of nominal die area.
- Net effect: $\sim 1.5\text{--}2\times$ FLOPS/W at the same process node, no chip redesign.

Long-term wins (chips designed around photonic cooling):

- Thermally viable 3D logic stacks (currently forbidden).
- Higher-density wire layouts and larger on-die memories.
- Rebalanced $V_{\text{dd}}/V_{\text{th}}$ cell selection.
- Net-generation operating regime on 600 K compute zones at the Tier-III ($\eta_{\text{ASF}} \geq 0.6$) design point; break-even at Tier II; high-recovery hot-spot razor at Tier I.

The ROI model (Section 10.6) translates all of the above into a measurable $\eta_{\text{cool}}/\eta_{\text{conv}}$:

- Performance gain decomposes into boost, timing-guardband, throttle-avoidance, TDP-uplift and latency terms (10.21); cost is $\Delta P_{\text{inj}} + Q_{\text{c}}/\text{COP} - \Delta P_{\text{leak}} - \Delta P_{\text{PDN}}$ (10.22).
- Architecture-class predictions (Table 10.6) put the FOM at $1.5\text{--}4\times$ for mobile, $1.1\text{--}2.5\times$ for desktop CPU, $1.2\text{--}3\times$ for compute-bound GPU, $1.5\text{--}5\times$ for HBM / 3D stacks — with the high end of each range applying when the workload is hotspot-bound.
- Even $\text{COP} = 0.3$ is ROI-positive when applied as a hotspot razor: the GPU tensor example (10.25) gives $\text{FOM} \approx 3.0$ with $Q_{\text{c}} = 2\text{ W}$, $\Delta T = 8\text{ K}$, unlocking 5% of 700 W chip performance.
- Time-forward, (10.26) guarantees $\eta_{\text{cool}}/\eta_{\text{conv}}$ rises on the architectures industry is converging toward (AI accelerators, HBM, chiplets).

What photonic cooling does *not* do. Move the Landauer floor; move the workload-entropy floor; reduce wire delay; reduce silicon cost; redesign the ISA. The cooling cap is moved; the fundamental limits of computation are not.

Bottom line. Hybrid ($0 < s < 1$) is the near-term operating sweet spot, delivering $\sim 3\text{--}5\times$ chip power at fixed package; pure laser ($s \rightarrow 1$) is the long-horizon endpoint, bounded by photonic-network throughput and feasibility binding; net generation ($s > 1$ effective) makes the loop pay for itself when $T_h \gg T_0$ and the extractor has climbed to the Tier-III rung of Table 8.2 ($\eta_{\text{ASF}} \geq 0.6$, $\eta_P \eta_{\text{cpl}} \geq 0.77$). The cooling-cap improvement dominates the per-operation-energy floor for the next several chip generations and is the most practical architectural lever currently available.

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